EE 2004 DSP Lecture 1

Note Title 11-08-2017

Dictionary definition of a signal : A gesture, action, or sound that is used to convey information or instruction
That is, a signal is used for conveying information

For us, a signal means a function, commonly denoted as $x(t) \in \mathbb{C}$

The independent variable 't' is called "time" but can be any one-dimensional variable, eg, space. $x(t)$ is called as a "continuous-time signal" even if the independent variable is not time.

In this course, in the time-domain we will mainly be dealing with discrete-time signals or sequences, denoted by $x[n] \in \mathbb{C}$

Both $x(t)$ and $x[n]$ can take on, in general, complex values, with the real-valued signal being a special case

Most real-world signals are real-valued, eg., speech, image, pressure, height, etc.

$x(t)$ and $x[n]$ take on a single value for a given t or n, ie., they are scalar valued

A signal can also be vector-valued. For example, a multi-lead ECG measures signals simultaneously via electrodes placed at various locations in the body. The recorded signal is a vector:

$$\mathbf{x}(t) = (x_1(t)\ x_2(t)\ \ldots\ x_N(t))$$

$x_i(t)$ represents the signal captured by the i-th lead. The independent variable is still only one-dimensional, ie., time

The independent variable can also be a vector

An image is a simple example where the independent variable is 2-dimensional
In this case, the value of the function---the pixel value---is a scalar. That is,
$f(x,y)$ is a scalar

Another classfication of signals is along the lines of whether it is deterministic or random

The mains supply voltage is sinusoidal and hence deterministic.
Whereas, speech is a non-deterministic waveform

Signals can also be classified as being periodic or non-periodic

In this course we will be dealing with functions whose arguments and values are salars, ie., $\in \mathbb{C}$

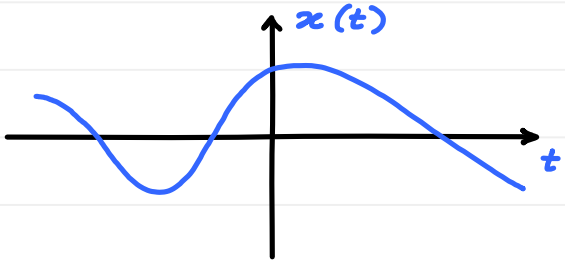
We will also not consider the effects any quantization, ie., we will assume that the values have infinite precision

Review of CT and DT Signals

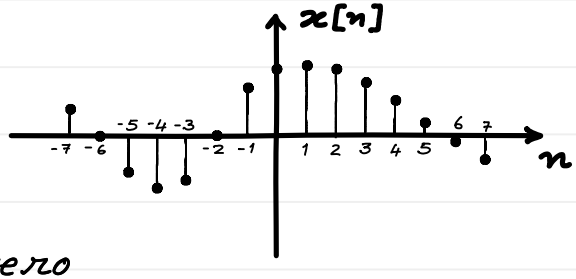
CT Signal $x(t) \in \mathbb{C}$ where $t \in \mathbb{R}$ DT Signal $x[n] \in \mathbb{C}$ where $n \in \mathbb{Z}$

The fact that "n" can take on only integer values introduces differences in behaviour from the CT counterpart

A CT real-valued signal is shown in the figure. The value of $x(1.5) = 2$ (say)



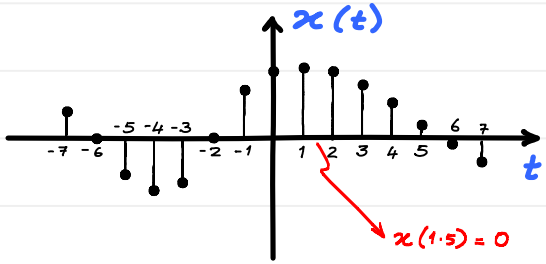
A DT real-valued signal is shown in the figure. The value of $x[1.5]$ = undefined



It is wrong to think that the value of $x[1.5]$ is zero

$x[a] = \text{undefinded} \quad \text{for} \quad a \notin \mathbb{Z}$

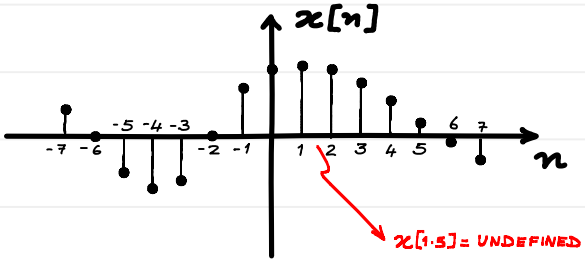
Consider the CT signal shown on the right.
It appears identical to the DT sequence shown earlier



The similarity is deceptive

The important difference is that $x(t) = 0$ whenever $t \notin \mathbb{Z}$

Whereas, $x[a] = \text{undefinded} \quad \text{for} \quad a \notin \mathbb{Z}$



Thus, $x(1.5) =$ 0, whereas $x[1.5]$ is undefined

A sequence can be thought of as an array, which stores values at various array locations. These locations are inherently integer-valued.

Just as an array location of 1.5 is meaningless, which in turn means we cannot enquire about the value at location 1.5, it is meaningless to talk of the value of $x[1.5]$

Energy and Power Signals

If $\|x\|_2^2 = \int_{-\infty}^{\infty} |x(t)|^2 \, dt < \infty$, then $x(t)$ is an ENERGY signal.

$x(t) = e^{-t}$ $(t \geq 0)$ is an energy signal

$x(t) = e^{t}$ $(t \geq 0)$ is NOT an energy signal

Discrete-time counterpart:

$$\|x\|_2^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty \Rightarrow x[n] \text{ is an ENERGY signal.}$$

$x[n] = a^{n}$ $(n \geq 0)$ is an energy signal if $|a| < 1$

$x[n] = a^{n}$ $(n \geq 0)$ is NOT an energy signal if $|a| \geq 1$

If $\lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt < \infty$, then $x(t)$ is called as a POWER signal

$x(t) = A\cos(\Omega_0 t + \theta)$ is a power signal

Periodic signals (both CT and DT) are typically power signals.

Give an example of a periodic signal that is not a power signal.

Operations on the Independent Variable

We will demonstrate some of the common operations by using both continuous- and discrete-time signals, to see the similarities and differences.

We'll mainly consider Scaling and Shifting

Nonlinear transformation is also possible but we will not consider such transformations

Scaling + Shifting = Affine Transform

$x(at+b)$ involves both scaling and shifting

Can be done in two ways:

(a) Shift first and then scale :

(b) Scale first and then shift

Given $y(t) = x(at+b)$ can we get back $x(t)$?

Consider $y\left(\frac{t-b}{a}\right) = x\left(a \cdot \frac{t-b}{a} + b\right)$

$= x(t)$

That is, we can recover $x(t)$ from $y(t)$

As an example, suppose $y(t) = x(2t)$, then $y(t/2) = x(t)$

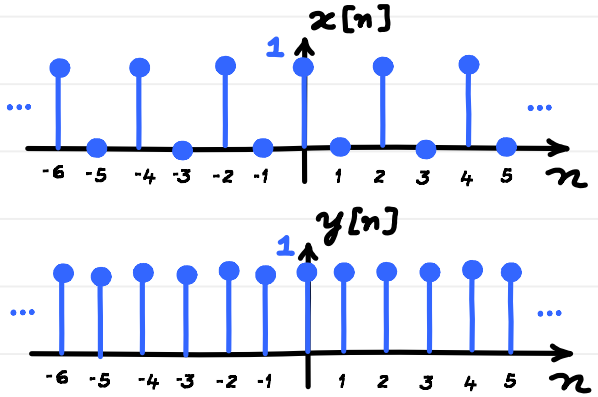
That is, compression by a factor of 2 can be undone by subsequent expansion by a factor of 2

The affine transform counterpart for the DT sequence is $y[n] = x[Mn+N]$

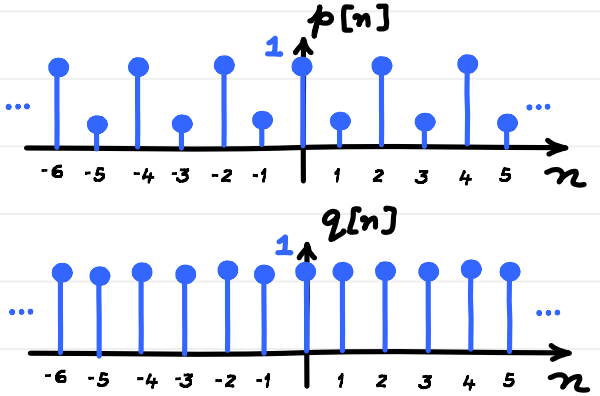
As an example, consider $y[n] = x[2n]$ [Appears similar to $y(t) = x(2t)$]

That is $y[0] = x[0]$, $y[1] = x[2]$, $y[-1] = x[-2]$, $y[2] = x[4]$, and so on

This operation picks out the even-indexed samples



Because the odd-indexed samples are discarded, two sequences that differ only at the odd-indexed samples, will give the same result.



This illustrates an important difference in behaviour between CT and DT signals

In the CT case, the affine transform can be undone, ie, we can get back $x(t)$ given $x(2t)$

In the DT case, the affine transform cannot be undone, in general, ie, we cannot get back, in general, $x[n]$ given $x[2n]$

Consider $y[n] = x[n/2]$. Clearly, $y[0] = x[0]$, $y[2] = x[1]$, $y[-2] = x[-1]$, etc.

What about $y[1] = x[1/2] = ?$

Is it zero? From our earlier discussion, we know it is not zero but undefined. Hence

$$y[1] = x[1/2] = \text{undefined}$$

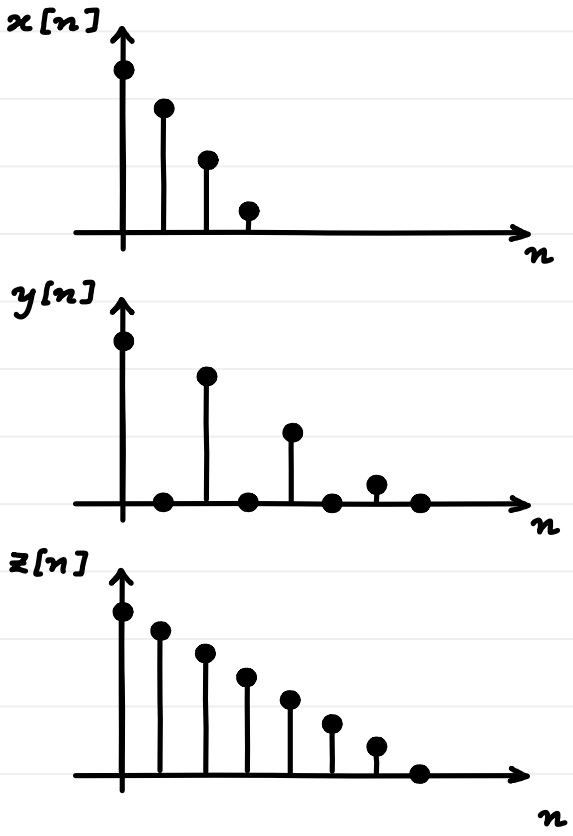
In the above example, it is common to set the odd-indexed samples to be zero.

If this is not given or understood, the odd-indexed samples are, strictly speaking, undefined.

Unlike in the continuous-time case, replacing "n" by "n/2" is not "pure expansion", as illustrtated in the figures on the right.

$$y[n] = \begin{cases} x[n/2] & n \text{ even} \\ 0 & \text{otherwise} \end{cases}$$

More processing needs to be done if we want to get something like this.



EE 2004 DSP Lecture 2

Note Title 17-01-2019

Even & Odd Signals

Let $x[n] \in \mathbb{R}$

$x[n]$ is an **even symmetric** signal if $x[n] = x[-n]$.

$x[n]$ is an **odd symmetric** signal if $x[n] = -x[-n]$.

A signal that is neither even nor odd can be split into its even and odd parts:

$$x[n] = x_e[n] + x_o[n]$$

where

$$x_e[n] = \frac{x[n] + x[-n]}{2} \quad \text{and} \quad x_o[n] = \frac{x[n] - x[-n]}{2}$$

Suppose $x[n] = 1$ for $0 \le n \le 3$. It is easy to see that $x_e[\cdot] = \{1/2, 1/2, 1/2, \underset{\downarrow}{1}, 1/2, 1/2, 1/2\}$

$x_o[\cdot] = \{-1/2, -1/2, -1/2, \underset{\uparrow}{0}, 1/2, 1/2, 1/2\}$

In the CT case, consider 1) $x_1(t) = 1$ for $0 \le t \le 3$ and 0 otherwise

2) $x_2(t) = 1$ for $0 < t \le 3$ and 0 otherwise

Find $Ev\{x_1(t)\}$ and $Ev\{x_2(t)\}$. How do these two cases differ?

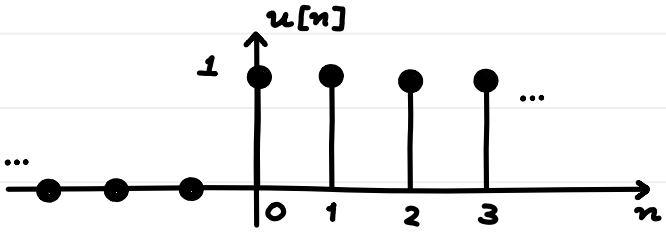
Let $x[n] \in \mathbb{C}$. The sequence (or signal) is said to be conjugate symmetric if $x[n] = x^*[-n]$ and conjugate antisymmetric if $x[n] = -x^*[-n]$. It can easily be seen that $e^{j\omega_0 n}$ is a conjugate even signal. However, $Re\{e^{j\omega_0 n}\}$ is even and $Im\{e^{j\omega_0 n}\}$ is odd.

Elementary Signals

Let us look at some well-known DT signals and compare them with their CT counterparts to see the similarities and differences

The Unit Step Sequence

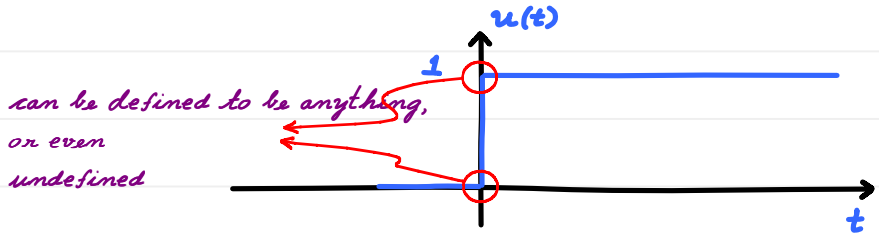
$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



$n < 0$ is the same as $n \leq -1$. There is no ambiguity in the definition at $n = 0$

The CT Unit Step Function:

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$



The function at $t = 0$ is undefined

Other common definitions:

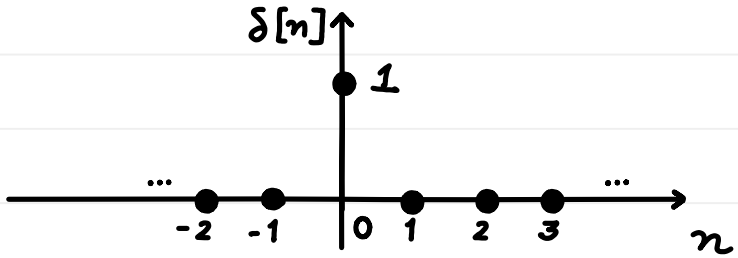
$$u_1(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \qquad u_2(t) = \begin{cases} 1 & t > 0 \\ 0 & t \leq 0 \end{cases} \qquad u_3(t) = \begin{cases} 1 & t > 0 \\ \frac{1}{2} & t = 0 \\ 0 & t < 0 \end{cases} \qquad u_4(t) = \begin{cases} 1 & t > 0 \\ 13 & t = 0 \\ 0 & t < 0 \end{cases}$$

They all differ at only one point, viz, at $t = 0$. Do these differences matter?

Discrete-Time Unit Impulse

It is a simple, well-defined, and well-behaved function!

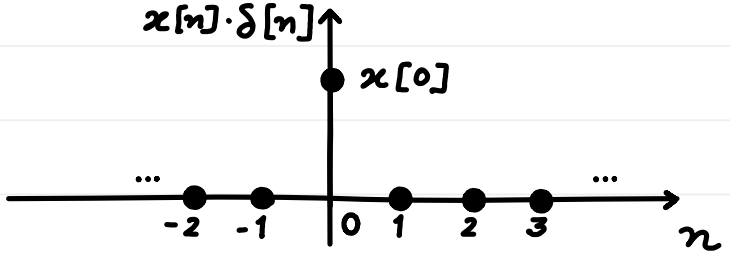
$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



$$x[n] \cdot \delta[n] = \begin{cases} x[0] & n = 0 \\ 0 & n \neq 0 \end{cases}$$

$$= x[0]\delta[n]$$

Sifting property



Common mistake:

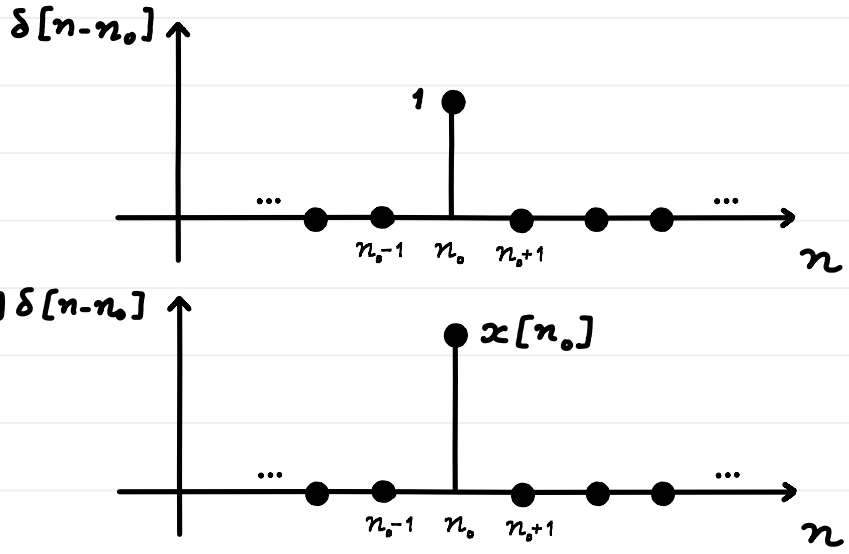
$$y[n] = x[n] \cdot \delta[n] = x[0]$$

$y[n] = x[0]$ means $y[n] = x[0] \ \forall n$ ie, a constant sequence

Shifted Impulse:

n_0 has been assumed positive in the figure

$$x[n] \cdot \delta[n - n_0] = x[n_0]\delta[n - n_0]$$
$$\neq x[n_0]$$

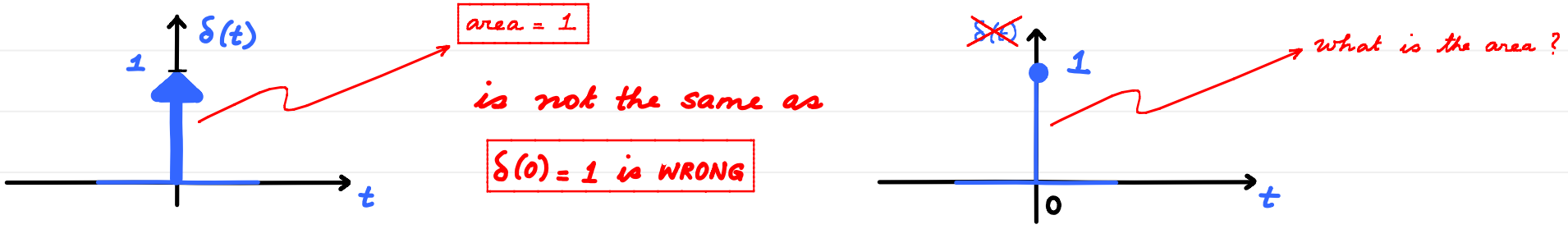


The CT Impulse Function

It is commonly called as the Dirac delta function. In this course the Dirac delta function will appear in the transform domain. Hence we need to understand this signal properly for our usage.

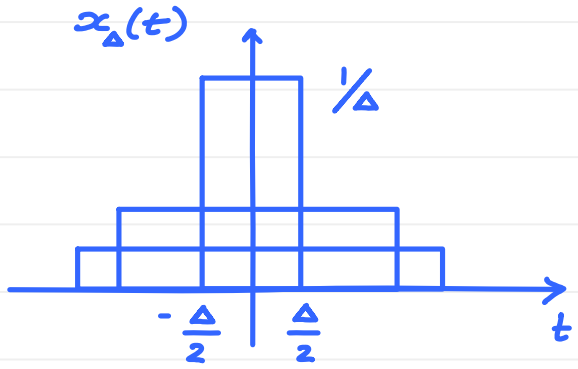
It is commonly defined as: $\delta(t) = 0 \text{ for } t \neq 0$ and $\int_{-\infty}^{\infty} \delta(t)\, dt = 1$

In plots it is shown with an arrowhead and height proportional to area



One viewpoint about $\delta(t)$ is that it is a shorthand notation for limiting arguments

In engineering books it is defined as the limit of sequence of functions such as:

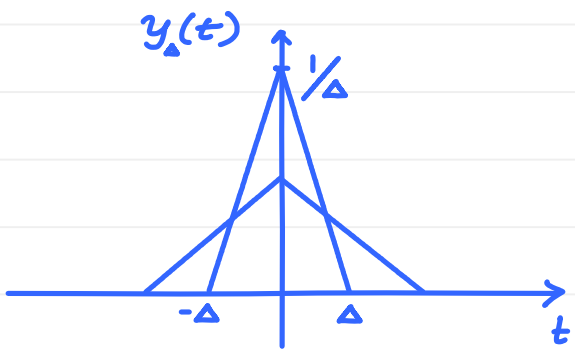


$$x_\Delta(t) = \begin{cases} \frac{1}{\Delta} & -\frac{\Delta}{2} < t < \frac{\Delta}{2} \\ 0 & \text{otherwise} \end{cases}$$

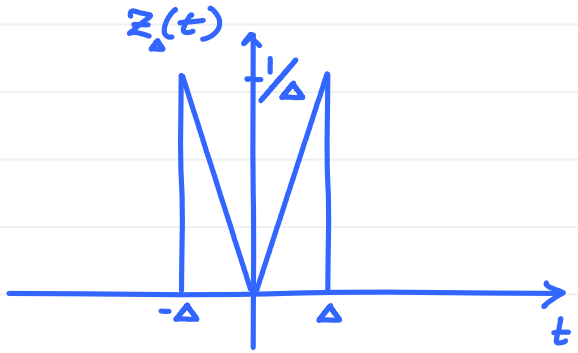
Clearly, $\lim_{\Delta \to 0} x_\Delta(t) = \delta(t)$, i.e., the limit

vanishes for $t \neq 0$ while preserving the area

There are several different sequences of functions whose limiting form satisfies the property of the Dirac delta.



$$\lim_{\Delta \to 0} y_\Delta(t) = \delta(t)$$



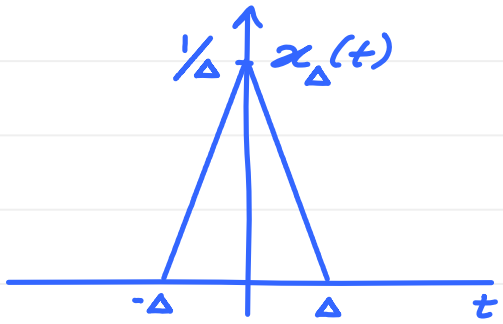
$$\lim_{\Delta \to 0} z_\Delta(t) = \delta(t)$$

Is $\delta(0) = \infty$? Clearly, $z_\Delta(0) = 0$. Hence $\lim_{\Delta \to 0} z_\Delta(0) = 0$!

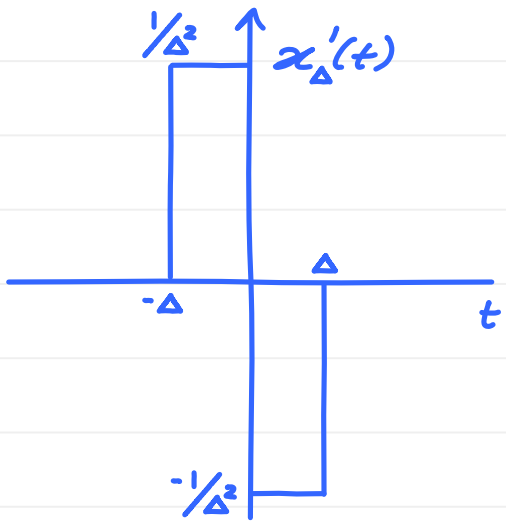
EE 2004 DSP Lecture 3

Note Title 21-01-2019

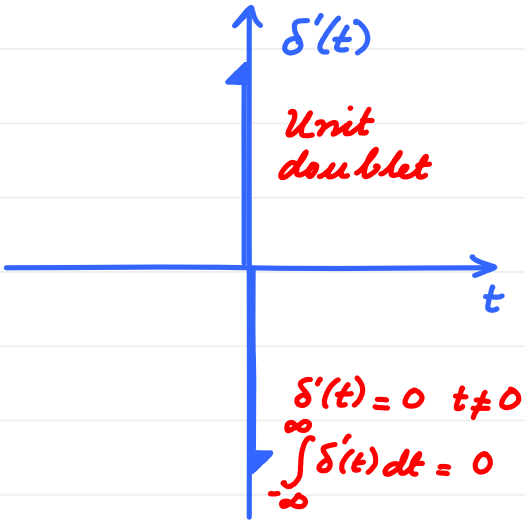
Do the two equations $\delta(t) = 0 \text{ for } t \neq 0$ and $\int_{-\infty}^{\infty} \delta(t)\, dt = 1$ uniquely define $\delta(t)$?



sequence that converges to $\delta(t)$ as $\Delta \to 0$



as $\Delta \to 0$ converges to



$\delta'(t) = 0 \quad t \neq 0$

$\int_{-\infty}^{\infty} \delta'(t)\, dt = 0$

Hence, $\delta(t) + \delta'(t) = 0$ for $t \neq 0$ and $\int_{-\infty}^{\infty} [\delta(t) + \delta'(t)]\, dt = 1$, i.e., $\delta(t) + \delta'(t)$ satisfies the two equations!

The proper definition of $\delta(t)$ is where $f(t)$ continuous at $t = 0$

$$\int_{-\infty}^{\infty} f(t)\,\delta(t)\,dt = f(0)$$

$\delta(t)$ is a mathematical model for a very brief pulse with a certain area, but whose shape does not matter.

All practical measuring devices have a finite resolving power. Making a brief pulse briefer will not make a difference in the measured output because of the finite resolution.
Such pulses can be modeled as $\delta(t)$ Thus, if a pulse is brief, independent of its shape, as long as the area is the same, the output will be a very close approximation to the ideal impulse response.

Never forget that the Dirac delta "function" is not an ordinary function.

It will be more appropriate to call it as a "generalized function" or as a "distribution".

Calling it as a function will tend to make you treat it as an ordinary function, and lead to mistakes in interpretation and wrong results.

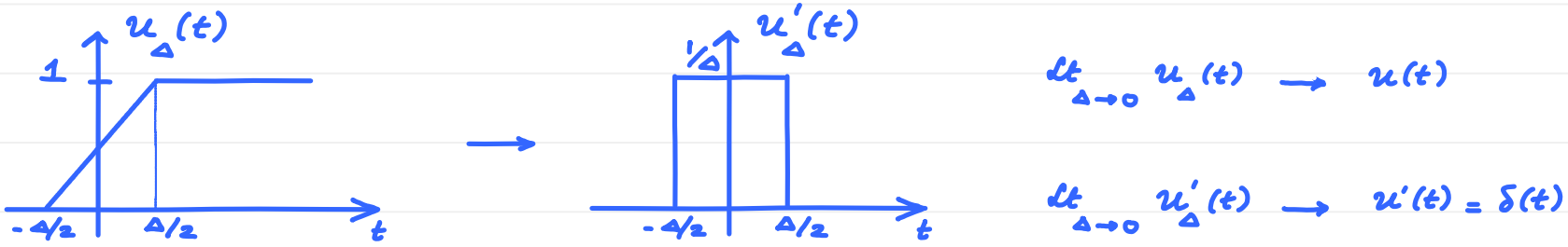
Inside an integral, when it appears as a product with a function that is continuous at the location of the impulse, the sifting property can be used without any fear of getting wrong results. In other contexts, one has to be very careful in dealing with it!

Observe that $\int_{-\infty}^{\infty} \delta(t)\,dt = \int_{0^-}^{0^+} \delta(t)\,dt = 1.$

Hence, $$\int_{-\infty}^{t} \delta(\tau)\,d\tau = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} = u(t)$$

Thus, $$\frac{d}{dt}u(t) = \delta(t)$$

A sequence of functions that captures the above is:



$\lim_{\Delta \to 0} u_\Delta(t) \to u(t)$

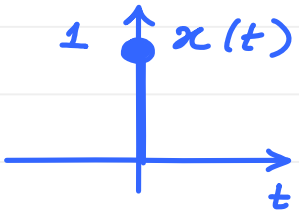
$\lim_{\Delta \to 0} u'_\Delta(t) \to u'(t) = \delta(t)$

Note that $\sum_{m=-\infty}^{n} \delta[m] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases} = u[n]$ and $u[n] - u[n-1] = \delta[n]$

are the counterparts of

$$\int_{-\infty}^{t} \delta(\tau)\, d\tau = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} = u(t) \quad \text{and} \quad \frac{d}{dt}\, u(t) = \delta(t)$$

Exercise:



$R = 1\,\Omega$

$C = 1\,F$

$$x(t) = \begin{cases} 1 & t = 0 \\ 0 & t \neq 0 \end{cases}$$

What is $y(t)$ when $x(t)$ is as shown above?

Recall that the impulse response $h(t)$ is $\frac{1}{RC} e^{-t/RC} u(t)$

The answer to this question will help you see if there is any signifance to the differences between the various versions of $u(t)$.

The *Complex Exponential* is one of the most important basic signals. The CT and DT versions share similarities but there are fundamental differences that have to be understood clearly

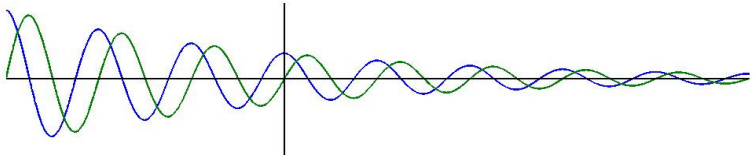
$x(t) = A\,e^{s_0 t}$ *where* $A,\ s_0 \in \mathbb{C}$

$A = |A|\,e^{j\phi}$ and $s_0 = \sigma_0 + j\Omega_0$ *cartesian form*

Hence, $x(t) = |A|e^{\sigma_0 t}\,e^{j(\Omega_0 t+\phi)}$

$\mathrm{Re}\{x(t)\} = |A|e^{\sigma_0 t}\,\cos(\Omega_0 t + \phi)$

$\mathrm{Im}\{x(t)\} = |A|e^{\sigma_0 t}\,\sin(\Omega_0 t + \phi)$



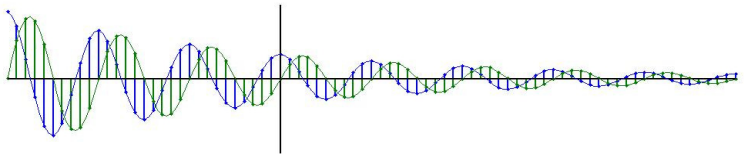
$x[n] = A\,z_0^n$ *where* $A,\ z_0 \in \mathbb{C}$

$A = |A|\,e^{j\phi}$ and $z_0 = r_0 e^{j\omega_0}$ *polar form*

Hence, $x[n] = |A|r_0^n\,e^{j(\omega_0 n+\phi)}$

$\mathrm{Re}\{x[n]\} = |A|r_0^n\,\cos(\omega_0 n + \phi)$

$\mathrm{Im}\{x[n]\} = |A|r_0^n\,\sin(\omega_0 n + \phi)$



Consider $x(t) = e^{j\Omega_0 t} \implies x(t+T) = x(t)$ for $T = \dfrac{2\pi}{|\Omega_0|}$ ie, is always periodic.

Also, as $|\Omega_0|$ increases, T decreases.
That is, the sinusoid oscillates more and more rapidly as $|\Omega_0|$ increases.
There is no upper bound on $|\Omega_0|$

Do these hold for the DT sinusoids as well?

Specifically, we ask:
(1) Does the DT sinusoid oscillate more and more rapidly as frequency increases?
(2) Is the DT sinusoid always periodic?
Examining $e^{j\omega_0 n}$ will provide us the answers

If we increase ω_0 by 2π, we get the same signal because

$$x[n] = e^{j\omega_0 n} \implies e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}\, e^{j2\pi n} = e^{j\omega_0 n}$$

i.e., ω_0 and $\omega_0 + 2\pi$ are indistinguisable!

Note that $\omega_0+2\pi$ is a *higher frequency* than ω_0

Thus, we can restrict the range of ω_0 to the interval $[0, 2\pi]$ or, equally well, $[-\pi, \pi]$

Observe that for the CT exponential, $e^{j\Omega_0 t}$ and $e^{j(\Omega_0+2\pi)t}$ are *not the same*

The latter is a more rapidly oscillating complex sinusoid

Because we can restrict the range of ω_0 to the interval $[0, 2\pi]$ or, equally well, $[-\pi, \pi]$,

$$\omega_0 \equiv \langle \omega_0 \rangle_{2\pi} := \omega_0 \bmod 2\pi$$

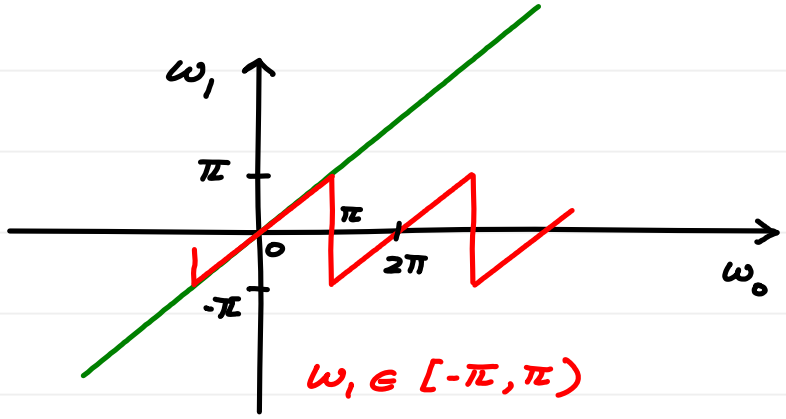
Example: $\omega_0 = 0$ "DC sequence" i.e., $x[n] = 1 \quad \forall n$

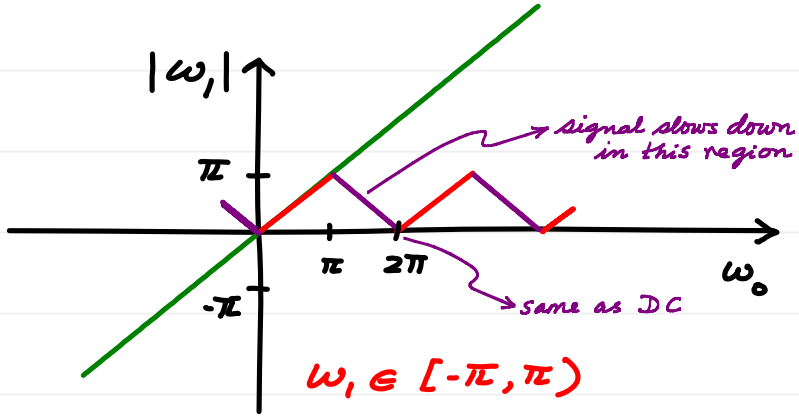
$\omega_0 = 2\pi k, \; k \in \mathbb{Z}$ is also $x[n] = 1 \; \forall n$

EE 2004 DSP Lecture 4

Note Title 22-01-2019

Recall that for $x[n] = e^{j\omega_0 n}$, it is enough if we let $\omega_0 \in [0, 2\pi)$, or equivalently, $\omega_0 \in [-\pi, \pi)$. We also saw that $\omega_0 = 0$ and $\omega_0 = 2\pi$ both yield the 'DC sequence'. Hence, it appears that, as ω_0 increases from 0 to 2π, the signal's rapidity of oscillations must increase, but eventually slow down as it reaches 2π. The following plot illustrates this:





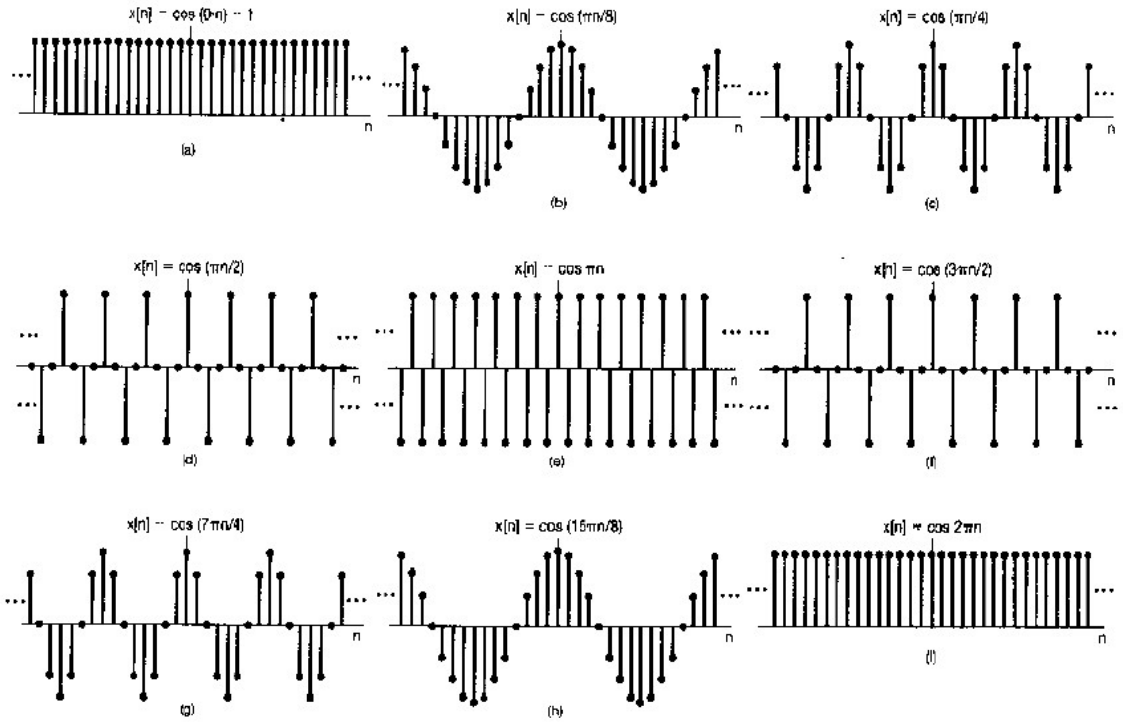


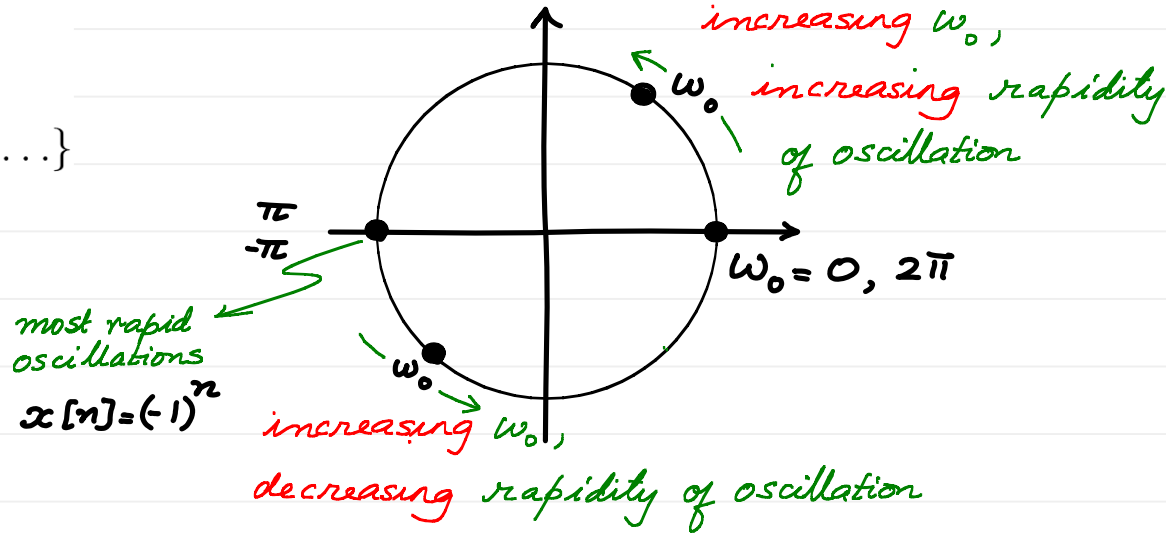
Figure 1.27 Discrete-time sinusoidal sequences for several different frequencies.

As the frequency increases from π to 2π, the signal slows down, reaching DC frequency at 2π as seen from the plot shown on the left.

From, "Signals & Systems" by Oppenheim and Willsky, Prentice-Hall, 1997, 2nd edition.

The plots illustrate the behaviour wherein the rapidity of the oscillations increases as ω_0 increases, reaches a maximum when $\omega_0 = \pi$ and then decreases as approaches 2π When $\omega_0 = 2\pi$, it becomes identical to $\omega_0 = 0$.

$$\text{For } \omega_0 = \pi, \quad x[n] = e^{j\pi n}$$
$$= e^{-j\pi n} \qquad n=0 \downarrow$$
$$= \{\ldots, -1, 1, -1, \underset{\downarrow}{1} - 1, 1, \ldots\}$$



Rapidity of oscillations increases as ω_0 goes from 0 to π

Rapidity of oscillations decreases as ω_0 goes from π to 2π

Highest frequency of oscillations occurs when $\omega_0 = \pm\pi$

$n = 0$ ↓

Observe that the sign alternates $x[n] = \{\ldots, -1, 1, -1, 1 - 1, 1, \ldots\}$

The other question was, Is $e^{j\omega_0 n}$ always periodic ?

Recall that $e^{j\Omega_0 t}$ was always periodic with period $\frac{2\pi}{|\Omega_0|}$

Let $x[n] = e^{j\omega_0 n}$ be such that $x[n] = x[n+N]$
Periodicity demands $e^{j\omega_0 n} = e^{j\omega_0 (n+N)}$
That is, we require,

$$e^{j\omega_0 N} = 1 \implies \omega_0 N = 2\pi k$$

$$\frac{\omega_0}{2\pi} \equiv f_0 = \frac{k}{N}$$

Thus, if and only if f_0 is a rational number will $e^{j\omega_0 n}$ be periodic.
Otherwise it will not be.

Hence, e^{jn} is not periodic because $\omega_0 = 1 \implies f_0 = \frac{1}{2\pi}$
which is not a rational number

Unlike in the CT case, there are complex DT sinusoids that are not periodic.

For periodic complex exponentials with period N, what is the range of k we need to consider in $f_0 = \frac{k}{N}$?

Recall ω_0 is actually $\langle \omega_0 \rangle_{2\pi}$

$$\omega_0 = \langle \omega_0 \rangle_{2\pi} \implies f_0 = \langle f_0 \rangle_1$$

This means that, since $f_0 = \frac{\omega_0}{2\pi}$

$$\omega_0 \in [0, 2\pi) \implies f_0 \in [0, 1)$$

Hence,

$$f_0 = \frac{k}{N} \in [0, 1) \implies k = \{0, 1, \ldots, N-1\}$$

$k = 0$ is the same as $k = N$, ie, $k \equiv \langle k \rangle_N$

In particular, the set of periodic complex exponentials with period N contains precisely N distinct signals:

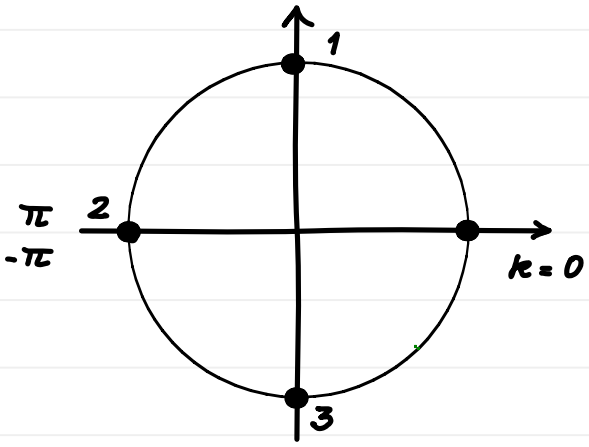
$$x_k[n] = \exp\left(j\,2\pi\frac{k}{N}n\right), \quad k = 0, 1, \ldots, N-1$$

$$\omega_o \in \left\{0, \frac{2\pi}{N}, \frac{4\pi}{N}, \ldots, \frac{2\pi(N-1)}{N}\right\}$$

$$f_o \in \left\{0, \frac{1}{N}, \frac{2}{N}, \ldots, \frac{N-1}{N}\right\}$$

When N is even, for $k = \frac{N}{2}$, $\omega_0 = \pi$ Thus, an exponential with the highest possible frequency of π is present in the N-element set of harmonically related sinusoids. For N odd, the highest frequencies are $\pi\left(1 - \frac{1}{N}\right)$ and $\pi\left(1 + \frac{1}{N}\right)$

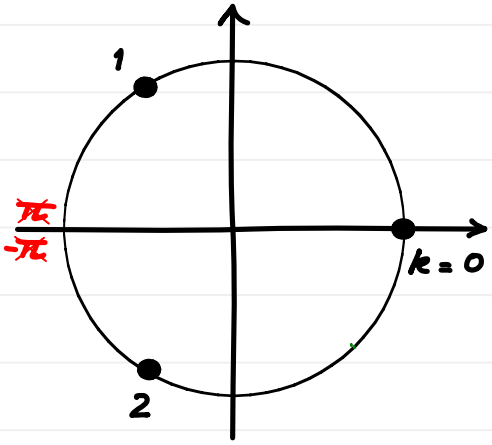
N = 4



$$\omega_k = k \cdot \frac{2\pi}{4} \qquad k = 0, 1, 2, 3$$

$$= \left\{ 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2} \right\}$$

N = 3



$$\omega_k = k \cdot \frac{2\pi}{3} \qquad k = 0, 1, 2$$

$$= \left\{ 0, \frac{2\pi}{3}, \frac{4\pi}{3} \right\} = \left\{ 0, \pi\left(1 - \frac{1}{3}\right), \pi\left(1 + \frac{1}{3}\right) \right\}$$

EE 2004 DSP Lecture 5

Note Title 24-01-2019

Next consider $x[n] = e^{j\omega_0 n}$, where $x[n]$ is not necessarily periodic. As ω_0 varies continuously from 0 to 2π, distinct sinusoids are produced. Stated more precisely, $e^{j\omega_1 n}$ and $e^{j\omega_2 n}$ are independent signals whenever $\omega_1 \neq \omega_2$. That is,

$$c_1 e^{j\omega_1 n} + c_2 e^{j\omega_2 n} = 0 \Rightarrow c_1 = c_2 = 0$$

Hence, as ω_0 varies from 0 to 2π, an uncountably infinite set of complex exponentials are generated. From this set there is a finite subset containing N elements of harmonically related sinusoids with period N, i.e., signals satisfying the property $x[n] = x[n+N]$. We saw earlier that these exponentials have the form $\exp\left(j\frac{2\pi k}{N}n\right)$, $k = 0, 1, 2, \ldots, N-1$

A simple consequence of independence of $e^{j\omega_1 n}$ and $e^{j\omega_2 n}$ is that $e^{j\omega_0 n}$ and $e^{j(-\omega_0)n}$ are independent. That is, there does not exist $k \in \mathbb{C}$ such that $e^{j\omega_0 n} = k e^{-j\omega_0 n}$.
This result is a special case of the general one with $\omega_1 = \omega_0$ and $\omega_2 = -\omega_0$.

On the other hand $\cos(\omega_0 n) = \cos(-\omega_0 n)$

$\sin(\omega_0 n) = -\sin(-\omega_0 n)$

That is $\cos(\omega_0 n)$ and $\cos(-\omega_0 n)$ are dependent signals; similarly for $\sin(\omega_0 n)$.
Hence, for the real-valued counterparts, it is enough if we restrict ω_0 to the interval $[0, \pi]$. Any signal $\cos(\omega_0 n + \phi)$ where $\omega_0 \in (\pi, 2\pi)$ can be written as $\cos(\omega_1 n + \theta)$ where $\omega_1 \in (0, \pi)$.

A real-valued sinusoid with $\omega_0 \in [\pi, 2\pi]$ can be converted to a sinusoid with frequency $\omega_1 \in [0, \pi]$

$$
\begin{aligned}
x[n] &= \cos(\overbrace{1.6\pi}^{\omega_0 > \pi} n + \theta) \\
&= \cos(1.6\pi n - 2\pi n + \theta) \\
&= \cos(-0.4\pi n + \theta) \\
&= \cos(\underbrace{0.4\pi}_{\omega_1 < \pi} n - \theta)
\end{aligned}
$$

Exercise: Suppose $x[n] = e^{j(\omega_0 n + \theta)}, \qquad \omega_0 \in [\pi, 2\pi]$.
Can it be expressed as $x[n] = e^{j(\omega_1 n + \phi)}, \qquad \omega_1 \in [0, \pi]$?

If we require $x[n] = x[n+N]$ and $x[n] = \cos(\omega_0 n)$ & $y[n] = \sin(\omega_0 n)$
the number of independent signals is still only N (exercise)

Thus, in total there are N independent signals, similar to the complex case.

Even though there are N independent signals in each of the sets

$$\left\{\exp\left(j\frac{2\pi k}{N}n\right)\right\} \quad \text{and} \quad \left\{\cos\left(\frac{2\pi k}{N}n\right), \sin\left(\frac{2\pi k}{N}n\right)\right\}$$

the latter can be used for representing real-valued periodic signals only.
No such restrictions are in place for the complex exponentials case.

If we restrict the frequency to be in the range $[0, \pi]$, then $\cos(\omega_1 n)$ and $\cos(\omega_2 n)$ are independent signals, ie,

$$c_1 \cos(\omega_1 n) + c_2 \cos(\omega_2 n) = 0 \implies c_1 = c_2 = 0 \quad (\omega_1 \neq \omega_2)$$

Similar arguments hold for sin.

cos and sin are independent even when the frequency is the same:

$$c_1 \cos(\omega_0 n) + c_2 \sin(\omega_0 n) = 0 \implies c_1 = c_2 = 0$$

Another difference between CT and DT sinusoids: Let

$$\begin{aligned} x(t) &= \cos(\Omega_0 t) \\ y(t) &= \cos(\Omega_0 t + \theta) \\ \Longrightarrow \quad y(t) &= x\left(t + \frac{\theta}{\Omega_0}\right) \end{aligned}$$

ie, one is a *shifted version* of the other. Now consider two DT sequences:

$$\begin{aligned} x[n] &= \cos\left(\frac{\pi n}{5}\right) \\ y[n] &= \cos\left(\frac{\pi n}{5} - \frac{\pi}{2}\right) \\ \text{Is} \quad y[n] &= x[n - n_0] \ ? \end{aligned}$$

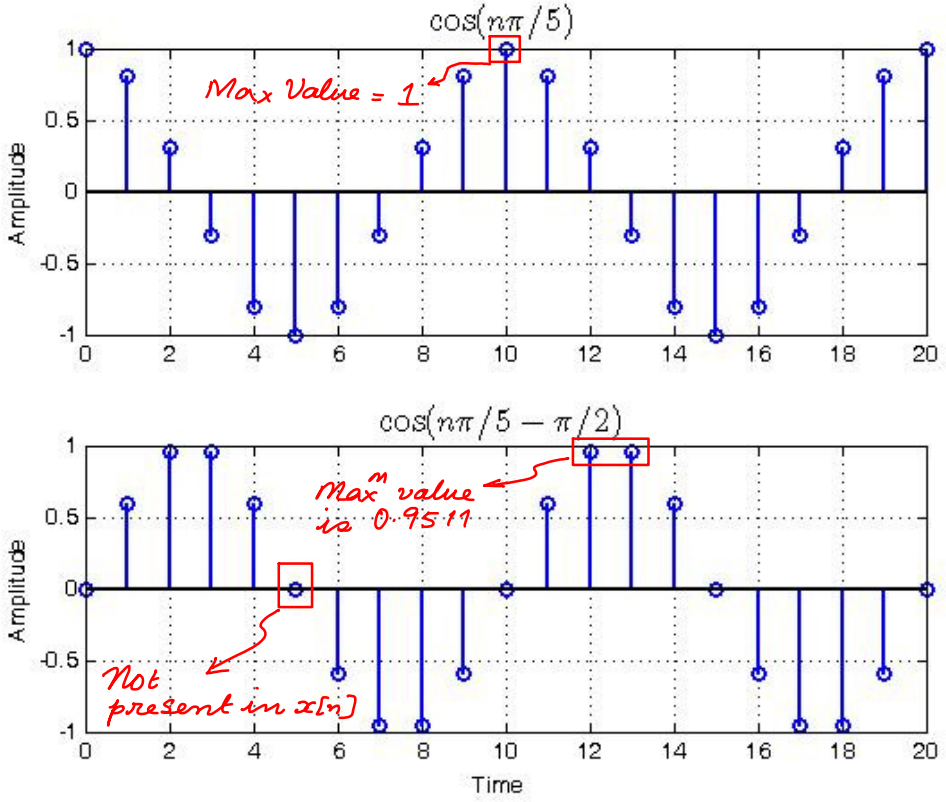
Are these two shifted versions of each other? Clearly they are not.

Remember, by the term "shifted version" we mean $y[n] = x[n - n_0]$ where $n - n_0 \in \mathbb{Z}$

$y[n]$ is actually

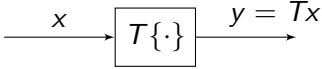
$$y[n] = \text{“}x[n - 2.5]\text{”}$$
$$= \cos\left[\frac{\pi(n - 2.5)}{5}\right]$$

Later we will give meaning to the term "fractional delay".



System: Definition and Representation

- A *system* takes an *input*, operates on it or transforms it to produce an *output*
- The block diagram representation is



- The notation "x" is general: stands for $x(t)$ for CT systems and $x[n]$ for DT systems
- Typically, $y = Tx$ is specified in the form of an input-output relation such as

$$y[n] = 0.8\,y[n-1] + \pi \qquad y(t) + 2\frac{dy}{dt} = x(t) + \frac{d^2x}{dt^2}$$

$$y[x] = x^2[n] \qquad y(t) = |x(t)|$$

System Properties

- A system can be either
  - Linear or Nonlinear
  - Time-Invariant or Time-Variant
    - Time-Invariance is also known as *Shift-Invariance*
  - Causal or Non-Causal
  - Memoryless or Non-Memoryless
    - Memoryless systems are also known as *Static* systems
    - Non-memoryless systems are also known as *Dynamic* systems
  - Stable or Unstable
- We'll examine each of these properties in more detail

Linearity

- A system is said to be linear if it satisfies *both* the following properties:
  - Additivity
  - Homogeneity

Linearity: Additivity

- Suppose the input x_1 produces the output y_1

  Suppose x_2 produces y_2

  If $x_1 + x_2$ produces $y_1 + y_2$ then the system is said to be additive
- In terms of equations

  Let $x_1 \xrightarrow{T} y_1$ and $x_2 \xrightarrow{T} y_2$

  If $x_1 + x_2 \xrightarrow{T} y_1 + y_2$, then the system is *additive*
- Notation: $y = Tx$ can also be written as $x \xrightarrow{T} y$
  - $y = T\{x\}$ is another variant

Linearity: Additivity

- Let $y[n] = x^*[n]$

  $x_1[n] \xrightarrow{T} y_1[n] = x_1^*[n]$

  $x_2[n] \xrightarrow{T} y_2[n] = x_2^*[n]$
- If the input is $x_3[n] = x_1[n] + x_2[n]$, then the output $y_3[n]$ is

$$\begin{aligned} y_3[n] &= x_3^*[n] \\ &= (x_1[n] + x_2[n])^* \\ &= x_1^*[n] + x_2^*[n] \\ &= y_1[n] + y_2[n] \end{aligned}$$

  That is, $T\{x_1 + x_2\} = T\{x_1\} + T\{x_2\}$
- The system is therefore additive

Linearity: Additivity

- Let $y[n] = x^2[n]$

  $x_1[n] \xrightarrow{T} y_1[n] = x_1^2[n]$

  $x_2[n] \xrightarrow{T} y_2[n] = x_2^2[n]$
- If the input is $x_3[n] = x_1[n] + x_2[n]$, then the output $y_3[n]$ is

$$\begin{aligned} y_3[n] &= x_3^2[n] \\ &= (x_1[n] + x_2[n])^2 \\ &= x_1^2[n] + x_2^2[n] + 2\,x_1[n]x_2[n] \\ &\neq y_1[n] + y_2[n] \end{aligned}$$

  That is, $T\{x_1 + x_2\} \neq T\{x_1\} + T\{x_2\}$
- The system is therefore not additive $\implies$ not linear

Linearity: Homogeneity

- Suppose the input x produces the output y
- If $c \cdot x$ produces $c \cdot y$ where c is an arbitrary complex constant, then the system is homogeneous
- In terms of equations

  Let $x \xrightarrow{T} y$

  If $c \cdot x \xrightarrow{T} c \cdot y \qquad \forall c \in \mathbb{C},$

  then the system is *homogeneous*

Linearity: Homogeneity

- Let $y[n] = x^*[n]$ (recall that this system is additive)

  i.e., $x[n] \xrightarrow{T} y[n] = x^*[n]$
- If the input is $x_1[n] = c \cdot x[n]$, then the output $y_1[n]$ is

$$\begin{aligned} y_1[n] &= x_1^*[n] \\ &= (c \cdot x[n])^* \\ &= c^* \cdot x^*[n] \\ &\neq c \cdot y[n] \end{aligned}$$

  That is, $T\{c \cdot x\} \neq c \cdot T\{x\}$
- The system is therefore not homogeneous $\implies$ not linear

Linearity: Homogeneity

- Let $y[n] = \begin{cases} \frac{x[n-1] \cdot x[n+1]}{x[n]} & x[n] \neq 0 \\ 0 & x[n] = 0 \end{cases}$
- If the input is $x_1[n] = c \cdot x[n]$, then the output $y_1[n]$ equals

$$y_1[n] = \begin{cases} \frac{(c \cdot x[n-1])(c \cdot x[n+1])}{c \cdot x[n]} & x_1[n] \neq 0 \\ 0 & x_1[n] = 0 \end{cases}$$
$$= c \cdot y[n]$$

  That is, $T\{c \cdot x\} = c \cdot T\{x\}$
- The system is therefore homogeneous
- Is the system additive?

Additivity & Homogeneity are Independent!

- A system can be

  additive but not homogeneous
  homogeneous but not additive

- From the previous examples you must have noticed that

  $y[n] = x^*[n]$ is additive but not homogeneous

  $$y[n] = \begin{cases} \frac{x[n-1] \cdot x[n+1]}{x[n]} & x[n] \neq 0 \\ 0 & x[n] = 0 \end{cases}$$

  is homogenous but not additve

- A system is linear only if it is additive and homogeneous

Linearity: The Principle of Superposition

- Recall that additivity and homogeneity are:

$$T\{x_1 + x_2\} = T\{x_1\} + T\{x_2\}$$
$$T\{c \cdot x\} = c \cdot T\{x\}$$

- If a system satisfies both properties, they can be combined into the following single equation:

$$T\{a_1 \cdot x_1 + a_2 \cdot x_2\} = a_1 \cdot T\{x_1\} + a_2 \cdot T\{x_2\}$$

- It is called as the Principle of Superposition

Linearity: The Principle of Superposition

- Using *mathematical induction* we can extend the principle of superposition as follows:

$$T\left\{\sum_{k=1}^{N} a_k \cdot x_k\right\} = \sum_{k=1}^{N} a_k \cdot T\{x_k\}$$

- The term $\sum_{k=1}^{N} a_k \cdot x_k$ is called as a linear combination

  By *linear combination* we always mean finite linear combination

- $T\left\{\sum_{k=1}^{N} a_k \cdot x_k\right\} = \sum_{k=1}^{N} a_k \cdot T\{x_k\} \implies T\left\{\sum_{k=1}^{\infty} a_k \cdot x_k\right\} = \sum_{k=1}^{\infty} a_k \cdot T\{x_k\}$

  Additional conditions have to be met for superposition to hold as $N \to \infty$

  We'll assume that these conditions are satisfied

An Important Consequence of Homogeneity

- Recall that a *linear* system must obey the principle of *homogeneity*. That is,

$$x \xrightarrow{T} y \implies c \cdot x \xrightarrow{T} c \cdot y \qquad \forall c \in \mathbb{C}$$

  In particular, the above is true for $c = 0$
- Hence, if the input is $0 \cdot x = 0$, the output must also be $0 \cdot y = 0$
- "Zero input produces zero output"
- Zero input means that the input $= 0$ for all time
- The above must not be misunderstood to mean that the output will be zero *whenever* the input becomes zero

An Important Consequence of Homogeneity

- Let $y[n] = 3x[n] + 2$

  Is this system linear?

  It appears so because $x[n]$ and $y[n]$ are linearly related

- The system fails the linearity test because homogeneity does not hold
- Recall that if the input is zero, the output must also be zero

  If $x[n] = 0$, the output $y[n] = 2$, which is non-zero

  The system is therefore not linear

  It cannot be called *nonlinear* in the usual sense but is said to be incrementally linear

  The difference between two outputs obeys linearity

Systems with Initial Conditions

- In the continuous-time case, suppose we have an RLC network with initial conditions

  $\Longrightarrow$ non-zero $v_c(0^-)$ and/or $i_L(0^-)$

- Suppose the input is set to zero

  The output will be non-zero due to transients caused by the intial conditions

  The system is, strictly speaking, not linear

  (counter intuitive!)

- *However, if we treat the initial conditions also as input, the output is the sum of the responses due to both the inputs*

  Recall that the initial conditions would be set to zero when computing the ouput due to the conventional input

Systems with Initial Conditions

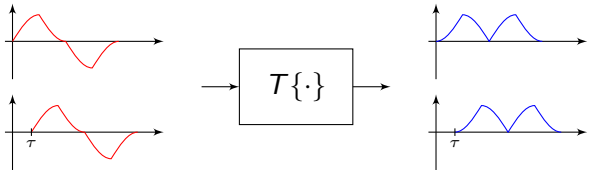
- Therefore, depending on the point of view, an RLC circuit with initial conditions can be considered either as linear or incrementally linear
- If the system is treated like a black box, i.e., initial conditions are not accessible, then the system is incrementally linear
- If the initial conditions are also viewed as inputs that can be zeroed out when considering other sources, the the system is linear
- While referring to textbooks, one has to be clear as to what viewpoint has been adopted
- The discrete-time counterpart will involve linear difference equations with non-zero initial conditions

Time-Invariance

- Suppose the input $x[n]$ produces the output $y[n]$

  If $x[n - n_0]$ produces $y[n - n_0]$, the system is time-invariant
- What's the intuition behind this principle?
- If we delay the input by a certain amount, the output also gets delayed by the *same amount*
  - Intuitively speaking, we expect the "same output" when we apply an input now or later, except for a delay
  - Output delay must be identical to input delay
- Loosely speaking, systems whose parameters don't change with time will be time-invariant
- Recall that another name for time-invariance is *shift*-invariance

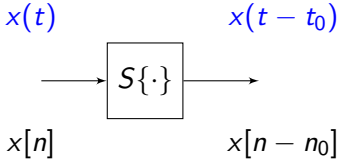
The idea can be pictured along the following lines:



- Continuous-time curve is shown, but same idea holds for the discrete-time case too

Time-Invariance

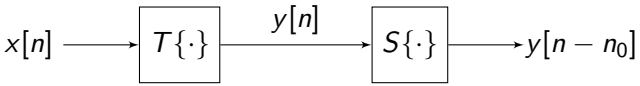
- Another way to look at time-invariance is in conjunction with the shift operator $S\{\cdot\}$:



- That is, $x(t) \xrightarrow{S} x(t - t_0)$
  $x[n] \xrightarrow{S} x[n - n_0]$
- The notation $S_\tau\{\cdot\}$ is also commonly used to denote explicitly the delay introduced
- The shift operator is sometimes called as the delay operator
  - If the delay is negative, it advances the output signal

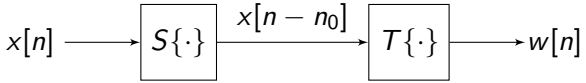
Time-Invariance

- Consider the following two cases:



That is, we first form the output and then shift it

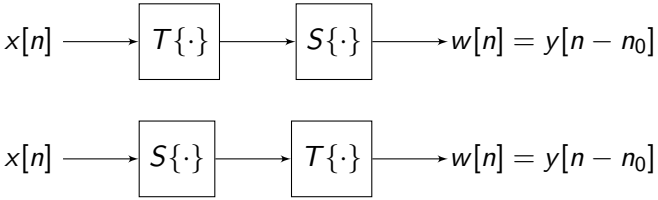
Suppose we instead did the following:



That is, we first shift and then form the output

- If the system were time-invariant, $x[n-n_0]$ must produce $y[n-n_0]$
  $\implies w[n] = y[n-n_0]$

Time-Invariance



That is,

$$\begin{aligned} w[n] &= ST\{x[n]\} \\ &= TS\{x[n]\} \\ &= y[n - n_0] \end{aligned}$$

- For a time-invariant system, T and S operators commute

Time-Invariance

- Let $y(t) = x(2t)$

  Shifted output: $y(t - t_0)$

  But, since $y(t) = x(2t)$, the shifted output is

$$\begin{aligned} y(t - t_0) &= x(2\overline{t - t_0}) \\ &= x(2t - 2t_0) \end{aligned}$$

- If the shifted input $x(t - t_0)$ is applied, the output will be

$$x(t - t_0) \xrightarrow{T} y(2t - t_0)$$

- The output is not the same as $y(2t - 2t_0)$

  $\implies$ $y(t) = x(2t)$ is a time-variant system

Time-Invariance

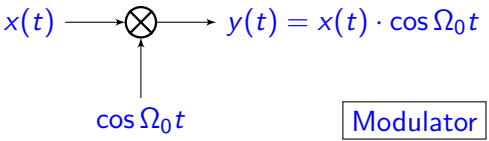
- Exercise:

  Let $y[n] = x[Mn]$ where $M \in \mathbb{Z}$

  Is the system linear?

  Is it time-invariant?

- Exercise:



  Is the system linear?

  Is the system time-invariant?

Linearity & Time-Invariance are Independent

- Let $y[n] = |x[n]|^2$

  Shifted output: $y[n - n_0] = |x[n - n_0]|^2$

  If the shifted input $x[n - n_0]$ is applied, the output is $|x[n - n_0]|^2$
- The system is clearly time-invariant
- It is easy to see that the system is nonlinear
- This is an illustration of the fact that linearity and time-invariance are independent

  A system can be linear but time-variant

  A system can be nonlinear but time-invariant

  — a total of 4 combinations are possible

Systems With and Without Memory

- If $y[n_0]$ depends *only on* $x[n_0]$, then the system is memoryless ("static")

  In other words, current output depends only on the current input
- Otherwise, the system has memory (non-memoryless or "dynamic")
- $y[n] = x^2[n]$ is clearly a memoryless system

  $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$ is a CT system that has memory

  $y[n] = x[n+1]$ is a DT system that has memory
- The last example illustrates that by "memory" we don't mean dependence only on past values but possibly future ones too

Causality

- If $y[n]$ depends *only on* present and past inputs and past outputs then the system is causal

  In other words, $y[n_0]$ depends only on *inputs* for $n \leq n_0$ and *outputs* for $n < n_0$
- Otherwise, the system is called non-causal
- $y[n] = x[n-1]$ causal

  $y[n] = x[n+1]$ non-causal

  $y[n] = x[n] \cdot \cos(n+1)$ causal
- The last example illustrates that causality is with respect to the *input/output* and not on other factors

Causality

- Let $y[n] = x[-n]$

  Is the system causal?

  Superficially it appears so:

  $y[0] = x[0]$, $y[1] = x[-1]$, $y[2] = x[-2]$, and so on
- However, causality must hold true *for all* n:

  $y[-1] = x[1]$ and test for causality fails!
- Make sure that you carefully check the conditions for all n
- Exercise:

  Is the system $y[n] = x[Mn]$, where $M \in \mathbb{Z}$, causal?

  Is the system $y(t) = x(2t)$ causal?

Stability

- Whether or not a system is *stable* depends on the *stability criterion*
- One of the most commonly used criterion is bounded input bounded output (BIBO) stability
- Suppose the input is bounded, i.e., $|x[n]| < \infty$
  If the output is also bounded, i.e., $|y[n]| < \infty$,
  then the system is BIBO stable
- $y[n] = x^2[n]$ — BIBO stable

  $y[n] = \sum_{m=0}^{n} x[m]$ — unstable

  because $u[n] \longrightarrow (n+1)\,u[n]$

  ($u[n]$ is bounded, whereas $(n+1)\,u[n]$ is not)

  CT counterpart: $y(t) = \int_0^t x(\tau)\,d\tau$ — unstable

System Properties are Independent

- Recall that we mentioned that linearity and time-invariance are *independent* properties
- The same is, in general, true for all the properties

  A system can be linear or nonlinear, causal or non-causal, be memoryless or have memory, time-invariant or time-variant, and stable or unstable
- However, if the system is memoryless, then it is necessarily causal

  Can you conclude anything about the stability of a memoryless system?

  What about time-invariance?

Example

- Let $y[n] = \begin{cases} x[n], & n \geq 1 \\ 0, & n = 0 \\ x[n+1] & n \leq -1 \end{cases}$

  Check if the system is (a) linear, (b) time-invariant, (c) memoryless, (d) causal, and (e) stable

- To prove that a property holds, a *general proof* must be given. To show that it does not hold, a *single counterexample* is enough

Example

- Let $y[n] = \begin{cases} x[n], & n \geq 1 \\ 0, & n = 0 \\ x[n+1] & n \leq -1 \end{cases}$

  Clearly, $y[1] = x[1], \quad y[2] = x[2], \quad y[3] = x[3], \quad \ldots$

  $y[0] = 0$ always

  $y[-1] = x[0], \quad y[-2] = x[-1], \quad y[-3] = x[-2], \quad \ldots$

  i.e., $x[n]$ for $-\infty < n \leq 0$ is shifted by one sample to the left at the output

Example

- For linearity, superposition must hold, i.e., we must check if

$$\underbrace{a_1\, x_1[n] + a_2\, x_2[n]}_{x_3[n]} \xrightarrow{T} \underbrace{a_1\, y_1[n] + a_2\, y_2[n]}_{y_3[n]} \quad \text{holds}$$

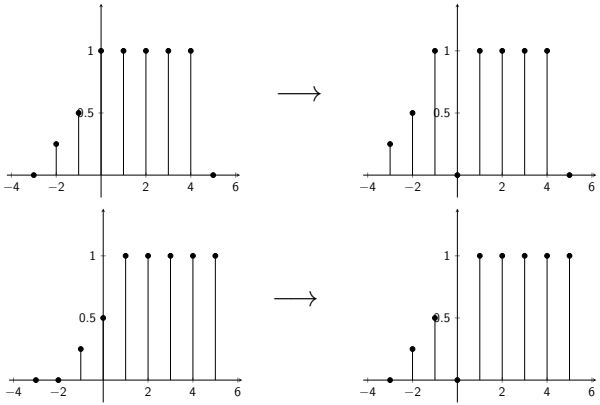
- $y_3[n] = \begin{cases} x_3[n], & n \geq 1 \\ 0, & n = 0 \\ x_3[n+1] & n \leq -1 \end{cases}$

- $y_3[n] = \begin{cases} a_1\, x_1[n] + a_2\, x_2[n], & n \geq 1 \\ 0, & n = 0 \\ a_1\, x_1[n+1] + a_2\, x_2[n+1] & n \leq -1 \end{cases}$

That is, $y_3[n] = a_1\, y_1[n] + a_2\, y_2[n]$ for all n

Example

- For time-invariance, we must check if $x[n - n_0]$ produces $y[n - n_0]$. E.g., does $x[n - 1]$ produce $y[n - 1]$?



Shifted input does not produce shifted output

The system is time variant

Example

Since $y[-1] = x[0]$, the system is non-causal

For the same reason, the system has memory

Since $y[n] = \begin{cases} x[n], & n \geq 1 \\ 0, & n = 0 \\ x[n+1] & n \leq -1 \end{cases}$

if $x[n]$ is bounded, then it is easy to see that $y[n]$ is also bounded, i.e., the system is BIBO stable

- In summary, the given system is (a) linear, (b) time-variant, (c) non-memoryless, (d) non-causal, and (e) stable

Linear Time-Invariant (LTI) Systems

- Systems that are both *linear* and *time-invariant* are a very important subclass

  They are abbreviated as LTI systems

  A number of very useful and practical systems can be modelled as LTI

- Recall that an LTI system must satisfy the following:

$$\sum_{k=1}^{N} a_k \cdot x_k[n] \xrightarrow{T} \sum_{k=1}^{N} a_k \cdot y_k[n] \qquad \text{(linearity)}$$

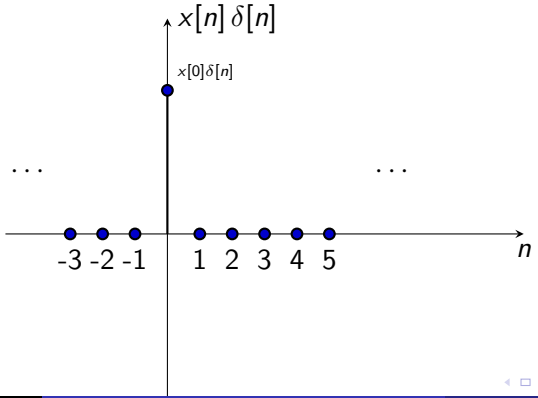
$$x[n - n_0] \xrightarrow{T} y[n - n_0] \qquad \text{(time-invariance)}$$

- We will further assume that superposition holds for *infinite* linear combination as well:

$$\sum_{k=1}^{\infty} a_k \cdot x_k[n] \xrightarrow{T} \sum_{k=1}^{\infty} a_k \cdot y_k[n]$$

Linear Time-Invariant (LTI) Systems

- The starting point of LTI systems theory development stems from the following:
  Recall the *sifting property*: $x[n] \cdot \delta[n-k] = x[k]\delta[n-k]$
  Hence, any signal $x[n]$ can be expressed as a linear combination of scaled and delayed impulses



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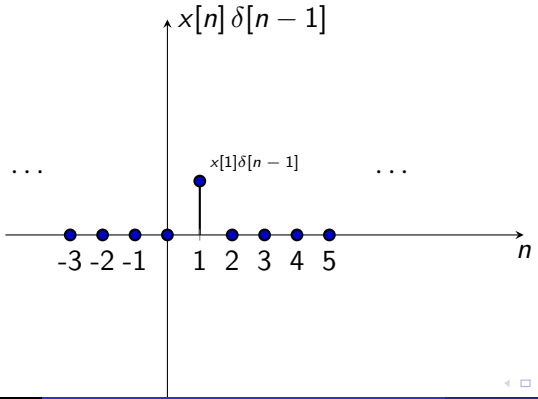

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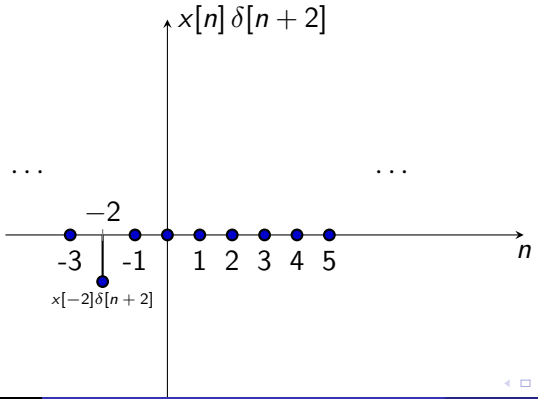

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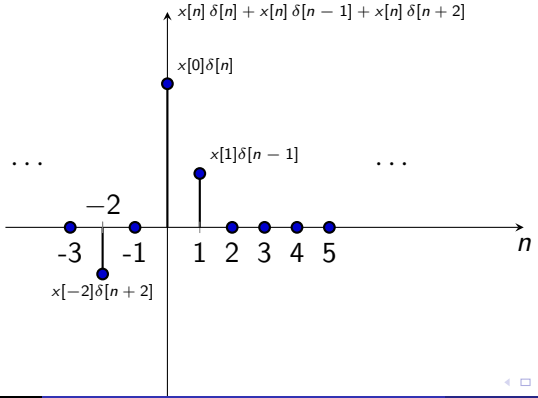

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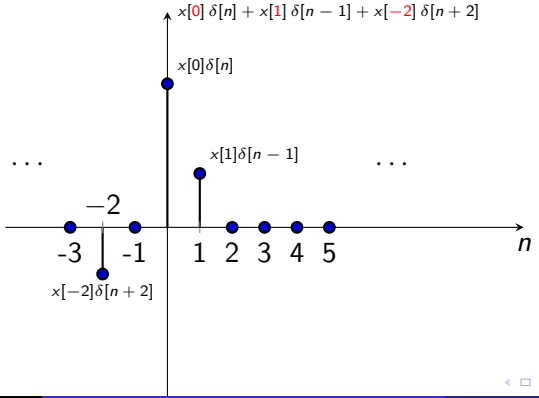
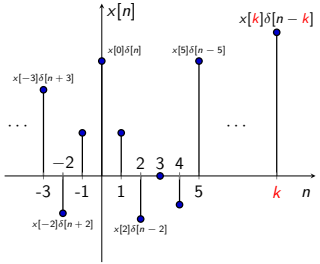


Linear Time-Invariant (LTI) Systems

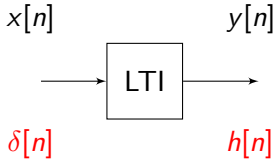


Thus, the sequence $x[n]$ can be thought of as

$$\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

i.e., $x[n]$ is, in general, an *infinite* linear combination of scaled and delayed impulses

LTI Systems: Impulse Response



For an arbitrary input $x[n]$, we denote the output as $y[n]$

- If the input is the unit impulse function $\delta[n]$, we denote the output as $h[n]$

  $h[n]$ is called as the impulse response and plays a central role in LTI systems theory

- We will see that knowledge of $h[n]$ will help us find the output to *any* input $x[n]$

LTI Systems: Impulse Response

$$\delta[n] \longrightarrow h[n]$$

$$\delta[n-k] \longrightarrow h[n-k] \qquad \text{time-invariance}$$

$$x[k]\,\delta[n-k] \longrightarrow x[k]\,h[n-k] \qquad \text{homogeneity}$$

$$\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] \longrightarrow \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] \qquad \text{additivity}$$

$$x[n] \longrightarrow \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

i.e., for an LTI system, given an arbitrary input $x[n]$, the output $y[n]$ is given by

$$\boxed{x[n] \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]}$$

Convolution

- The following is called as the convolution sum:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

  It is central to LTI systems theory and commonly denoted as $y[n] = x[n] * h[n]$

- The corresponding relationship for a CT system with impulse response $h(t)$ is given by the convolution integral:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

  and denoted as $y(t) = x(t) * h(t)$

- $y = x * h$ can be used for both CT and DT convolution

Convolution

- Since convolution involves summation (integration), it is a smoothing operation
- The '$*$' is just notation and should not be misinterpreted:

  $y(t) = x(t) * h(t) \not\Longrightarrow y(ct) = x(ct) * h(ct)$

  What is true is $y(ct) = \int_{-\infty}^{\infty} x(\tau)\, h(ct - \tau)\, d\tau$

  Exercise: Show that the correct answer is $y(ct) = c \cdot x(ct) * h(ct)$

  Similar remarks apply to the DT case (but remember that there are differences in the way scaling works in the DT case)
- Better notation: $y(t) = (x * h)(t)$ or $y[n] = (x * h)[n]$

  — found more commonly in mathematics textbooks

Convolution

1. Change the independent variable to τ because we want the final answer to be in terms of t

$$x(t) \rightsquigarrow x(\tau) \text{ and } h(t) \rightsquigarrow h(\tau)$$

2. Keep one signal as it is, and flip the other

$$x(\tau) \text{ remains as is and } h(\tau) \rightsquigarrow h(-\tau)$$

3. Shift the flipped signal $w(\tau) = h(-\tau)$ by t_0 units:

$$\begin{aligned} w(\tau - t_0) &= h(-(\tau - t_0)) \\ &= h(t_0 - \tau) \end{aligned}$$

4. Multiply point by point and integrate over the entire range:

$$y(t_0) = \int_{-\infty}^{\infty} x(\tau)\, h(t_0 - \tau)\, d\tau$$

where t_0 is arbitrary

Convolution

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$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t - \tau)\, d\tau$$

where $-\infty < t < \infty$

Convolution

1. Change the independent variable to k because we want the final answer to be in terms of n

$$x[n] \rightsquigarrow x[k] \text{ and } h[n] \rightsquigarrow h[k]$$

2. Keep one signal as it is, and flip the other

$$x[k] \text{ remains as is and } h[k] \rightsquigarrow h[-k]$$

3. Shift the flipped signal $w[k] = h[-k]$ by n units:

$$\begin{aligned} w[k-n] &= h[-(k-n)] \\ &= h[n-k] \end{aligned}$$

4. Multiply point by point and sum over the entire range:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

where $-\infty < n < \infty$

An Important Identity

- Recall we expressed an arbitrary sequence $x[n]$ as a sum of scaled and delayed impulses

  The expression was

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

  This is precisely in the form of a convolution sum:

$$\boxed{x[n] = x[n] * \delta[n]}$$

  i.e., convolution with an impulse leaves a signal unchanged

- CT counterpart: $x(t) = x(t) * \delta(t)$

An Important Identity

- $x[n] * \delta[n - n_0] = x[n - n_0]$

Proof:

$x[n] \rightsquigarrow x[k]$ and $\delta[n - n_0] \rightsquigarrow \delta[k - n_0]$

$\delta[k - n_0] \rightsquigarrow \delta[-k - n_0] \rightsquigarrow \delta[-(k - n) - n_0] \rightsquigarrow \delta[n - n_0 - k]$

Hence,

$$x[n] * \delta[n - n_0] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n - n_0 - k]$$
$$= x[n - n_0]$$

That is,

$$\boxed{x[n] * \delta[n - n_0] = x[n - n_0]}$$

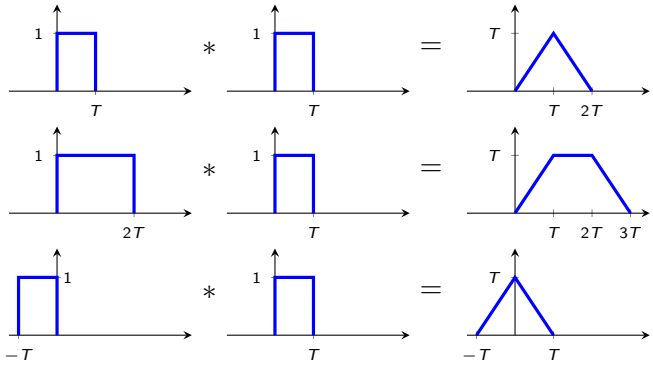
- CT counterpart: $x(t) * \delta(t - t_0) = x(t - t_0)$

Difference Between Convolution and Sifting

Notice carefully the difference between convolution and the sifting property:

- Convolution:
  - $x[n] * \delta[n - n_0] = x[n - n_0]$
  - $x(t) * \delta(t - t_0) = x(t - t_0)$
- Sifting:
  - $x[n] \cdot \delta[n - n_0] = x[n_0] \cdot \delta[n - n_0]$
  - $x(t) \cdot \delta(t - t_0) = x(t_0) \cdot \delta(t - t_0)$

Examples



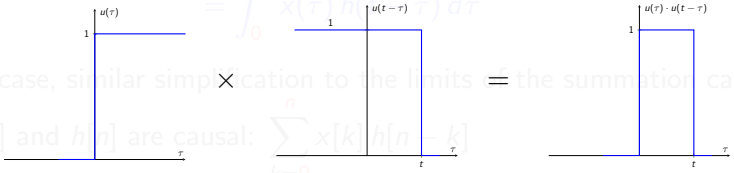
Convolution for Causal Signals

- Suppose both $x(t)$ and $h(t)$ are *causal*, i.e., zero for $t < 0$

  Hence, $x(t) = x(t)\,u(t)$ and $h(t) = h(t)\,u(t)$, where $u(t)$ is the unit step function

  We can then simplify the limits of the convolution integral:

$$\begin{aligned}(x * h)(t) &= \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau \\ &= \int_{-\infty}^{\infty} x(\tau)u(\tau)\,h(t-\tau)u(t-\tau)\,d\tau\end{aligned}$$



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  i.e., $u(\tau)\cdot u(t-\tau) = 1 \qquad \text{for } 0 < \tau < t$

- In the DT case, similar simplification to the limits of the summation can be made if both $x[n]$ and $h[n]$ are causal: $\sum_{k=0}^{n} x[k]\,h[n-k]$

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Convolution for Causal Signals

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- In the DT case, similar simplification to the limits of the summation can be made if both $x[n]$ and $h[n]$ are causal: $\sum_{k=0}^{n} x[k]\,h[n-k]$

CT Convolution Exercises

Since CT and DT convolution are fundamentally the same, exercises in CT convolution will help to cement the basic concepts

- Suppose $x(t) = 0$ outside $t \in [a_1, b_1]$ and $h(t) = 0$ outside $t \in [a_2, b_2]$. Show that $x(t) * h(t) = 0$ outside $t \in [a_1 + a_2, b_1 + b_2]$

  Hint: $x(t) = x(t)\,[u(t - a_1) - u(t - b_1)]$ and $h(t) = h(t)\,[u(t - a_1) - u(t - b_1)]$. Reason out as in the case of convolution of causal signals.

- Let $a,\ b > 0$ and consider the cases $a \neq b$ and $a = b$. Find the simplified expressions for:

  (a) $e^{-at}\,u(t) * e^{-bt}\,u(t)$

  (b) $e^{-at}\,u(t) * e^{bt}\,u(-t)$

DT Convolution of Finite Duration Sequences

- Suppose $x[n]$ and $h[n]$ are *finite duration* sequences with lengths P and Q respectively

  What is the length of $x[n] * h[n]$?

  Without loss of generality, assume $P \geq Q$

  As $h[n]$ slides into $x[n]$ and remains within $x[n]$, P samples of the output are generated

  As $h[n]$ starts to come out of $x[n]$, output samples get generated till there is overlap

  This overlap lasts for $Q - 1$ samples

- Hence the length of $y[n]$ equals $\boxed{P + Q - 1}$

Properties of Convolution

1. Commutativity: $x[n] * y[n] = y[n] * x[n]$

   Consider $\sum_{k=-\infty}^{\infty} x[k]\, y[n-k]$

   Let $l = n - k \implies k = n - l$. Hence,

$$\begin{aligned}\sum_{k=-\infty}^{\infty} x[k]\, y[n-k] &= \sum_{l=\infty}^{-\infty} x[n-l]\, y[l] \\ &= \sum_{l=-\infty}^{\infty} y[l]\, x[n-l] \\ &= y[n] * x[n]\end{aligned}$$

Properties of Convolution

2. Associativity: $(x[n] * y[n]) * z[n] = x[n] * (y[n] * z[n])$

   The above holds provided $x[n], y[n], z[n] \in \ell_1$

   ℓ_1 is the class of absolutely summable sequences: $\left\{ p[n] \;\middle|\; \sum_n |p[n]| < \infty \right\}$

- If the sequences do not belong to ℓ_1, then associativity will fail, as shown in the following example:

  $x[n] = \alpha^n, \;\; y[n] = \delta[n] - \alpha\,\delta[n-1], \;\; z[n] = \alpha^n\, u[n]$

  It should be easy to verify that $(x[n] * y[n]) * z[n] \neq x[n] * (y[n] * z[n])$

  even though the individual convolutions are well-defined

  $x[n] \notin \ell_1$, causing associativity to fail ($z[n] \in \ell_1$ provided $|\alpha| < 1$)

- Proof is very simple in the transform domain, which we will see later

Properties of Convolution

2. Associativity: $[x(t) * y(t)] * z(t) = x(t) * [y(t) * z(t)]$

   The above holds provided $x(t), y(t), z(t) \in L^1$

   L^1 is the class of absolutely integrable functions: $\left\{ p(t) \;\middle|\; \int_{-\infty}^{\infty} |p(t)|\, dt < \infty \right\}$

- If the functions do not belong to L^1, then associativity will fail, as shown in the following example:

  $x(t) = u(t), \;\; y(t) = -t\, e^{-t^2/2}, \;\; z(t) = u(-t)$

  It should be easy to verify that $[x(t) * y(t)] * z(t) \neq x(t) * [y(t) * z(t)]$

  even though the individual convolutions are well-defined

  $x(t), z(t) \notin L^1$, causing associativity to fail

- Proof is very simple in the transform domain

Properties of Convolution

3. Distributivity: $x[n] * (y[n] + z[n]) = x[n] * y[n] + x[n] * z[n]$

Proof:

$$\begin{aligned} x[n] * (y[n] + z[n]) &= \sum_{k=-\infty}^{\infty} x[k] \cdot (y[n-k] + z[n-k]) \\ &= \sum_{k=-\infty}^{\infty} x[k]\, y[n-k] + \sum_{k=-\infty}^{\infty} x[n]\, z[n-k] \\ &= x[n] * y[n] + x[n] * z[n] \end{aligned}$$

- CT convolution is also distributive: $x(t) * [y(t) + z(t)] = x(t) * y(t) + x(t) * z(t)$

System Interconnections

- Convolution properties help us to break a complex system into lower order sub-systems

  There is no advantage in doing so when the precision is infinite

  When using *finite precision*, the lower order sub-systems are less affected by quantization

- Parallel Decomposition

  — distributivity

- Cascade Decomposition

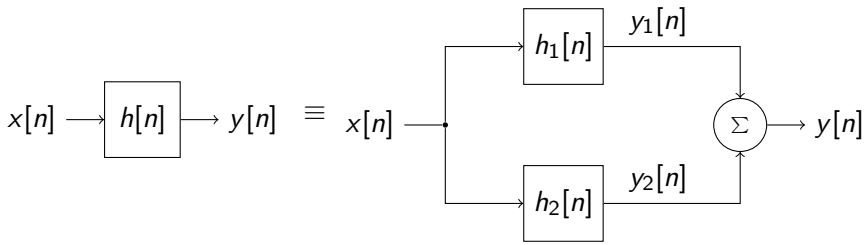
  — associativity and commutativity

Parallel Decomposition

Suppose $h[n] = h_1[n] + h_2[n] \implies x[n] * h[n] = x[n] * (h_1[n] + h_2[n])$

(distributivity) $= \underbrace{x[n] * h_1[n]}_{y_1[n]} + \underbrace{x[n] * h_2[n]}_{y_2[n]}$

In terms of block diagrams, we can depict the above as follows:



Cascade Decomposition

Suppose $h[n] = h_1[n] * h_2[n]$

$$
\begin{aligned}
y[n] &= x[n] * h[n] \\
&= x[n] * (h_1[n] * h_2[n]) \\
&= x[n] * (h_2[n] * h_1[n]) && \text{(commutativity)} \\
&= (x[n] * h_2[n]) * h_1[n] && \text{(associativity)}
\end{aligned}
$$

Order of interconnections does not matter!

DT Convolution Exercises

- $a^n\, u[n] * b^n\, u[n] \quad (a \neq b)$
- $a^n\, u[n] * a^n\, u[n]$
- $a^n\, u[n] * b^n\, u[-n]$

Are there any conditions on a and b for the convolution sums to exist?

Importance of Impulse Response for LTI Systems

- An LTI system is completely characterized by its impulse response, which is a signal

  Hence, conditions on the system can be restated as conditions on the signal

- If the system is causal, then $h[n] = 0$ for $n < 0$

  This is because the impulse is applied at $n = 0$ and a causal system cannot produce a response before an input is applied

  For CT systems, the corresponding condition is $h(t) = 0$ for $t < 0$

- Similarly, the BIBO stability criterion can be expressed as a condition on the impulse response

BIBO Stability for LTI Systems

- Assume $x[n]$ is *bounded*, i.e., $|x[n]| < B_x$
- The requirement of getting a *bounded output* can be translated into a condition on the impulse response:

We want $y[n] = \sum_{k=-\infty}^{\infty} x[n-k]\,h[k]$ to be bounded. Hence

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} x[n-k]\,h[k] \right| \leq \sum_{k=-\infty}^{\infty} |x[n-k]|\,|h[k]|$$

$$\leq B_x \sum_{k=-\infty}^{\infty} |h[k]|$$

$$|y[n]| < \infty \implies \boxed{\sum_{k=-\infty}^{\infty} |h[k]| < \infty} \text{ i.e., } h[n] \text{ must be absolutely summable}$$

Example

- Consider $y[n] = \sum_{k=0}^{n} x[k]$. Is this system BIBO stable?

  The impulse response is $h[n] = \sum_{k=0}^{n} \delta[k] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$

  Clearly, $h[n] = u[n]$

  Since $u[n]$ is *not absolutely summable*, the given system is not BIBO stable

- Alternatively, the output $y[n] = (n+1)\,u[n]$ when $u[n]$ is applied as the input. Clearly $(n+1)\,u[n]$ is not bounded

  The system cannot therefore be BIBO stable

EE 2004 DSP Lecture 8

Note Title 31-01-2019

LTI Systems Governed by LCCDE:

We will focus ourselves on the following class of input-output relationships:

$$y[n] = F\{x[n], x[n-1], \ldots, x[n-M], y[n-1], y[n-2], \ldots, y[n-N]\}$$

In particular, we will focus on the class having the form:

$$y[n] = -\sum_{k=1}^{N} a_k\, y[n-k] + \sum_{k=0}^{M} b_k\, x[n-k]$$

i.e., the class of systems described by Linear Constant Coefficient Difference Equations (LCCDE)

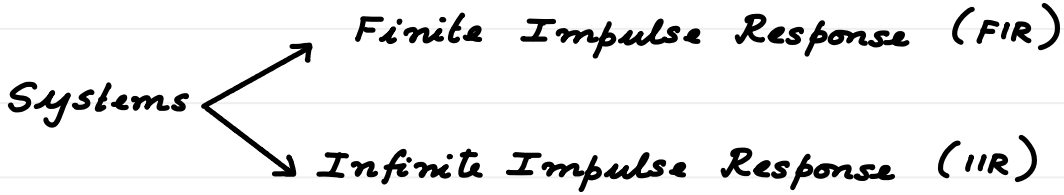
The equation $y[n] = a \cdot y[n-1] + b\,x[n]$ is an example of an LCCDE. OTOH, $y[n] = \underbrace{a[n]}_{\text{time-varying coeff.}} \cdot x[n-1] + \underbrace{x^2[n]}_{\text{nonlinear term}}$ is not an LCCDE.

From now on we will focus on systems whose I/O relationship is an LCCDE.

Under what conditions does an LCCDE represent (a) a system? (b) an LTI system? and (c) a causal LTI system?

See "Signals & Systems" by Oppenheim & Willsky, 2nd edition, Sec. 2.4 (pp. 116-127)

An important classification of systems is based on the duration of $h[n]$:



Recall that, an LCCDE system is given by

$$y[n] = -\sum_{k=1}^{N} a_k\, y[n-k] + \sum_{k=0}^{M} b_k\, x[n-k]$$

In particular, $y[n] = a \cdot y[n-1] + b \cdot x[n]$ gives $h[n] = b \cdot a^n\, u[n]$

The above system is IIR when $a \neq 0$.

OTOH, $y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2]$ is an FIR system because

$$h[n] = b_0 \delta[n] + b_1 \delta[n-1] + b_2 \delta[n-2] \equiv \{ \ldots, 0, 0, 0, \underset{\downarrow}{b_0}, b_1, b_2, 0, 0, \ldots \}$$

If the system is IIR, then at least one a_k will be non-zero.

If all the a_k's are zero, then the system will be FIR.

If some of a_k's are non-zero, does it mean that the system is necessarily IIR ?

EE 2004 DSP Lecture 9

Note Title 05-02-2019

If the system is IIR, then at least one a_k will be non-zero.

If all the a_k's are zero, then the system will be FIR.

If some of a_k's are non-zero, does it mean that the system is necessarily IIR ?

The answer is NO because there are recursive implementations of FIR systems.

Consider the following system: $y[n] = \frac{1}{N} \sum_{k=n-N+1}^{n} x[k]$

$$= \frac{1}{N} \left(x[n] + x[n-1] + x[n-2] + \dots + x[n-N+1] \right)$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} x[n-k] \qquad \text{(i)}$$

Comparing this with $y[n] = -\sum_{k=1}^{N} a_k\, y[n-k] + \sum_{k=0}^{M} b_k\, x[n-k]$

we see that $a_k = 0 \quad \forall k$ and $b_k = \frac{1}{N}$ for $0 \le k \le N-1$

Clearly, $h[n] = \frac{1}{N}$ for $0 \le n \le N-1$, i.e., it is an FIR filter

It is easy to see that Eq. (i) can also be written as

$$y[n] = y[n-1] + \frac{1}{N} x[n] - \frac{1}{N} x[n-N] \qquad \text{(ii)}$$

Eq. (ii) is a recursive implementation of a non-recursive system.

Even though $a_1 = -1$, the system is not IIR.

The recursive implementation requires only two additions, whereas the non-recursive implementation requires N-1 additions.

A simple difference equation can model a plucked string to a very good approximation! Consider $y[n] = a \cdot y[n-M] + x[n]$ with $a = 0.98$ and $M = 100$. For each 'n', $x[n]$ is a uniform random variable in the range $[-1, 1]$. The soln. with $y[-1] = 0$, produces a waveform that sounds like a plucked string when played at 44.1 kHz!

The following MATLB code snippet produces a plucked string sound:

```
>> x = 2*rand(100,1)-1;                % x is random vector of 100 samples uniformly distributed between in the range [-1, 1]
>> x = [x;zeros(44000,1)];             % x is appended with zeros because 'filter' produces an output of the same lenght as x
>> y = filter(1,[1;zeros(99,1);-0.98],x);   % filter the input through a system that feeds the output after a 100 sample delay
>> soundsc(y,44100)                    % play the filtered output using a sampling frequency of 44.1 kHz
```

Vary the delay M to vary the output frequency.

Lookup the Karplus Strong Algorithm for more details.

EE 2004 DSP Lecture 9

Note Title 05-02-2019

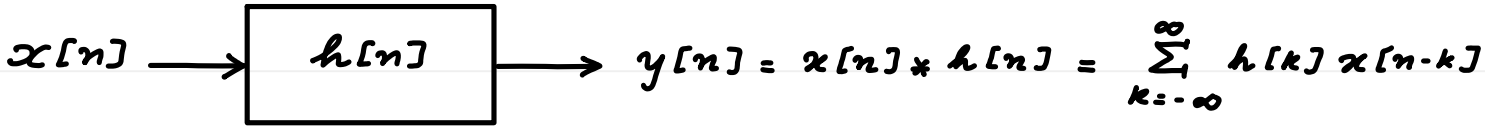
Z-Transform The **z-transform** $X(z)$ of a sequence $x[n]$ is **defined** as

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

Notation $x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$ ($\mathcal{Z}$ omitted if the context is clear)

denotes a transform pair

In the context of LTI systems, the above definition can be motivated as follows:



If $x[n] = z_0^n$, then $y[n] = \sum_{k=-\infty}^{\infty} h[k]\, z_0^{n-k} = z_0^n \sum_{k=-\infty}^{\infty} h[k]\, z^{-k} = z_0^n\, H(z_0)$

(z_0^n: Everlasting exponential (from $-\infty$ to ∞))

$H(z_0)$ is a complex constant. Note that if the input is z_0^n, then the output is **identical to**

the input except for a scale factor. This is reminiscent of $\underline{A}\,\underline{x} = \lambda\,\underline{x}$. Hence z_0^n is called as an eigensignal (with eigenvalue $H(z_0)$).

Example

$$x[n] = a^n u[n]$$

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

$$= \frac{1}{1 - az^{-1}}$$

provided $|z| > |a|$

Example

$$x[n] = -a^n u[-n-1]$$

$$X(z) = -\sum_{n=-\infty}^{\infty} a^n u[-n-1] z^{-n}$$

$$= -\sum_{n=1}^{\infty} (a^{-1}z)^n$$

$$= \frac{1}{1 - az^{-1}}$$

provided $|z| < |a|$

Thus, both $a^n u[n]$ and $-a^n u[-n-1]$ have identical $X(z)$, i.e., $\frac{1}{1-az^{-1}}$ but the regions in the z-plane where they are valid are different.

It should be clear that given $X(z) = 1/(1-az^{-1})$ alone, we cannot infer whether $x[n]$ is $a^n u[n]$ or $-a^n u[-n-1]$. Thus, RoC is an inseparable part of $X(z)$. Thus, $X(z)$ + RoC $\equiv$ Power Series

$x[n] = \{-1, \underset{\downarrow}{2}, 4, \pi\} \Rightarrow X(z) = -1 + 2z^{-1} + 4z^{2} + \pi z^{-3}$ $\quad z = 0 \notin$ RoC

$x[n] = \{-1, 2, 4, \underset{\downarrow}{\pi}\} \Rightarrow X(z) = -z^{3} + 2z^{2} + 4z + \pi$ $\quad z = \infty \notin$ RoC

$x[n] = \{-1, 2, \underset{\downarrow}{4}, \pi\} \Rightarrow X(z) = -z^{2} + 2z + 4 + \pi z^{-1}$ $\quad z = 0, \infty \notin$ RoC

Thus, for a finite duration sequence, the RoC is the entire z-plane except possibly 0 and/or ∞

EE 2004 DSP Lecture 10

Note Title 06-02-2019

Recall that $a^n u[n] \leftrightarrow \frac{1}{1-az^{-1}} \quad |z| > |a|$

$$X(z) = \frac{z}{z-a} = \frac{B(z)}{A(z)}$$ i.e., a ratio of polynomials

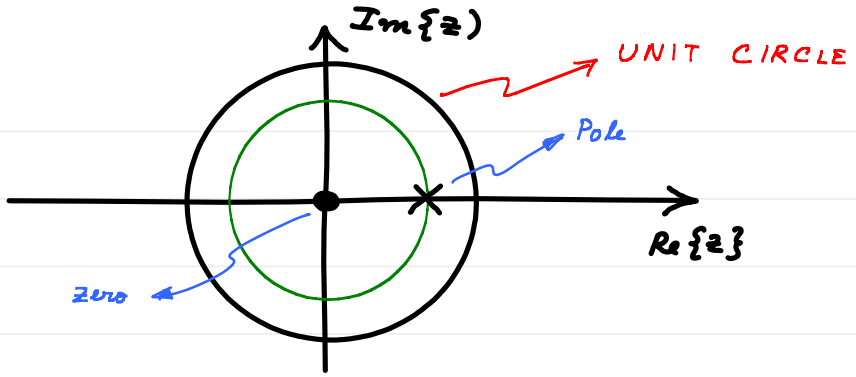
Roots of $B(z)$ are called as the **zeros** of $X(z)$

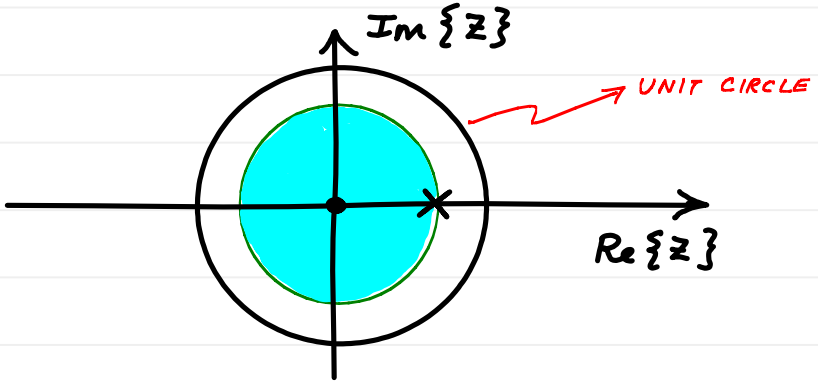
Roots of $A(z)$ are called as the **poles** of $X(z)$

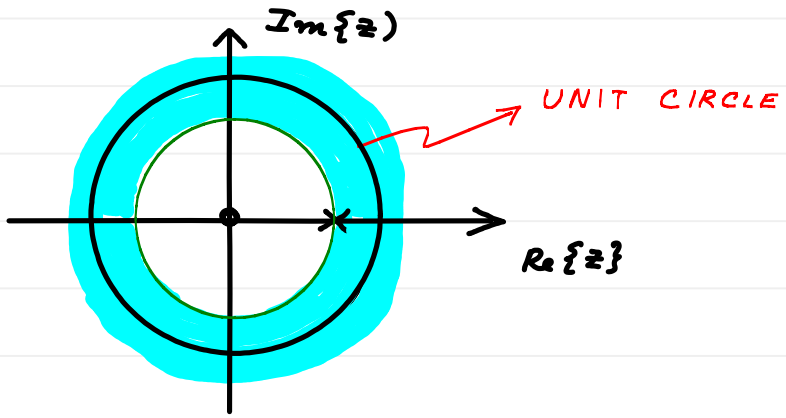
For this example, zeros: $z = 0$

poles: $z = a$

Pole/zero Plot :



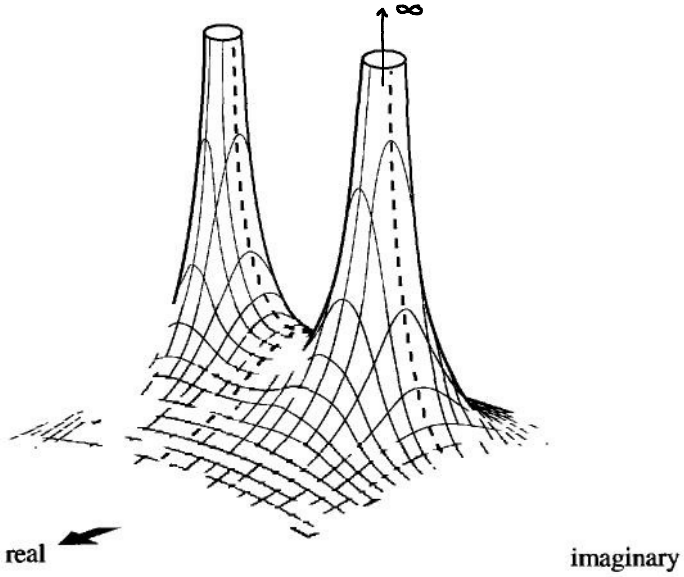




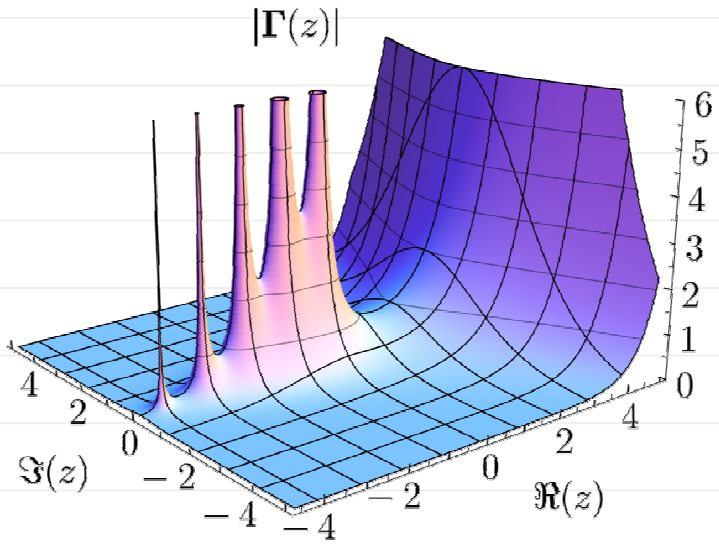
RoC: $|z| < |a| \Rightarrow x[n] = -a^n u[-n-1]$

RoC: $|z| > |a| \Rightarrow x[n] = a^n u[n]$

$|X(z)| \to \infty$ as we approach a pole



From *Visual Complex Analysis* by T. Needham, Oxford University Press, 1999, p. 66.



The absolute value of the Gamma function. This shows the function becomes infinite at the poles at n = -1, -2, -3, ... (Wikipedia, "Pole (complex analysis)")

Example :

$$x[n] = \underbrace{\left(\frac{1}{2}\right)^n u[n]}_{x_1[n]} + \underbrace{\left(\frac{-1}{3}\right)^n u[n]}_{x_2[n]}$$

both sequences are right sided

$$X_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \qquad X_2(z) = \frac{1}{1 + \frac{1}{3}z^{-1}}$$

$$|z| > \frac{1}{2} \qquad |z| > \frac{1}{3}$$

RoC for $X(z)$ is $RoC_{x_1} \cap RoC_{x_2}$, i.e., $|z| > 1/2 \cap |z| > 1/3$

i.e., the final RoC is $|z| > 1/2$

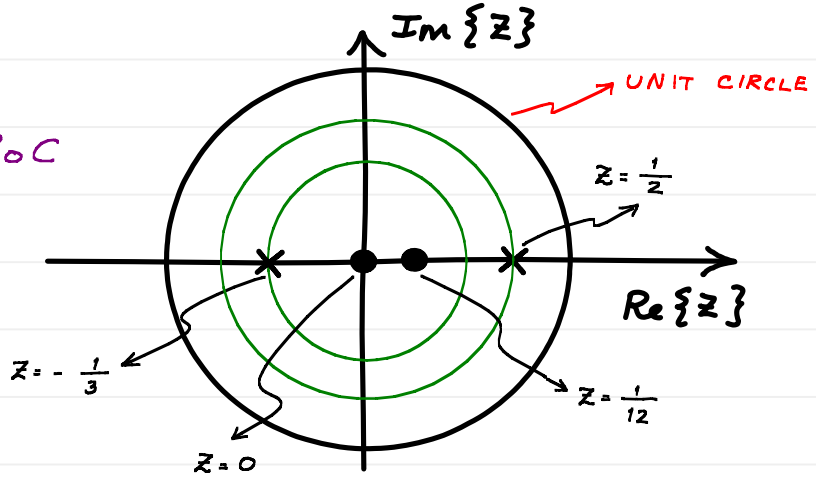
For the given $x[n]$, which is **right-sided**, the RoC is

outside of a certain circle (in this case it is a circle of radius 1/2)

The final expression for $X(z)$ is

$$X(z) = \frac{2\left(1 - \frac{1}{12}z^{-1}\right)}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{3}z^{-1}\right)}$$

RoC: $|z| > \frac{1}{2}$



Poles: $z = 1/2, -1/3$

Zeros: $z = 1/12, 0$

The zero at $z = 0$ will be missed if we look at $X(z)$ superficially

Note:

The expression for $X(z)$ is, **in powers of z**, $X(z) = \dfrac{2z\left(z - \frac{1}{2}\right)}{\left(z - \frac{1}{2}\right)\left(z + \frac{1}{3}\right)}$

In this form, the zero at $z = 0$ is clearly seen.
To calculate the poles and zeros, express $X(z)$ in powers of 'z'
Otherwise, you are likely to miss the poles and zeros at the origin.

Poles and zeros at $z = 0$ are called **TRIVIAL** poles and zeros.

Later when learn about magnitude frequency response we will see that poles and zeros at $z = 0$ do not contribute to the magnitude response. That is why they are called trivial poles/zeros.

Example

$$x[n] = \underbrace{-\left(\tfrac{1}{2}\right)^n u[-n-1]}_{x_1[n]} + \underbrace{\left(-\tfrac{1}{3}\right)^n u[n]}_{x_2[n]}$$

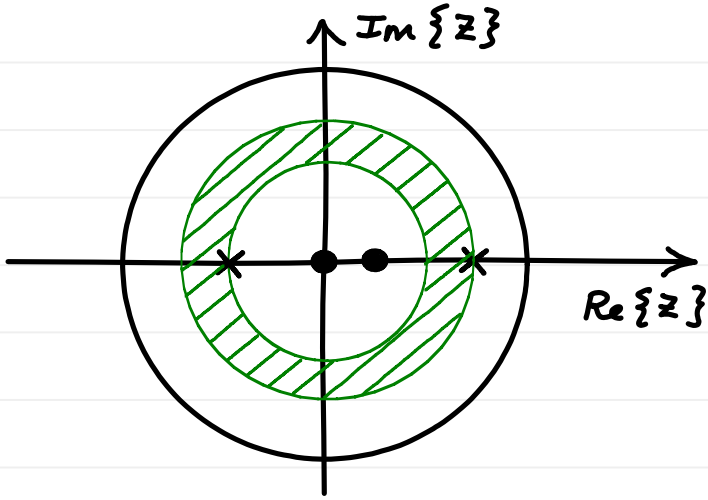
Left-sided seq. | Right sided seq.

Two-sided sequence

$RoC_{x_1}: |z| < 1/2 \qquad RoC_{x_2}: |z| > 1/3$

$RoC_x: RoC_{x_1} \cap RoC_{x_2} \Rightarrow RoC_x$ is the

annular region $1/3 < |z| < 1/2$



Example:

$$x[n] = \underbrace{-\left(\tfrac{1}{2}\right)^n u[-n-1]}_{x_1[n]} - \underbrace{\left(-\tfrac{1}{3}\right)^n u[-n-1]}_{x_2[n]}$$

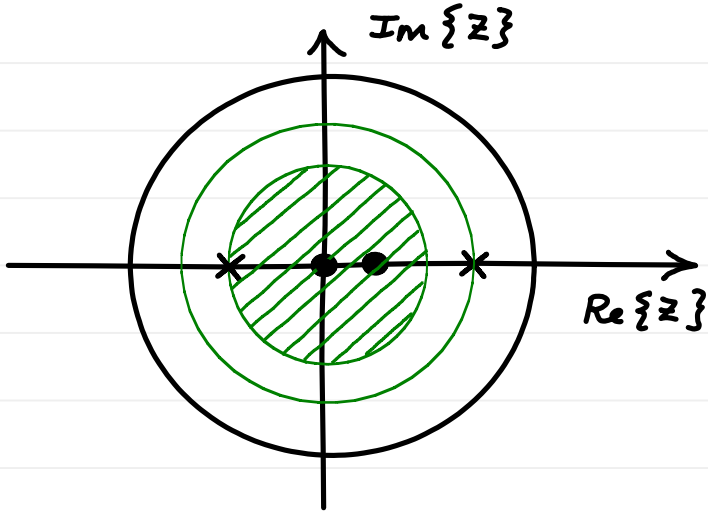
Left-sided seq. — Left-sided seq.

Left-sided sequence

$RoC_{x_1}: |z| < 1/2$ $\qquad$ $RoC_{x_2}: |z| < 1/3$

$RoC_x: RoC_{x_1} \cap RoC_{x_2} \Rightarrow$ RoC is the

circular region $|z| < 1/3$



Example:

$$x[n] = \overbrace{\left(\tfrac{1}{2}\right)^n u[n]}^{x_1[n]} - \overbrace{\left(-\tfrac{1}{3}\right)^n u[-n-1]}^{x_2[n]}$$

$x_1[n]$: Right-sided seq. $\quad$ $x_2[n]$: Left-sided seq.

$RoC_{x_1}: |z| > 1/2 \qquad RoC_{x_2}: |z| < 1/3$

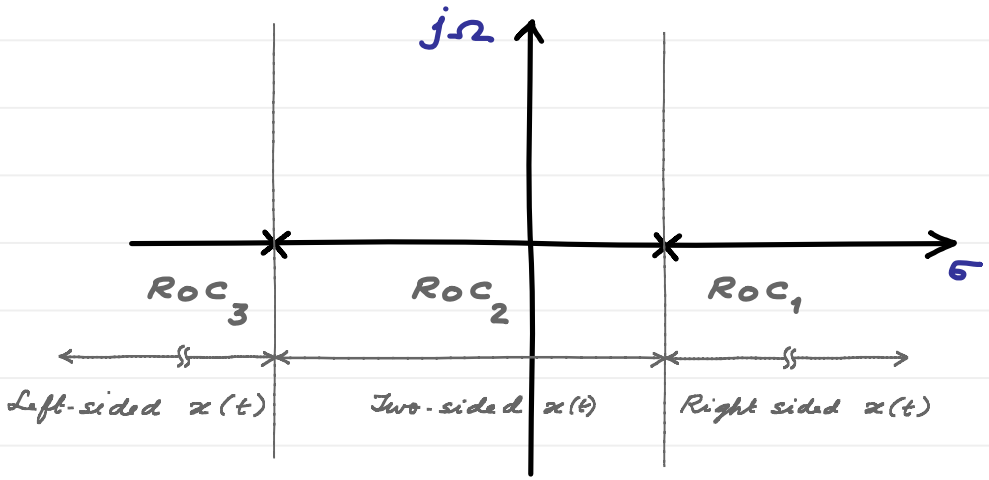
Two sided sequence that does NOT possess a Z-transform

$$RoC_x: RoC_{x_1} \cap RoC_{x_2} = \phi$$

The given two-sided sequence does not possess a z-transform because there is

no common region where both the individual z-transforms converge.

Compare the above with the adjoining s-plane pole-zero plot. The three regions give rise to (i) left-sided, (ii) two-sided, and (iii) right sided time-domain functions.



The behaviour between parallel lines (s-plane) is analogous to that between concentric circles (z-plane).

The UNIT CIRCLE in the z-plane plays the role of the jΩ axis of the s-plane

More on Poles and Zeros

	Zero	Pole
(1) $X(z) = \dfrac{1}{1 - az^{-1}} = \dfrac{z}{z-a}$	$z = 0$	$z = a$
(2) $X(z) = \dfrac{z}{(z-a)(z-b)}$	$z = 0, \infty$ (first order)	$z = a, b$

As $z \to \infty$, $X(z) \simeq \dfrac{1}{z} \to 0$, i.e., there is a zero at $z = \infty$. Hence, apart from the finite-plane zero, there is also a zero at $z = \infty$, which is a simple zero

(3) $X(z) = \dfrac{1}{(z-a)(z-b)}$	$z = \infty$ (second order)	$z = a, b$
(4) $X(z) = \dfrac{(z-a)(z-b)}{z}$	$z = a, b$	$z = 0, \infty$ (first order)

(5) $X(z) = (z-a)(z-b)$ $\quad z = a, b \quad z = \infty$ (second order)

(6) $X(z) = a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}$

$$= \frac{a_0 z^N + a_1 z^{N-1} + \cdots + a_N}{z^N}$$

$z = z_1, z_2, \ldots, z_N$ (finite-plane zeros) $\quad z = 0$ (N^{th} order)

$X(z)$ is said to be of the all-zero type because the only poles are the trivial ones

Note that the corresponding $x[n]$ is $\{\underset{\downarrow}{a_0}, a_1, a_2, \ldots, a_N\}$, which is of finite duration.

(7) $$X(z) = \frac{1}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}}$$

$$= \frac{z^N}{a_0 z^N + a_1 z^{N-1} + \cdots + a_N}$$

$z = 0$ (N^{th} order)

$z = p_1, p_2, \ldots, p_N$ (finite-plane poles)

$X(z)$ is said to be of the all-pole type because the only zeros are the trivial ones

Note that the corresponding $x[n]$ is of infinite duration.

(8) $$X(z) = \frac{p_0 + p_1 z^{-1} + \cdots + p_M z^{-M}}{q_0 + q_1 z^{-1} + \cdots + q_N z^{-N}}$$

$z = z_1, z_2, \ldots, z_M$ (finite-plane zeros)

$z = p_1, p_2, \ldots, p_N$ (finite-plane poles)

Can you say anything about trivial poles or zeros?

(9) $h[n] = \frac{1}{N} \quad 0 \le n \le N-1$

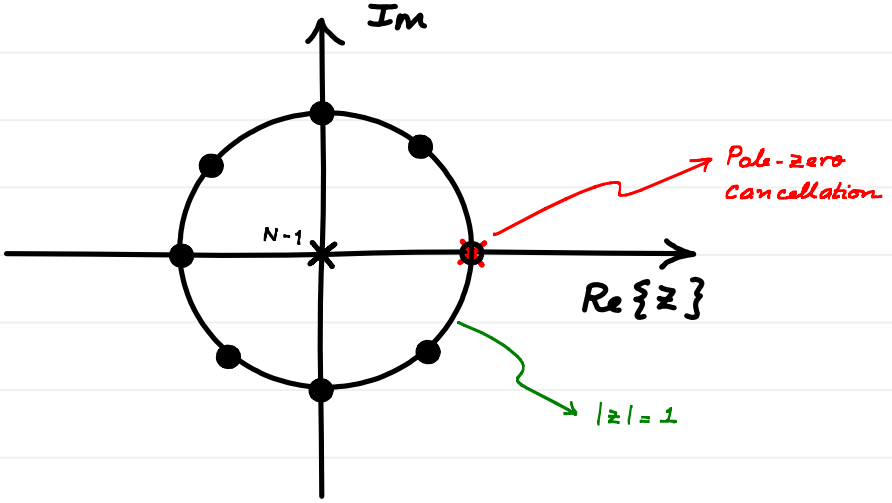
$$H(z) = \frac{1}{N}\left[1 + z^{-1} + z^{-2} + \dots + z^{-(N-1)}\right]$$

$$= \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$= \frac{1}{N} \frac{z^N - 1}{z - 1} \cdot \frac{1}{z^{N-1}}$$

$z = N^{th}$ roots of 1 $\qquad$ $z = 1, 0$ ($N-1$ order)

The pole at $z = 1$ is cancelled by the zero at $z = 1$. There are no uncancelled non-trivial poles. Any uncancelled non-trivial pole will give rise to an exponential (left-sided or right sided).



EE 2004 DSP Lecture 11

Note Title 07-02-2019

$$H(z) = \frac{1}{N} \frac{z^N - 1}{z - 1} \cdot \frac{1}{z^{N-1}}$$

Such a system cannot be FIR. No FIR system can have an uncancelled non-trivial pole. Later we will see that $H(z) = \frac{Y(z)}{X(z)}$ and that

(i) $\frac{Y(z)}{X(z)} = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}} \Rightarrow y[n] = y[n-1] - \frac{1}{N} x[n-N] + \frac{1}{N} x[n]$ Will be shown using the "Delay Property"

(ii) $\frac{Y(z)}{X(z)} = \frac{1}{N} \left[1 + z^{-1} + \dots + z^{-(N-1)} \right] \Rightarrow y[n] = \frac{1}{N} \left(x[n] + x[n-1] + \dots + x[n-N+1] \right)$

These are precisely the recursive and non-recursive implementations of the moving average FIR system considered earlier. The recursive implementation must necessarily

correspond to a system with pole-zero cancellation since it is FIR.

Convergence

Under what conditions does the z-transform converge? The answer depends on the criterion for convergence. We will consider absolute convergence. That is, we want $|X(z)| < \infty$.

$X(z)$ is said to converge absolutely to $X(z)\Big|_{z=z_0} = X(z_0)$ if $|X(z_0)| < \infty$.

Recall that,

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \qquad \text{where } z = r e^{j\omega}$$

$$|X(z)| = \left| \sum_{-\infty}^{\infty} x[n] z^{-n} \right| \leq \sum_{n=-\infty}^{\infty} \left| x[n] r^{-n} e^{-j\omega n} \right| = \sum_{n=-\infty}^{\infty} |x[n]| r^{-n}$$

$$= \underbrace{\sum_{n=0}^{\infty} |x[n]| r^{-n}}_{\text{causal part}} + \underbrace{\sum_{n=1}^{\infty} |x[-n]| r^{n}}_{\text{anti-causal part}}$$

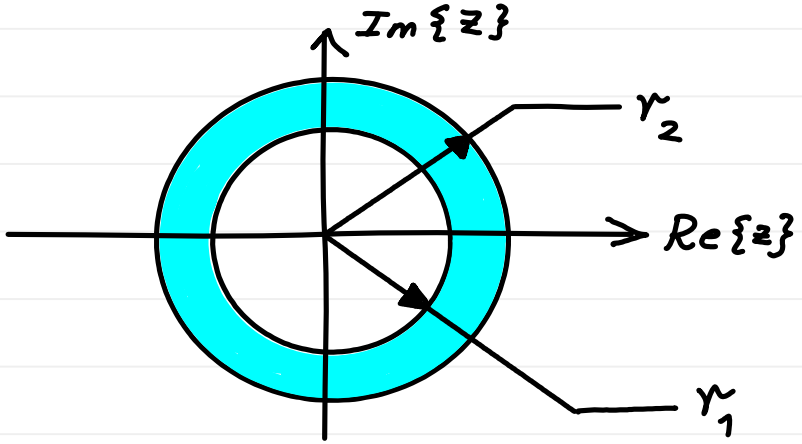
If $\exists r_0$ s.t. the above converges, then the sum converges $\forall r > r_0$ because $r^{-n} < r_0^{-n}$ for $r > r_0$

If $\exists r_0$ s.t. the above converges, then the sum converges $\forall r < r_0$ because $r^{n} < r_0^{n}$ for $r < r_0$

Note that convergence is dictated by the parameter r (radius) alone and does not depend on ω. Thus the convergence region is, in general, an ANNULAR REGION of the form $r_1 < |z| < r_2$

r_1 can be as small as zero

r_2 can be as large as infinity



Note:

If $z_0 = r_0 e^{j\omega_0} \in$ RoC, then $|z_0| \in$ RoC, i.e. if it converges at $\omega = \omega_0$, then it converges for all ω, i.e., $r_0 e^{j\omega} \in$ RoC.

Cauchy - Hadamard: $r_1 = \overline{\lim}_{n\to\infty} |x[n]|^{1/n}$ $\quad r_2 = \underline{\lim}_{n\to\infty} |x[-n]|^{-1/n}$

Not needed for rational $X(z)$; simpler methods exist

Example

$$a^n u[n] \longleftrightarrow \frac{1}{1-az^{-1}} \qquad |z|>|a| \Rightarrow r_1 = |a|, \; r_2 = \infty$$

$$-a^n u[-n-1] \longleftrightarrow \frac{1}{1-az^{-1}} \qquad |z|<|a| \Rightarrow r_1 = 0, \; r_2 = |a|$$

In general, the RoC is of the $r_1 < |z| < r_2$. For sequences with z-transform of the form $\frac{P(z)}{Q(z)}$, the *inequality is strict*.

There are sequences, such as $x[n] = \frac{1}{n^2} u[n-1]$,

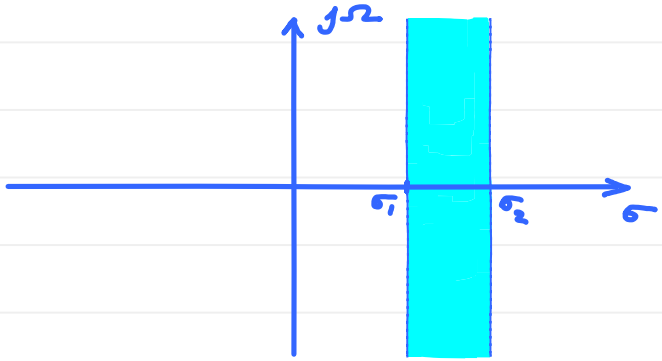
having RoC $|z| \geq 1$, whose z-transform is not of the form $\frac{P(z)}{Q(z)}$,
i.e., not a ratio of polynomials (outside our scope).

The sequence $x[n] = \frac{1}{|n|+1}$ does not possess a z-transform.

The sequence $x[n] = \frac{1}{n^2+1}$ possesses a z-transform that converges only for $|z| = 1$.

We will not explore such sequences in this course.

The RoC in the z-plane is an annular region because the convergence is is governed by 'r', which is the radius.



Consider $x(t) \leftrightarrow X(s)$ where $s = \sigma + j\Omega$

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt = \int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\Omega t}\, dt$$

Hence

$$|X(s)| \leq \int_{-\infty}^{\infty} |x(t)|\, e^{-\sigma t}\, dt$$

The factor governing convergence is σ the real part of 's' and not Ω.
This is why the RoC in the 's'-plane is bounded by vertical lines.

Eg. $e^{-at} u(t) \leftrightarrow \frac{1}{s-a}$ $Re\{s\} > Re\{a\}$ $\quad -e^{at} u(-t) \leftrightarrow \frac{1}{s-a}$ $Re\{s\} < Re\{a\}$

Properties of the RoC for rational $X(z)$

(1) RoC is, in general, of the form $r_1 < |z| < r_2$, where r_1 can be as small as 0 and r_2 can be as large as ∞.

(2) RoC cannot, by definition, contain poles.

(3) If $e^{j\omega} \in$ RoC, the DTFT can be obtained from $X(z)$ by replacing 'z' by '$e^{j\omega}$', i.e.,

$$X(e^{j\omega}) = X(z)\Big|_{z=e^{j\omega}}$$

(4) If $x[n]$ is right-sided, the RoC is a region outside a certain circle.

(5) If $x[n]$ is left-sided, the RoC is a region inside a certain circle.

(6) If $x[n]$ is two-sided, the RoC is a region between two circles, i.e., an annular region.

(7) If $x[n]$ is of finite duration, the RoC is entire z-plane, except possibly for $z = 0$ and/or infinity. ∞.

(8) The RoC has to be a connected region.

EE 2004 DSP Lecture 12

Note Title 11-02-2019

If $z = e^{j\omega}$ (the unit circle) is part of the RoC, then we can substitute $z = e^{j\omega}$ in the expression of $X(z)$. This gives rise to

$$X(z)\Big|_{z=e^{j\omega}} = X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

Discrete-Time Fourier Transform (DTFT)

If the $j\Omega \in$ RoC in the s-plane, the CTFT can be obtained from $X(s)$ by replacing 's' by '$j\Omega$'.

Analogously, if $e^{j\omega} \in$ RoC in the z-plane, the DTFT can be obtained from $X(z)$ by replacing 'z' by '$e^{j\omega}$'.

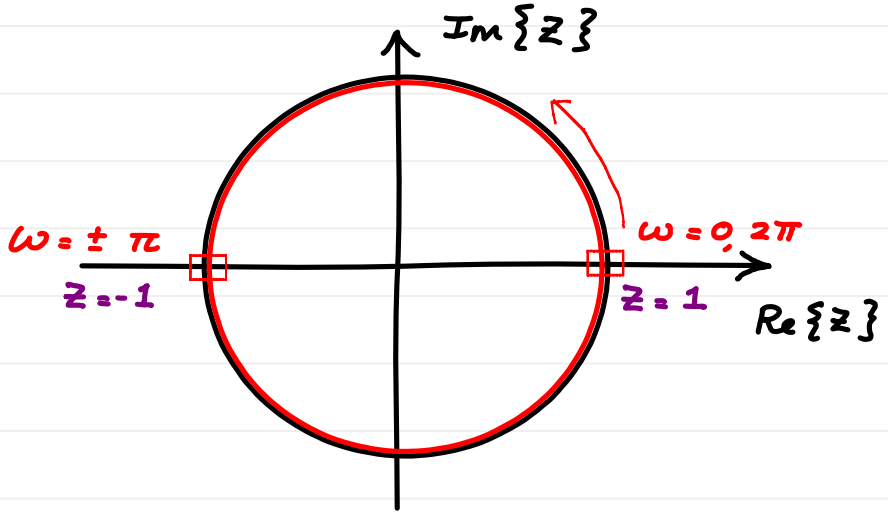
It is easy to see that the DTFT is 2π-periodic:

$$X(e^{j\overline{\omega+2\pi}}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j(\omega+2\pi)n} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

Hence

$$X(e^{j\overline{\omega+2\pi}}) = X(e^{j\omega})$$

That the DTFT is nothing but $X(z)$ evaluated along the unit circle gives another explanation for its 2π-periodic nature.



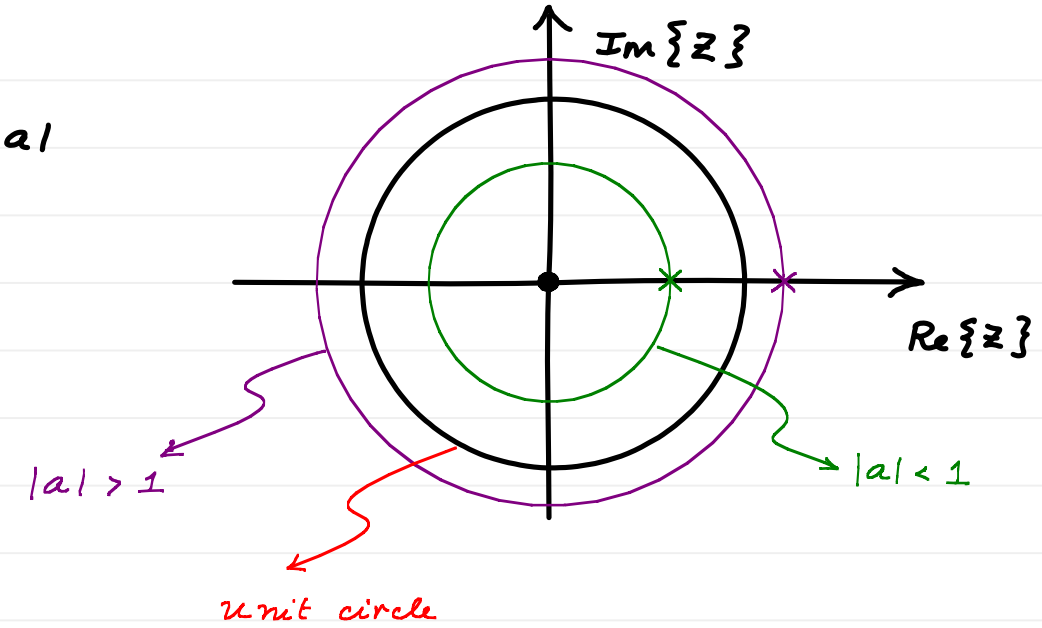
The notation $X(\omega)$ is also sometimes used to denote the DTFT.

In this notation, $X(0)$ denotes the value of the transform at $\omega = 0$. OTOH, in the notation $X(e^{j\omega})$, the value of the transform at $\omega = 0$ is

$$X(e^{j0}) = X(1) = X(z)\Big|_{z=1} .$$

If you are not careful, notation can trip you.

$$x[n] = a^n u[n] \longleftrightarrow \frac{1}{1-az^{-1}}, \qquad |z| > |a|$$



Case 1: $|a| > 1$

The RoC $|z| > |a|$ does not include the unit circle. DTFT does not exist.

Case 2: $|a| < 1$

The RoC $|z| > |a|$ does include the unit circle. DTFT does exist,

i.e.

$$X(e^{j\omega}) = \frac{1}{1-ae^{-j\omega}}$$

Note that $2^n u[-n-1]$ is $\{ \ldots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, \underset{\downarrow}{0}, 0, 0, \ldots \}$

Sum converges, i.e., $X(1)$ = finite $\Rightarrow e^{j\omega} \in$ RoC $\Rightarrow$ DTFT exists

OTOH,

$\left(\frac{1}{2}\right)^n u[-n-1]$ is $\{ \ldots, 8, 4 \quad 2, \underset{\downarrow}{0}, 0, 0, \ldots \}$

Sum diverges, i.e., $X(1) = \infty \Rightarrow e^{j\omega} \notin$ RoC $\Rightarrow$ DTFT does not exist

In general, $a^n u[-n-1]$ **decays** as $n \to -\infty$ if $|a| > 1$. Hence DTFT exists.

OTOH, $a^n u[-n-1]$ **blows up** as $n \to -\infty$ if $|a| < 1$. Hence DTFT does not exist.

Hence, for an anticausal exponential, the pole must lie outside the unit circle for the DTFT to exist. Only then the RoC of $X(z)$, the region inside the circle bounded by the pole, will contain the unit circle. Recall that $z = e^{j\omega}$ must, by definition, belong to the RoC for the DTFT to exist.

If $z = e^{j\omega} \in$ RoC, then the sequence is absolutely summable:

$$|X(e^{j\omega})| \leq \sum_{n=-\infty}^{\infty} |x[n]|\,|e^{-j\omega n}| \;(=1) < \infty$$

i.e., $x[n] \in \ell_1$ (the set of absolutely summable sequences)

Properties of the Z-Transform

(1) Linearity:

If $x_1[n] \longleftrightarrow X_1(z)$ RoC $_1$

$x_2[n] \longleftrightarrow X_2(z)$ RoC $_2$

then $a_1 x_1[n] + a_2 x_2[n] \longleftrightarrow a_1 X_1(z) + a_2 X_2(z)$ RoC $\supseteq$ RoC $_1 \cap$ RoC $_2$

The RoC can be larger if there are pole-zero cancellations

DTFT: $a_1 x_1[n] + a_2 x_2[n] \longleftrightarrow a_1 X_1(e^{j\omega}) + a_2 X_2(e^{j\omega})$

Example

$$x_1[n] = a^n u[n] \longleftrightarrow \frac{1}{1 - az^{-1}} \qquad |z| > |a|$$

$$x_2[n] = a^n u[n-N] \longleftrightarrow \frac{a^N z^{-N}}{1 - az^{-1}} \qquad |z| > |a|$$

$$a^n u[n] - a^n u[n-N] \longleftrightarrow \frac{1 - a^N z^{-N}}{1 - az^{-1}} \qquad \text{Entire } z\text{-plane except } z = 0$$

$$a^n \left(u[n] - u[n-N]\right) = a^n \quad 0 \le n \le N-1 \longleftrightarrow 1 + az^{-1} + a^2 z^{-2} + \dots + a^{-(N-1)} z^{-(N-1)}$$

Thus the RoC is larger than $|z| > |a|$ because the pole at $z = a$ is cancelled by the zero at $z = a$.

Example

Consider $x[n] = a^n \longleftrightarrow X(z) = ?$

Note that $a^n = a^n u[n] + a^n u[-n-1]$

$$a^n u[n] \longleftrightarrow \frac{1}{1-az^{-1}} \qquad |z| > |a|$$

$$a^n u[-n-1] \longleftrightarrow \frac{-1}{1-az^{-1}} \qquad |z| < |a|$$

Since $|z| > |a| \cap |z| < |a| = \phi$, a^n does NOT possess z-transform

This is similar to e^t not possessing (Bilateral) Laplace transform

($Re\{s\} > 1 \cap Re\{s\} < 1 = \phi$)

EE 2004 DSP Lecture 13

Note Title 12-02-2019

(2) Delay Property

If $x[n] \longleftrightarrow X(z)$, then $x[n-n_0] \longleftrightarrow z^{-n_0} X(z)$

RoC is the same except for addition/deletion of 0 and/or ∞.

DTFT:

$$x[n-n_0] \longleftrightarrow e^{-j\omega n_0} X(e^{j\omega})$$

Example

$$a^n u[n-N] = a^N a^{n-N} u[n-N]$$

Hence, since $a^n u[n] \longleftrightarrow \dfrac{1}{1-az^{-1}}$ $\quad |z|>|a|$

$$a^N \overbrace{a^{n-N} u[n-N]}^{\longleftrightarrow} \longleftrightarrow a^N \overbrace{\frac{z^{-N}}{1-az^{-1}}}^{} \qquad |z|>|a|$$

Therefore,

$$a^n u[n-N] \longleftrightarrow \frac{a^N z^{-N}}{1-az^{-1}} \qquad |z|>|a|$$

$$x[n] = 1 \qquad -N \le n \le N$$

$$X(z) = z^{N} + z^{N-1} + \cdots + z + 1 + z^{-1} + \cdots + z^{-N}$$

$$= \frac{z^{N}\left(1 - z^{-2N-1}\right)}{1 - z^{-1}} \qquad 0 < |z| < \infty$$

$$= \frac{z^{N} - z^{-N-1}}{1 - z^{-1}} = \frac{z^{N+1/2} - z^{-N-1/2}}{z^{1/2} - z^{-1/2}}$$

The transform of $y[n] = x[n-N] = 1$ for $0 \le n \le 2N$ is

$$Y(z) = \frac{1 - z^{-(2N+1)}}{1 - z^{-1}} = \frac{z^{-N}\left(z^{N+1/2} - z^{-N-1/2}\right)}{z^{1/2} - z^{-1/2}} \qquad (z \neq 0)$$

$$X(e^{j\omega}) = X(z)\Big|_{z=e^{j\omega}}$$

$$= \frac{e^{j\omega(N+1/2)} - e^{-j\omega(N+1/2)}}{e^{j\omega/2} - e^{-j\omega/2}}$$

$$= \frac{\sin(2N+1)\omega/2}{\sin \omega/2}$$

Dirichlet kernel

[See the command diric in MATLAB]

$$Y(e^{j\omega}) = e^{-j\omega N} \frac{\sin(2N+1)\omega/2}{\sin \omega/2}$$

Later we will encounter the following DTFT pair:

$$\cos \omega_0 n \longleftrightarrow \pi \left[\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right] \quad -\pi \leq \omega < \pi$$

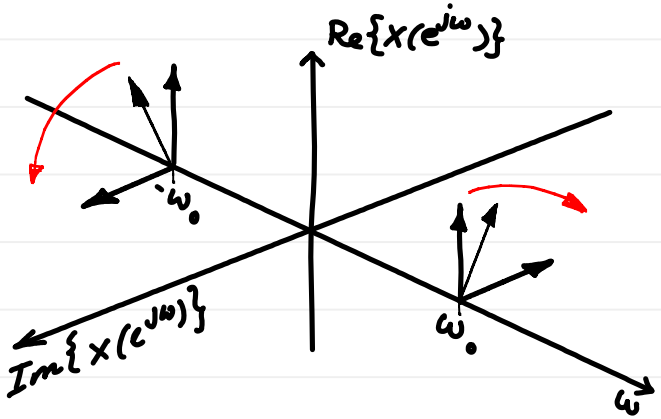
$$\sin \omega_0 n \longleftrightarrow \frac{\pi}{j} \left[\delta(\omega - \omega_0) - \delta(\omega + \omega_0) \right] \quad -\pi \leq \omega < \pi$$

By shifting $\cos \omega_0 n$ we can get $\sin \omega_0 n$. Shifting in the time domain rotates the impulses in the transform domain.

The following plot illustrates this.

$$x[n-n_0] \longleftrightarrow \underbrace{e^{-j\omega n_0}}\ X(e^{j\omega})$$

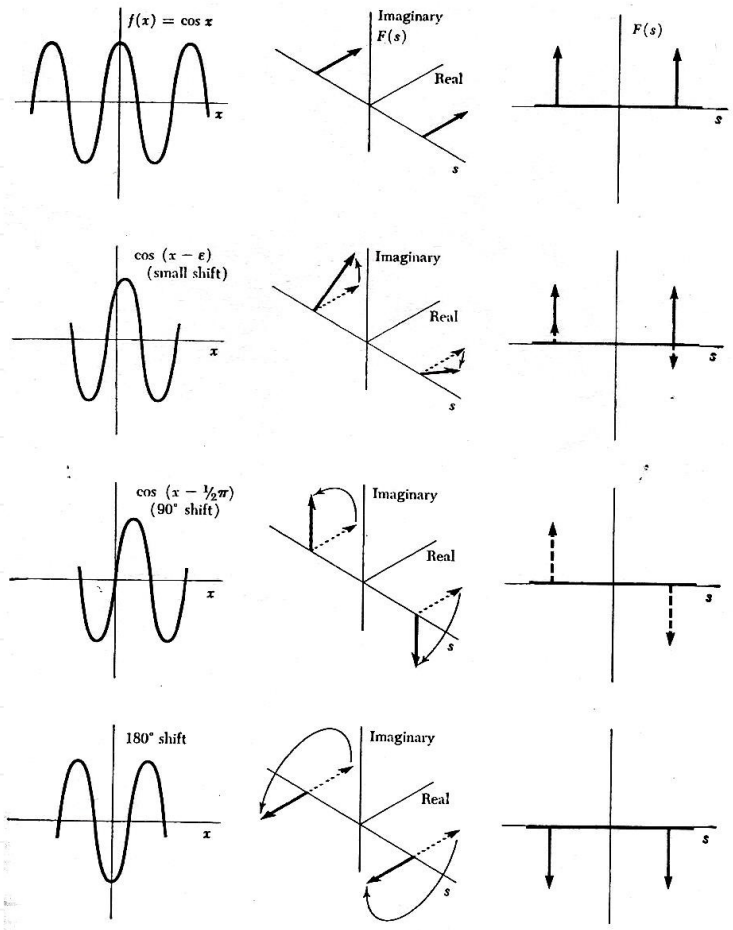
rotates $X(e^{j\omega})$ by an angle ωn_0



$$x(t-t_0) \longleftrightarrow e^{-st_0}X(s)$$

$$x(t-t_0) \longleftrightarrow e^{-j\Omega t_0}X(j\Omega)$$

Has the same geometric interpretation of rotation in the Fourier transform domain



Shifting in the time domain causing a rotation in the transform domain holds good in the CTFT case too. The figure shows how shifting a cosine causes the impulses in the transform domain to rotate.

$\cos \rightarrow \sin \rightarrow -\cos \rightarrow -\sin \rightarrow \cos$

$$x(t - t_0) \leftrightarrow e^{-j\Omega t_0} X(j\Omega)$$

The Fourier Transform and its Applications by R.N. Bracewell

The delay property also shows that associated with every Linear Constant Coefficient Difference Equation (LCCDE) there is a rational transfer function.

$$y[n] = -\sum_{k=1}^{N} a_k\, y[n-k] + \sum_{k=0}^{M} b_k\, x[n-k]$$

Transforming,

$$Y(z) = -\sum_{k=1}^{N} a_k z^{-k} Y(z) + \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$Y(z)\left[1 + \sum_{k=1}^{N} a_k z^{-k}\right] = X(z) \sum_{k=0}^{M} b_k z^{-k}$$

Let $H(z) = Y(z)/X(z)$

Hence

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \frac{B(z)}{A(z)}$$

The above result shows that associated with every LCCDE there is a rational transfer function.

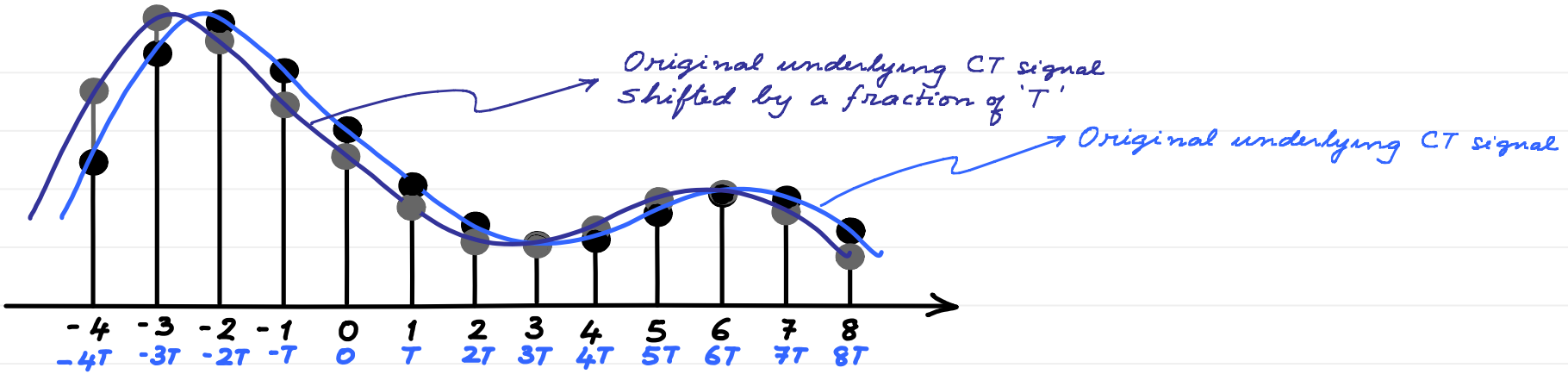
Conversely, associated with every rational transfer function there is a corresponding LCCDE.

Introducing Fractional Delay:

The inverse transform of $e^{-j\omega\alpha}X(e^{j\omega})$ when α is not an integer produces a sequence $y[n]$ that can be thought of as "$x[n-\alpha]$". This is interpreted as follows:

Associated with $x[n]$ is an underlying continuous-time function $x_c(t)$ which can be shifted by any amount, not necessarily by integer multiples of the sampling period. Thus, the new sequence $y[n] =$ "$x[n-\alpha]$" can be thought of as sampling $x_c(t-\alpha)$ at the original sampling instants.

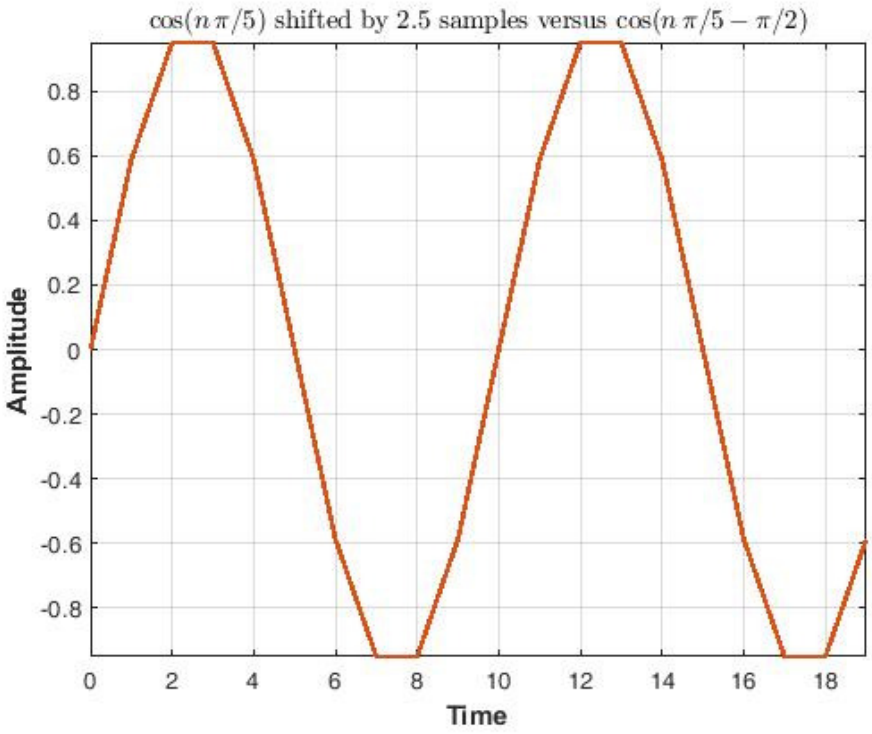
The figure below illustrates this.



The new sequence corresponds to the samples of the **fractionally delayed** [i.e., delayed by (int + fraction) x T] underlying continuous-time signal $x_c(t-\alpha)$.

This delay is imparted by working with the discrete-time sequence only, i.e., there is no need to generate either $x_c(t)$ or $x_c(t-\alpha)$.

```
>> n = (0:19)';
>> x = cos(n*pi/5);
X = fft(x);
>> X1 = X(1:11).*exp(-j*2*pi*2.5*(0:10)'/20);
>> X2 = [X1;conj(flipud(X1(2:end-1)))];
>> y = real(ifft(X2));
>> plot(n,[cos(pi*n/5-pi/2) y]);grid
```



We see that $\cos \frac{n\pi}{5}$ can be shifted by 2.5 samples by multiplying the transform of $\cos \frac{n\pi}{5}$ by $e^{-j2.5\omega}$ and then computing the inverse transform.

EE 2004 DSP Lecture 14

Note Title 14-02-2019

(3) **Exponential Multiplication**

If $x[n] \longleftrightarrow X(z), \quad r_1 < |z| < r_2$

then

$$y[n] = \gamma^n x[n] \longleftrightarrow Y(z) = X(z/\gamma) \qquad |\gamma| r_1 < |z| < |\gamma| r_2$$

Proof:

$$Y(z) = \sum_{n=-\infty}^{\infty} \gamma^n x[n] z^{-n} = \sum_{n=-\infty}^{\infty} x[n] (z/\gamma)^{-n}$$

$$= X(z/\gamma) \qquad r_1 < |z/\gamma| < r_2$$

$$\text{i.e., } |\gamma| r_1 < |z| < |\gamma| r_2$$

If $\gamma = e^{j\omega_0}$, then $e^{j\omega_0 n} x[n] \longleftrightarrow X(z/e^{j\omega_0})$

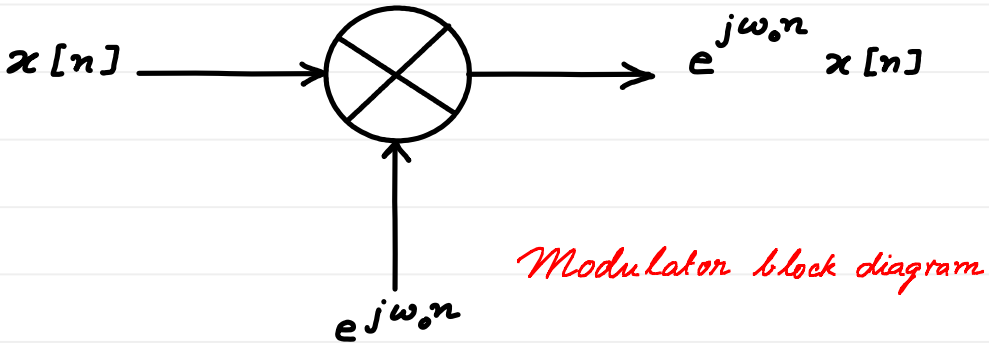
The corresponding DTFT property is: $e^{j\omega_0 n}\, x[n] \leftrightarrow X(e^{j\omega}/e^{j\omega_0}) = X(e^{j\overline{\omega-\omega_0}})$

This is commonly called as the **modulation property**. $\equiv X(\omega-\omega_0)$

If $\omega_0 = \pi$, then $e^{j\pi n} = (-1)^n$

Therefore, $(-1)^n x[n] \overset{z}{\leftrightarrow} X(-z)$

$(-1)^n x[n] \overset{DTFT}{\leftrightarrow} X(\omega-\pi) = X(\omega+\pi)$



Modulator block diagram

Starting from $u[n] \leftrightarrow \frac{1}{1-z^{-1}}$, $|z|>1$, we can derive the transform of $r^n \cos(\omega_0 n)\, u[n]$ as follows:

$$(r e^{j\omega_0})^n u[n] \leftrightarrow \frac{1}{1-(z/re^{j\omega_0})^{-1}} \qquad |z/re^{j\omega_0}|>1$$

$$= \frac{1}{1-re^{j\omega_0}z^{-1}} \qquad |z|>r$$

Similarly, $r^n e^{-j\omega_0 n} u[n] \leftrightarrow \frac{1}{1-re^{-j\omega_0}z^{-1}} \qquad |z|>r$

This leads to: $r^n \cos(\omega_0 n)\, u[n] \leftrightarrow \frac{1-r\cos\omega_0 z^{-1}}{1-2r\cos\omega_0 z^{-1}+r^2 z^{-2}} \qquad |z|>r$

Another consequence of exponential multiplication is the following:

Recall that $y[n] = \gamma^n x[n] \longleftrightarrow Y(z) = X(z/\gamma)$ $\quad |\gamma| r_1 < |z| < |\gamma| r_2$

That is

$$X(z) = Y(\gamma z)$$

Suppose z_0 is a zero of $X(z)$. Then $X(z_0) = 0 = Y(\gamma z_0)$

Thus γz_0 is a zero of $Y(z)$. In general, assuming $X(z) = \frac{P(z)}{Q(z)}$

$Y(z) = X(z/\gamma) = \frac{P(z/\gamma)}{Q(z/\gamma)}$, i.e., both poles and zeros get multiplied by a factor of $\gamma \Rightarrow$ scaling & rotation

(4) **Complex Conjugation**:

If $x[n] \leftrightarrow X(z)$, $r_1 < |z| < r_2$, then $x^*[n] \leftrightarrow X^*(z^*)$ with same RoC.

Proof:

$$\sum_{n=-\infty}^{\infty} x^*[n] z^{-n} = \left[\sum_{n=-\infty}^{\infty} x[n] (z^*)^{-n} \right]^*$$

$$= X^*(z^*) \qquad r_1 < |z^*| < r_2$$

$$= X^*(z^*) \qquad r_1 < |z| < r_2$$

The corresponding DTFT property is: $x^*[n] \leftrightarrow X^*(e^{-j\omega})$ $\left[= X^*(z^*)\Big|_{z=e^{j\omega}} \right]$

If $x[n] \in \mathbb{R}$, then $x[n] = x^*[n] \Rightarrow X(z) = X^*(z^*)$

The above leads to the following inference:

Suppose z_0 is a zero of $X(z)$. That is, $X(z_0) = 0 = X^*(z_0^*)$

$$\text{i.e., } X^*(z_0^*) = 0$$

$$\Rightarrow \quad X(z_0^*) = 0$$

From the above we see that z_0^* is also a zero. The same is true for poles as well:

$$X(z) = \frac{P(z)}{Q(z)} = X^*(z^*) = \frac{P^*(z^*)}{Q^*(z^*)}$$

Hence if q_0 is a root of $Q(z)$, then q_0^* is also a root because $Q(z) = Q^*(z^*)$

For a real-valued sequence, poles and zeros occur in complex conjugate pairs.

The corresponding DTFT property implies certain symmetries:

Recall $x[n] = x^*[n] \Rightarrow X(e^{j\omega}) = X^*(e^{-j\omega})$. Since $X(e^{j\omega}) = X_R(e^{j\omega}) + j\, X_I(e^{j\omega})$

$X^*(e^{-j\omega}) = X_R(e^{-j\omega}) - j\, X_I(e^{-j\omega})$. Hence $X_R(e^{j\omega}) + j\, X_I(e^{j\omega}) = X_R(e^{-j\omega}) - j\, X_I(e^{-j\omega})$

If $x[n] \in \mathbb{R}$, then $X_R(e^{j\omega}) = X_R(e^{-j\omega})$, i.e., it is even symmetric.
$X_I(e^{j\omega}) = -X_I(e^{-j\omega})$, i.e., it is odd symmetric. Therefore it follows that

$|X(e^{j\omega})| = \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})}$ is an even symmetric function

$\angle X(e^{j\omega}) = \tan^{-1}\dfrac{X_I(e^{j\omega})}{X_R(e^{j\omega})}$ is an odd symmetric function

Because of the above symmetry properties, the information about $X(e^{j\omega})$ in the range $[0, \pi]$ is enough to recover completely the corresponding inverse transform whenever the sequence is real-valued.

5) Time Reversal

If $x[n] \longleftrightarrow X(z)$ $\quad r_1 < |z| < r_2$

then $x[-n] \longleftrightarrow X(z^{-1})$ $\quad \dfrac{1}{r_2} < |z| < \dfrac{1}{r_1}$

This operation causes a causal sequence to become non-causal and vice-versa

Proof:

$$\sum_{n=-\infty}^{\infty} x[-n] z^{-n} = \sum_{\ell=\infty}^{-\infty} x[\ell] z^{\ell}$$

$$= \sum_{\ell=-\infty}^{\infty} x[\ell] (z^{-1})^{-\ell}$$

$$= X(z^{-1}) \qquad r_1 < |z^{-1}| < r_2 \Rightarrow \frac{1}{r_2} < |z| < \frac{1}{r_1}$$

Thus $x[-n] \leftrightarrow X(z^{-1})$ with RoC $\frac{1}{r_2} < |z| < \frac{1}{r_1}$

The above result can be used to derive the transform of $-a^n u[-n-1]$ starting from the transform of $a^n u[n]$.

Recall the pair $a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}} \quad |z| > |a|$

Hence $a^{-n}u[-n] \longleftrightarrow \frac{1}{1-az} \quad |z| < \frac{1}{|a|}$

$= \frac{-a^{-1}z^{-1}}{1-a^{-1}z^{-1}} \quad |z| < |a^{-1}|$ Multiplying numerator and denominator by $-a^{-1}z^{-1}$

Applying the delay property we get,

$$a^{-n-1}u[-n-1] \longleftrightarrow \frac{-a^{-1}}{1-a^{-1}z^{-1}} \quad |z| < |a^{-1}|$$

i.e., $$-a^{-n}u[-n-1] \longleftrightarrow \frac{1}{1-a^{-1}z^{-1}} \quad |z| < |a^{-1}|$$

Let $b = a^{-1}$, we get $-b^{n}u[-n-1] \longleftrightarrow \frac{1}{1-bz^{-1}} \quad |z| < |b|$ as before.

EE 2004 DSP Lecture 15

Note Title 18-02-2019

(6) Differentiation in the z-domain

If $x[n] \leftrightarrow X(z)$ $\quad r_1 < |z| < r_2$

then $n\,x[n] \leftrightarrow -z \frac{d}{dz} X(z)$ with same RoC, assuming rational transform

Proof: $X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$

$$\frac{d}{dz} X(z) = \frac{d}{dz} \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] (-n) z^{-n-1}$$ This is allowed because the series converges absolutely

Hence, $$-z \frac{d}{dz} X(z) = \sum_{n=-\infty}^{\infty} n\,x[n] z^{-n}$$

Hence,

$$n\,x[n] \longleftrightarrow -z\frac{d}{dz}X(z) \qquad r_1 < |z| < r_2$$ [assuming $X(z)$ is of the form $P(z)/Q(z)$]

Since $X(z)$ is analytic in the RoC, it can be differentiated any number of times. Thus, the above property can be applied repeatedly.

Example:

$$a^n u[n] \longleftrightarrow \frac{1}{1-az^{-1}} \qquad |z| > |a|$$

$$-z\frac{d}{dz}\frac{1}{1-az^{-1}} = \frac{(-z)(-1)(-a)(-z^{-2})}{(1-az^{-1})^2} = \frac{az^{-1}}{(1-az^{-1})^2} \qquad |z| > |a|$$

$$\therefore \quad n a^n u[n] \longleftrightarrow \frac{az^{-1}}{(1-az^{-1})^2}$$

$$(n+1)a^{n+1}u[n+1] \longleftrightarrow \frac{a}{(1-az^{-1})^2}$$

$$(n+1)a^{n}u[n+1] \longleftrightarrow \frac{1}{(1-az^{-1})^2} \qquad |z|>|a|$$

$$(n+1)a^{n}u[n] \longleftrightarrow \frac{1}{(1-az^{-1})^2} \qquad |z|>|a|$$

Exercise :

$$? \longleftrightarrow \frac{1}{(1-az^{-1})^2} \qquad |z|<|a|$$

At what index does the first non-zero sample begin ?

$$? \longleftrightarrow \frac{1}{(1-az^{-1})^M}$$

(i) $|z|>|a|$

(ii) $|z|<|a|$

At what index does the first non-zero sample begin ?

Corresponding DTFT property: $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

$$\frac{d}{d\omega} X(e^{j\omega}) \stackrel{?}{=} \sum_{n=-\infty}^{\infty} (-jn)\, x[n] e^{-j\omega n}$$

Hence, $n\,x[n] \longleftrightarrow j \frac{d}{d\omega} X(e^{j\omega})$

Exercise:

Starting from $n\,x[n] \longleftrightarrow -z \frac{d}{dz} X(z)$ derive the corresponding DTFT property by substituting $z = e^{j\omega}$.

(7) **Convolution in the time-domain:**

$$x[n] * h[n] \longleftrightarrow X(z) \cdot H(z) \qquad RoC \supseteq RoC_x \cap RoC_h$$

Proof:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\,z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x[k]\,h[n-k] \right] z^{-n}$$

$$\overset{?}{=} \sum_{k=-\infty}^{\infty} x[k] \sum_{n=-\infty}^{\infty} h[n-k]\,z^{-n}$$

$$= \underbrace{\sum_{k=-\infty}^{\infty} x[k]\,z^{-k}}_{X(z)} H(z)$$

$$= X(z)\,H(z) \qquad \text{RoC} \supseteq \text{RoC}_x \cap \text{RoC}_y$$

If $x[n] = a^n u[n]$

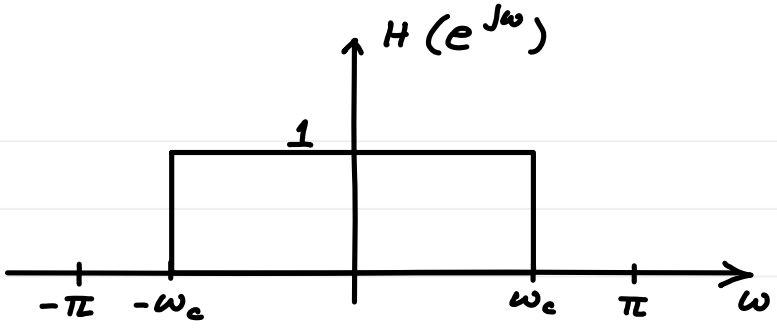
and $y[n] = b^n u[n]$

then $x[n] * y[n] \longleftrightarrow \dfrac{1}{(1-az^{-1})(1-bz^{-1})}$ $\quad |z| > \max\{|a|, |b|\}$

The corresponding DTFT property is: $x[n] * h[n] \longleftrightarrow X(e^{j\omega}) H(e^{j\omega})$

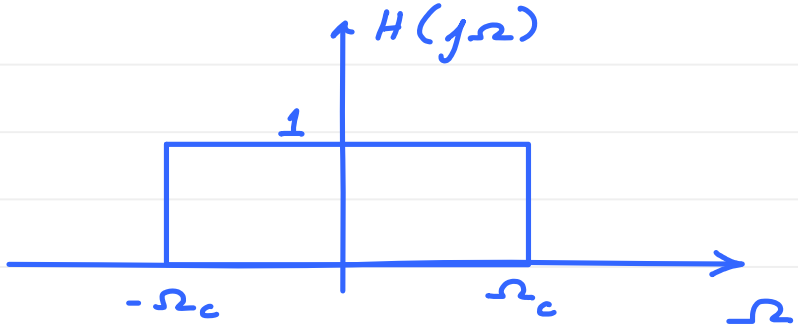
This property forms the basis for *frequency domain filtering*.

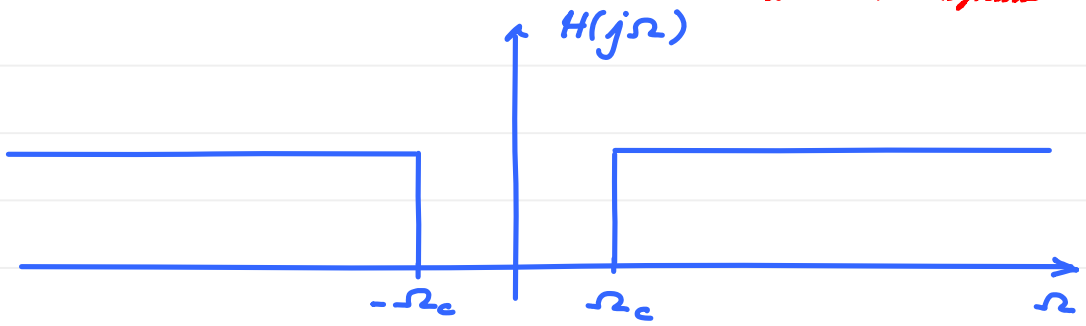
If $H(e^{j\omega}) = 0$ for $|\omega| > \omega_c$, then $Y(e^{j\omega}) = 0$ for $|\omega| > \omega_c$. Thus, components outside $[-\omega_c, \omega_c]$ in $x[n]$ are *filtered out*. Similar remarks apply for highpass, bandpass, and bandstop filters.



lowpass region: $|\omega| < \omega_c$ highpass region: $\omega_c < |\omega| < \boxed{\pi}$

→ Highest frequency for discrete-time signals





lowpass region: $|\Omega| < \Omega_c$ highpass region: $\Omega_c < |\Omega| < \boxed{\infty}$

→ Highest frequency for continuous-time signals

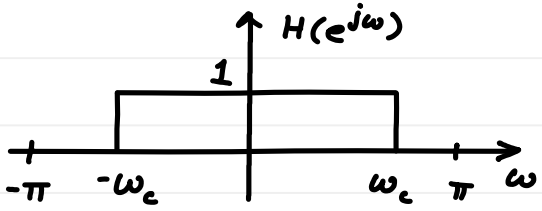
(8) If $z = 1 \in RoC$, then $\left. X(z) \right|_{z=1} = \sum_{n=-\infty}^{\infty} x[n]$

The corresponding DTFT property is: $\sum_{n=-\infty}^{\infty} x[n] = \left. X(e^{j\omega}) \right|_{\omega=0} = X(1)$

which will be the same as $X(0)$ in the $X(\omega)$ notation.

Later we will encounter the pair

$$h[n] = \frac{\sin(\omega_c n)}{\pi n} \longleftrightarrow$$



Does $|z| = 1$ belong to the RoC ?

This implies, $\sum_{n=-\infty}^{\infty} \frac{\sin(\omega_c n)}{\pi n} = \left. H(e^{j\omega}) \right|_{\omega=0} = 1$

which is difficult to show without using the above property.

(9) **Initial Value Theorem**

Let $x[n] = 0$ for $n < 0$. Then

$$X(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots$$

$$\lim_{z \to \infty} X(z) = x[0]$$

If $x[n] = 0$ for $n < 1$, then $X(z) = x[1]z^{-1} + x[2]z^{-2} + \dots$

$$\lim_{z \to \infty} z X(z) = \lim_{z \to \infty} x[1] + x[2]z^{-1} + \dots = x[1]$$ and so on.

EE 2004 DSP Lecture 16

Note Title 19-02-2019

10) **Final Value Theorem:**

Let $x[n] = 0$ for $n < M$. Then $\lim_{z \to 1} (1 - z^{-1}) X(z) = x[\infty]$

Proof:

Let $v[n] = x[n] - x[n-1] \Rightarrow V(z) = (1 - z^{-1}) X(z)$

$$V(z) = \sum_{n=-\infty}^{\infty} v[n] z^{-n} = \sum_{n=M}^{\infty} (x[n] - x[n-1]) z^{-n}$$

$$\lim_{z \to 1} V(z) = \lim_{z \to 1} \sum_{n=M}^{\infty} (x[n] - x[n-1]) z^{-n}$$

$$\stackrel{?}{=} \sum_{n=M}^{\infty} (x[n] - x[n-1]) \lim_{z \to 1} z^{-n}$$

$$= \lim_{N \to \infty} \sum_{n=M}^{N} (x[n] - x[n-1])$$

$$= \lim_{N \to \infty} \Big(x[M] - x[M-1] + x[M+1] - x[M] + x[M+2] - x[M+1] + \cdots + x[N-1] - x[N-2] + x[N] - x[N-1] \Big)$$

$$= \lim_{N \to \infty} x[N]$$

$$= x[\infty]$$

Since $V(z) = (1 - z^{-1}) X(z)$,

$$\lim_{z \to 1} (1 - z^{-1}) X(z) = x[\infty]$$

Example:

$$x[n] = u[n] \longleftrightarrow \frac{1}{1-z^{-1}} \qquad |z|>1$$

$$\lim_{z\to 1} (1-z^{-1})X(z) = \lim_{z\to 1} \frac{1-z^{-1}}{1-z^{-1}} = 1 = u[\infty]$$

However,

$$(-1)^n u[n] \longleftrightarrow \frac{1}{1+z^{-1}} \qquad |z|>1$$

and applying the above theorem yields $\lim_{z\to 1} (1-z^{-1}) \frac{1}{1+z^{-1}} = 0$, which is an incorrect result

For the theorem to hold, both limits must exist. If so, the theorem establishes their equality.

11) **Multiplication in the time-domain :**

Let $x[n] \leftrightarrow X(\omega)$ and $y[n] \leftrightarrow Y(\omega)$

[Note: $X(\omega)$ notation is used in this property, rather than $X(e^{j\omega})$ because of convenience]

Then

$$x[n] \cdot y[n] \leftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta)\, Y(\omega-\theta)\, d\theta$$

$$\equiv \frac{1}{2\pi} X(\omega) \circledast Y(\omega)$$

$\circledast$ denotes CIRCULAR CONVOLUTION

The corresponding z-domain property is :

$$x[n]\, y[n] \leftrightarrow \frac{1}{2\pi j} \oint X(\gamma)\, Y(z/\gamma)\, \frac{d\gamma}{\gamma}$$

Proof of this property requires knowledge of the inversion integral and will be provided later.

12) **Parseval's Theorem**:

$$\sum_{n=-\infty}^{\infty} x[n] \cdot y^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) Y^*(\omega)\, d\omega$$

Proof:

Recall the following two properties: (i) $\sum_{n=-\infty}^{\infty} x[n] = X(\omega)\Big|_{\omega=0}$

(ii) $x[n] \cdot y[n] \longleftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta)\, Y(\omega-\theta)\, d\theta$

Therefore,

$$\sum_{n=-\infty}^{\infty} x[n] \cdot y^*[n] = \frac{1}{2\pi} \left[X(\omega) \circledast Y^*(-\omega) \right]\Big|_{\omega=0}$$

$$\frac{1}{2\pi}\left[X(\omega) \circledast Y^*(-\omega)\right]\Big|_{\omega=0} = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)\, Y^*(-\overline{\omega-\theta})\, d\theta \Bigg|_{\omega=0}$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)\, Y^*(\theta)\, d\theta$$

Thus,

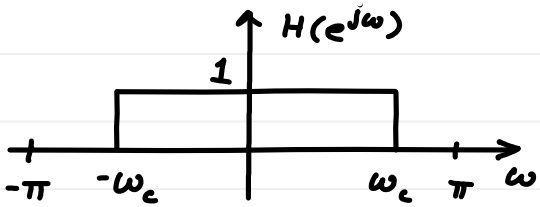
$$\sum_{n=-\infty}^{\infty} x[n]\, y^*[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)\, Y^*(\theta)\, d\theta$$

In particular, if $y[n] = x[n]$, we get

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\omega)|^2\, d\omega$$

Parseval's theorem helps us to compute sums such as $\sum_{n=-\infty}^{\infty}\left[\frac{\sin \omega_c n}{\pi n}\right]^2$, which is difficult to do in the time domain.

Recall the transform pair $h[n] = \frac{\sin(\omega_c n)}{\pi n} \leftrightarrow$



Hence

$$\sum_{n=-\infty}^{\infty}\left[\frac{\sin \omega_c n}{\pi n}\right]^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi}|H(e^{j\omega})|^2 d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} d\omega = \frac{\omega_c}{\pi}$$

The theorem shows that the DTFT is a *norm-preserving transform* because

$$\sum_{n=-\infty}^{\infty}|x[n]|^2 \equiv \|x\|_2^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi}|X(e^{j\omega})|^2 d\omega \equiv \|X\|_2^2$$, i.e., it is a UNITARY transform.

EE 2004 DSP Lecture 17

Note Title 25-02-2019

Inverse Z-transform

We will only consider Z-transforms that are **rational**, i.e., of the form $\frac{P(z)}{Q(z)}$.

Recall that a system whose i/p - o/p relationship is in the form of an **LCCDE**, has a transfer function that is **rational**, i.e., $H(z) = \frac{B(z)}{A(z)}$

Let $X(z) = \frac{P(z)}{Q(z)} = \frac{p_0 + p_1 z^{-1} + \cdots + p_M z^{-M}}{1 + q_1 z^{-1} + \cdots + q_N z^{-N}}$ ← If the leading coefficient is not 1, it can always be normalized

If the first $r-1$ terms are zero in $P(z)$, z^{-r} can be factored out. That is, $X(z) = z^{-r} \frac{P_1(z)}{Q(z)}$

The inverse transforms of $\frac{P(z)}{Q(z)}$ and $\frac{P_1(z)}{Q(z)}$ differ only by an r-sample delay.

Hence we will consider $X(z)$ to be of the form $\frac{P(z)}{Q(z)}$ without loss of generality.

1) Partial Fraction Expansion:

$$X(z) = \frac{P(z)}{Q(z)} = \frac{p_0 + p_1 z^{-1} + \cdots + p_M z^{-M}}{1 + q_1 z^{-1} + \cdots + q_N z^{-N}}$$

(a) $M < N$, simple roots

Let $Q(z) = \prod_{k=1}^{N} (1 - q_k z^{-1})$.

Then $\frac{P(z)}{Q(z)} = \sum_{k=1}^{N} \frac{A_k}{1 - q_k z^{-1}}$ → A_k Residue

where $A_k = \left. \frac{P(z)}{\prod_{\substack{l=1 \\ l \neq k}}^{N} (1 - q_l z^{-1})} \right|_{z = q_k}$

Lookup the residuez command in MATLAB

Example:

$$X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = \frac{1}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})}$$

$$= \frac{A_1}{1 - z^{-1}} + \frac{A_2}{1 - \frac{1}{2}z^{-1}}$$

$$A_1 = \left.\frac{1}{(1 - \frac{1}{2}z^{-1})}\right|_{z=1} = 2 \qquad A_2 = \left.\frac{1}{(1 - z^{-1})}\right|_{z=1/2} = -1$$

Hence $$\frac{1}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})} = \frac{2}{1 - z^{-1}} + \frac{-1}{1 - \frac{1}{2}z^{-1}}$$

We cannot proceed further unless we have RoC information.

Three different RoC's are possible: (1) $|z| < 1/2$ (2) $1/2 < |z| < 1$ (3) $|z| > 1$

Left-sided Two-sided Right-sided

(1) $x[n] = -2u[-n-1] + (1/2)^n u[-n-1]$

(2) $x[n] = -2u[-n-1] - (1/2)^n u[n]$

(3) $x[n] = 2u[n] - (1/2)^n u[n]$

The final answer depends on which particular RoC has been specified.

(b) $M \geq N$, simple roots

We must divide to get quotient and remainder. The previous method can be applied to the remainder term.

Thus $X(z)$ is transformed to the form

$$X(z) = \sum_{r=0}^{M-N} c_r z^{-r} + \underbrace{\sum_{k=1}^{N} \frac{A_k}{1 - q_k z^{-1}}}_{\text{num. degree } M' < N}$$

The first terms inverse transform is $\sum_{r=0}^{M-N} c_r \delta[n-r]$.

Example

$$X(z) = \frac{1+2z^{-1}+z^{-2}}{1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}} = 2 + \underbrace{\frac{-1+5z^{-1}}{1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}}}_{\text{simple roots, } M<N} = 2 + \frac{-9}{1-\frac{1}{2}z^{-1}} + \frac{8}{1-z^{-1}}$$

As before, the final answer depends on the RoC (of the 2nd and 3rd terms).

(i) $|z|>1$: $2\delta[n] - 9(1/2)^n u[n] + 8u[n]$

(ii) $\frac{1}{2} < |z| < 1$: $2\delta[n] - 9(1/2)^n u[n] - 8u[-n-1]$

(iii) $|z| < \frac{1}{2}$: $2\delta[n] + 9(1/2)^n u[-n-1] - 8u[-n-1]$

(c) $M<N$, repeated roots

$$X(z) = \frac{G\prod_{i=1}^{M}(1-a_i z^{-1})}{\prod_{q=1}^{Q}(1-b_q z^{-1})\prod_{l=1}^{R}(1-\gamma_l z^{-1})^{\sigma_l}}$$

where $M < N = Q + \sum_{l=1}^{R}\sigma_l$

It can be decomposed as

$$X(z) = \sum_{q=1}^{Q} \frac{A_q}{1 - b_q z^{-1}} + \sum_{l=1}^{R} \sum_{k=1}^{\sigma_l} \frac{C_{l,k}}{(1 - \gamma_l z^{-1})^k}$$

γ_l = root

σ_l = multiplicity

The residues are:

$$A_q = X(z)(1 - b_q z^{-1}) \Big|_{z = b_q}$$

$$C_{l,k} = \frac{1}{(-\gamma_l)^{\sigma_l - k} (\sigma_l - k)!} \frac{d^{\sigma_l - k}}{d\xi^{\sigma_l - k}} \left[X(\xi^{-1})(1 - \gamma_l \xi)^{\sigma_l} \right] \Bigg|_{\xi = \frac{1}{\gamma_l}} \qquad k = 1, 2, \ldots, \sigma_l$$

Example

$$X(z) = \frac{12 - 22z^{-1} + 16z^{-2}}{(1-2z^{-1})^3} \qquad R = 1, \quad \sigma_1 = 3$$

$$C_{1,3} = \frac{1}{(-2)^0 0!} \frac{d^0}{d\xi^0} \left[\frac{12 - 22\xi + 16\xi^2}{(1-2\xi)^3} (1-2\xi)^3 \right] \Bigg|_{\xi = 1/2} = 5$$

$$C_{1,2} = \frac{1}{(-2)^1 1!} \frac{d}{d\xi} \left[\frac{12 - 22\xi + 16\xi^2}{(1-2\xi)^3} (1-2\xi)^3 \right] \Bigg|_{\xi = 1/2} = 3$$

$$C_{1,3} = \frac{1}{(-2)^2 2!} \frac{d^2}{d\xi^2} \left[\frac{12 - 22\xi + 16\xi^2}{(1-2\xi)^3} (1-2\xi)^3 \right] \Bigg|_{\xi = 1/2} = 4$$

As before, we need RoC information to proceed further.

You will need results of the form

$$\frac{(n+1)(n+2)\cdots(n+M-1)}{(M-1)!}\, a^n u[n] \leftrightarrow \frac{1}{(1-az^{-1})^M} \qquad |z|>|a|$$

What is the corresponding expression if RoC is $|z|<|a|$?

EE 2004 DSP Lecture 18

Note Title 26-02-2019

2) Power Series Method

(i) $X(z) = z^2\left(1-\frac{1}{2}z^{-1}\right)\left(1+z^{-1}\right)\left(1-z^{-1}\right) = z^2 - \frac{1}{2}z - 1 + \frac{1}{2}z^{-1}$

$\Rightarrow x[n] = \left\{1, -\frac{1}{2}, \underset{\downarrow}{-1}, \frac{1}{2}\right\}$

(ii) $X(z) = \frac{1}{1-az^{-1}} \quad |z| > |a|$

$= 1 + az^{-1} + a^2z^{-2} + \dots$

$\longleftrightarrow \{0, 0, \underset{\downarrow}{1}, a, a^2, \dots\}$

$$
\begin{array}{r|l}
1-az^{-1} & 1 + az^{-1} + a^2z^{-2} + a^3z^{-3} + \cdots \\
\hline
 & 1 \\
 & 1 - az^{-1} \\
\hline
 & az^{-1} \\
 & az^{-1} - a^2z^{-2} \\
\hline
 & a^2z^{-2} \\
 & a^2z^{-2} - a^3z^{-3} \\
\hline
 & \vdots
\end{array}
$$

(iii) $X(z) = \dfrac{1}{1-az^{-1}} \qquad |z| < |a|$

$= \dfrac{-a^{-1}z}{1-a^{-1}z}$

$= -a^{-1}z\,(1 + a^{-1}z + a^{-2}z^2 + \dots)$

$= -a^{-1}z - a^{-2}z^2 - a^{-3}z^3 + \dots$

$\longleftrightarrow \{\dots, -a^{-3}, -a^{-2}, -a^{-1}, \underset{\downarrow}{0}, 0, 0, 0, \dots\}$

$-az^{-1}+1 \;\overline{)\;1}$

$-a^{-1}z - a^{-2}z^2 - a^{-3}z^3$

$1 - a^{-1}z$

$a^{-1}z$

$a^{-1}z - a^{-2}z^2$

a^2z^2

$a^2z^2 - a^{-3}z^3$

$\vdots$

(iv) $X(z) = \dfrac{1+2z^{-1}}{1-2z^{-1}+z^{-2}} \qquad |z| < 1$

$z^{-2} - 2z^{-1} + 1 \;\overline{)\;2z + 5z^2 + 8z^3 + \dots}$

$2z^{-1} + 1$

$2z^{-1} - 4 + 2z$

$5 - 2z$

$5 - 10z + 5z^2$

$8z - 5z^2$

Hence, $x[n] = \{\dots, 8, 5, 2, \underset{\downarrow}{0}, 0, 0, \dots\}$

An alternate method is use to previously discussed ideas:

Let $X_1(z) = \dfrac{1}{(1 - a z^{-1})^2} \quad \underset{|z|<1}{\longleftrightarrow} \quad -(n+1)\,u[-n-2]$

$X_2(z) = 1 + 2z^{-1} \longleftrightarrow \delta[n] + 2\,\delta[n-1]$

Hence,

$$x[n] = -(n+1)\,u[-n-2] * (\delta[n] + 2\,\delta[n-1])$$

$$= -(n+1)\,u[-n-2] - 2n\,u[-n-1]$$

$$= -(n+1)\,u[-n-1] - 2n\,u[-n-1]$$

$$= -(3n+1)\,u[-n-1] = \{\ldots, 8, 5, 2, \underset{\downarrow}{0}, 0, 0, \ldots\}$$, as before

(v) $X(z) = \dfrac{1-a^2}{1+a^2-a(z+z^{-1})}$

If the RoC is $|a| < |z| < \dfrac{1}{|a|}$, then $x[n] = a^{|n|}$, which is a **two-sided** sequence.

$a^{|n|} = a^n u[n] + a^{-n} u[-n-1]$

OTOH, if the RoC is $|z| > \dfrac{1}{|a|}$, then the sequence is **right-sided**; for $|z| < |a|$, the sequence is **left-sided**.

Using long-division, we can get either the left-sided sequence or the right-sided one (depending on whether the quotient is in powers of z or z^{-1}. However the method of long-division cannot give a two-sided inverse z-transform

To get a two-sided sequence, we must proceed as follows:

$$X(z) = \frac{1-a^2}{1+a^2-a(z+z^{-1})} = \underbrace{\frac{1}{1-az^{-1}}}_{\text{expand this in -ve powers of } z} + \underbrace{\frac{az}{1-az}}_{\text{expand this in +ve powers of } z}$$

The first term gives the causal part, whereas the second term gives the anti-causal part.

$$X(z) = 1 + az^{-1} + a^2z^{-2} + \dots \qquad \text{causal part}$$

$$+ az\,(1 + az + a^2z^2 + \dots) \qquad \text{anti-causal part}$$

$$= 1 + az^{-1} + a^2z^{-2} + \dots$$

$$+ az + a^2z^2 + a^3z^3 + \dots$$

$$\leftrightarrow \{\dots, a^3, a^2, a, \underset{\downarrow}{1}, a, a^2, a^3, \dots\}$$

Note: Both terms can be expanded in +ve powers of z or in negative powers of z. If we do so, the final answer will be the same as what would result if $X(z)$ is expanded in positive or negative powers of z.

The advantage of doing so after breaking $X(z)$ into two simple terms is that we may be able to see the individual patterns that may help us in writing the closed-form expression for the n^{th} term.

(vi) $X(z) = e^{z} = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots \qquad |z| < \infty$

By inspection,

$$x[n] = \{ \ldots, \frac{1}{3!}, \frac{1}{2!}, \frac{1}{1!}, \underset{\downarrow}{1}, 0, 0, 0, \ldots \}$$

The above shows that the power series method works even when $X(z)$ is **not rational**.

<u>Exercise</u>: Using the power series expansion of $\ln(1+x)$, find the inverse z-transform of $\ln(1+az^{-1})$ where $|z| > |a|$. Verify the answer with that obtained using the differentiation method.

3) Contour Integral Method

Given $X(z)$, the **inversion formula** to get $x[n]$ is:

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} \, dz \qquad \text{where } C \text{ is a closed curve in the RoC}$$

$$= \sum [\text{residues of } X(z)z^{n-1} \text{ evaluated at poles encircled by } C]$$

If there is an m^{th}-order pole at $z = z_0$, $X(z)$ can be written as $\frac{\Gamma(z)}{(z-z_0)^m}$. The residue is given by

$$\frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \Gamma(z) \Big|_{z=z_0}$$

How does this compare with the residue formula in the partial fraction method?

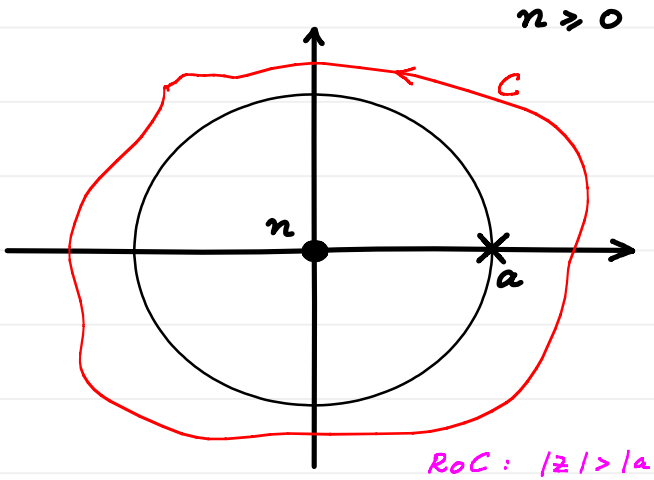
EE 2004 DSP Lecture 19

Note Title 28-02-2019

Example

Let $X(z) = \frac{1}{1 - az^{-1}}$ $\quad |z| > |a|$

$$x[n] = \frac{1}{2\pi j} \oint_C \frac{z^{n-1}}{1 - az^{-1}} \, dz = \frac{1}{2\pi j} \oint_C \frac{z^n}{z - a} \, dz$$

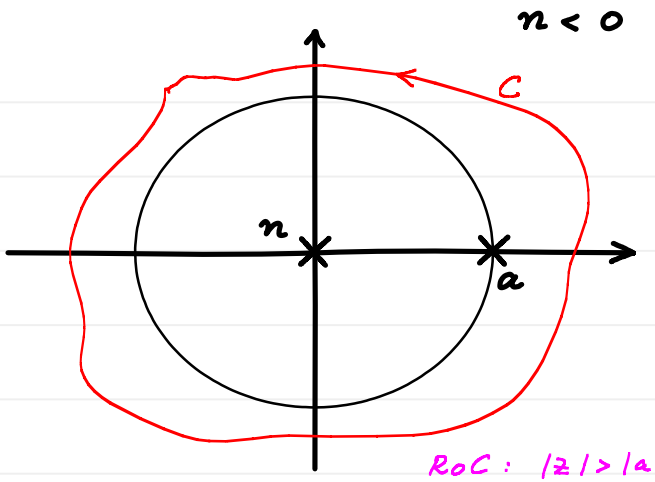


The pole-zero plot for $n \geq 0$ is shown above. The contour 'C' is a closed curve in the region of convergence.

The only pole encircled by C is at $z = a$. Residue at $z = a$ is :

$$(z - a) \frac{z^n}{z - a} \Bigg|_{z=a} = a^n$$

For $n<0$, the pole-zero plot changes, because the trivial n^{th} order zero now becomes a trivial n^{th} order pole. Hence, residues have to be evaluated at $z=a$ (simple pole) and at $z=0$ (n^{th} order)



Hence,

$$X(z)\,z^{n-1} = \frac{1}{z^n(z-a)} \quad [\text{where } n \geq 1]$$

$$\text{Residue at } z=a : \left.\frac{(z-a)}{z^n(z-a)}\right|_{z=a} = \frac{1}{a^n}$$

$$\text{Residue at } z=0 : \left.\frac{1}{(n-1)!}\,\frac{d^{n-1}}{dz^{n-1}}\,\frac{1}{z-a}\right|_{z=0} = \frac{1}{(n-1)!}\,\frac{(-1)^{n-1}(n-1)!}{(-a)^n} = \frac{-1}{a^n}$$

Hence, for $n<0$, sum of residues $= \frac{1}{a^n} - \frac{1}{a^n} = 0$. Therefore, $x[n] = \begin{cases} a^n & n \geq 0 \\ 0 & n<0 \end{cases}$

i.e., $a^n u[n]$, which is what it should be.

To verify the inversion integral does indeed give back $x[n]$, we proceed as follows:

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz = \frac{1}{2\pi j} \oint_C \left[\sum_{k=-\infty}^{\infty} x[k] z^{-k} \right] z^{n-1} dz$$

$$= \sum_{k=-\infty}^{\infty} x[k] \frac{1}{2\pi j} \oint_C z^{n-1-k} dz$$

But $$\frac{1}{2\pi j}\oint_C z^n\,dz = \begin{cases} 1 & n=-1 \\ 0 & \text{otherwise} \end{cases}$$

Hence the only non-zero term in the summation is for $k=n$. Therefore

$$\sum_{k=-\infty}^{\infty} x[k]\,\frac{1}{2\pi j}\oint_C z^{n-1-k}\,dz = x[n]$$

Thus

$$\frac{1}{2\pi j}\oint_C X(z)\,z^{n-1}\,dz = x[n]$$

as claimed.

From the z-transform inversion formula, we can derive the corresponding formula for DTFT inversion.

Let $z = e^{j\omega} \in$ RoC and the closed curve C be the unit circle.

$$z = e^{j\omega} \Rightarrow dz = j\underbrace{e^{j\omega}}_{z} d\omega \quad \text{i.e.,} \quad \frac{dz}{jz} = d\omega$$

Hence,
$$x[n] = \frac{1}{2\pi} \oint_C X(z) z^n \frac{dz}{jz}$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

since 'C' is the unit circle and the integration is w.r.t. 'ω'

Thus we have the following pair:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

Inverse DTFT

The DTFT/IDTFT pair of equations are not new but merely represent the Fourier Series expansion of the 2π-periodic frequency domain function $X(e^{j\omega})$.

Recall the property $x[n] \cdot y[n] = \frac{1}{2\pi j} \oint X(\gamma) Y(z/\gamma) \frac{d\gamma}{\gamma}$. This can proved using the inversion integral.

Proof:

$$p[n] = x[n] \cdot y[n] \quad \leftrightarrow \quad P(z) = \sum_{n=-\infty}^{\infty} p[n] \cdot z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] \cdot y[n] \, z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \left[\frac{1}{2\pi j} \oint X(\gamma) \gamma^{n-1} d\gamma \right] y[n] z^{-n}$$

$$= \frac{1}{2\pi j} \oint X(\gamma) \left[\sum_{n=-\infty}^{\infty} y[n] \left(\frac{z}{\gamma} \right)^{-n} \right] \frac{d\gamma}{\gamma}$$

$$= \frac{1}{2\pi j} \oint X(\gamma)\, Y(z/\gamma)\, \frac{d\gamma}{\gamma}$$

Exercise: What is the RoC ? Hint: It must be valid for both $X(\gamma)$ and $Y(z/\gamma)$.

The DTFT inversion formula can be used to obtain transform pairs that are difficult to obtain directly.

Let $X(e^{j\omega}) = 2\pi\, \delta(\omega) \qquad -\pi \le \omega < \pi$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} [2\pi\, \delta(\omega)]\, e^{j\omega n}\, d\omega = 1 \quad \forall n$$

Thus we have the pair $1 \leftrightarrow 2\pi\delta(\omega)$ $[-\pi \le \omega < \pi]$. Note that $x[n] = 1\ \forall n$ does not possess a z-transform. Thus absolute summability is sufficient but not necessary for the existence of the DTFT.

The above is similar to $1 \overset{CTFT}{\longleftrightarrow} 2\pi\delta(\Omega)$; note that $x(t) = 1$ does not possess bilateral Laplace transform.

Since $X(e^{j\omega})$ is periodic, $x[n] = 1 \leftrightarrow 2\pi\delta(\omega)$ $[-\pi \le \omega < \pi]$ is periodic in the frequency domain. If we remove the restriction that the transform expression is limited to the interval $-\pi \le \omega < \pi$, then $\delta(\omega)$ must be replaced by $\tilde{\delta}(\omega)$, its periodic counterpart.

$$\tilde{\delta}(\omega) = \sum_{k=-\infty}^{\infty} \delta(\omega + 2\pi k)$$

1
n
DTFT
$2\pi\,\tilde{\delta}(\omega)$
-2π
$-\pi$
0
π
2π
ω

EE 2004 DSP Lecture 20

Note Title 05-03-2019

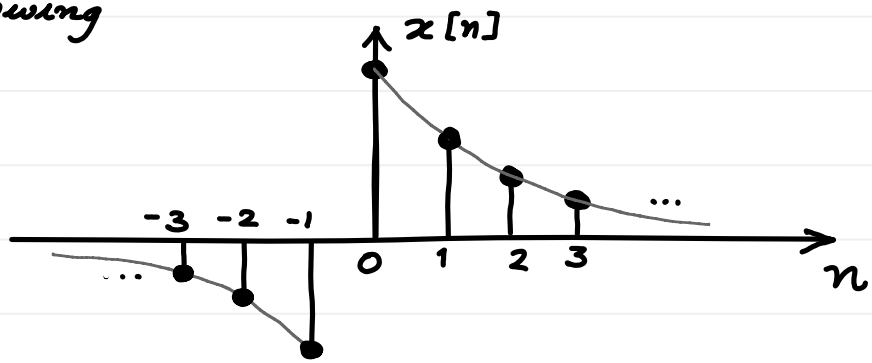
Using the modulation property, we get $e^{j\omega_0 n} \leftrightarrow 2\pi\,\delta(\omega-\omega_0)$, whence

$$\cos(\omega_0 n) \leftrightarrow \pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right] \qquad \sin(\omega_0 n) \leftrightarrow \frac{\pi}{j}\left[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\right]$$

To get the transform of $u[n]$, we consider the following sequence:

$$x[n] = a^n u[n] - \bar{a}^n u[-n-1]$$

$$\leftrightarrow \frac{1}{1-a\bar{e}^{j\omega}} + \frac{1}{1-a^{-1}\bar{e}^{j\omega}} \qquad |a|<1$$



$\lim_{a \to 1} x[n] = \text{sgn}[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$. Hence, $\lim_{a \to 1} X(e^{j\omega}) = \frac{2}{1 - e^{-j\omega}}$

We thus have the pair, $\text{sgn}[n] \overset{\text{DTFT}}{\longleftrightarrow} \frac{2}{1 - e^{-j\omega}}$

$\text{sgn}[n]$ and $u[n]$ are related as follows: $u[n] = \frac{1}{2}(1 + \text{sgn}[n])$. Hence,

$$U(e^{j\omega}) = \frac{1}{2}\left[2\pi\delta(\omega) + \frac{2}{1 - e^{-j\omega}}\right] \qquad -\pi \leq \omega < \pi$$

Therefore,

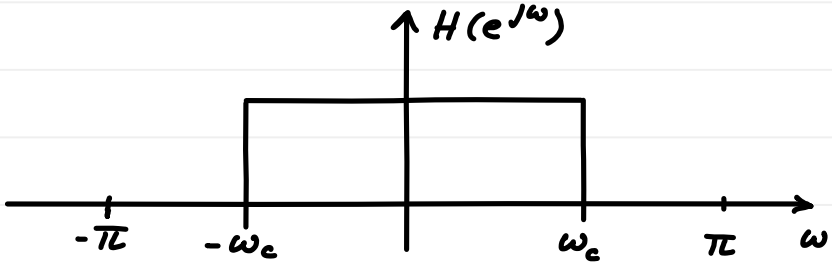
$$u[n] \longleftrightarrow \pi\delta(\omega) + \frac{1}{1 - e^{-j\omega}}$$

The above is reminiscent of: $1 \longleftrightarrow 2\pi\delta(\Omega)$ $\qquad u(t) \longleftrightarrow \pi\delta(\Omega) + \frac{1}{j\Omega}$

Ideal Discrete-Time Lowpass Filter:

$$H_{LP}(e^{j\omega}) = \begin{cases} 1 & |\omega| \le \omega_c \\ 0 & \text{otherwise} \end{cases}$$

↑ in the range $(-\pi, \pi)$

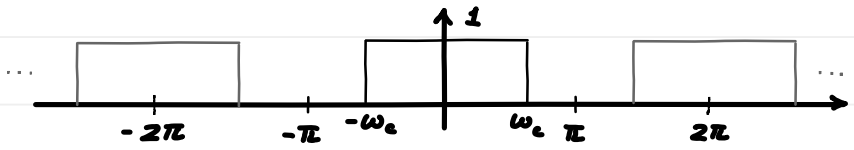


$$h_{LP}[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_{LP}(e^{j\omega})\, e^{j\omega n}\, d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega n}\, d\omega$$

$$= \frac{\sin \omega_c n}{\pi n}$$

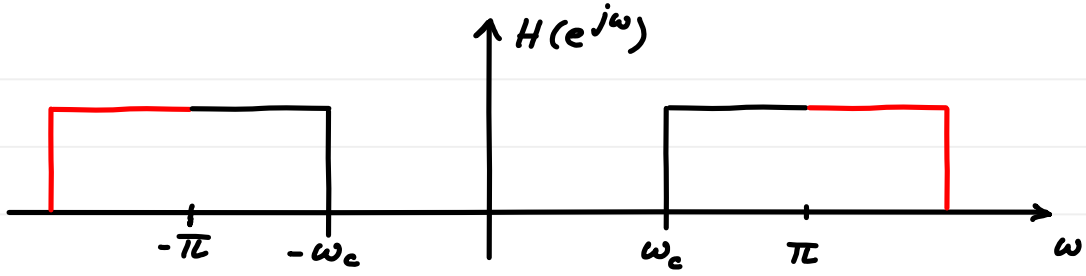
Thus,

$$\frac{\sin \omega_c n}{\pi n} \longleftrightarrow$$



Ideal Highpass Filter:

$$H_{HP}(e^{j\omega}) = 1 - H_{LP}(e^{j\omega})$$



Hence

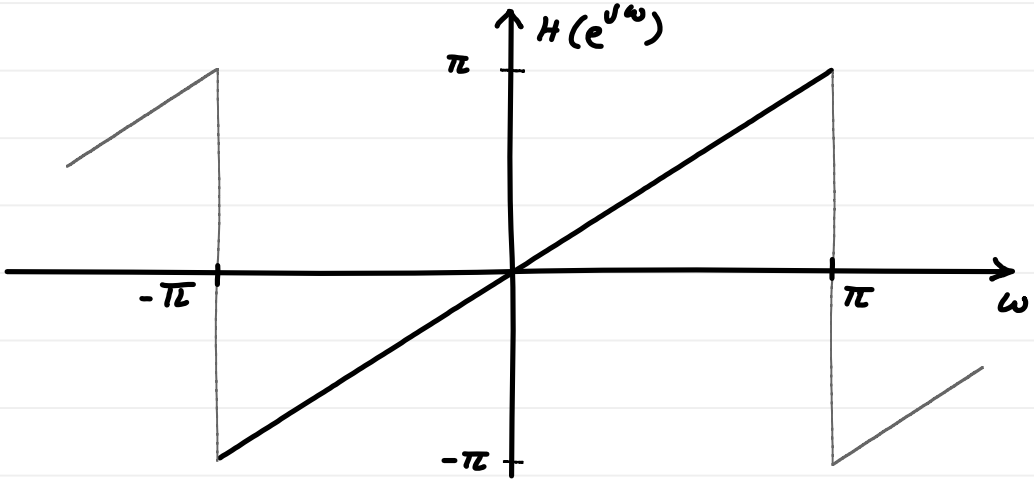
$$h_{HP}[n] = \delta[n] - \frac{\sin \omega_c n}{\pi n}$$

Exercise:

Consider the ideal differentiator with

$H(e^{j\omega}) = j\omega \qquad -\pi \leq \omega < \pi$

Find the corresponding $h[n]$.



$$x[n] = \frac{A}{2} e^{j\phi} e^{j\omega_0 n} + \frac{A}{2} e^{-j\phi} e^{-j\omega_0 n}$$

Suppose $h[n] \in \mathbb{R}$. Then $H^*(e^{j\omega}) = H(e^{-j\omega})$. Recall that this implies **even symmetry** for the **magnitude** response and **odd symmetry** for the **phase** response. We also know that **complex exponentials** are **eigensignals**. Hence

$$\frac{A}{2} e^{j\phi} e^{j\omega_0 n} \longrightarrow \frac{A}{2} e^{j\phi} e^{j\omega_0 n} H(e^{j\omega_0}) = \frac{A}{2} e^{j\phi} e^{j\omega_0 n} \left|H(e^{j\omega_0})\right| e^{j \angle H(e^{j\omega_0})}$$

Similarly,

$$\frac{A}{2} e^{-j\phi} e^{-j\omega_0 n} \longrightarrow \frac{A}{2} e^{-j\phi} e^{-j\omega_0 n} H(e^{-j\omega_0}) = \frac{A}{2} e^{-j\phi} e^{-j\omega_0 n} \left|H(e^{-j\omega_0})\right| e^{j \angle H(e^{-j\omega_0})}$$

$$= \frac{A}{2} e^{-j\phi} e^{-j\omega_0 n} \left|H(e^{j\omega_0})\right| e^{-j \angle H(e^{j\omega_0})}$$ because of symmetry properties

Hence the combined response is

$$\frac{A}{2} e^{j\phi} e^{j\omega_0 n} \left|H(e^{j\omega_0})\right| e^{j \angle H(e^{j\omega_0})} + \frac{A}{2} e^{-j\phi} e^{-j\omega_0 n} \left|H(e^{j\omega_0})\right| e^{-j \angle H(e^{j\omega_0})}$$

$$= A \left|H(e^{j\omega_0})\right| \cos[\omega_0 n + \phi + \angle H(e^{j\omega_0})]$$

i.e., the magnitude is scaled by $|H(e^{j\omega_0})|$ and the phase shifted by $\angle H(e^{j\omega_0})$.

EE 2004 DSP Lecture 21

Note Title 07-03-2019

When is a causal system stable ?

Assuming $H(z)$ is rational, if $h[n]$ is causal, then the RoC is of the form $|z| > r_{max}$ where r_{max} is the radius of the outermost pole.

If the system is stable, then $e^{jw} \in$ RoC because $h[n] \in l_1$

Thus, if we want both causality and stability, then it is to see that we need $r_{max} < 1$.

We know that a simple pole at $z = p_k$ will give rise to an exponential.

For causal systems, the corresponding exponential is :

$$\frac{A_k}{1-p_k z^{-1}} \quad \leftrightarrow \quad A_k \, (p_k)^n \, u[n]$$

If p_k is the outermost pole, then $\sum_{n=0}^{\infty} |h[n]| < \infty$ iff $|p_k| < 1$

$\Rightarrow$ **all poles must lie inside the unit circle**

If p_k is not a simple pole, $\frac{(n+1)(n+2)\cdots(n+M-1)}{(M-1)!} \, p_k^n \, u[n] \leftrightarrow \frac{1}{(1-p_k z^{-1})^M}$ $\quad |z| > |p_k|$

Since $\sum_{n=0}^{\infty} n^l |p_k|^n < \infty$ iff $|p_k| < 1$ for any 'l', we once again see that all poles must strictly lie inside the unit circle for the system to be Stable.

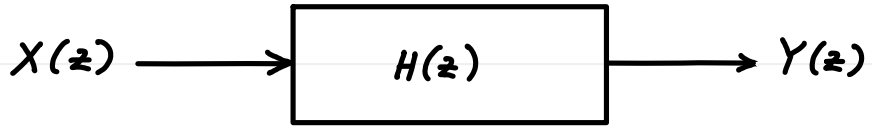
The above is analogous to the conditions required for causal CT systems to be stable. For a causal system with impulse response $h(t)$, the corresponding $H(s)$ has RoC to the right of a certain vertical line. If the system is also stable, then $j\Omega \in$ RoC.

Hence, if we want both causality and stability, then all the poles must lie strictly in the left-half plane. Only then will $h(t)$ be absolutely integrable.

What is the condition that has to be met for an anti-causal signal to be stable?

Response to suddenly applied inputs:

Suppose $x[n]$ is applied to an LTI system at $n = 0$. What can be said about the nature of the output $y[n]$, in general?



Let $X(z)$ be of the form $X(z) = \frac{P(z)}{Q(z)}$ and $H(z) = \frac{B(z)}{A(z)}$. For illustrative purposes, assume simple poles throughout.

$$Y(z) = X(z) \cdot H(z) = \frac{P(z)\,B(z)}{Q(z)\,A(z)}$$

$$= \frac{P(z) \cdot B(z)}{\underbrace{\prod_{k=1}^{N_a} (1 - \xi_k z^{-1})}_{Q(z)} \underbrace{\prod_{k=1}^{N_b} (1 - p_k z^{-1})}_{A(z)}}$$

Using partial fraction expansion,

$$Y(z) = \sum_{k=1}^{N_a} \frac{C_k}{1 - \xi_k z^{-1}} + \sum_{k=1}^{N_b} \frac{D_k}{1 - p_k z^{-1}}$$

Assuming causality

$$y[n] = \underbrace{\sum_{k=1}^{N_a} C_k (\xi_k)^n u[n]}_{\text{input modes}} + \underbrace{\sum_{k=1}^{N_b} D_k (p_k)^n u[n]}_{\text{system modes, a.k.a. natural modes}}$$

If the system is stable, the natural modes will eventually die out. Only the input modes will be left. Even these will die out if $|\xi_k| < 1 \ \forall k$.

> Whenever a system is excited suddenly, the output response will contain, in general, both input modes and natural modes (a.k.a. system modes)

The input modes represent the PARTICULAR SOLUTION and the natural modes represent the COMPLEMENTARY FUNCTION.

Let $x[n] = b^n u[n]$ and $h[n] = a^n u[n]$, where $a \neq b$. The output $y[n]$ is:

$$Y(z) = \frac{1}{1-az^{-1}} \cdot \frac{1}{1-bz^{-1}} = \frac{1}{a-b}\left[\frac{a}{1-az^{-1}} - \frac{b}{1-bz^{-1}}\right]$$

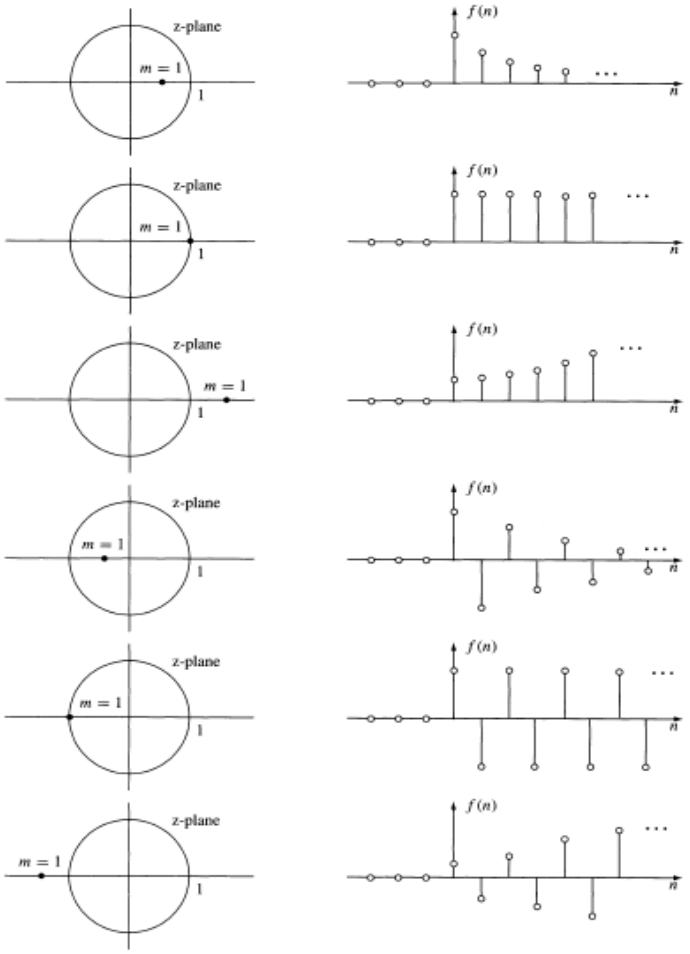
Inverse transforming,

$$y[n] = \underbrace{\frac{a}{a-b}a^n u[n]}_{\text{system mode}} - \underbrace{\frac{b}{a-b}b^n u[n]}_{\text{input mode}}$$

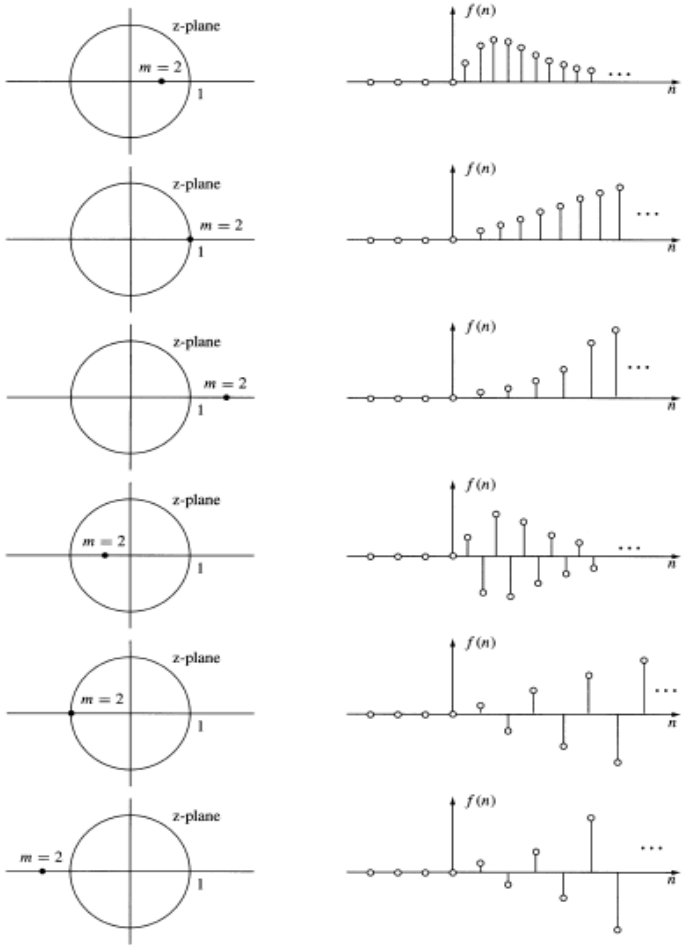
If the input is $x[n] = a^n u[n]$, the o/p is $\mathcal{Z}^{-1}\left[\frac{1}{(1-az^{-1})^2}\right] = (n+1)a^n u[n]$

↑ RESONANCE!

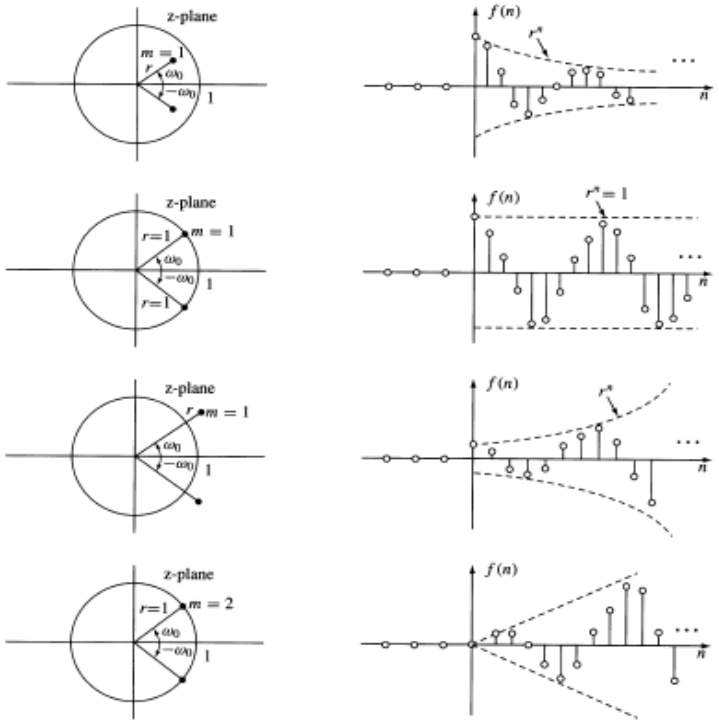
Single Real Poles—Causal Signals



Double Real Poles—Causal Signals



Complex-Conjugate Poles—Causal Signals



Transforms and Applications Handbook by A. D. Poularikas (Ed.) CRC Press, 2010, pp. 6-12 to 6-14

EE 2004 DSP Lecture 21

Note Title 07-03-2019

Frequency Response of Systems with Rational Transfer Functions:

Systems with rational transfer functions are very easy to realize in practice and also model a wide range of very useful transfer functions.

If the system is stable, then the unit circle is part of the RoC, i.e., $e^{j\omega} \in$ RoC.
Almost always we will also require causality. Hence we will focus on causal and stable LTI systems with rational transfer functions.

Some important frequency responses are: lowpass, highpass, bandpass, bandstop, differentiator, Hilbert transformer, etc.

Recall that if $z = e^{j\omega} \in$ RoC, then we can replace 'z' by '$e^{j\omega}$' to get the frequency response $H(e^{j\omega})$.

The ideal filters such LPF, HPF, etc. (also called as brickwall filters because of their shape) have impulse responses that are non-causal. Brickwall filters that are causal cannot be realized because they violate the Paley-Wiener theorem on realizability. Hence we have to settle for realizable filters that are approximations to the ideal filters. We will restrict ourselves to the class of systems whose i/p - o/p relationship takes the form of an LCCDE. We know that such systems have transfer functions that are rational, i.e., have the form $H(z) = B(z)/A(z)$ where $B(z)$ and $A(z)$ are polynomials in z (or z^{-1}).

EE 2004 DSP Lecture 22

Note Title 11-03-2019

Consider $$y[n] = -\sum_{k=1}^{N} a_k\, y[n-k] + \sum_{k=0}^{M} b_k\, x[n-k]$$

Hence $$\frac{Y(z)}{X(z)} = H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = b_0 \frac{\prod_{k=1}^{M} (1 - z_k z^{-1})}{\prod_{k=1}^{N} (1 - p_k z^{-1})} = b_0 \underbrace{z^{N-M}} \frac{\prod_{k=1}^{M} (z - z_k)}{\prod_{k=1}^{N} (z - p_k)}$$

trivial pole/zero (depending on the sign of N-M)

The frequency response $H(e^{j\omega})$ now becomes

$$H(e^{j\omega}) = b_0\, e^{j\omega(N-M)} \frac{\prod_{k=1}^{M} (e^{j\omega} - z_k)}{\prod_{k=1}^{N} (e^{j\omega} - p_k)}$$

Clearly, $H(e^{j\omega})$ is a function of both poles and zeros. How do poles and zeros influence the frequency response?

$$H(e^{j\omega}) = b_0 \, e^{j\omega(N-M)} \, \frac{\prod_{k=1}^{M} (e^{j\omega} - z_k)}{\prod_{k=1}^{N} (e^{j\omega} - p_k)} = \underbrace{|H(e^{j\omega})|}_{\text{Magnitude response (non-negative)}} \, e^{j \underbrace{\angle H(e^{j\omega})}_{\text{Phase response}}}$$

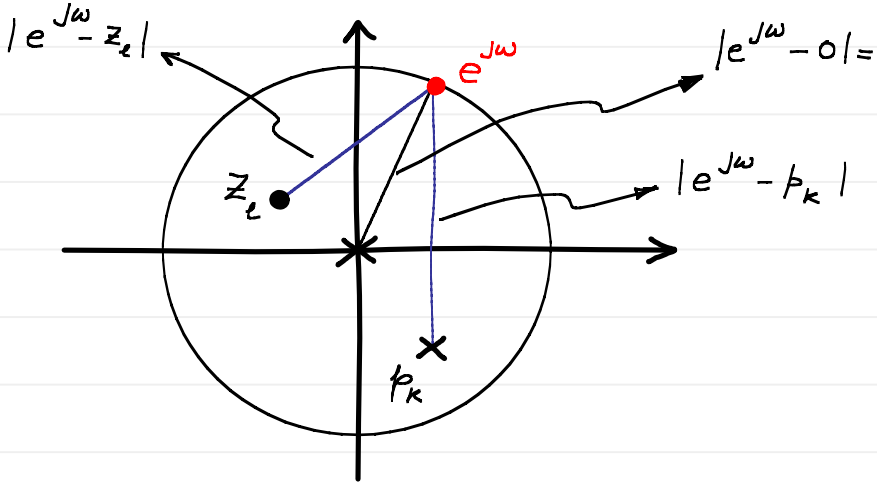
Therefore

$$|H(e^{j\omega})| = |b_0| \, \cancel{|e^{j\omega(N-M)}|}^{\,1} \, \frac{\prod_{k=1}^{M} |e^{j\omega} - z_k|}{\prod_{k=1}^{N} |e^{j\omega} - p_k|}$$

Poles or zeros at $z = 0$ do not contribute to the magnitude response. That is why they are called as trivial poles or zeros. They do, however, contribute to the phase response, as we shall see later.

Consider $|e^{j\omega} - z_\ell|$. Geometrically this term represent the distance between z_ℓ (a zero at $z = z_\ell$) and $e^{j\omega}$ (which is a point on the unit circle for a particular value of ω). Thus, the **numerator** is the product of all the distances from a particular point on the unit circle to all the **zeros**. Similarly, the **denominator** is the product of all the distances from the same point on the unit circle to all the **poles**. Finally, $|H(e^{j\omega})|$ is the ratio of these two product of distances scaled by the gain term $|b_0|$. $|H(e^{j\omega})|$ changes as 'ω' varies, i.e., as we traverse along the unit circle.

It is usual to plot the frequency response as $20 \log_{10} |H(e^{j\omega})|$. The gain term $|b_0|$ merely shifts the curve up or down the log plot.



$\Rightarrow$ Trivial pole (or zero) does not contribute to the magnitude response

If a zero is on the unit circle at $\omega = \omega_0$, then $H(e^{j\omega_0}) = 0$ because that particular distance becomes zero.

When there is a **zero close to the unit circle**, when we are near this zero, i.e., the point on the unit circle is near this zero, $|e^{j\omega} - z_\ell|$ is small. This in turn brings down the resulting product of all the distances in the numerator. Thus, $|H(e^{j\omega})|$ is **small** for

values of ω that make $|e^{j\omega} - z_\ell|$ small (i.e., neighbourhood of the zero). In particular, if there is a **zero on the unit circle**, i.e. at $z = e^{j\omega_0}$, then $H(e^{j\omega_0}) = 0$ since the distance to that zero becomes zero when $\omega = \omega_0$. Thus, in general, **zeros provide attenuation**, with the degree of attenuation increasing as the zero moves closer to the unit circle (with complete attenuation at $\omega = \omega_0$ if the zero is on the unit circle at $z = e^{j\omega_0}$).

By similar reasoning, we see that **poles provide gain** because the product of distances to all the poles becomes small when we are near to a pole that is close to the unit circle. Hence $|H(e^{j\omega})|$ becomes large as we go near a pole. Note that the pole cannot lie on the unit circle for a stable system (unlike a zero).

Both poles and zeros shape the frequency response.
Poles provide gain, whereas zeros provide attenuation.

For systems with poles and zeros, generally speaking, poles govern passband behaviour whereas zeros govern stopband behaviour.

This is similar to the frequency response of CT systems:

$$H(s) = \frac{B(s)}{A(s)} = \frac{\prod_{k=1}^{M}(s - z_k)}{\prod_{k=1}^{N}(s - p_k)} \Rightarrow H(j\Omega) = \frac{\prod_{k=1}^{M}(j\Omega - z_k)}{\prod_{k=1}^{N}(j\Omega - p_k)}$$

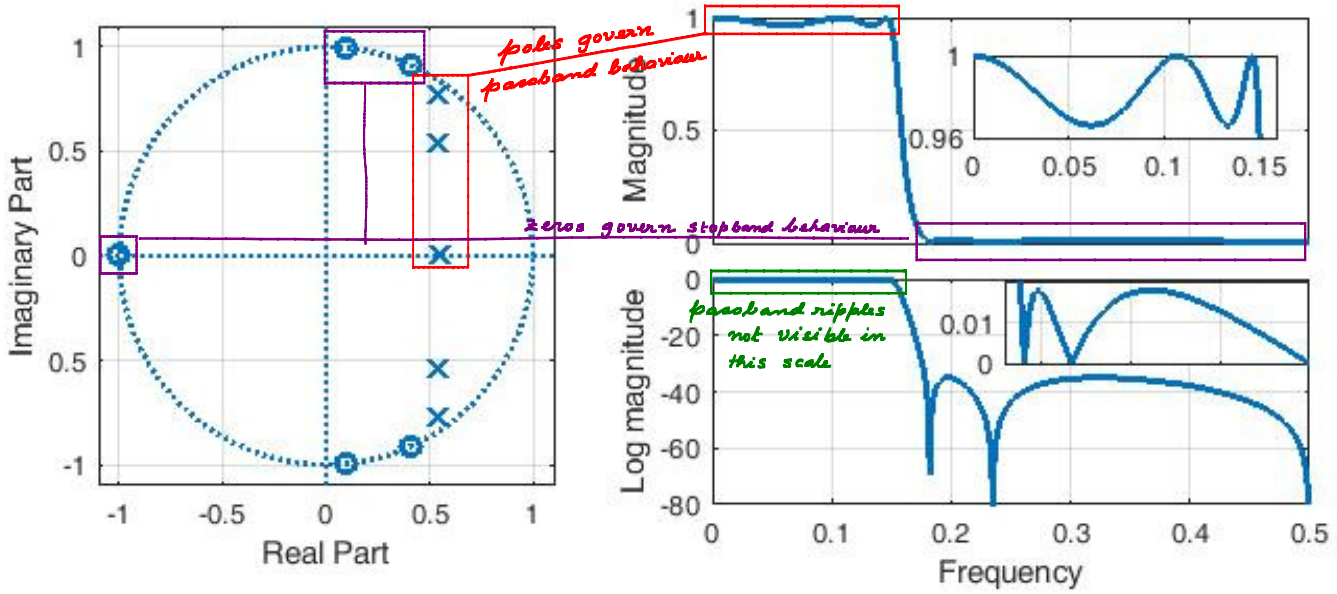
Similar to the DT System case, poles and zeros influence the frequency response

The same geometric interpretation applies to the CT frequency response $H(j\Omega)$. If the $j\Omega$ axis is part of RoC, then $H(s)\Big|_{s=j\Omega} = H(j\Omega)$.

Hence

$$|H(j\Omega)| = |b_0| \frac{\prod_{k=1}^{M} |j\Omega - z_k|}{\prod_{k=1}^{N} |j\Omega - p_k|}$$

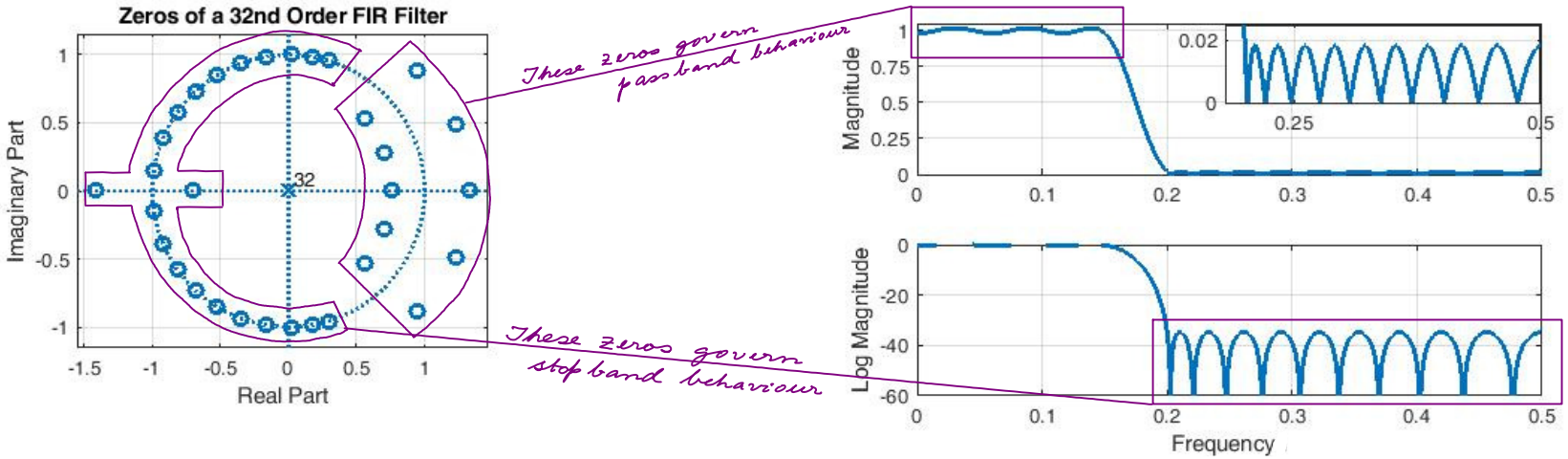
The frequency response magnitude is the ratio of the product of all the distances from a particular point on the $j\Omega$ axis to all the zeros to the product of all the distances from the same point on the $j\Omega$ axis to all the poles, scaled by $|b_0|$.



Pole-Zero plot and magnitude response (both linear and log scale) for a 5th order elliptic filter. Poles govern passband and zeros govern stopband.

EE 2004 DSP Lecture 23

Note Title 13-03-2019



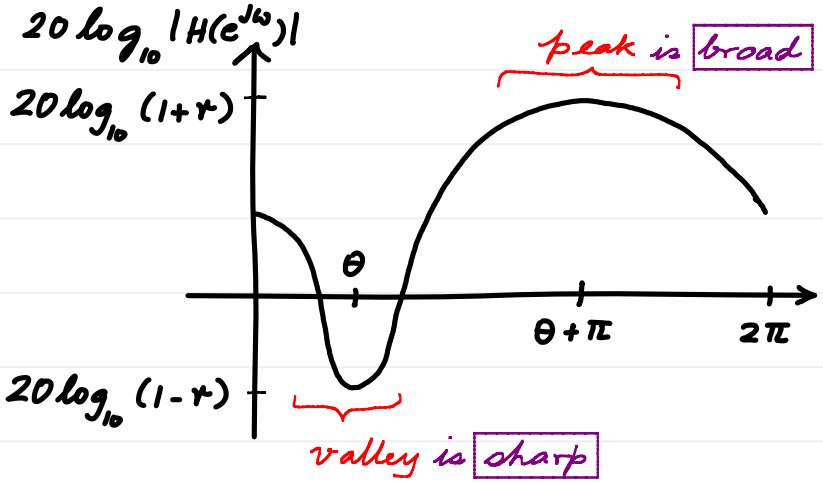
Since FIR filters do not have non-trivial poles, zeros shape both passband and stopband behaviour.

Response of a Single Complex Zero:

Let $H(z) = 1 - re^{j\theta} z^{-1}$

$$|H(e^{j\omega})|^2 = |1 - re^{j\theta} e^{-j\omega}|^2 = 1 + r^2 - 2r\cos(\omega - \theta)$$

replacing 'ω' by '$-\omega$' gives a different response as $h[n]$ is complex, except when $\theta = 0$ or $\theta = \pi$



Minimum occurs at $\omega = \theta$: $|H(e^{j\omega})|^2_{min} = (1-r)^2$

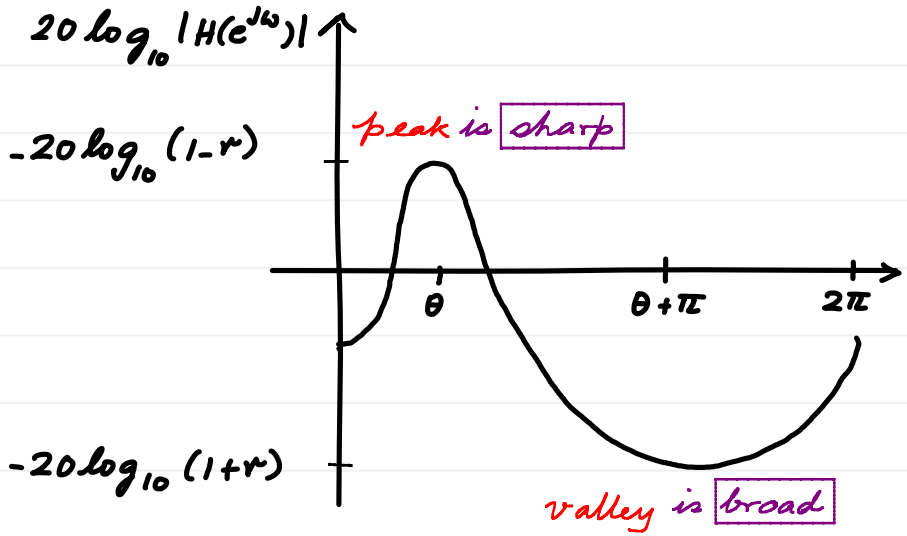
Maximum occurs at $\omega = \theta + \pi$: $|H(e^{j\omega})|^2_{max} = (1+r)^2$

For $r = 0.9$, $|H(e^{j\omega})|^2_{min} = 0.01$

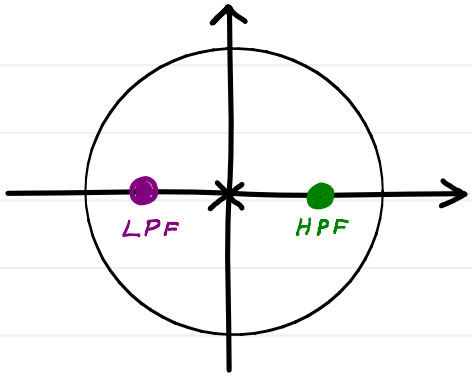
$|H(e^{j\omega})|^2_{max} = 3.61$

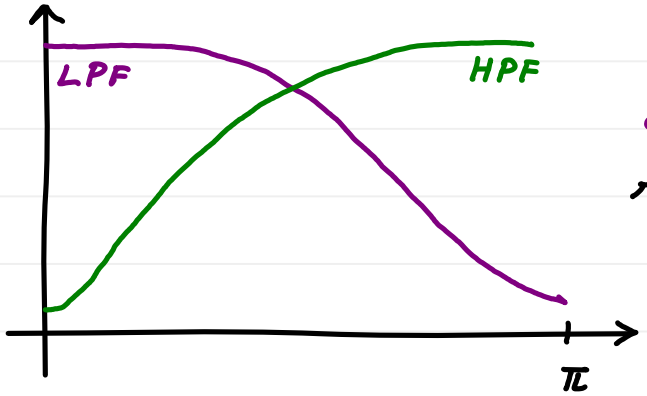
For a single complex pole,

$$H(z) = \frac{1}{1 - a z^{-1}} \Rightarrow$$ the log. magnitude plot is the negative of the plot shown earlier.

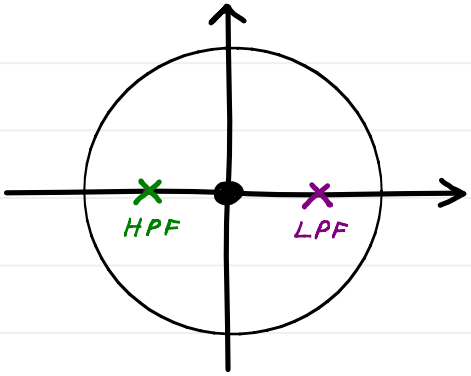


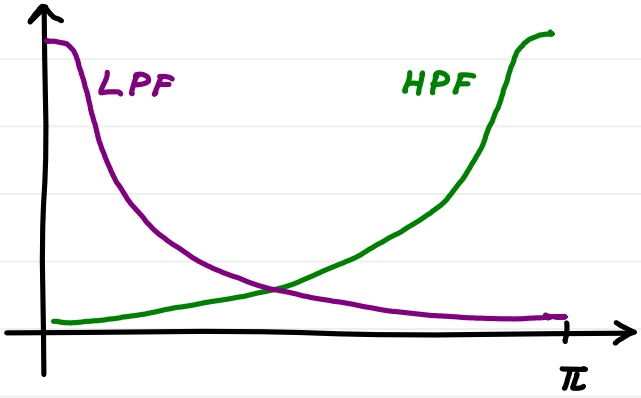
Recall that poles near the unit circle boost the frequency response.
Zeros near the unit circle attenuate the frequency response.





Lowpass and highpass filters realized using a single zero





Lowpass and highpass filters realized using a single pole

The main difference between an LPF or HPF realized using a pole versus another realized using a zero is the narrowness of the passband.

> Zeros cause sharp valleys and broad peaks in the frequency response
> Poles cause sharp peaks and broad valleys in the frequency response

Consider the following two LPFs: (i) realized using a pole, and (ii) realized using a zero

(i) $H(z) = \frac{1}{1 - az^{-1}}$ (ii) $H(z) = 1 + az^{-1}$ where $0 < a < 1$ in both cases.

$$|H(e^{j0})|^2 = \frac{1}{(1-a)^2}$$

$$= 100 \ (20\,dB) \quad \text{if } a = 0.9$$

$$|H(e^{j0})|^2 = (1+a)^2$$

$$= 3.61 \ (5.58\,dB) \quad \text{if } a = 0.9$$

$$|H(e^{j\pi/2}|^2 = \frac{1}{1+a^2}$$

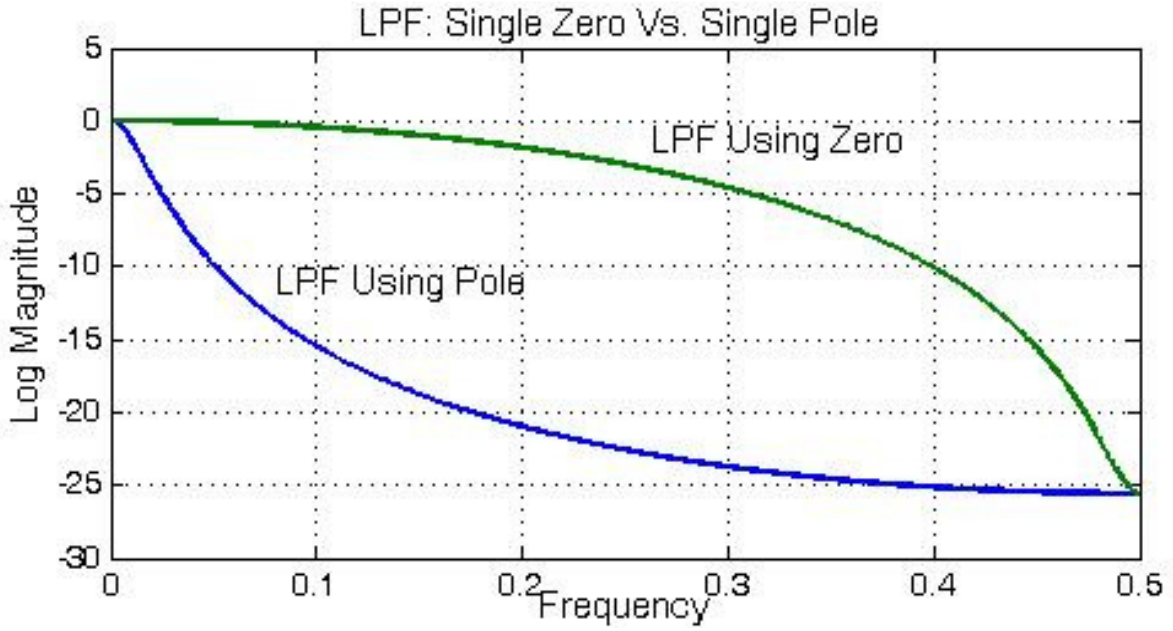
$$= 0.55 \ (-5.15\,dB) \quad \text{if } a = 0.9$$

$$|H(e^{j\pi/2}|^2 = 1+a^2$$

$$= 1.81 \ (2.58\,dB) \quad \text{if } a = 0.9$$

Thus the **3 dB bandwidth** of an LPF realized using a single zero is $\frac{\pi}{2}$ if $a = 0.9$. The 3dB bandwidth of an LPF realized using a single pole can be shown to be $\frac{\pi}{30}$ if $a = 0.9$, i.e., it *fifteen times narrower!*

Exercise: Derive the 3-dB bandwidth formula for $H(z) = \frac{1}{1-az^{-1}}$ where $-1 < a < 1$

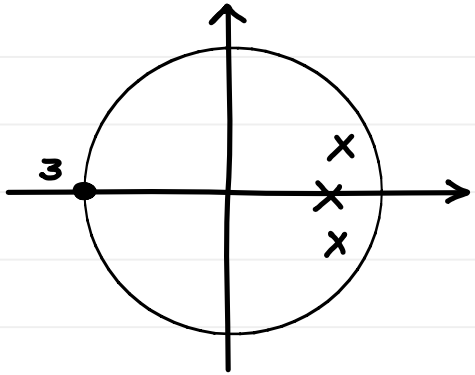


$H_{LPF}(z)$ vs. $H_{LPF}(z)$ where

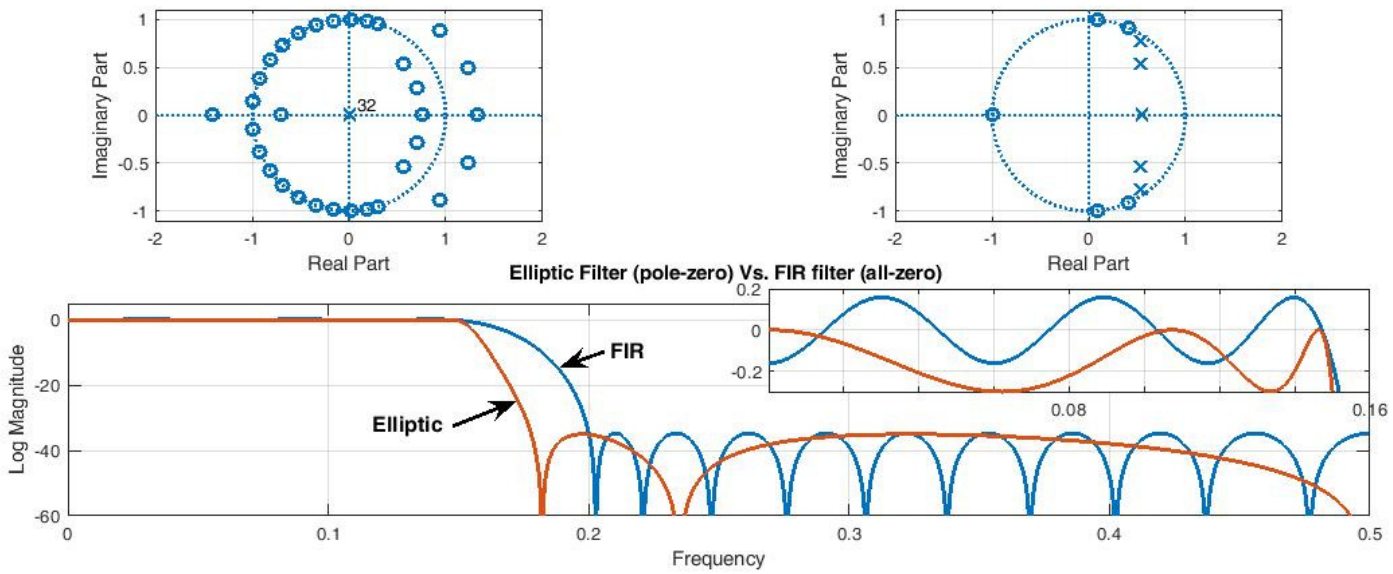
$H_{LPF}(z) = \frac{1}{1-az^{-1}}$ and $H_{LPF}(z) = 1 + az^{-1}$

for $a = 0.9$. The peak gain has been normalized to unity (i.e., 0 dB) in both cases in the plots. Observe that the 3-dB bandwidth is 15-times smaller for $H_{LPF}(z)$ compared to $H_{LPF}(z)$

One can add more poles to get a flatter passband:



Systematic procedures for designing digital filters will be taught in the Digital Filter Design course. Classical filters: Butterworth, Chebyshev, Elliptic.



To meet the same specifications, a pole-zero filter requires a smaller order than an all-zero filter. A 5^{th} order elliptic filter meets the same specifications than a 32^{nd} order (linear phase) FIR filter

EE 2004 DSP Lecture 24

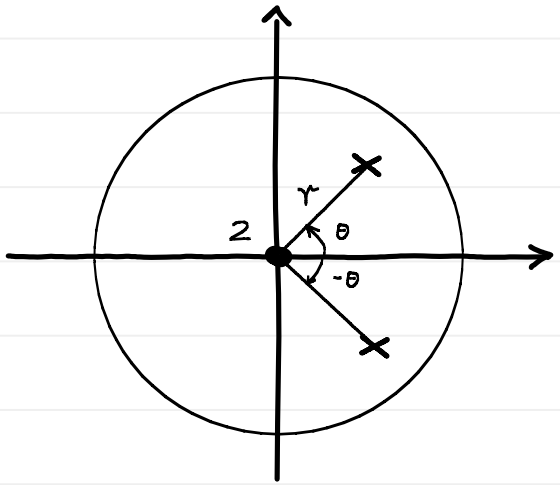
Note Title 14-03-2019

Second Order Filter

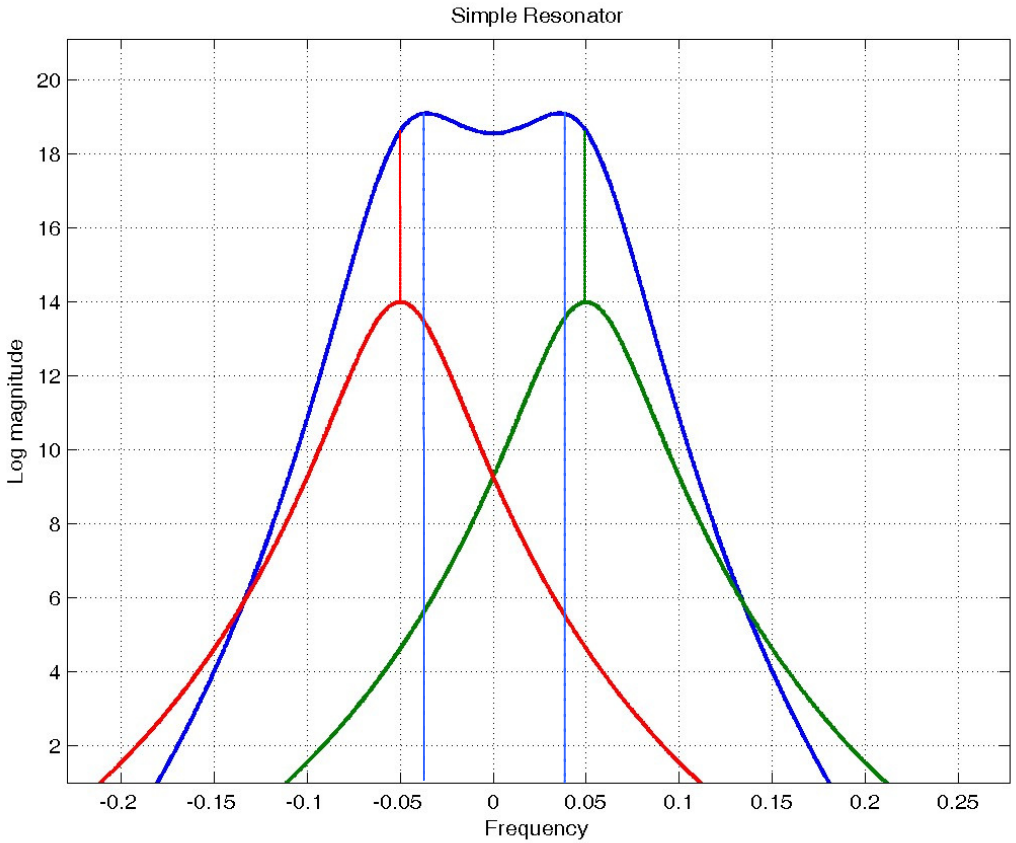
$$h[n] = r^n \frac{\sin(n+1)\theta}{\sin\theta} u[n]$$

$$H(z) = \frac{1}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}} = \frac{1}{(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1})}$$

$$|H(e^{j\omega})|^2 = \frac{1}{[1 - 2r\cos(\omega-\theta) + r^2][1 - 2r\cos(\omega+\theta) + r^2]}$$



This is also called as a RESONATOR



The tail of the response due to the pole at $re^{-j\theta}$ will interfere and cause a shift in the peak that would have occurred at $\omega = \theta$ if $re^{j\theta}$ were the only pole present. Similarly, the peak location around $\omega = -\theta$ is affected by the tail of the response due to the pole at $re^{j\theta}$.

$r = 0.8, \quad \theta = \frac{\pi}{10}$

The expression for peak location is obtained by solving

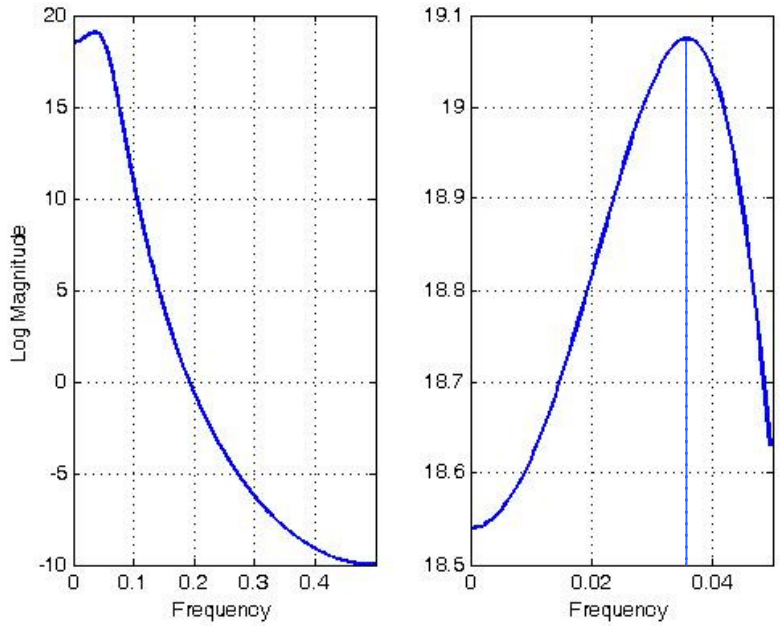
$$\underset{\omega}{\operatorname{argmax}} \frac{1}{[1-2r\cos(\omega-\theta)+r^2][1-2r\cos(\omega+\theta)+r^2]}$$

Show that $\omega_0 = \cos^{-1}\left[\frac{1+r^2}{2r}\cos\theta\right]$

Interference is reduced if the poles move farther apart. The farthest apart they can be is $\theta = \frac{\pi}{2}$. For this value of θ, $\omega_0 = \theta = \frac{\pi}{2}$ independent of r, there is no shift in peak location!

Interference also reduces as $r \to 1$. We require $-1 \leq \frac{1+r^2}{2r}\cos\theta \leq 1$.

For a peak to occur at $\omega_0 \neq 0$ or π,



The plots show peak-shifting due to interference when $r = 0.8$ and $\theta = \frac{\pi}{10}$.
Because of tail interference, the peak does not occur at $\frac{\omega_0}{2\pi} = \frac{1}{20} = 0.05$.

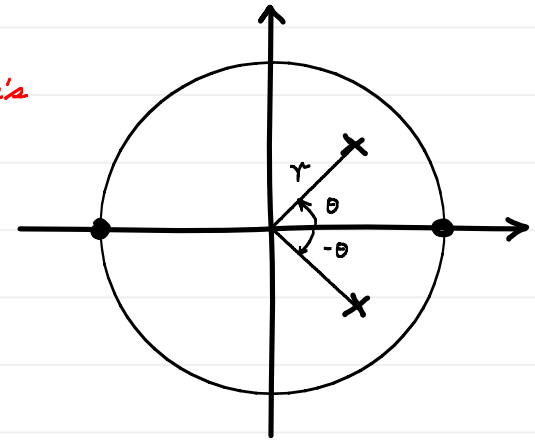
Instead it occurs at $\frac{\omega_0}{2\pi} = 0.0358$, as can be seen from the plots.

Improved Resonator: The resonator considered previously is a crude BPF. A canonic BPF must reject $\omega = 0$ and $\omega = \pi$. This aspect can be realized if zeros are added at $z = \pm 1$. Hence,

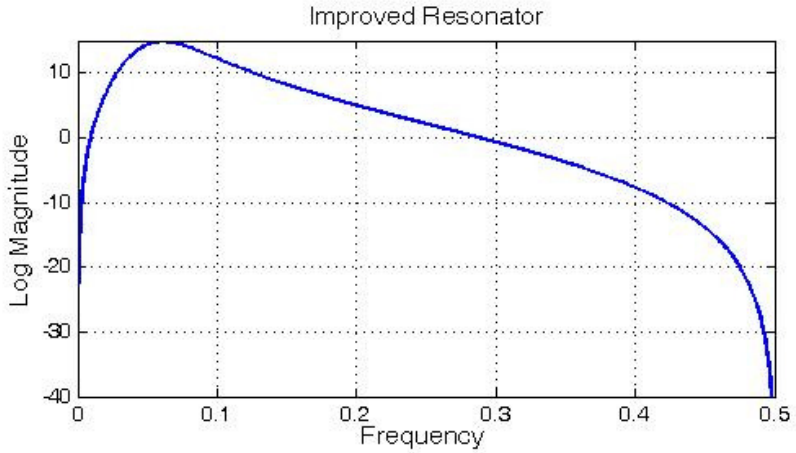
$$H(z) = \frac{(1 - z^{-1})(1 + z^{-1})}{(1 - r e^{j\theta})(1 - r e^{-j\theta})}$$

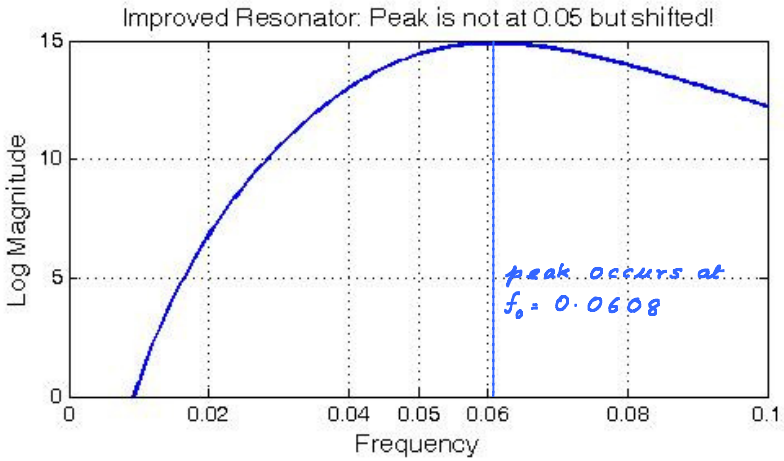
$$= \frac{1 - z^{-2}}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}}$$

Improved resonator's pole-zero plot. Note that the zeros are now at $z = \pm 1$.



The improved resonator also suffers from peak shifting due to tail interference.





Freq. response of the improved resonator for $r = 0.8$ and $\theta = \frac{\pi}{10}$. The frequency is forced to go to zero at $\omega = 0$ and π. This also suffers from peak-shifting.

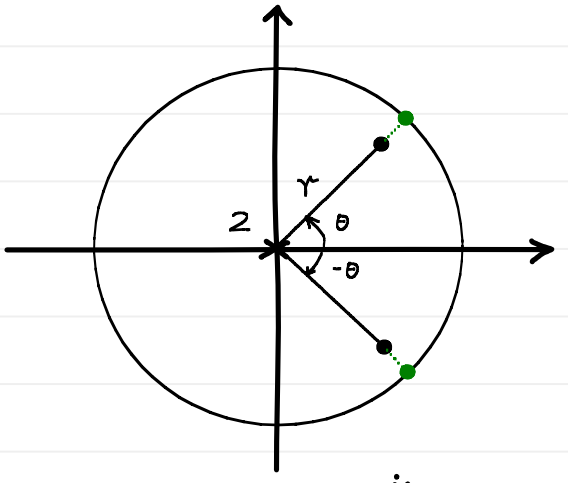
Show that the peak occurs at $\omega_0 = \cos^{-1}\left[\frac{2r}{1+r^2}\cos\theta\right]$. Note that $\frac{2r}{1+r^2}\cos\theta \in [-1, 1]$.

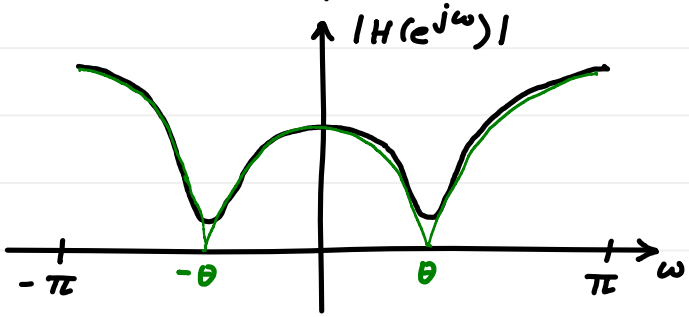
Notch Filter: Used for removing one or more sinusoids.

Consider $H(z) = 1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}$

If $r = 1$, then the zeros lie on the unit circle. Hence $H(e^{j\omega}) = 0$ when $\omega = \pm\theta$. Unlike in the resonator's case, where the pole cannot lie on the unit circle, we can have $r = 1$ in the above expression. When $r = 1$, sinusoids with frequencies $\pm\omega_0$ are completely removed.

$$H(z) = 1 - 2\cos\theta\, z^{-1} + z^{-2}$$

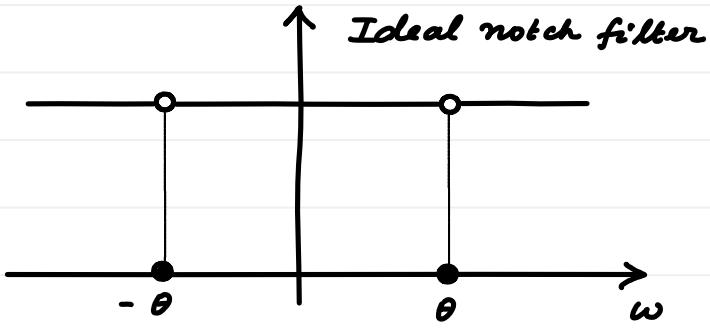




Minimum occurs at $\omega_0 = \cos^{-1}\left[\frac{1+r^2}{2r}\cos\theta\right]$

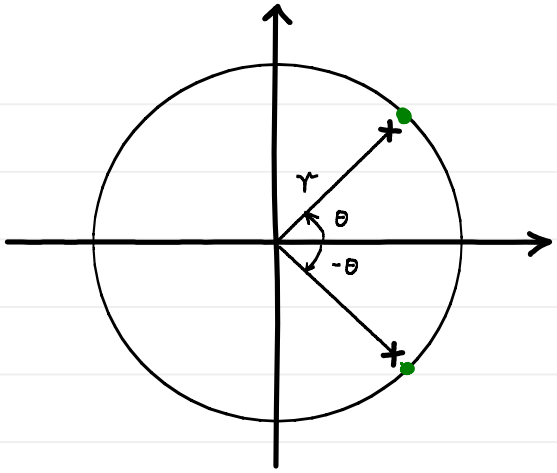
While the frequency components at $\omega = \pm\theta$ are completely eliminated, the response is *far from ideal*. An ideal notch filter has unity gain whenever $\omega \neq \pm\theta$.

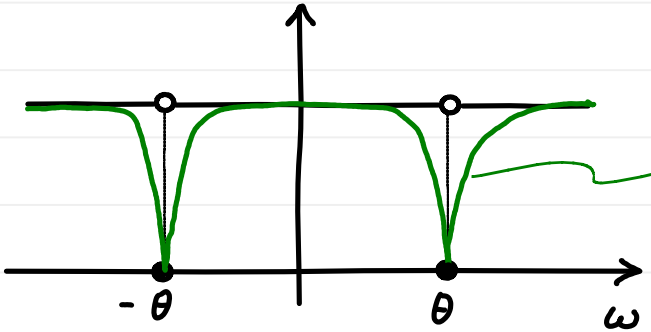
The non-ideal notch filter's response can be considerably improved by *placing poles at* $re^{\pm j\theta}$ *where* 'r' *is close to 1*. That is,



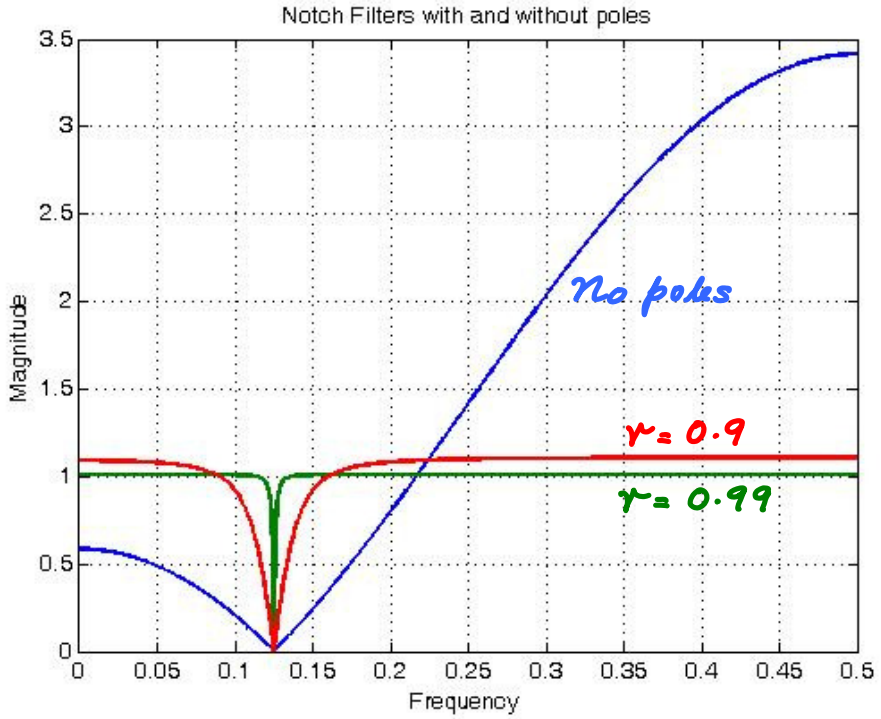
$$H(z) = \frac{1 - 2\cos\theta\, z^{-1} + z^{-2}}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}}$$

The poles at $re^{\pm j\theta}$ *help in providing gain rapidly as we move away from* $\omega = \pm\theta$.





Improved notch filter



Adding poles at $re^{j\theta}$ significantly improves the frequency response of the simple notch filter.

Improved notch filter: $H(z) = \dfrac{1 - 2\cos\theta\, z^{-1} + z^{-2}}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}}$

In practice 'r' cannot be made too close to 1 because of finite precision effects.

Moving Average Filter:

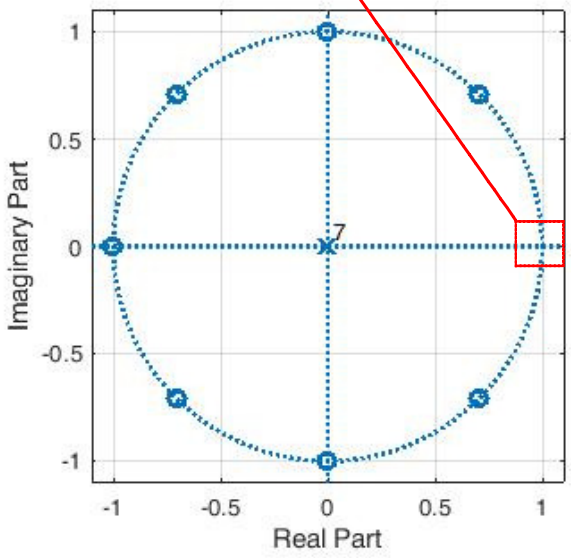
$$h[n] = \frac{1}{N} \quad 0 \le n \le N-1 \quad \longleftrightarrow \quad H(z) = \frac{1}{N}\left(1 + z^{-1} + z^{-2} + \cdots + z^{-(N-1)}\right) = \frac{1}{N}\,\frac{1 - z^{-N}}{1 - z^{-1}}$$

No. of zeros depends on the value of N

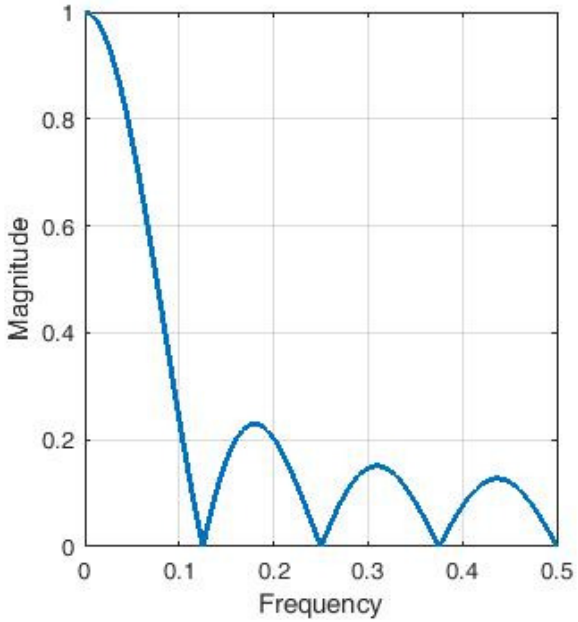
The zero at $z = 1$ is cancelled by the pole at $z = 1$. Hence there are $N-1$ non-trivial zeros and $N-1$ trivial poles.

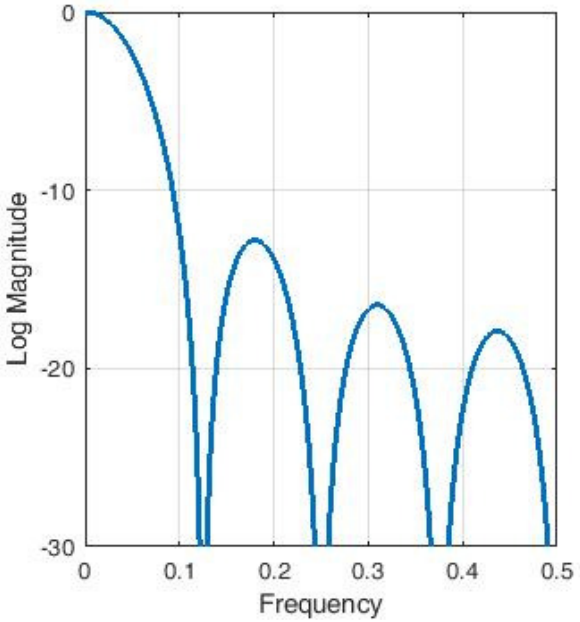
If $N = 8$, then $H(z) = \frac{1}{8}\,\frac{1 - z^{-8}}{1 - z^{-1}}$. Magnitude response plots are shown next.

pole-zero cancellation



Verify that this plot is consistent with the geometric evaluation of the $|H(e^{j\omega})|$





EE 2004 DSP Lecture 25

Note Title 18-03-2019

Comb Filters

Consider the simple LPF $H(z) = \frac{1+z^{-1}}{2}$. ← ensures unity gain at $\omega = 0$

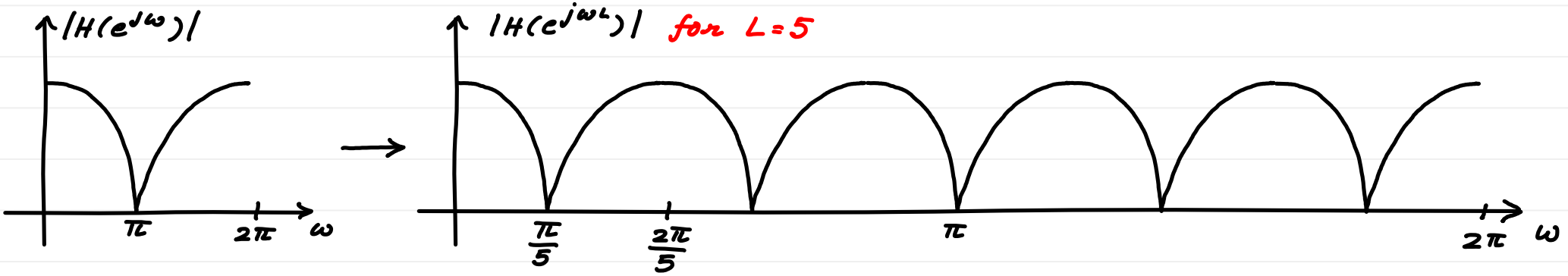
The comb-filter $G(z)$ is obtained by replacing z by z^L. That is, $G(z) = H(z^L)$. Thus

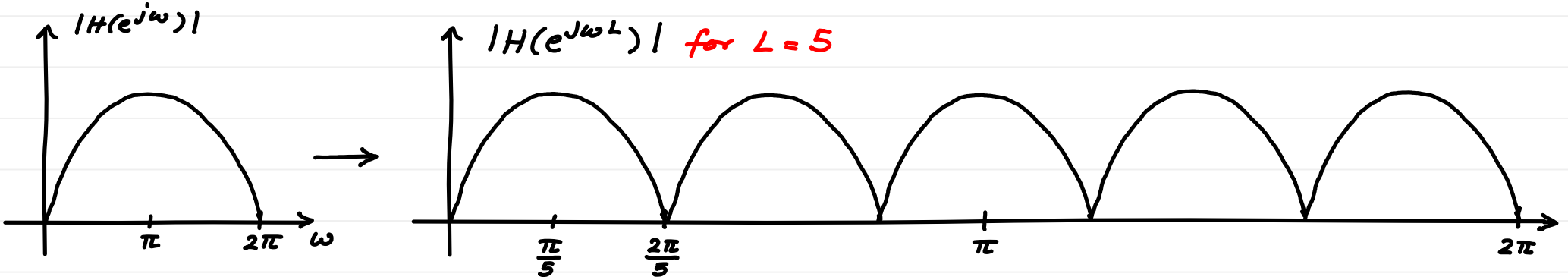
$$G(z) = \frac{1+z^{-L}}{2} \longleftrightarrow \{ \tfrac{1}{2}, \underbrace{0,0,0,\ldots,0}_{L-1 \text{ zeros}}, \tfrac{1}{2} \}$$

> In general, replacing z by z^L introduces $L-1$ zeros between each time-domain sample

The zeros of $G(z)$ are the L^{th} roots of -1, i.e., $e^{j(2k+1)\pi/L}$. The peaks occur at $\omega = \frac{2\pi k}{L}$.

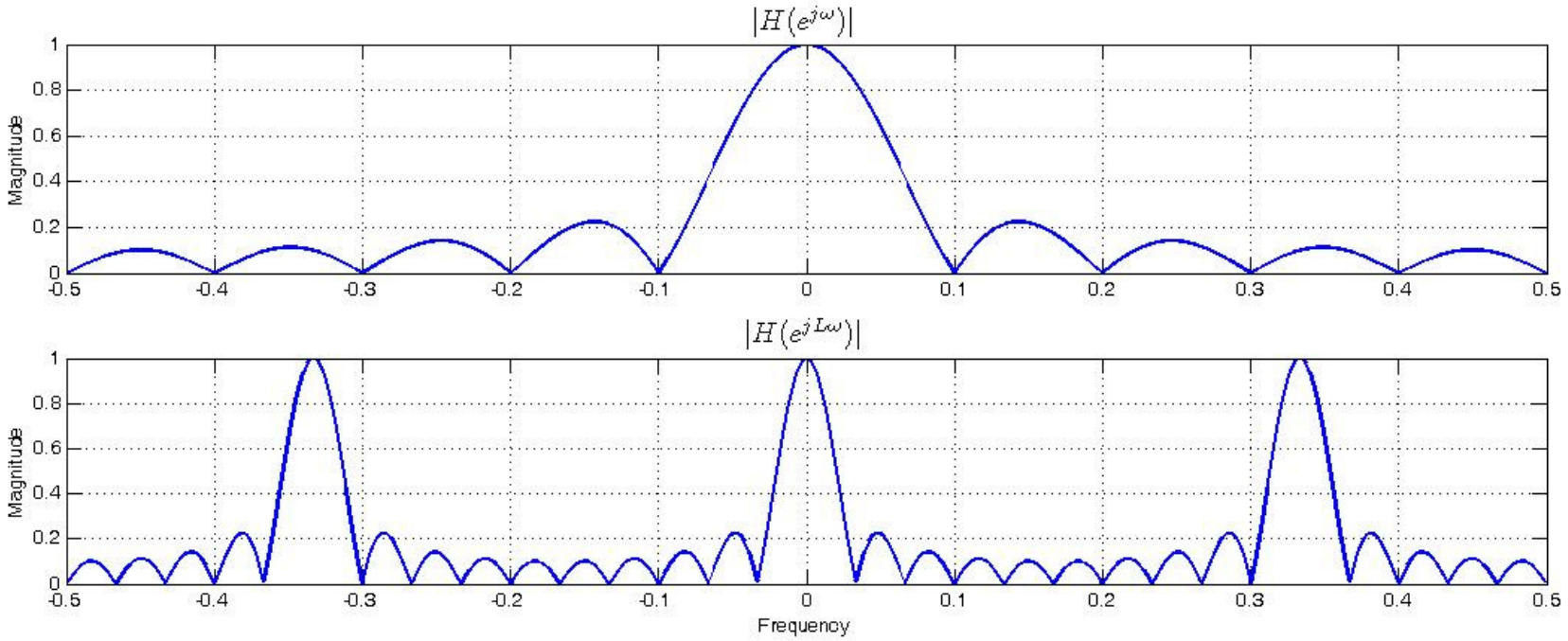
Similarly, for a highpass filter, we start off with $H(z) = \frac{1-z^{-1}}{2}$ to get $G(z) = \frac{1-z^{-L}}{2}$. The zeros are the L^{th}-root of 1.


$|H(e^{j\omega})|$
π
2π
ω
$|H(e^{j\omega L})|$ for $L=5$
$\frac{\pi}{5}$
$\frac{2\pi}{5}$
π
2π
ω


$|H(e^{j\omega})|$
π
2π
ω
$|H(e^{j\omega L})|$ for $L=5$
$\frac{\pi}{5}$
$\frac{2\pi}{5}$
π
2π

If $H(z) = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$, then $|H(e^{j\omega})| = \frac{1}{N} \left| \frac{\sin N\omega/2}{\sin \omega/2} \right|$

The plots for $N = 10$ and $L = 3$ are given below:



Phase Response

Recall that standard form of the transfer function of a system governed by an LCCDE is

$$H(z) = b_0 \frac{\prod_{k=1}^{M} (1 - z_k z^{-1})}{\prod_{k=1}^{N} (1 - p_k z^{-1})}$$

Hence,

$$\angle H(e^{j\omega}) = \arg\{b_0\} + \sum_{k=1}^{M} \arg\{1 - z_k e^{-j\omega}\} - \sum_{k=1}^{N} \arg\{1 - p_k e^{-j\omega}\}$$

The form of a typical term is

$$\arg\{1 - r e^{j\theta} e^{-j\omega}\} = \arg\{1 - r\cos(\omega - \theta) + j\, r\sin(\omega - \theta)\} = \tan^{-1} \frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)}$$

Note: $\tan^{-1}\left(\frac{3}{4}\right) \neq \tan^{-1}\left(\frac{-3}{-4}\right)$

We are interested in the overall phase response, which is contributed by individual terms such as the above. For a system with **real-valued coefficients**, the phase response will be **odd-symmetric**.

Consider the case where $r < 1$ and $\theta = 0$.

$H(z) = 1 - r z^{-1}$

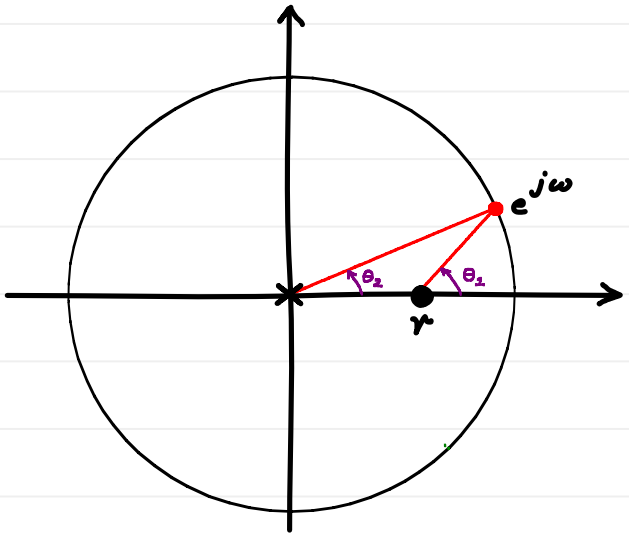
$\angle H(e^{j\omega}) = \arg\{1 - r e^{-j\omega}\} = \arg\{e^{j\omega} - r\} - \arg\{e^{j\omega}\}$

Geometric interpretation

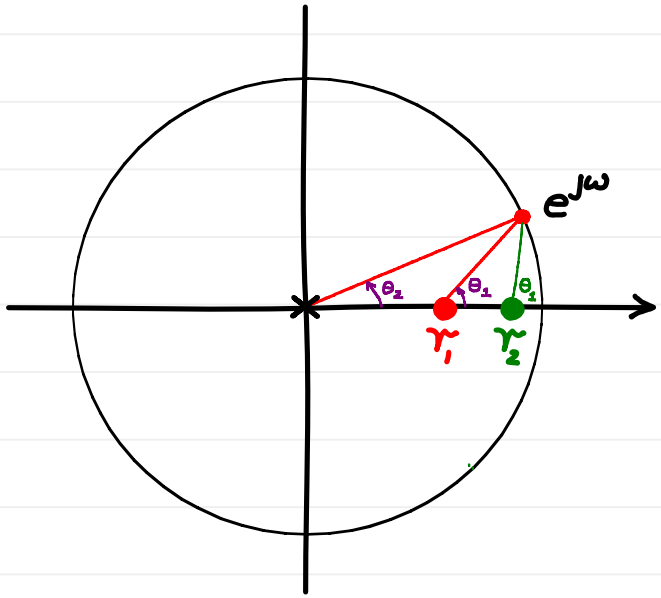
θ_1: angle of the line joining $e^{j\omega}$ and r

θ_2: angle of the line joining $e^{j\omega}$ and 0

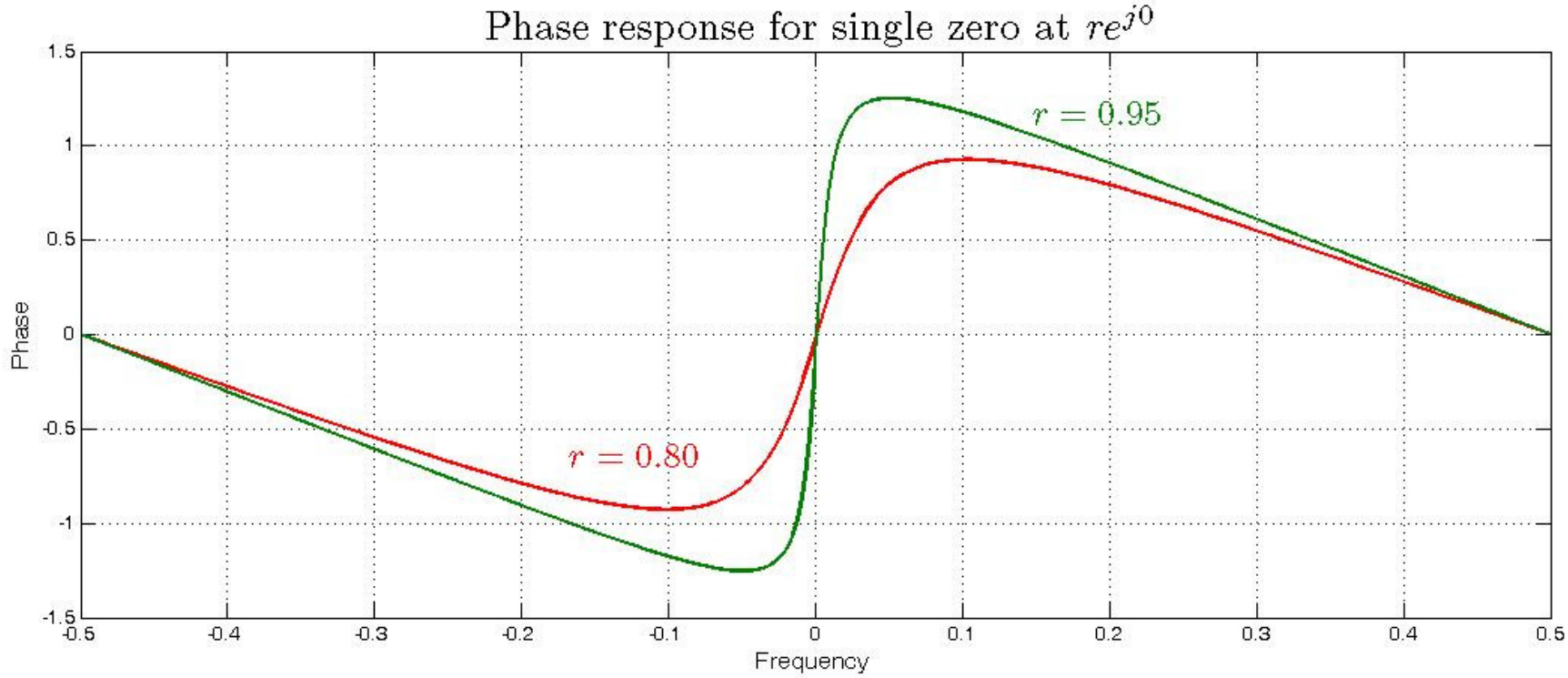
Overall angle, i.e., phase response, is $\theta_1 - \theta_2$.



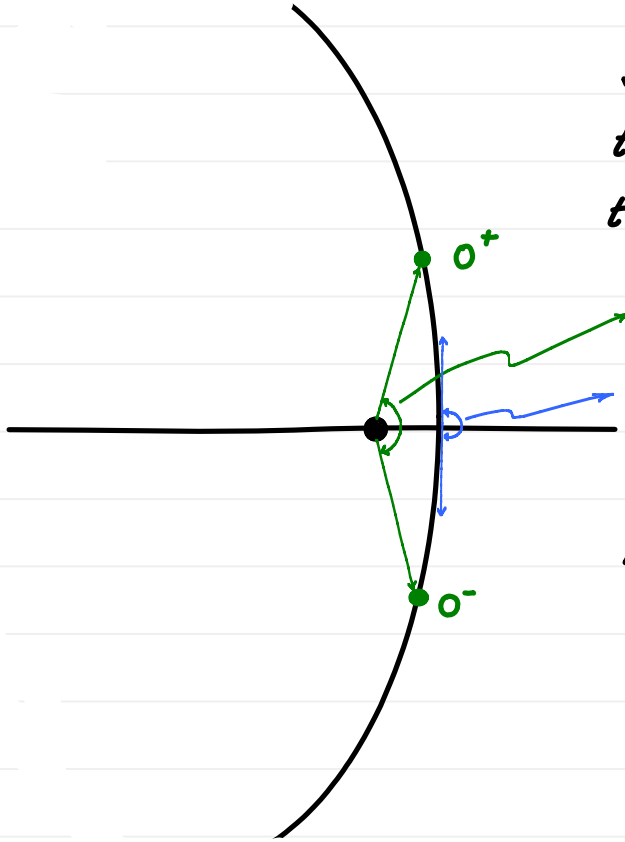
Note: Trivial pole or zero will contribute to the phase response



For $0 < r_1 < r_2 < 1$, the shape of phase response is shown below. At $\omega = 0$, $\theta_1 = \theta_2 = 0$, i.e., $\theta_1 - \theta_2 = 0$. At $\omega = \pi$, $\theta_1 = \theta_2 = \pi$, i.e., $\theta_1 - \theta_2 = 0$. Just beyond $\omega = 0$, θ_1 increases more rapidly than θ_2, which means that $\theta_1 - \theta_2 > 0$. As ω increases further, the value of $\theta_1 - \theta_2$ reaches a maximum, and then eventually reduces to 0 at $\omega = \pi$.



If $\theta \neq 0$, the phase response will be centred at $\omega = \theta$ rather than at $\omega = 0$.



As $r \to 1$, the change the phase tends to π as we cross the zero while moving along the unit circle. When $r = 1$ the change in phase is exactly π.

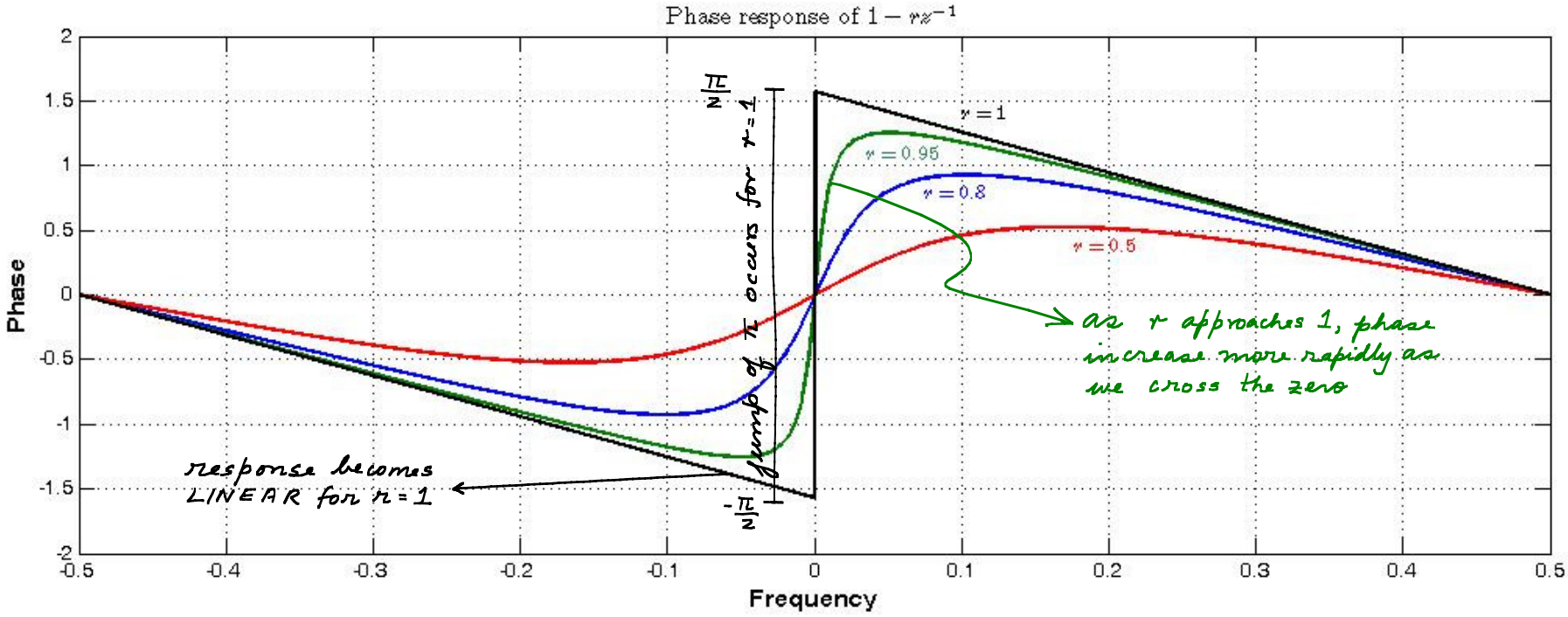
As $r \to 1$, the change in phase tends to π

In the limit, when $r = 1$, the change equals $\boxed{\pi}$

For $r = 1$, the expression for the phase is

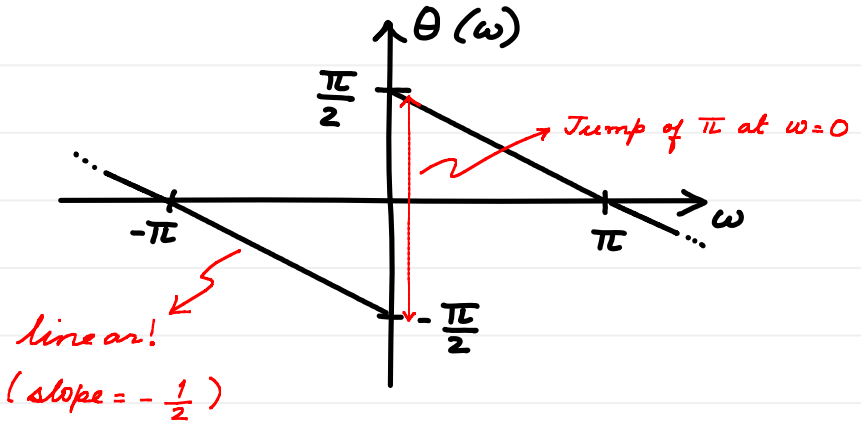
$$\phi(\omega) = \tan^{-1} \frac{\sin \omega}{1 - \cos \omega} = \tan^{-1} \tan\left(\frac{\pi}{2} - \frac{\omega}{2}\right)$$

$$= \frac{\pi}{2} - \frac{\omega}{2}$$


Phase response of $1 - rz^{-1}$
Phase
Frequency
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-0.5
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2
0.3
0.4
0.5
$r = 1$
$r = 0.95$
$r = 0.8$
$r = 0.5$
$\frac{\pi}{2}$
jump of π occurs for $r = 1$
$-\frac{\pi}{2}$
as r approaches 1, phase increase more rapidly as we cross the zero
response becomes LINEAR for r = 1

Thus, since $\phi(\omega)$ is odd symmetric,

$$\angle H(e^{j\omega}) = \begin{cases} \frac{\pi}{2} - \frac{\omega}{2} & \omega > 0 \\ -\frac{\pi}{2} - \frac{\omega}{2} & \omega < 0 \end{cases}$$



Important Features:

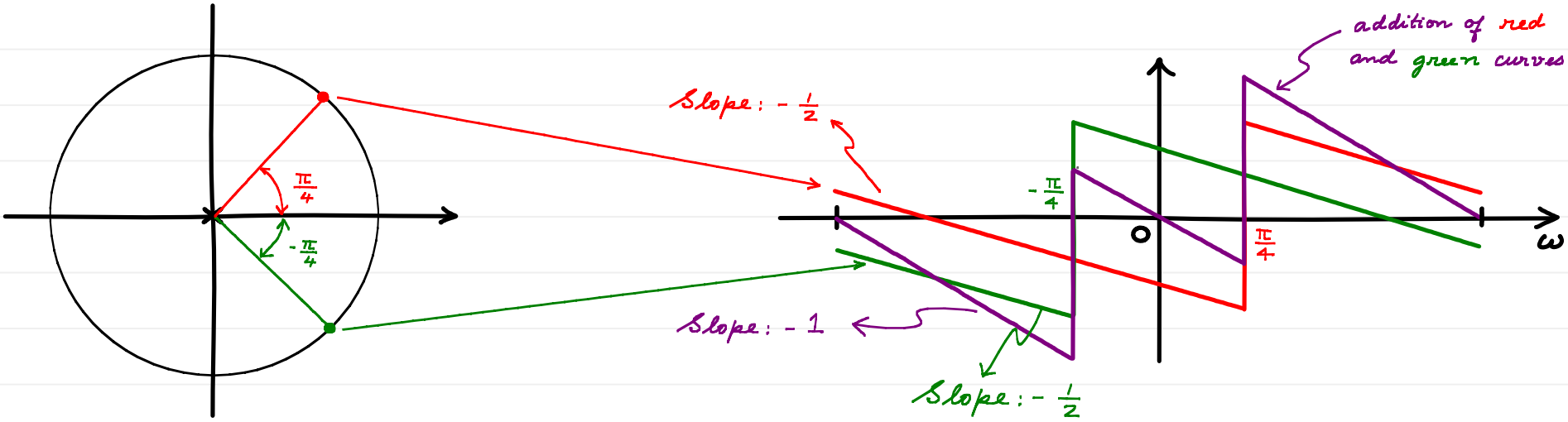
(1) Phase jumps by π at $\omega = 0$

(2) The phase is linear

(3) The slope of the linear part is $-1/2$.

If $r = 1$ and $\theta \neq 0$, the phase response will be shifted to $\omega = \theta$, i.e., the jump of π will occur at $\omega = \theta$. The slope of the linear region remains unchanged.

Any collection of zeros on the unit circle will give rise to a phase response that is linear overall. The slope of the linear region will be $-\frac{N}{2}$ (assuming there are N zeros). Jumps of π will occur at the location of the zeros. If the zeros occur in complex conjugate pairs, the overall response will be odd-symmetric.



EE 2004 DSP Lecture 26

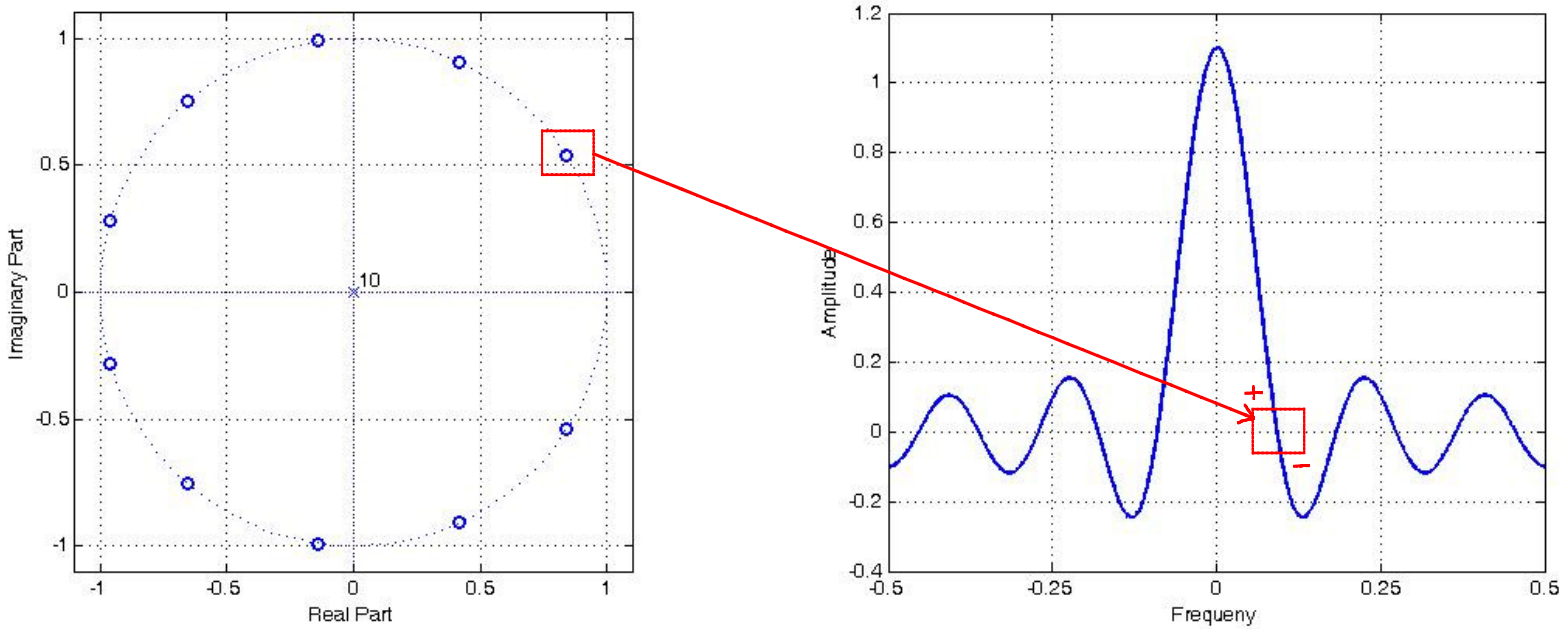
Note Title 20-03-2019

Zero on the unit circle causes a sign change:

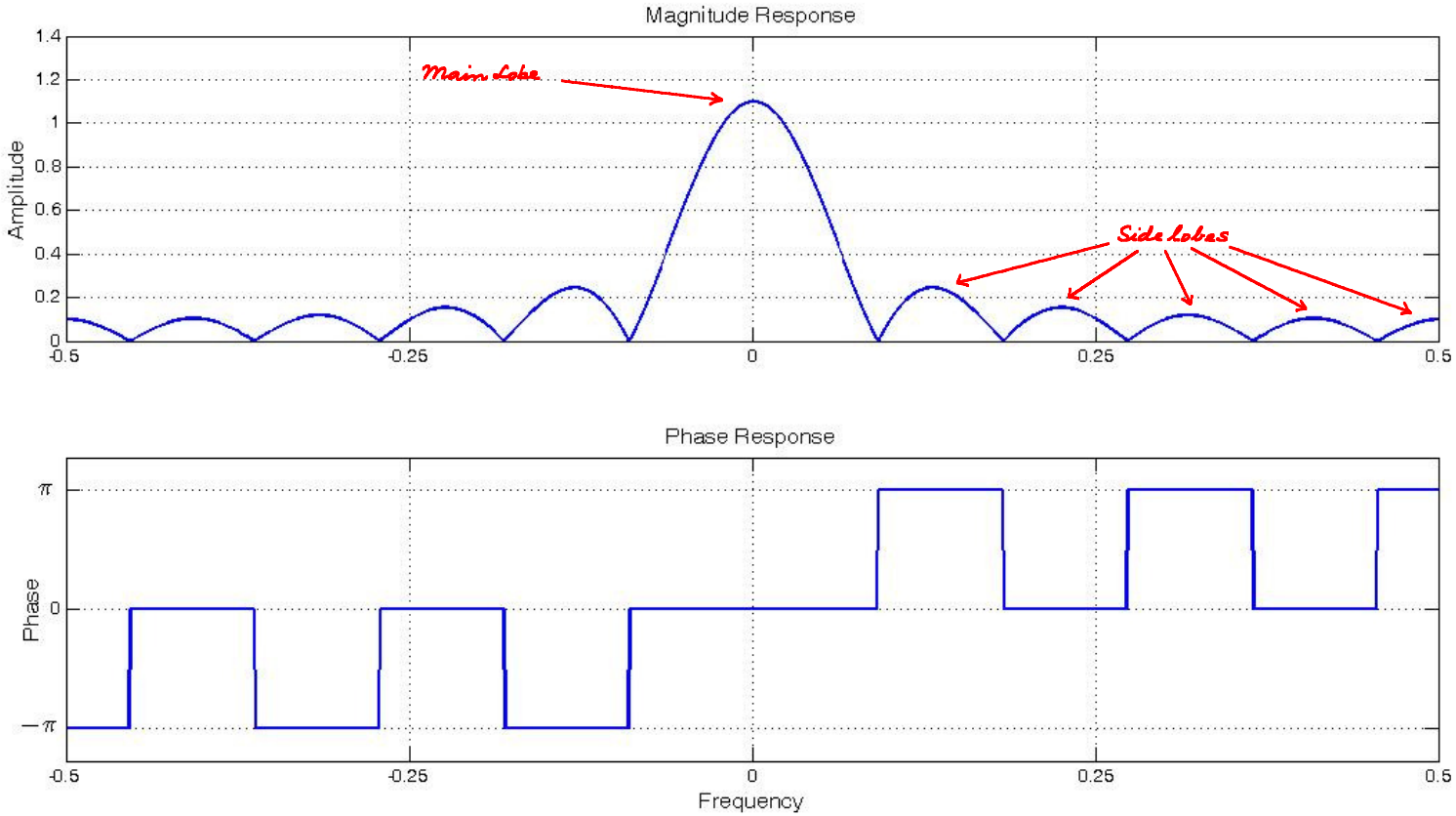
Suppose there is a zero on the unit at $\omega = \theta$. Let $H(e^{j\theta^-}) = H_1$. At $\omega = \theta^+$, the change in contribution due to all the remaining poles and zeros is negligible since the distance and angle change is negligible. However, the zero at $\omega = \theta$ contributes a phase change of π when we go from $\omega = \theta^-$ to $\omega = \theta^+$. Hence, $H(e^{j\theta^+}) = H_1 . e^{j\pi}$, i.e., $H(e^{j\theta^+}) = -H(e^{j\theta^-})$. Thus crossing a zero on the unit circle causes a sign change.

Example: $x[n] = 1 \quad -N \le n \le N \quad \leftrightarrow \quad \frac{\sin (2N+1)\omega/2}{\sin \omega/2}$

If $N = 5$, then there 10 zeros on the unit circle. Crossing each zero will cause a sign change.



If the above frequency response is plotted as two separate plots, i.e., as separate magnitude and phase plots, they will be as shown below:


Magnitude Response
Main Lobe
Side lobes
Amplitude
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-0.5
-0.25
0
0.25
0.5
Phase Response
π
0
−π
Phase
-0.5
-0.25
0
0.25
0.5
Frequency

A representation that avoids phase jumps of π :

$\tau_g(\omega)$ is computed from the unwrapped phase, which is **continuous** when $H(z)$ is a **rational** transfer function. However it still has **jumps of π** caused by zeros on the unit circle.

We can **get rid of jumps of π** from $\phi(\omega)$ if we express $H(e^{j\omega})$ as $A(\omega)e^{j\theta(\omega)}$, where $A(\omega) \in \mathbb{R}$, i.e., it is **real-valued** $\Rightarrow$ it can take on negative values also. Hence, causing a zero on the unit circle causes a **sign change in $A(\omega)$** and **avoids** the discontinuous **jump of π** in the phase response.

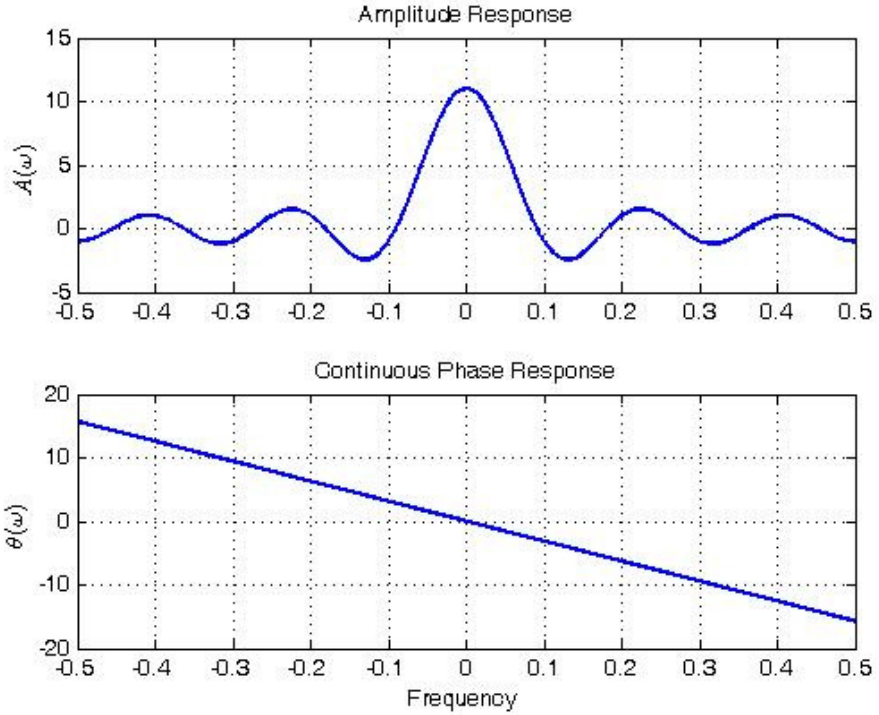
$A(\omega)$ is called as the "amplitude response" or "zero phase" response.

Hence if $h[n] = 1 \quad 0 \le n \le N-1$, then

$$H(e^{j\omega}) = e^{-j\omega\frac{N-1}{2}} \underbrace{\frac{\sin N\omega/2}{\sin \omega/2}}$$

$\theta(\omega) = -\omega\frac{N-1}{2}$ $\qquad A(\omega) \in \mathbb{R}$

Since $A(\omega)$ is allowed to take on negative values, $\theta(\omega)$ is continuous, with even jumps of π being absent. The sign change when crossing the zeros on the unit circle are absorbed by $A(\omega)$ [which is allowed to take on -ve values also, unlike $|H(e^{j\omega})|$].

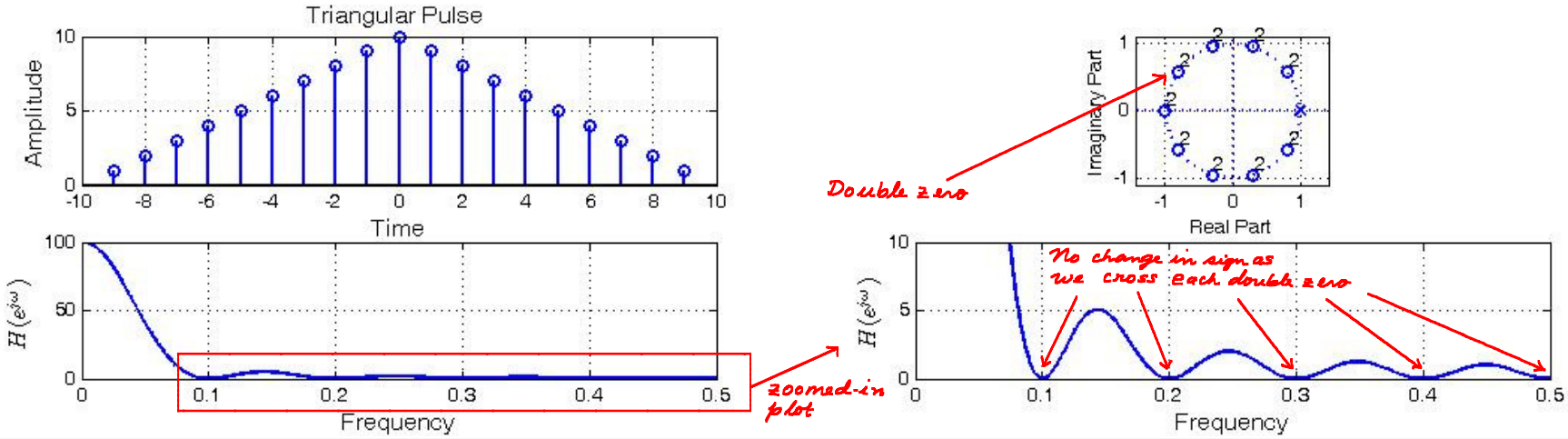


When we cross a first order zero on the unit circle, the phase changes by π, as was explained earlier. If we cross an N^{th} order zero on the unit circle, the phase changes by $N\pi$. Hence, crossing a 2^{nd} order zero causes a change of 2π, which causes no sign change.

Let $x[n]$ be the sequence considered earlier, i.e., $x[n] = 1 \quad -N \le n \le N$. Then $y[n] = x[n] * x[n] \longleftrightarrow X^2(z)$. This causes all the unit circle zeros to become as double zeros. The frequency response is

$$Y(e^{j\omega}) = \left[\frac{\sin (2N+1)\omega/2}{\sin \omega/2} \right]^2$$

has no sign change because of the double zeros

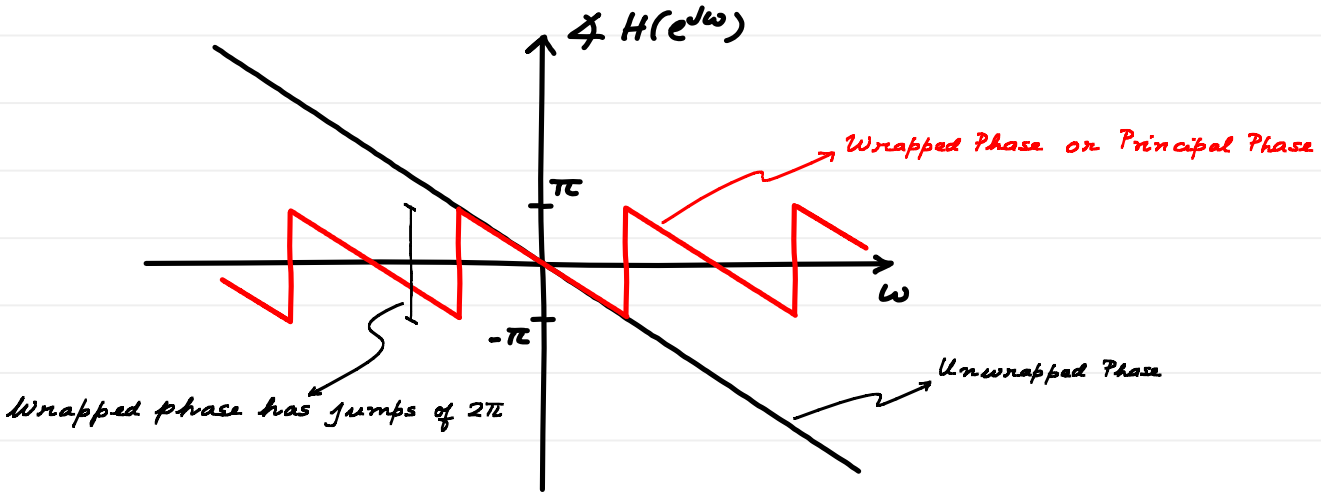

Triangular Pulse
Amplitude
Time
$H(e^{j\omega})$
Frequency
Double zero
Imaginary Part
Real Part
zoomed-in plot
No change in sign as we cross each double zero
$H(e^{j\omega})$
Frequency

If the phase response $\angle H(e^{j\omega})$ is constrained to be in $[-\pi, \pi)$, then it is called as the PRINCIPAL PHASE (or the WRAPPED PHASE) and is denoted as $\boxed{ARG}\{H(e^{j\omega})\}$. When $\angle H(e^{j\omega})$ is not constrained, then it is called as the UNWRAPPED PHASE and denoted by $\boxed{arg}\{H(e^{j\omega})\}$.

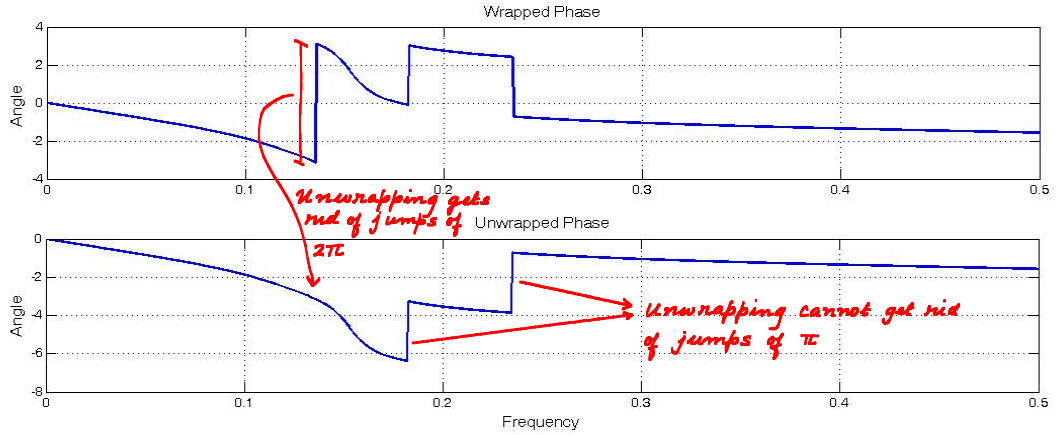
To obtain $ARG\{H(e^{j\omega})\}$ from $arg\{H(e^{j\omega})\}$: if $arg\{H(e^{j\omega})\} < -\pi$, add $k \cdot 2\pi$ by choosing 'k' such that the result lies in the range $[-\pi, \pi]$. If $arg\{H(e^{j\omega})\} > \pi$, subtract $k \cdot 2\pi$ by choosing 'k' such that the result lies in the range $[-\pi, \pi]$.

Unwrapped phase is used in cepstral processing, a topic that is outside the scope of this course.

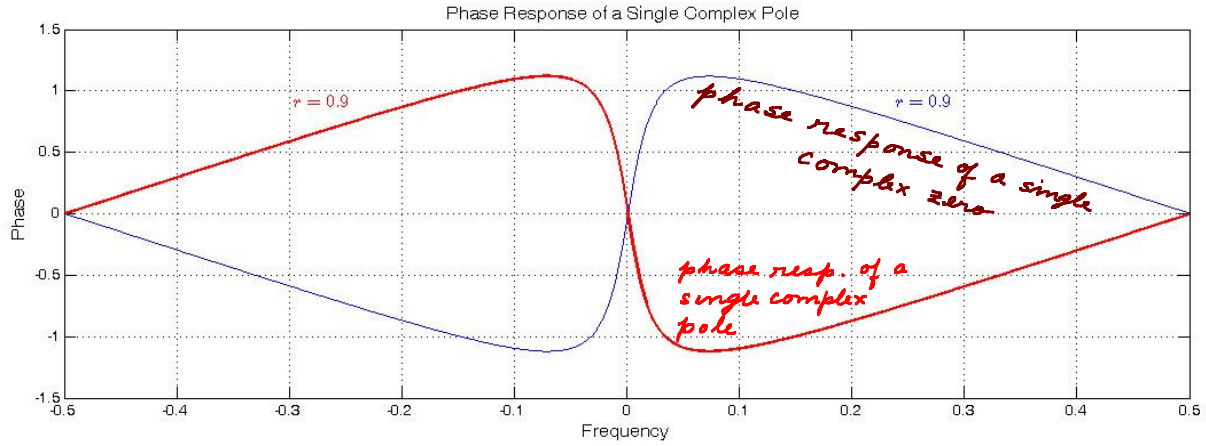
The plot below shows a linear phase term and its wrapped version.



One can go, in principle, from the wrapped phase to the unwrapped phase. This is called **phase unwrapping**. Note that any **jumps of π still remain after phase unwrapping**. Thus, phase unwrapping cannot get rid of these jumps, unlike jumps of 2π.



Phase Response of a Single Complex Pole: The phase response of a single complex pole is the negative of the phase response of a single complex pole.



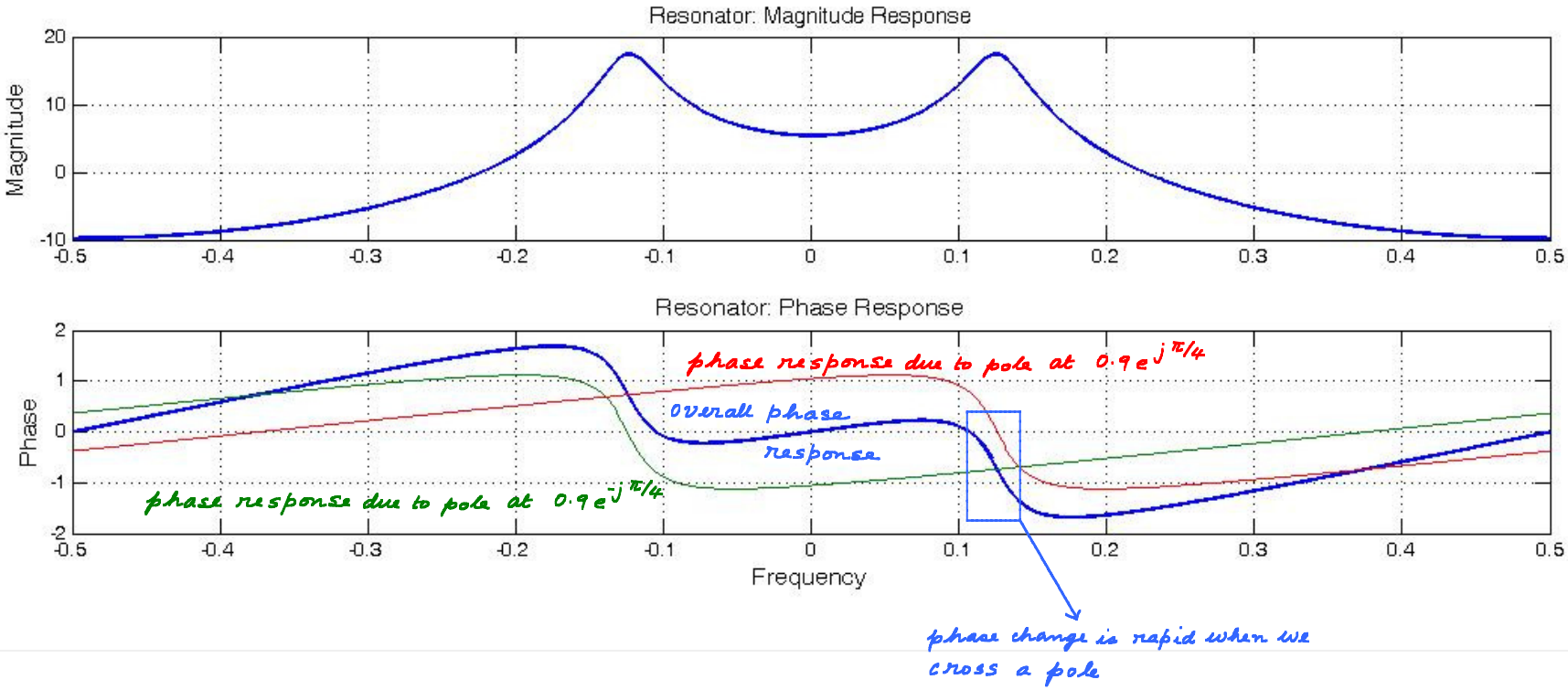
Phase response of a single complex zero is shown for comparitive purposes.

The phase response of a resonator is the sum of the phase responses due to each pole.

Recall that the transfer function of a resonator is

$$H(z) = \frac{1}{(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1})}$$

The magnitude and phase plots are shown below. Note that phase response changes rapidly as we cross a pole. The plots correspond to $r = 0.9$ and $\theta = \frac{\pi}{4}$


Resonator: Magnitude Response
Magnitude
20
10
0
-10
-0.5
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2
0.3
0.4
0.5
Resonator: Phase Response
Phase
2
1
0
-1
-2
Frequency
phase response due to pole at $0.9e^{j\pi/4}$
overall phase response
phase response due to pole at $0.9e^{-j\pi/4}$
phase change is rapid when we cross a pole

EE 2004 DSP Lecture 27

Note Title 22-03-2019

Now let us consider another aspect of phase response by considering a transfer function related to $H(z) = 1 - az^{-1}$. Recall that the corresponding magnitude squared response is $|H(e^{j\omega})|^2 = 1 - 2r\cos(\omega - \theta) + r^2$. Now consider,

$$G(z) = -a^* + z^{-1} = -re^{-j\theta} + z^{-1} \Rightarrow G(e^{j\omega}) = -re^{-j\theta} + e^{-j\omega}$$

Coeff. Vector associated with $H(z)$ $[1, -a]$. That of $G(z)$: $[-a^*, 1]$ i.e., the coeff. vector is complex conjugated and time-reversed

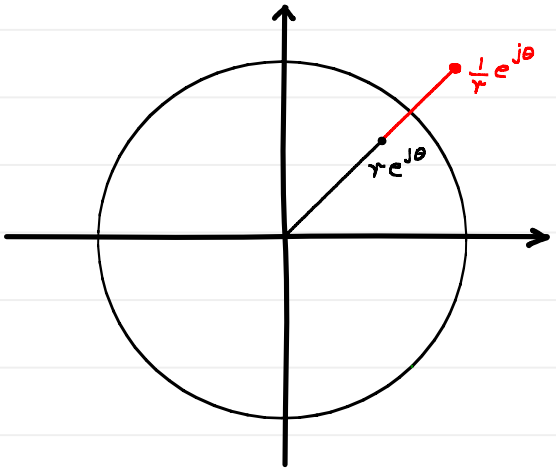
Observe that $|G(e^{j\omega})|^2 = (-re^{-j\theta} + e^{-j\omega})(-re^{j\theta} + e^{j\omega}) = 1 - 2r\cos(\omega - \theta) + r^2 = |H(e^{j\omega})|^2$ i.e., the **magnitude squared response remains unchanged**. However, the **phase responses are different**.

$$\phi_H(\omega) = \tan^{-1}\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)} \qquad \phi_G(\omega) = \tan^{-1}\frac{r\sin\theta - \sin\omega}{\cos\omega - r\cos\theta}$$

Since

$$G(z) = -a^* + z^{-1} = \underbrace{-a^*}\left[1 - \frac{1}{a^*} z^{-1}\right]$$

factor needed to keep the mag. squared response IDENTICAL after reflection.



the zero of $G(z)$ is at $\frac{1}{r} e^{j\theta}$, i.e., it is reflected about the unit circle.

The phase response of $1 - \frac{1}{2} z^{-1}$ and $-\frac{1}{2} + z^{-1}$ are very different.

$\theta_1(\omega)$ of $H_1(z) = 1 - \frac{1}{2}z^{-1}$ and

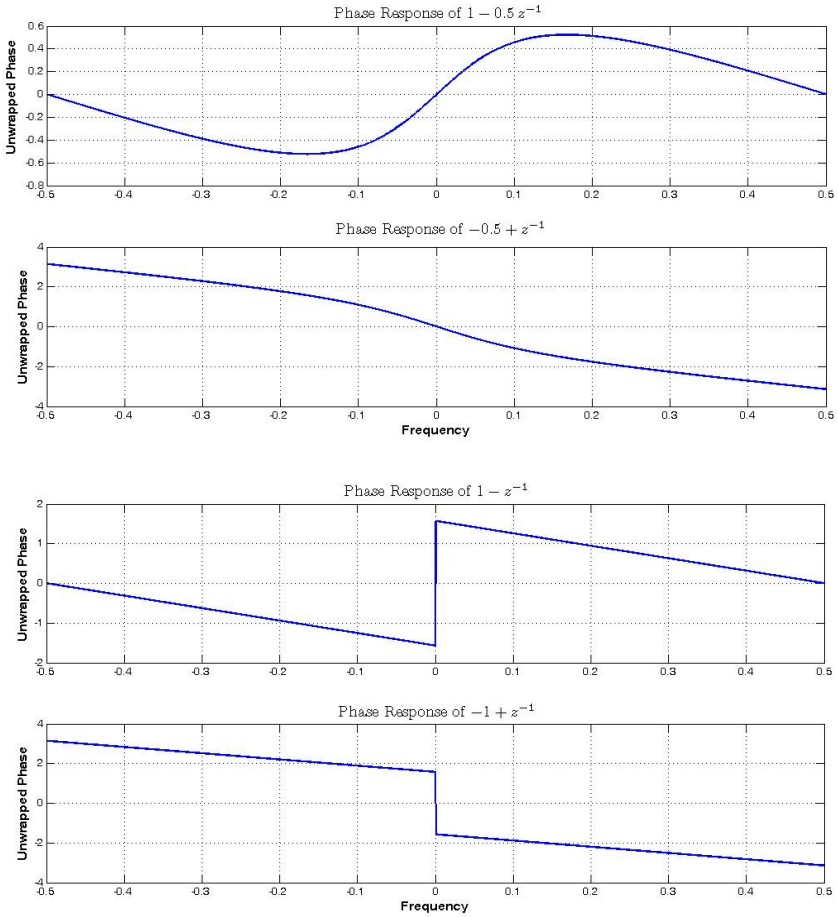
$\theta_2(\omega)$ of $H_2(z) = -\frac{1}{2} + z^{-1}$

are shown on the right top.

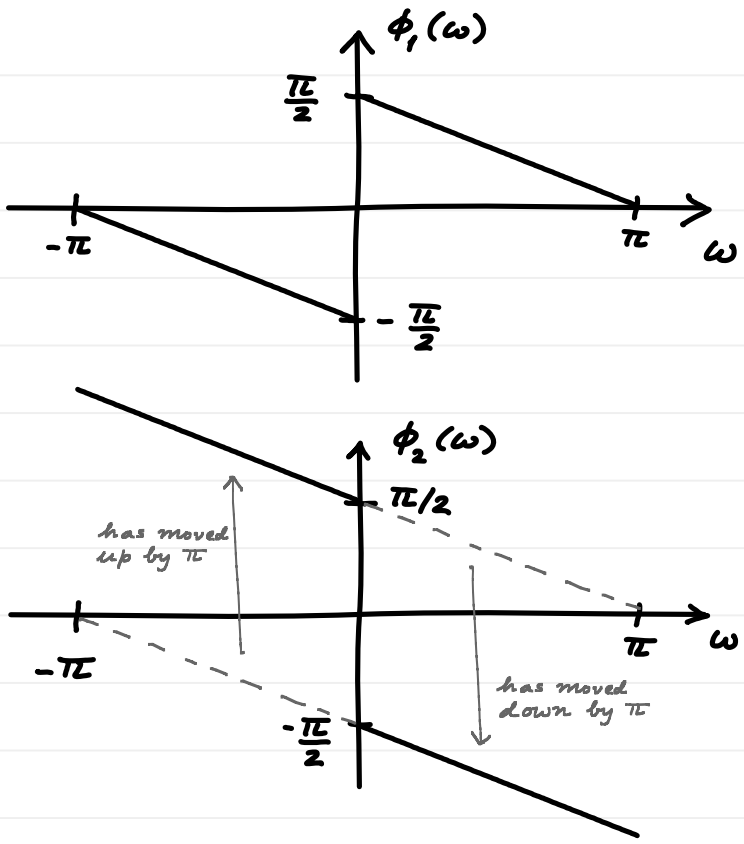
$\phi_1(\omega)$ of $G_1(z) = 1 - z^{-1}$ and

$\phi_2(\omega)$ of $G_2(z) = -1 + z^{-1}$

are shown on the right bottom.



Since $G_2(z) = -G_1(z)$, the multiplication by -1 corresponds to adding a phase of π. The phase for $\omega > 0$ in $\phi_1(\omega)$ shifts down by π to yield the curve $\phi_2(\omega)$ in that region. Since $\phi_2(\omega)$ must be odd-symmetric, the final curve is as shown.



We saw that $1-az^{-1}$ and $-a^{*}+z^{-1}$ have identical magnitude response. Hence

$$H(z) = \frac{-a^{*}+z^{-1}}{1-az^{-1}}$$

has unit magnitude response for all ω. Such a filter is called as an allpass filter.

$$|H(e^{j\omega})| = \left|\frac{-a^{*}+e^{-j\omega}}{1-ae^{-j\omega}}\right| = \left|\frac{-a^{*}+e^{-j\omega}}{e^{-j\omega}(-a+e^{j\omega})}\right| = \left|\frac{(-a+e^{j\omega})^{*}}{e^{-j\omega}(-a+e^{j\omega})}\right| = 1$$

Allpass filter's pole is at $re^{j\theta}$, whereas its zero is at $\frac{1}{r}e^{j\theta}$, i.e., at its reflected location.

An N-th order allpass filter has the form

$$H(z) = \frac{a_N^{*} + a_{N-1}^{*}z^{-1} + \cdots + a_1^{*}z^{-(N-1)} + z^{-N}}{1 + a_1z^{-1} + a_2z^{-2} + \cdots + a_Nz^{-N}} = \prod_{k=1}^{N} \frac{-p_k^{*}+z^{-1}}{1-p_kz^{-1}}$$

An N^{th} order allpass section can be thought of as a cascade of 1^{st} order sections

The inverse of an allpass filter is also all pass, i.e., if $H(z)$ is all pass, then $G(z) = \frac{1}{H(z)}$ is also allpass. If we want causal and stable all pass filters, then $|p_k| < 1 \ \forall k$

Since allpass filters do not affect the magnitude response, they are used for **phase compensation**.

There are allpass filter realization that are robust to coefficient quantization. Such sections can be used to realize stable all-pole filters with less sensitivity to quantization.

In general, the allpass section is an important building block in many signal processing applications.

We now introduce an important idea related to the phase response, i.e., group delay.

Group Delay:

$$\tau_g(\omega) = -\frac{d}{d\omega}\phi(\omega)$$

where $\phi(\omega)$ is the continuous phase function

For a single complex zero, we have $\phi(\omega) = \tan^{-1}\frac{r\sin(\omega-\theta)}{1-r\cos(\omega-\theta)}$

leading to

$$\tau_g(\omega) = \frac{r-\cos(\omega-\theta)}{r+\frac{1}{r}-2\cos(\omega-\theta)}$$

If r is less than 1, then it is easy to see that reflecting the zero outside the unit circle INCREASES $\tau_g(\omega)$. Units of $\tau_g(\omega)$ are samples. If $h[n] \in \mathbb{R}$, then $\phi(\omega)$ is odd-symmetric and therefore $\tau_g(\omega)$ is even-symmetric.

$\tau_g(\omega)$ can take on any value, not necessarily an integer. It can be shown that $\tau_g(\omega) > 0$ in the passband of causal and stable filters. In general, $\tau_g(\omega)$ is a nonlinear function because the phase response $\phi(\omega)$ is itself nonlinear.

A rapid change in phase, caused by a pole or zero near the unit circle, in turn causes a spike in $\tau_g(\omega)$.

If $\phi(\omega)$ is (piecewise) linear, then $\tau_g(\omega)$ is constant.

We will attach meaning to group delay and then see what a desirable phase response is.

Recall that poles and zeros close to the unit circle cause a rapid change in $\theta(\omega)$ and a corresponding spike in $\tau_g(\omega)$.

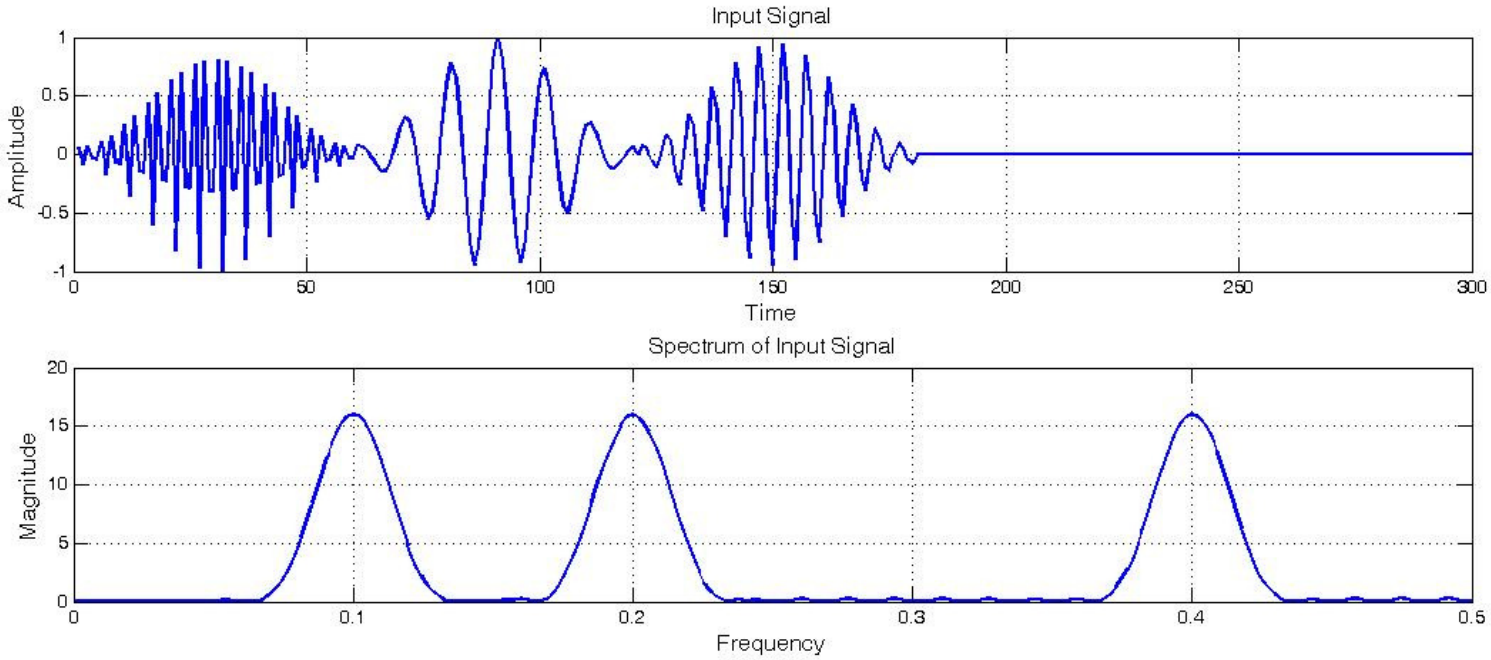
Example

Let $x[n] = w[n]\, x_3[n] + w[n]\, x_1[n-M-1] + w[n]\, x_2[n-2M-2]$

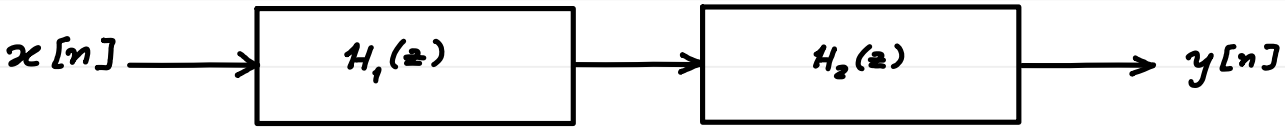
where $w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{M}\right)$, $0 \le n \le M-1$

$x_1[n] = \cos(0.2\pi n)$ $\quad$ $x_2[n] = \cos(0.4\pi n - \pi/2)$ $\quad$ $x_3[n] = \cos(0.8\pi n + \pi/5)$

M has been chosen to be 60.



The signal $x[n]$ is filtered first by $H_1(z)$ and then by $H_2(z)$.



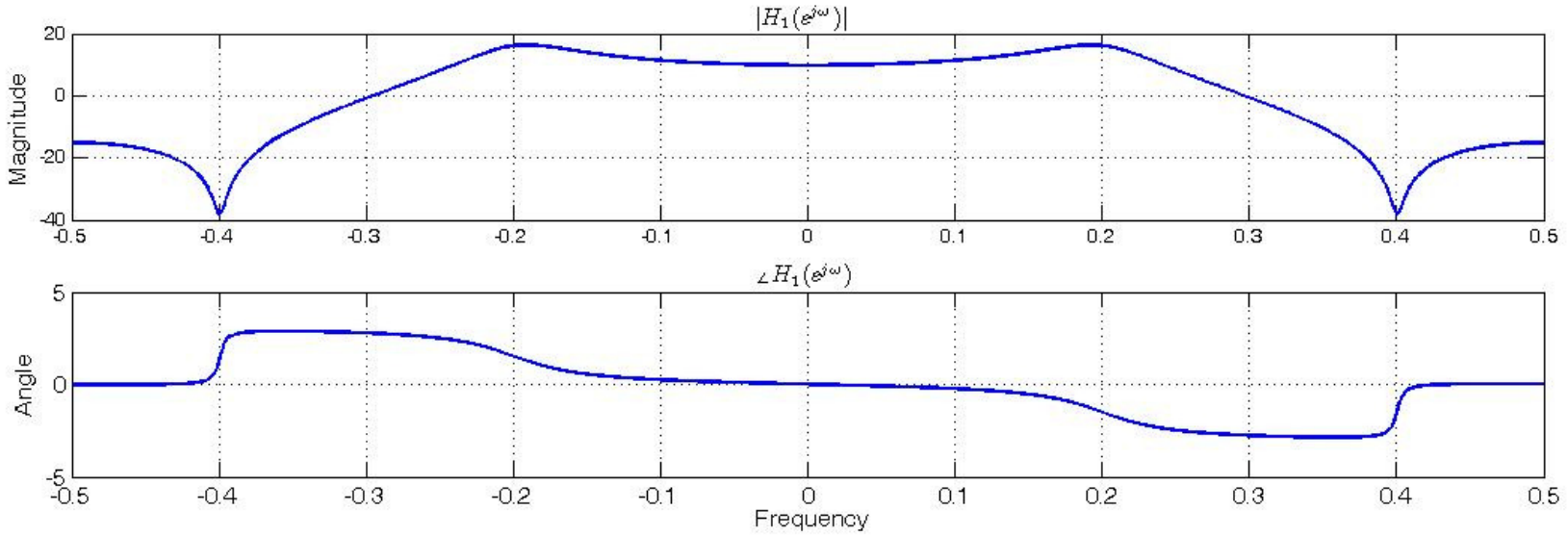
$$H_1(z) = \frac{(1 - 0.98\, e^{j0.8\pi} z^{-1})(1 - 0.98\, e^{-j0.8\pi} z^{-1})}{(1 - 0.8\, e^{j0.4\pi} z^{-1})(1 - 0.8\, e^{-j0.4\pi} z^{-1})}$$

$$H_2(z) = \prod_{k=1}^{4} \left[\frac{(c_k^* - z^{-1})(c_k - z^{-1})}{(1 - c_k^* z^{-1})(1 - c_k z^{-1})} \right]^2$$

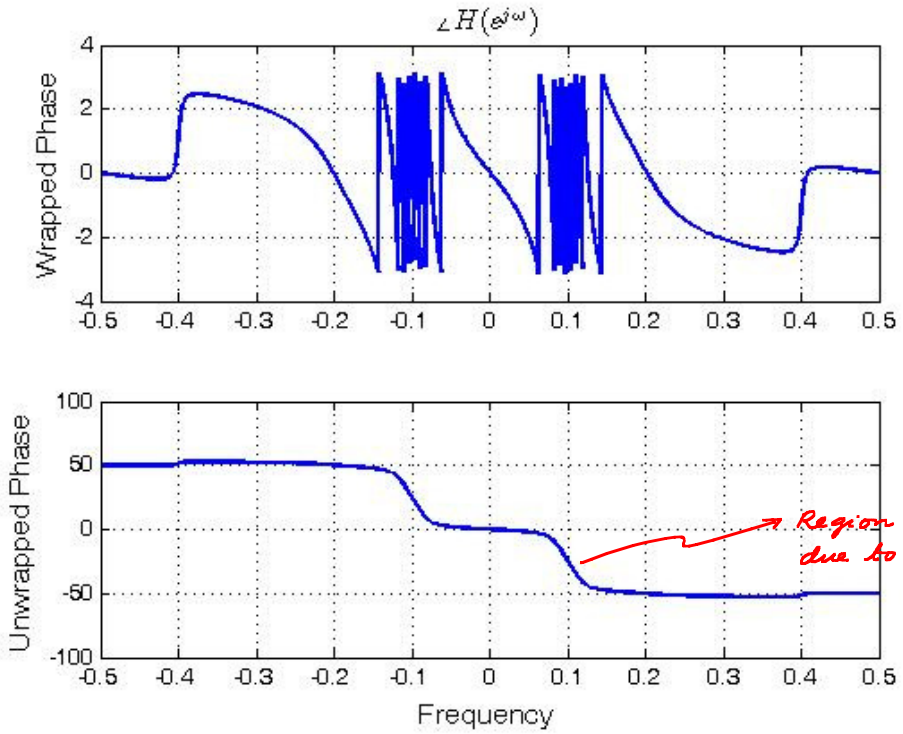
All Pass section : Does not affect magnitude response but contributes to phase response & hence to $\tau_g(\omega)$ ↓ GROUP DELAY

where $c_k = 0.95 \exp\left(j \overline{0.15\pi + 0.02\pi k}\right)$ for $k = 1, 2, 3, 4$

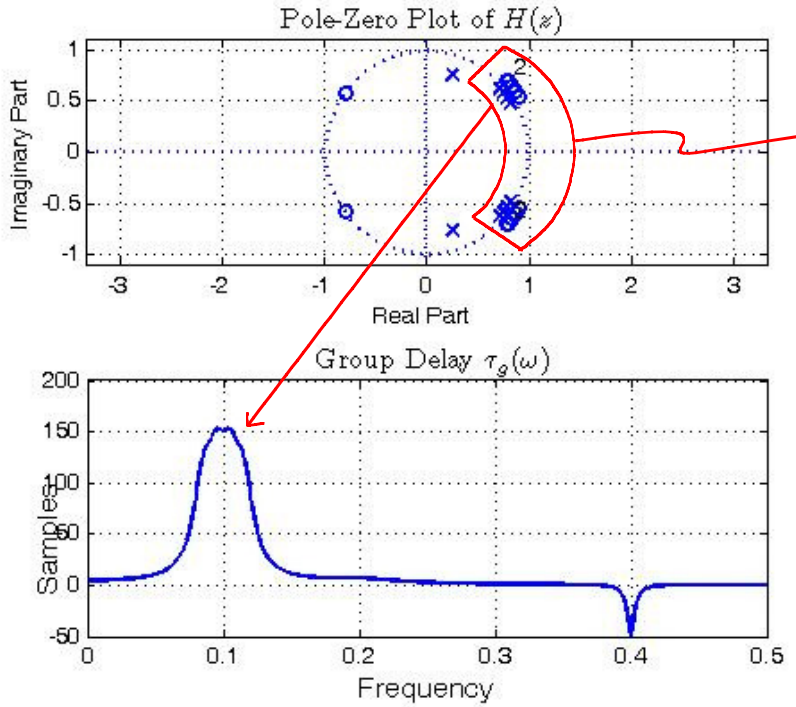
Magnitude and phase response of $H_1(z)$:



The wrapped and unwrapped phase of $H(z) = H_1(z) \cdot H_2(z)$

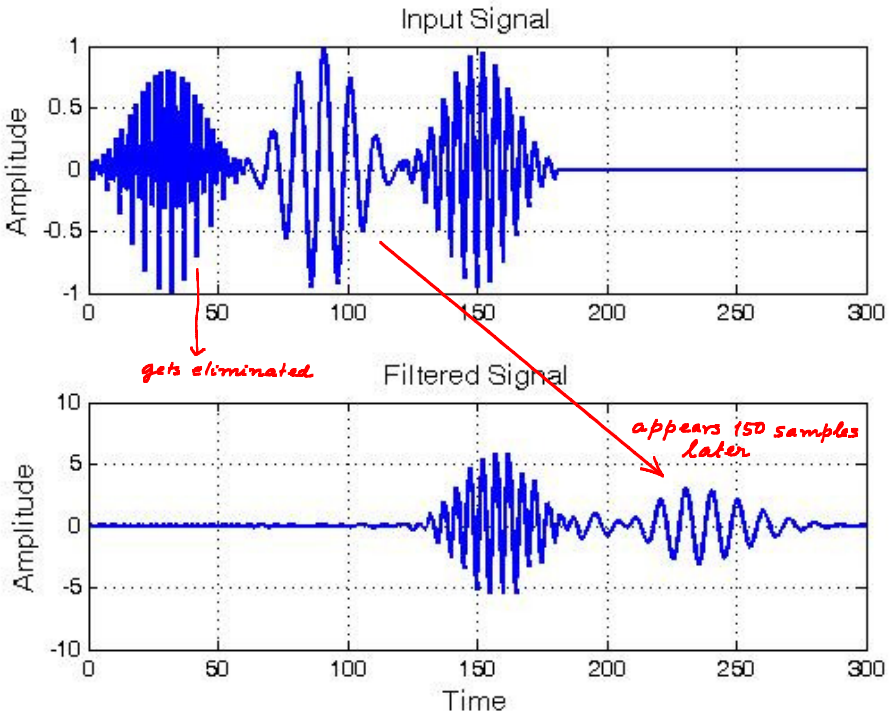


Region of rapid phase change due to the all-pass section $\Rightarrow$ large $\tau_g(\omega)$



all pass section : contributes to group delay

Components around $f = 0.1$ get delayed by 150 samples



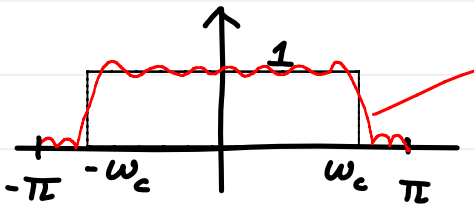
The first pulse gets filtered out due to the null in the frequency response around $f = 0.4$.

The order of appearance of the 2nd and 3rd pulses changes due to the group delay of 150 samples around $f = 0.1$.

EE 2004 DSP Lecture 28

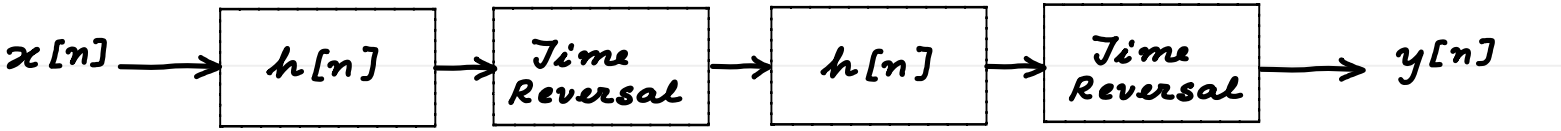
Note Title 22-03-2019

Since causal ideal filters are not realizable (as they violate the Paley-Wiener theorem) we ask the question, "Are filters with rational TF having **zero phase** realizable?"



Rational Transfer function approximation

Zero-phase filtering can be obtained as follows:



Therefore, $Y(e^{j\omega}) = X(e^{j\omega})\,H(e^{j\omega})\,H(e^{-j\omega})$

For real-valued $h[n]$, $H(e^{-j\omega}) = H^*(e^{j\omega})$. Hence, $Y(e^{j\omega}) = X(e^{j\omega}) \cdot \underbrace{|H(e^{j\omega})|^2}_{\text{zero-phase transfer function}}$

However, the above sequence of operations make the overall system non-causal and hence not realizable.

Hence, realizable filters cannot have zero-phase.

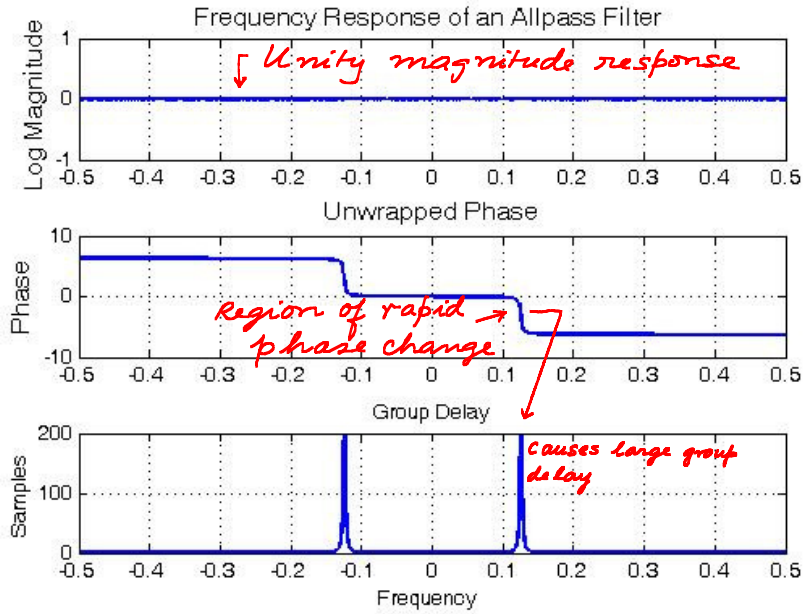
Let us now look at the effect of phase response on a signal. Consider the input

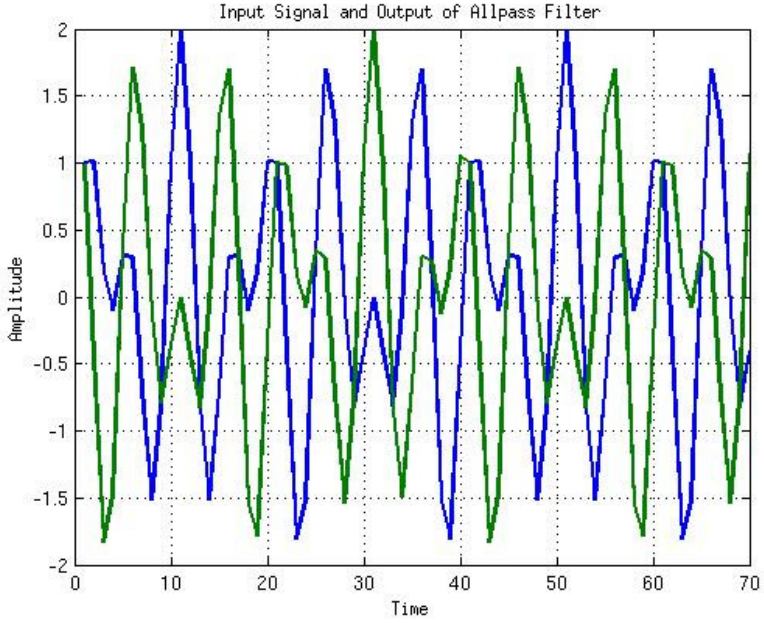
$$x[n] = \cos(0.25\pi n) + \cos(0.4\pi n)$$

It is passed through an allpass filter $H(z) = \dfrac{0.9801 - 1.4001 z^{-1} + z^{-2}}{1 - 1.4001 z^{-1} + 0.9801 z^{-2}} \Rightarrow |H(e^{j\omega})| = 1$

The magnitude and phase response are shown below. The phase response is highly nonlinear. Therefore, even though the magnitude response is unity, the nonlinear phase response causes waveshape distortion, i.e., the waveshape is not preserved. The given $x[n]$ and the filtered output are shown alongside. The distortion is severe and, in particular, the output is not delayed version of the input.

Thus $|H(e^{j\omega})| = 1$ at the signal frequencies is not enough to preserve the waveshape at the output. Nonlinear phase causes waveshape distortion despite unity gain.


Frequency Response of an Allpass Filter
Log Magnitude
Unity magnitude response
Unwrapped Phase
Phase
Region of rapid phase change
Group Delay
Samples
causes large group delay
Frequency


Input Signal and Output of Allpass Filter
Amplitude
Time

If the phase response is zero and $|H(e^{j\omega})| = 1$ at the frequencies of interest, there will not be any distortion. But we saw earlier that zero-phase filtering is unrealizable.

The next best alternative is **delay distortion**, i.e., $y[n] = x[n-n_0]$. That is, the o/p is merely a delayed version of the input. For the example at hand,

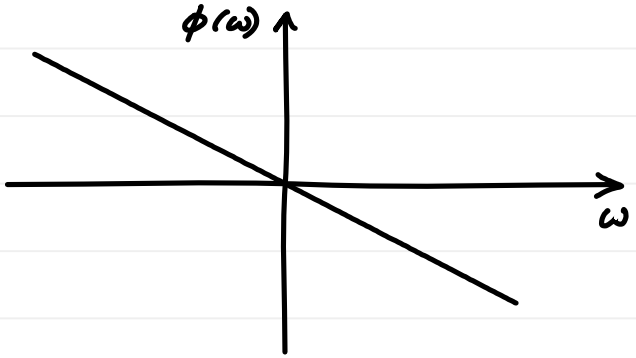
$$y[n-n_0] = \cos(\omega_1 \overline{n-n_0}) + \cos(\omega_2 \overline{n-n_0}) = \cos(\omega_1 n - \omega_1 n_0) + \cos(\omega_2 n - \omega_2 n_0)$$

is an acceptable output because the delay distortion is a **benign** one. Notice that in such a case **phase is proportional to frequency**. Therefore, if the **phase response is LINEAR**, only the benign delay distortion is introduced. **Most importantly, WAVESHAPE IS PRESERVED.**

Can filters with rational transfer functions with **linear phase** be realized?

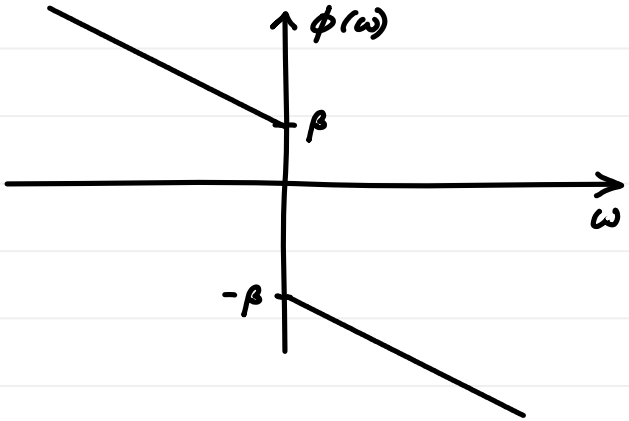
Linear Phase Filters:

A pure delay introduces a linear phase. Since delay distortion is a benign one, we examine what restrictions, if any, are imposed if we enforce linear phase.



$$\phi(\omega) = -\alpha\omega$$

Linear Phase



$$\phi(\omega) = -\alpha\omega + \beta$$

Generalized Linear Phase

Linear phase is a special case of Generalized Linear Phase with $\beta = 0$.

Note that if $\theta(\omega) = -\alpha\omega + \beta$, then $\tau_g(\omega) = -\frac{d}{d\omega}\theta(\omega) = \alpha$, a constant.

Let us now enforce the generalized linear phase condition on $H(e^{j\omega})$.

$$H(e^{j\omega}) = |H(e^{j\omega})|\, e^{j(-\alpha\omega+\beta)} = \sum_n h[n]\, e^{-j\omega n}$$

Equating the tangent of the angle on both sides, we get

$$\frac{-\sum_n h[n]\,\sin(\omega n)}{\sum_n h[n]\,\cos(\omega n)} = \frac{\sin(-\alpha\omega+\beta)}{\cos(-\alpha\omega+\beta)}$$

If we cross multiply and simplify, we get

$$\sum_n h[n]\,\sin\left(\overline{-\alpha+n}\,\omega+\beta\right) = 0$$

This is the condition that the impulse response $h[n]$ must satisfy if we impose the condition $\theta(\omega) = -\alpha\omega + \beta$.

Clearly $h[n]=0$ satisfies the above equation. Does a non-trivial solution exist ?
To simplify matters we will assume that $h[n]$ is a causal FIR filter. Initially we will set $\beta = 0$. That is we need a non-trivial solution for

$$\sum_{n=0}^{N-1} h[n] \sin(\overline{n-\alpha}\,\omega) = 0$$

We will explore if cancellations are possible. In particular, are pairwise cancellations possible ? To this end, consider $h[0]\sin(-\alpha\omega)$ and $h[N-1]\sin(\overline{N-1-\alpha}\,\omega)$

Suppose $N-1-\alpha = \alpha$ i.e., $\alpha = \frac{N-1}{2}$ and $h[0] = h[N-1]$, these terms cancel ! That is,

$$h[0]\sin(-\alpha\omega) + h[N-1]\sin(\overline{N-1-\alpha}\,\omega) = 0 \qquad \text{where} \quad h[0]=h[N-1] \;\&\; \alpha = \frac{N-1}{2}$$

Next consider $h[1]\,\mathrm{Sin}(\overline{1-\alpha}\,\omega)$ and $h[N-2]\,\mathrm{Sin}(\overline{N-2-\alpha}\,\omega)$. Once again $\alpha = \frac{N-1}{2}$ and $h[1] = h[N-2]$ causes pairwise cancellation! Proceeding similarly,

$$\boxed{\alpha = \frac{N-1}{2} \quad \text{and} \quad h[n] = h[N-1-n]} \qquad \text{for } n = 0, 1, 2, \ldots$$

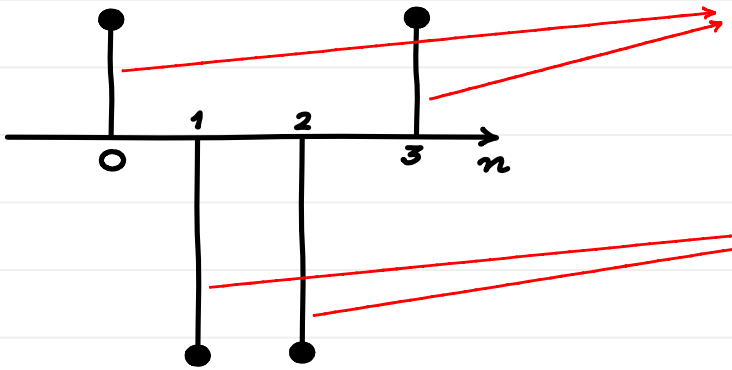
gives rise to pairwise cancellations and thus leads to a non-trivial solution of

$$\sum_{n=0}^{N-1} h[n]\,\mathrm{Sin}(\overline{n-\alpha}\,\omega) = 0$$

Observe that the above conditions are also valid if $\beta = \pi$. Observe that $h[n]$ is even-symmetric about $n = \frac{N-1}{2}$.

Example Sequence for N = 4 :

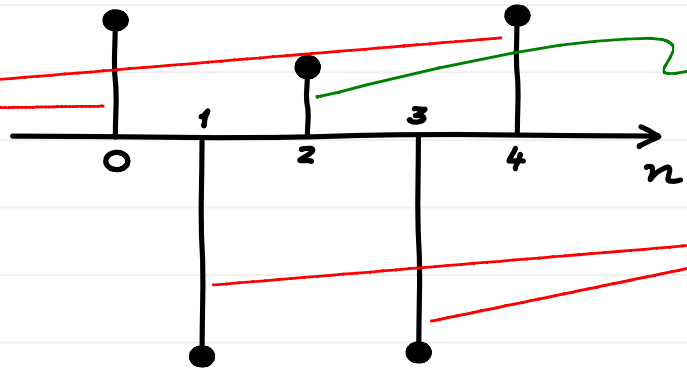
$$\alpha = \frac{N-1}{2} = 1.5$$



pairwise cancel each other

pairwise cancel each other

Example Sequence for N = 5 :



pairwise cancel each other

Does not need a pair to cancel because, since $\alpha = 2$, $\sin(\overline{n-\alpha}\,\omega) = 0$ for $n = 2$ independent of $x[2]$.

pairwise cancel each other

$$\alpha = \frac{N-1}{2} = 2$$

EE 2004 DSP Lecture 29

Note Title 26-03-2019

Suppose $\beta = \frac{\pi}{2}$. Then,

$$\sum_n h[n] \sin\left(-\alpha n \omega + \frac{\pi}{2}\right) = \sum_n h[n] \cos\left(-\alpha n \omega\right) = 0$$

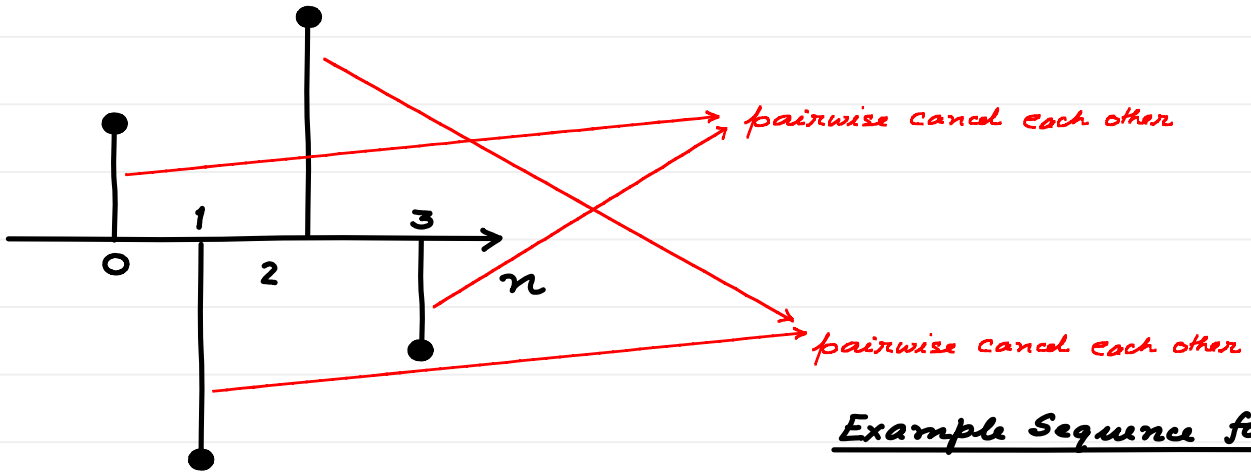
Arguing similarly, we see that, since cosine is an even function, we require

$$\alpha = \frac{N-1}{2} \quad \text{and} \quad h[n] = -h[N-1-n]$$

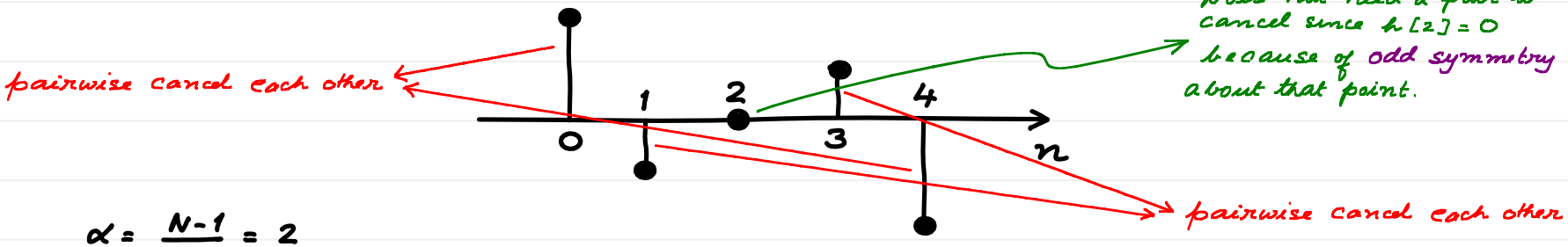
The non-trivial solution now demands that $h[n]$ be odd-symmetric about $n = \frac{N-1}{2}$.

Example Sequence for N = 4 :

$$\alpha = \frac{N-1}{2} = 1.5$$



Example Sequence for N = 5 :



$$\alpha = \frac{N-1}{2} = 2$$

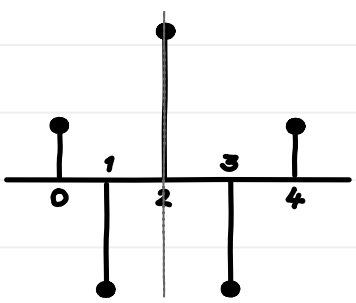
Thus the symmetry about $\frac{N-1}{2}$ can be either EVEN ($\beta = 0$ or π) or ODD ($\beta = \frac{\pi}{2}$ or $\frac{3\pi}{2}$).
The filter length N can be even or odd. FIR filters can therefore be classified as follows:

Length	Symmetry		Group Delay		Name
Odd	Even	$\beta = 0$	↑	int	Type I
Even	Even	$\beta = 0$	$\frac{N-1}{2}$	int + $\frac{1}{2}$	Type II
Odd	Odd	$\beta = \frac{\pi}{2}$		int	Type III
Even	Odd	$\beta = \frac{\pi}{2}$	↓	int + $\frac{1}{2}$	Type IV

$\beta = 0$ requires even symmetry and results in linear phase. For $\beta = \frac{\pi}{2}$ we require odd symmetry and results in generalized linear phase. The sequence is real-valued and odd, which means that the transform is purely imaginary, i.e. $|H(e^{j\omega})| e^{-j\alpha\omega} \cdot \underbrace{e^{j\pi/2}}_{=j} \leftarrow \beta$

If N is odd, the point of (anti-)symmetry falls on a sample value.

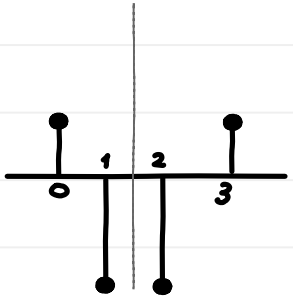
If N is even, the point of (anti-)symmetry falls midway between the centre two samples, which is a imaginary point. $\tau_g(\omega) = \frac{N-1}{2}$, a constant (independent of $h[n]$).



Type I

N = 5

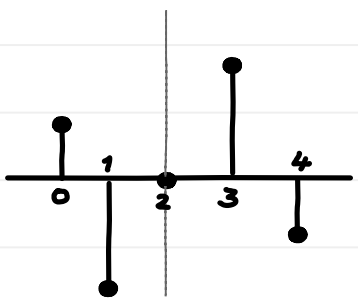
$\tau_g(\omega) = \frac{5-1}{2} = 2$



Type II

N = 4

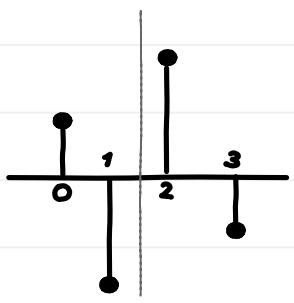
$\tau_g(\omega) = \frac{4-1}{2} = 1.5$



Type III

N = 5

$\tau_g(\omega) = \frac{5-1}{2} = 2$



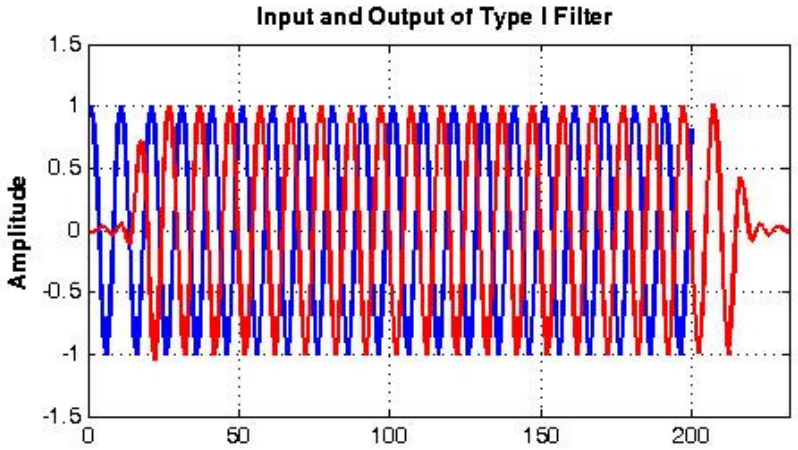
Type IV

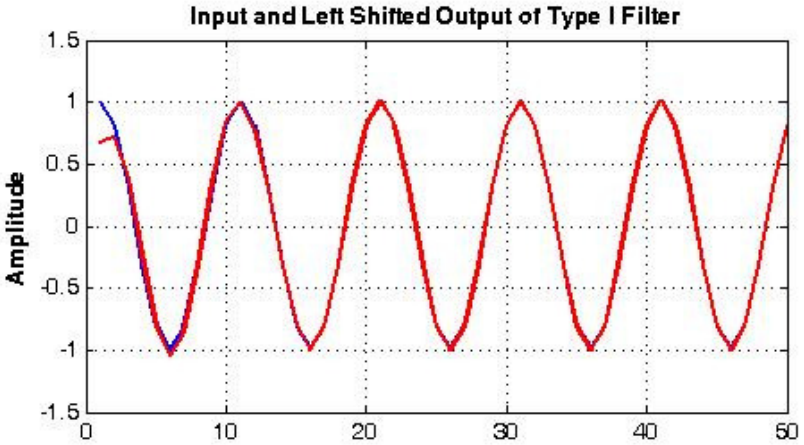
N = 4

$\tau_g(\omega) = \frac{4-1}{2} = 1.5$

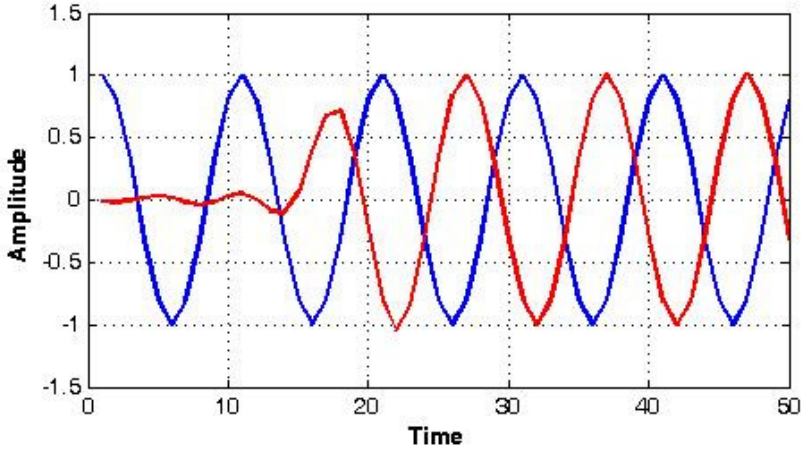
If the group delay introduced by a linear phase filter is an integer, the output can be shifted by that many samples to the left to align the output and the input (which is useful in those cases where such alignment can lead to meaningful comparisons).

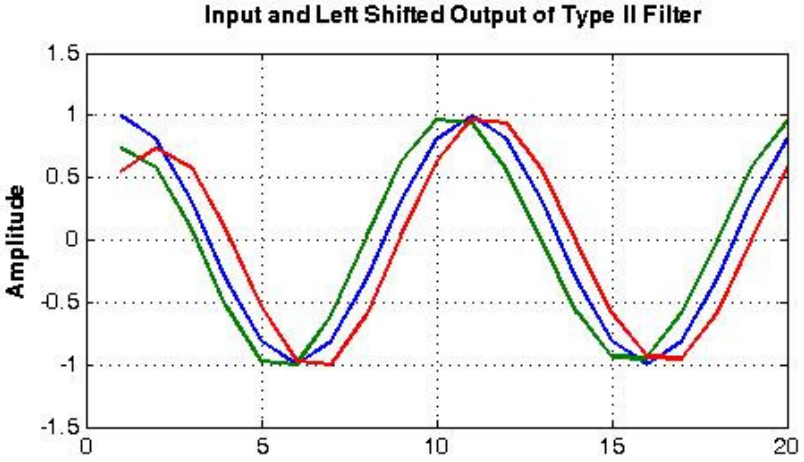
$\tau_g(\omega) = \frac{N-1}{2}$ is an integer for Types I and III. For Types II and IV $\tau_g(\omega)$ is integer $+ \frac{1}{2}$, i.e., these introduce half-sample delays. In applications such as image processing, half-sample delays are undesirable. Hence Types II and IV cannot be used in applications where such delays cannot be tolerated.





Integer shift equal to 0.5 x (N-1) produces exact alignment with the input





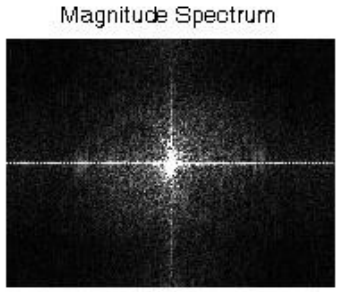
Neither of the integer shifts closest to 0.5 x (N-1) produces exact alignment with the input

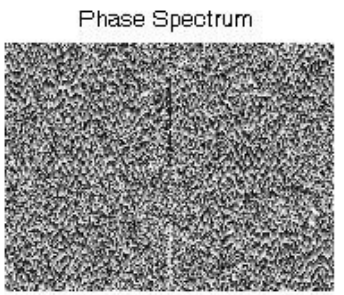
The eye is very sensitive to phase, whereas the ear is completely insensitive to fixed changes in phase.

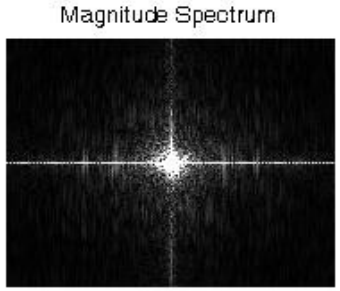
Suppose x is a vector containing an audio clip. Playing x and $-x$ (i.e., a signal and its 180° out-of-phase version) will sound *identical* to the ear. OTOH, if x is an image, displaying $-x$ will be perceived as the *negative* of the original image. Thus the eye is extremely sensitive to phase.

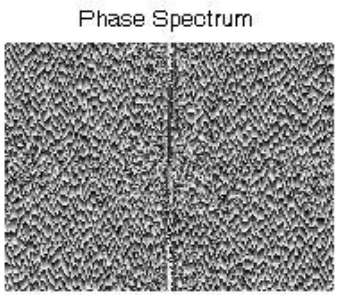
Consider two images X and Y, and their transforms $\hat{X}$ and $\hat{Y}$. If we form

$|\hat{X} e^{j \angle \hat{Y}}|$, i.e., take the magnitude from one image's spectrum and the phase from the other, and then inverse transform, the resulting image will *appear closer to the image that contributed to the phase!*









See http://matlabgeeks.com/tips-tutorials/how-to-do-a-2-d-fourier-transform-in-matlab/

Church

ifft2(|church|*exp(j*angle(face)))

Face

ifft2(|face|*exp(j*angle(church)))

When the magnitude and phase come from different images, the resultant image, even though it is garbled, looks very similar to the image whose phase function was used!

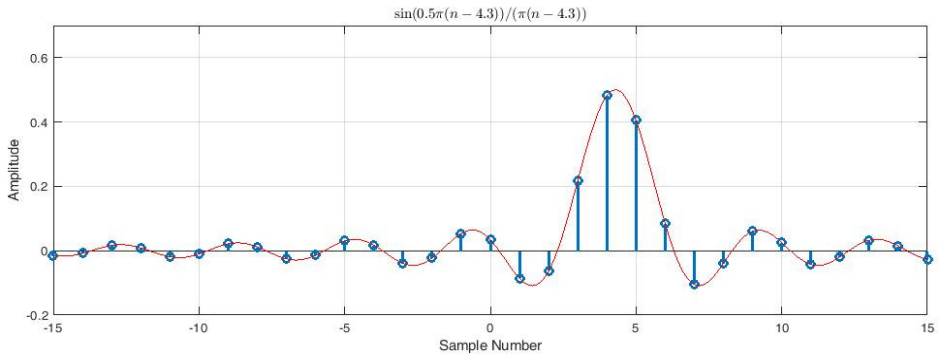
EE 2004 DSP Lecture 30

Note Title 27-03-2019

For FIR filters symmetry is both necessary and sufficient for linear phase. Note that the symmetry can either be even or odd. For IIR filters, symmetry is sufficient but not necessary for a filter to have linear phase. To see this, consider

$$h[n] = \frac{\sin \omega_c (n-\alpha)}{\pi (n-\alpha)}$$

The above is symmetric only when α is an integer or integer + ½. For other values of α, $h[n]$ is not symmetric (see plot) but the filter has linear phase

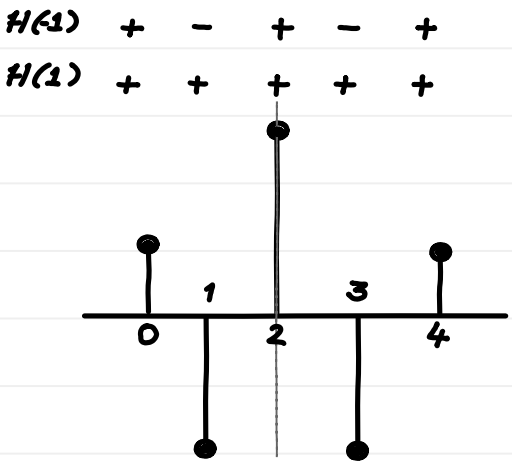


Constrained Zeros of Linear Phase FIR filters:

We will show that the symmetry conditions (even or odd) force some zeros to be present in the transform domain. These constrained zeros occur at $z = 1$ and/or $z = -1$.

$$H(1) = \sum_{n=0}^{N-1} h[n] \quad \text{and} \quad H(-1) = \sum_{n=0}^{N-1} h[n](-1)^{-n} = \sum_{n=0}^{N-1} h[n](-1)^{n}$$

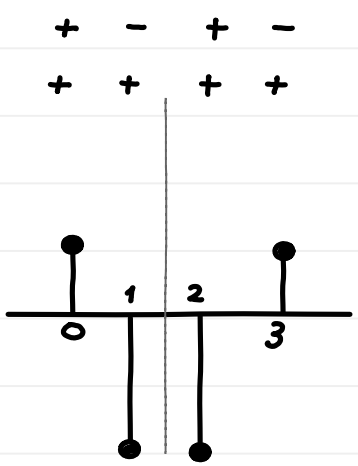
By considering examples of length 4 and 5 we can easily see the pattern for the general case.



Type I

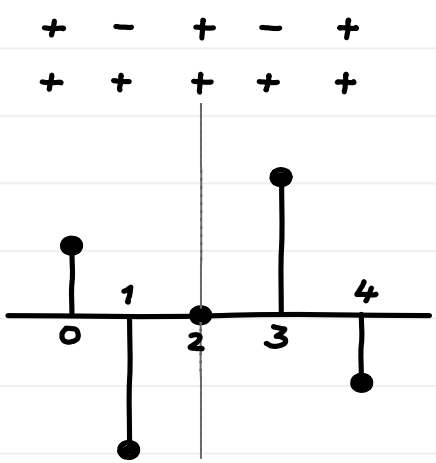
H(1) need not always be 0

H(-1) need not always be 0



Type II

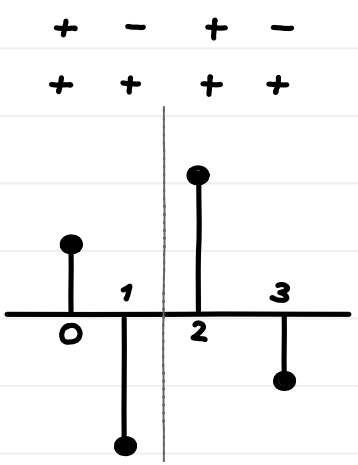
H(-1) = 0 always



Type III

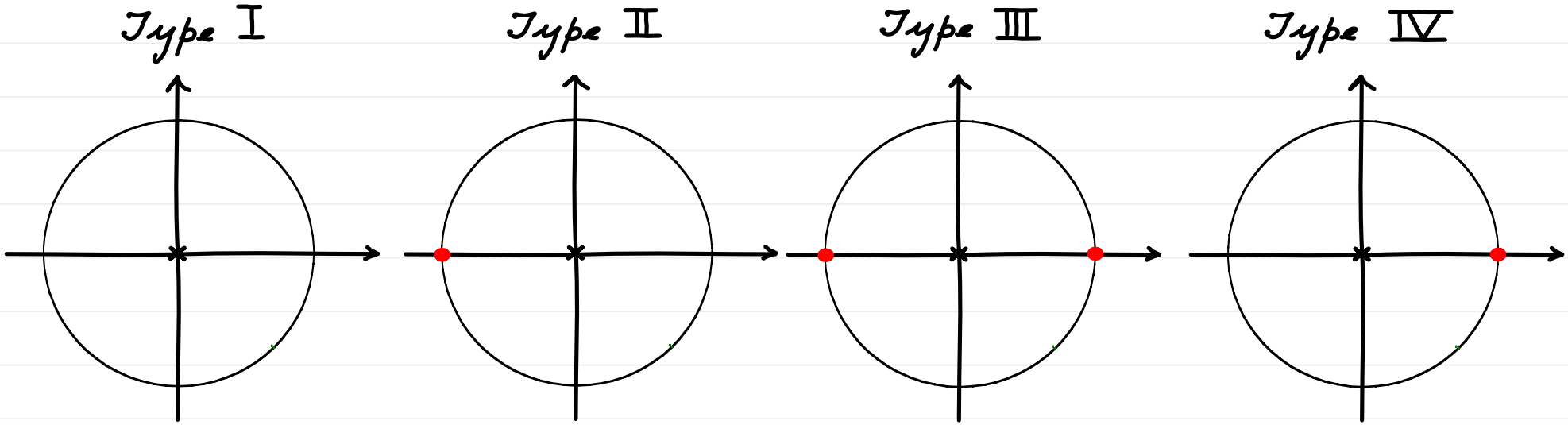
H(1) = 0 always

H(-1) = 0 always



Type IV

H(1) = 0 always



Cannot be used for building
HPF

Cannot be used for building
LPF, HPF

Cannot be used for building
LPF

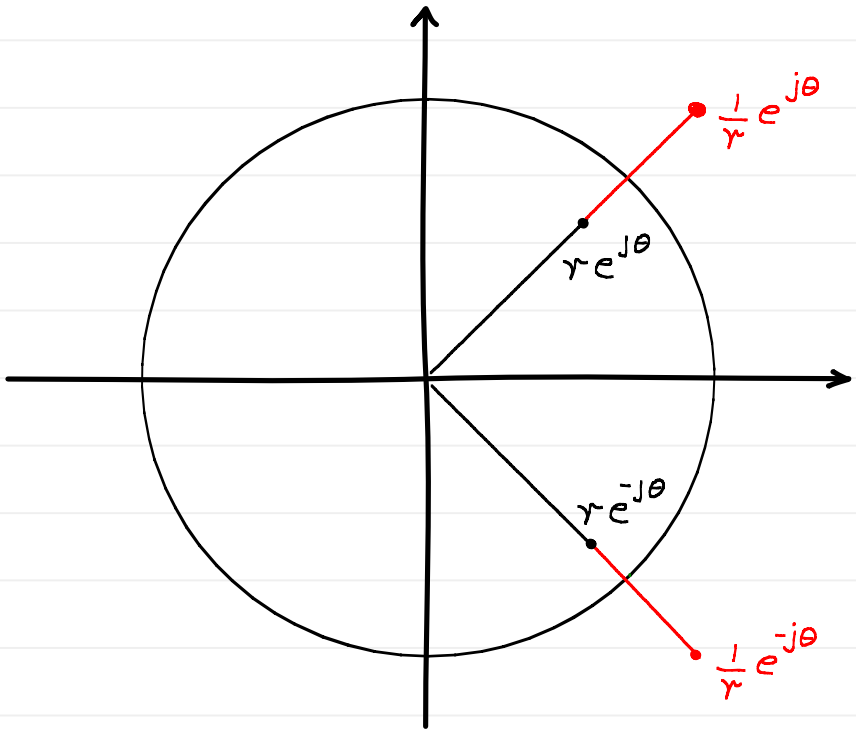
Relationship Between the zeros of a linear phase FIR filter:

Recall that the impulse response of a linear phase FIR filter must satisfy $h[n] = \pm h[N-1-n]$. This implies

$$H(z) = \pm z^{-(N-1)} H(z^{-1})$$

In addition, if $h[n] \in \mathbb{R}$, then $h[n] = h^*[n]$, which implies $H(z) = H^*(z^*)$. This means that the roots of $H(z)$ occur in complex conjugate pairs. Symmetry implies that, if z_0 is a zero, z_0^{-1} is also a zero. Thus, for real-valued sequences with symmetry, if z_0 is a zero, then z_0^*, $1/z_0$, and $1/z_0^*$ are also zeros. Thus

$$H(z_0) = H(z_0^*) = H(1/z_0) = H(1/z_0^*) = 0$$



Hence if $z_0 = re^{j\theta}$, the related zeros are

$re^{-j\theta}$, $\frac{1}{r}e^{j\theta}$, and $\frac{1}{r}e^{-j\theta}$

If $r = 1$, then just two roots are enough to satisfy all four conditions, assuming $\theta \neq 0, \pi$. If $\theta = 0$ or π, just one root satisfies all 4 conditions.

Exercise: Show that when $h[n] \in \mathbb{C}$, for (generalized) linear phase we need $h[n] = \pm h^*[N-1-n]$

EE 2004 DSP Lecture 31

Note Title 28-03-2019

Earlier we saw an example of a linear phase IIR filter. However it is not realizable because it is non-causal. There exists a causal, linear phase IIR filter, but it does not have a rational TF, and hence not realizable. A method for realizing this non-rational TF exists but it turns out to be non-causal. What about IIR filters with rational TF that have exact linear phase? Note that if $H(z)$ is a linear phase FIR filter, then $G(z) = 1/H(z)$ is a linear phase IIR filter. If it is to be stable, it must be non-causal.

Hence, if we want realizable, causal, and stable linear phase filters, then FIR filters are the only choice.

Frequency Response of Linear Phase FIR Filters:

Let $H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n] e^{-j\omega n}$. It is usual to let $M = N-1$.

For Type I filter, $h[0] = h[M]$, $h[1] = h[M-1]$, and so on. Hence,

$$H(e^{j\omega}) = h[0] + h[1]e^{-j\omega} + h[2]e^{-j2\omega} + \dots + h\left[\frac{M}{2}-1\right]e^{-j\omega\frac{M}{2}-1} + h\left[\frac{M}{2}\right]e^{-j\omega\frac{M}{2}}$$

$$\underbrace{h[M]}_{=h[0]}e^{-jM\omega} + \underbrace{h[M-1]}_{=h[1]}e^{-j(M-1)\omega} + \dots + \underbrace{h\left[\frac{M}{2}+1\right]}_{=h\left[\frac{M}{2}-1\right]}e^{-j\omega\frac{M}{2}+1}$$

Hence, collecting similar terms and simplifying, we get

$$H(e^{j\omega}) = e^{-j\omega M/2}\overbrace{\left[h\left[\frac{M}{2}\right] + \sum_{n=0}^{\frac{M}{2}-1} 2h[n]\cos\left(\frac{M}{2}-n\,\omega\right)\right]}^{A(\omega)\in\mathbb{R},\ \text{zero-phase response}}$$

[fill in the missing steps]

Now consider $g[n] = h\left[n + \frac{N}{2}\right]$. It is easy to see that $G(e^{j\omega}) = A(\omega)$ since $G(e^{j\omega}) = e^{j\omega M/2} H(e^{j\omega})$. Thus $g[n]$ is the sequence whose DTFT equals $A(\omega)$.

For Type II filter, the only difference is that $N-1 = M$ is an odd number. Hence,

$$H(e^{j\omega}) = e^{-j\omega M/2} \overbrace{\left[\sum_{n=0}^{\frac{M-1}{2}} 2h[n] \cos\left(\frac{M}{2} - n \, \omega \right) \right]}^{A(\omega) \in \mathbb{R}, \text{ zero-phase response}}$$

[fill in the missing steps]

Note that in this case there does not exist a sequence with DTFT $A(\omega)$ [Explain why this is so.]

For Types III and IV, the derivation is similar. The final expressions are:

[fill in the missing steps]

Type III

$$H(e^{j\omega}) = j\, e^{-j\omega M/2} \underbrace{\left[\sum_{n=0}^{\frac{M-1}{2}} 2h[n] \operatorname{Sin}\left(\tfrac{M}{2} - n \;\omega \right) \right]}_{A(\omega) \in \mathbb{R},\ \text{zero-phase response}}$$

$j = e^{j\frac{\pi}{2}}$

Type IV

$$H(e^{j\omega}) = j\, e^{-j\omega M/2} \underbrace{\left[\sum_{n=0}^{\frac{M-1}{2}} 2h[n] \operatorname{Sin}\left(\tfrac{M}{2} - n \;\omega \right) \right]}_{A(\omega) \in \mathbb{R},\ \text{zero-phase response}}$$

Similar to Type II, there is no sequence with DTFT $A(\omega)$ for Type IV.

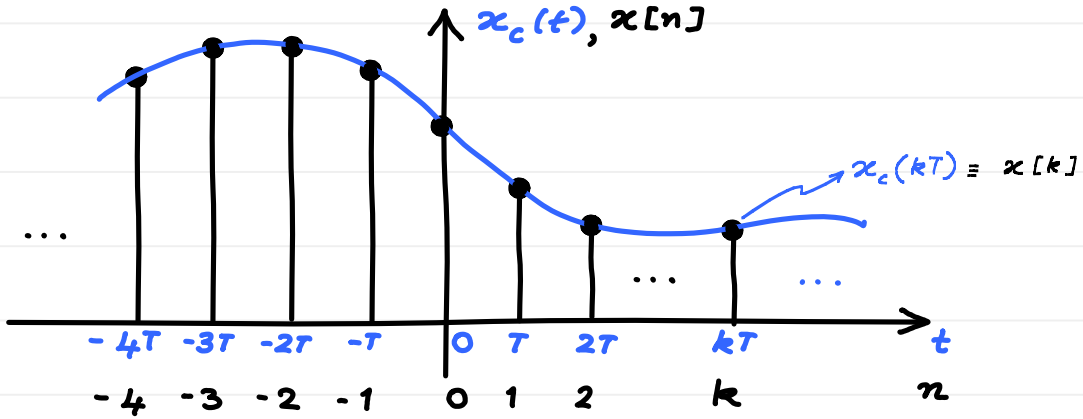
EE 2004 DSP Lecture 31

Note Title 28-03-2019

Sampling:

Sampling provides the bridge that connects the continuous-time and discrete-time domains.

Consider the continuous-time signal $x_c(t)$ shown on the right. $x_c(kT)$ are the values at $t = kT$. Suppose we consider the sequence of numbers $\{x[n]\}$ where $x[n] \equiv x_c(nT)$. The DTFT of $x[n]$ is $X(e^{j\omega})$.



What is the connection, if any, between $X_c(\Omega)$ (the CTFT of $x_c(t)$) and $X(e^{j\omega})$ (the DTFT of $x[n]$)?

Since we have to deal with the frequency domain for both CTFT and DTFT, the following notation will be used:

CTFT: $\frac{\Omega}{2\pi} = F \qquad \Omega \in (-\infty, \infty)$

DTFT: $\frac{\omega}{2\pi} = f \qquad \omega \in [-\pi, \pi)$

The theoretical idea of impulse train sampling will be used to establish the connection. Let $p(t)$ be the impulse train.

$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$ where $\delta(t)$ is the continuous-time impulse with period T.

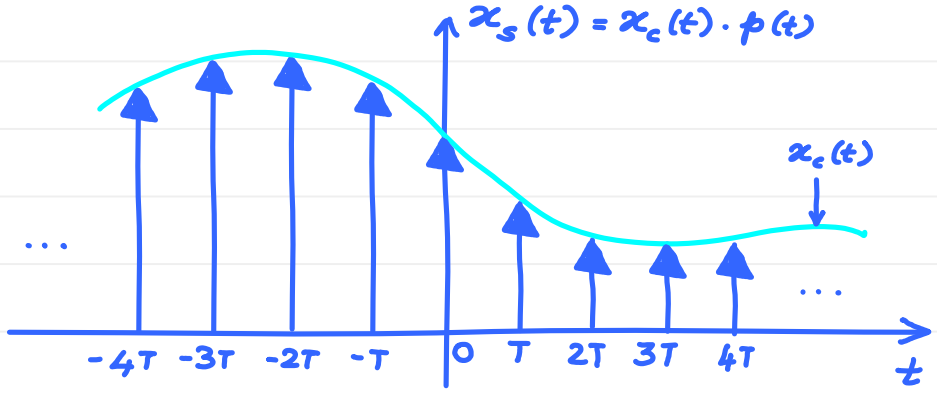
The *impulse train sampled signal* $x_s(t)$ is

$$x_s(t) = x_c(t) \cdot p(t)$$

$$= x_c(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$= \sum_{n=-\infty}^{\infty} x_c(nT)\, \delta(t - nT)$$

[since $x(t) \cdot \delta(t - t_0) = x(t_0)\, \delta(t - t_0)$]



The CTFT of the sampled signal $x_s(t)$ is

$$X_s(\Omega) = \int_{-\infty}^{\infty} x_s(t)\, e^{-j\Omega t}\, dt$$

$$= \int_{-\infty}^{\infty} \left[\sum_{n=-\infty}^{\infty} x_c(nT)\,\delta(t-nT) \right] e^{-j\Omega t}\,dt$$

$$\stackrel{?}{=} \sum_{n=-\infty}^{\infty} x_c(nT) \int_{-\infty}^{\infty} \delta(t-nT)\,e^{-j\Omega t}\,dt$$

$$= \sum_{n=-\infty}^{\infty} x_c(nT)\,e^{-j\Omega nT}$$

i.e.,

$$X_s(\Omega) = \sum_{n=-\infty}^{\infty} x_c(nT)\,e^{-j\frac{\Omega}{F_s}n}$$

since $F_s = \frac{1}{T}$, the sampling frequency

Thus,

$$X_s(\Omega) = \sum_{n=-\infty}^{\infty} x_c(nT)\,e^{-j\frac{\Omega}{F_s}n}$$

EE 2004 DSP Lecture 32

Note Title 01-04-2019

Recall that we derived the following:

$$X_s(\Omega) = \sum_{n=-\infty}^{\infty} x_c(nT)\, e^{-j\frac{\Omega}{F_s}n}$$

We'll now derive another expression for $X_s(\Omega)$ in terms of $X_c(\Omega)$. We will then show that the above expression can be related to $X(e^{j\omega})$. Thus, we will see the connection between $X_c(\Omega)$ and $X(e^{j\omega})$, since $X_s(\Omega)$ is related to both.

To this end, consider

$$x_s(t) = x_c(t) \cdot p(t) \longleftrightarrow \frac{1}{2\pi} X_c(\Omega) * P(\Omega)$$

But $P(\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$ where $\Omega_s = 2\pi F_s = \frac{2\pi}{T}$

Hence,

$$X_s(\Omega) = \frac{1}{2\pi} X_c(\Omega) * \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s).$$

Sampling in one domain causes PERIODIC REPETITION in the other domain

Thus,

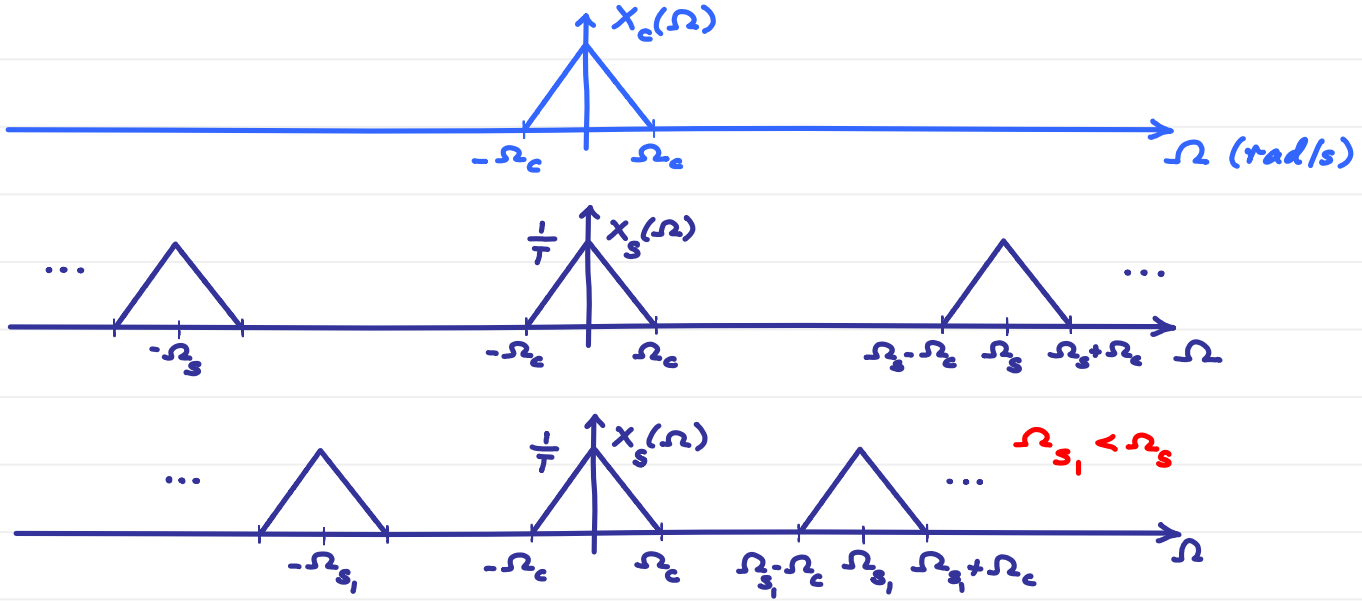
$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s)$$

(1) Periodic repetition

(2) Gain term : $\frac{1}{T}$

$$X_s(\Omega) = X_s(\Omega + \Omega_s)$$

periodic with period Ω_s



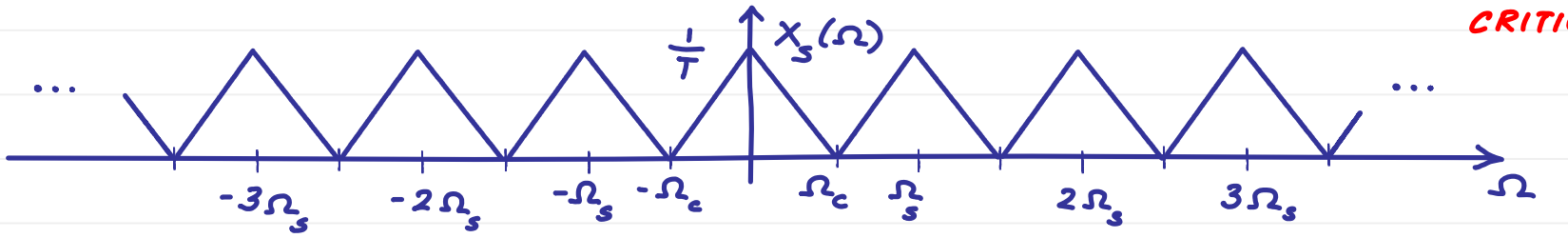
If we want the repetitions to not overlap, we require $\Omega_s - \Omega_c \geq \Omega_c$, i.e., $\Omega_s \geq 2\Omega_c$.

When $\Omega_s \geq 2\Omega_c$, no distortion happens. When $\Omega_s = 2\Omega_c$ (critical sampling), the repetitions just abut each other. If $\Omega_s < 2\Omega_c$, overlap happens and leads to aliasing errors.

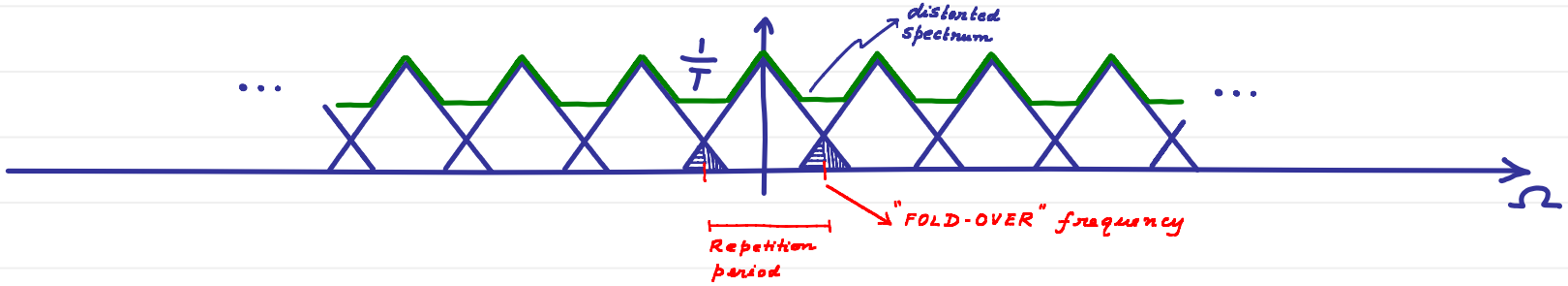
→ also called as the NYQUIST rate

When $\Omega_s = 2\Omega_c$

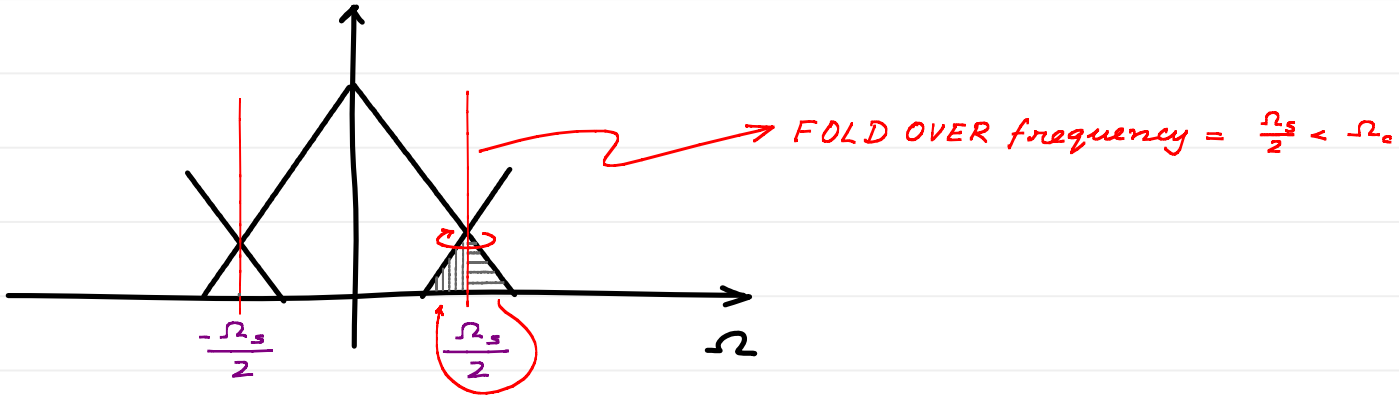
CRITICAL SAMPLING



If $\Omega_s < 2\Omega_c$, the repetitions overlap and distortion occurs. This is called ALIASING.



Undersampling causes aliasing. When aliasing happens, high frequency components masquerade as low frequency ones.



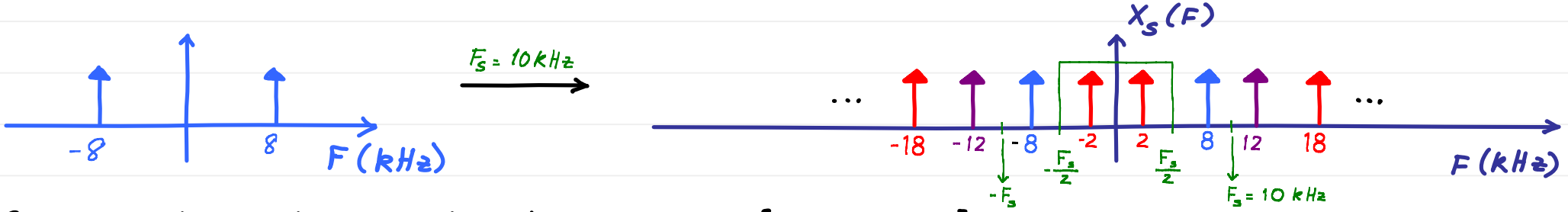
To avoid distortion in the sampled signal's spectrum, aliasing must be avoided, i.e., $x_c(t)$ must be sampled adequately.

If $x_c(t)$ is not bandlimited, then aliasing cannot be avoided because the repetitions in the frequency domain will always overlap for any Ω_s. One possibility is to introduce a bandlimiting filter or an anti-aliasing filter. If this filter's cutoff frequency is Ω_c, then $\Omega_s \geq 2\Omega_c$ will cause no aliasing error. However, we now suffer distortion in the form of loss of high frequency components.

Hence, if a signal is not bandlimited, either loss of high frequency components or aliasing will result when it is sampled.

Perceptual studies have shown that loss of high frequency components is less objectionable than aliasing. Hence, an anti-aliasing filter is an integral part of all sampling circuitry.

Aliasing happens due to "fold over", wherein high frequency components masquerade as low frequency ones. This is easy to see in the case of sinusoids. Let $x_c(t) = \cos 2\pi F_0 t$, where $F_0 = 8$ kHz and $F_s = 10$ kHz. This is *undersampling* and will cause *aliasing*. This is most easily seen in the $X_s(F)$ domain.

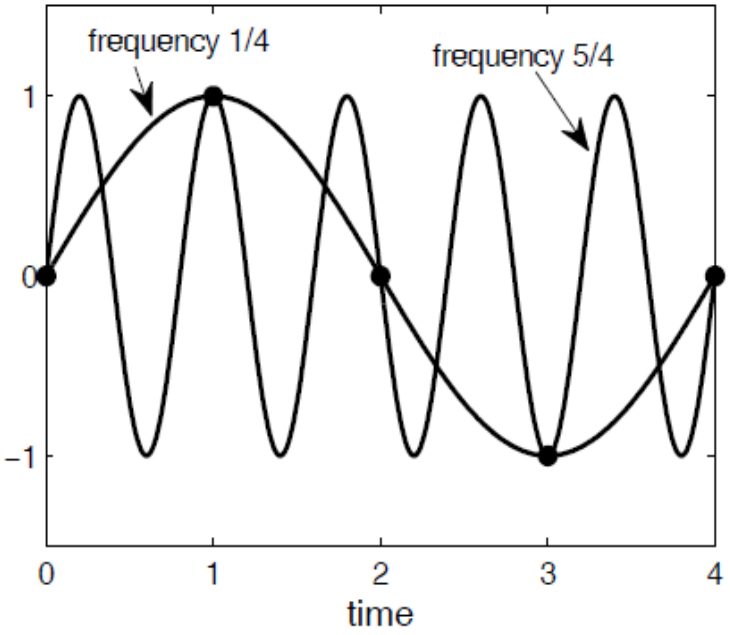


Since $F_s = 10$ kHz, the impulses in the range [-5 kHz, 5 kHz] are:
$-8 + 10 = 2$ and $8 - 10 = -2$. The impulses at -2 kHz and 2 kHz repeat with a periodicity of 10 kHz, as shown in the plot on the right.

12. **Computer assignment**

(a) The figure on the right shows two sinusoids with frequencies $f_1 = 1/4$ and $f_2 = 5/4$. They coincide when $t \in \mathbb{Z}$. Reproduce this plot.

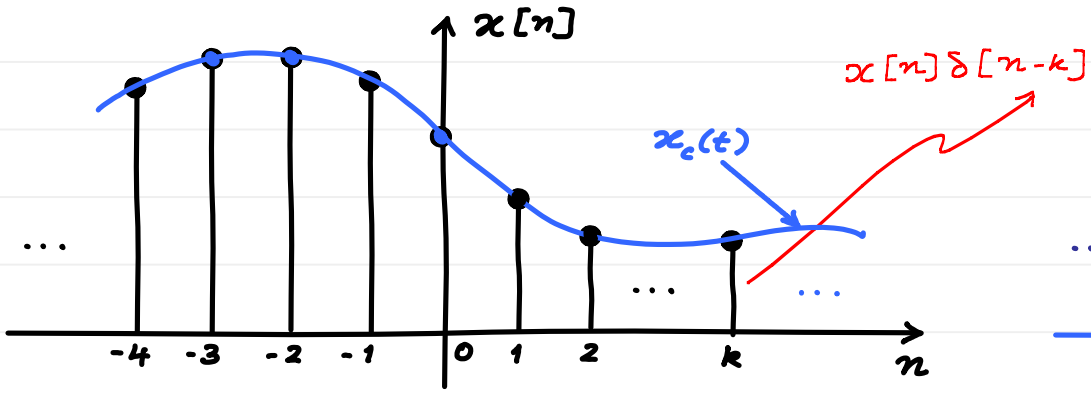
(b) Let $x(t) = \cos(2\pi f t)$ where $f = \pi$ and $T = 1$ (sampling interval). Is the sampled signal periodic? Plot it over the interval $[0, 50]$.

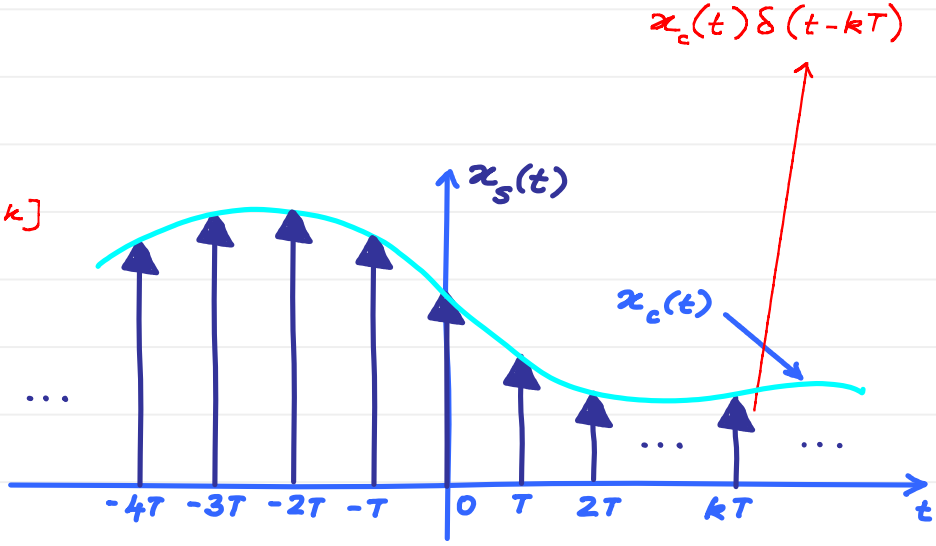


The signal $\cos\left(2\pi \frac{5}{4} t\right)$ when sampled at $T = 1s$, i.e., when undersampled, becomes $\cos\left(2\pi \frac{1}{1+\frac{1}{4}} n\right)$, which is the same as $\cos\left(2\pi \frac{1}{4} n\right)$. Thus is nothing but the samples of $\cos\left(2\pi \frac{1}{4} t\right)$ taken at $T = 1s$ apart.

Thus $\frac{5}{4}$ Hz has aliased to $\frac{1}{4}$ Hz

How are $X(e^{j\omega})$ and $X_s(\Omega)$ related ?





To relate the spectra of these two signals, define $x[n] \equiv x_c(nT)$

Hence

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_c(nT)\, e^{-j\omega n}$$

But we previously showed that

$$X_s(\Omega) = \sum_{n=-\infty}^{\infty} x_c(nT)\, e^{-j\frac{\Omega}{F_s}n}$$

Let

$$Y_s(\Omega) = X_s(F_s\Omega)$$

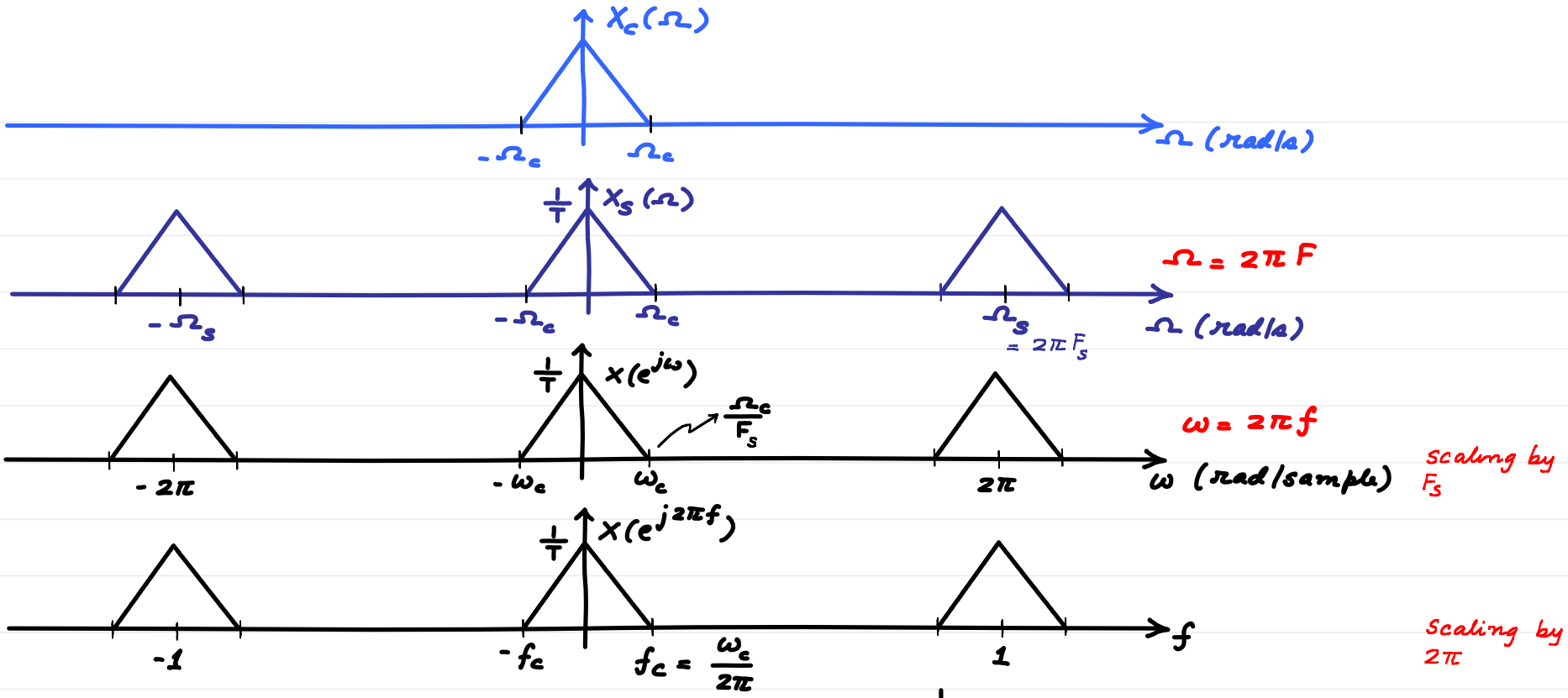
i.e., $X_s(\Omega)$ is scaled by a factor of F_s to give $Y_s(\Omega)$. If $F_s > 1$ (which is typical), this can be interpreted as compression by a factor of F_s.

Note that $Y_s(2\pi) = X_s(2\pi F_s) = X_s(\Omega_s)$, i.e., $Y_s(\Omega)$ is 2π-periodic. Let us change the independent variable to 'ω' and relabel $Y_s(\Omega)$ as $X(e^{j\omega})$.

Hence to obtain $X(e^{j\omega})$ from $X_s(\Omega)$ we just need to replace $\frac{\Omega}{F_s}$ by ω.

$$X_s(\Omega)\Big|_{\Omega \to \omega F_s} = \sum_{n=-\infty}^{\infty} x_c(nT)\, e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n} = X(e^{j\omega})$$

Thus, the Ω_s-periodic function $X_s(\Omega)$ becomes the 2π-periodic function $X(e^{j\omega})$ after scaling the frequency axis by F_s. That is we scale independent axis by F_s and rename the independent variable as 'ω'; $X_s(\omega F_s)$ is written as $X(e^{j\omega})$.



The change of variable $\Omega \rightarrow \omega F_s$ means that $X(e^{j\omega})\Big|_{\omega = 2\pi} = X_s(\Omega_s)$

EE 2004 DSP Lecture 33

Note Title 03-04-2019

Scaling the frequency axis by F_s causes the analog frequency Ω_0 to get mapped to $\frac{\Omega_0}{F_s} = \frac{2\pi F_0}{F_s} = 2\pi f_0 = \omega_0$

Scaling the independent axis by F_s in the transform causes the time domain signal to be scaled by $T = \frac{1}{F_s}$. Hence the samples are now spaced '1' apart (previously they were 'T' apart). Thus this scaling is consistent with $x[n]$ being '1' apart.

Example

$$x_c(t) = \cos \Omega_0 t = \cos 2\pi F_0 t$$

$$x[n] = x_c(nT) = \cos 2\pi F_0 nT = \cos 2\pi \frac{F_0}{F_s} n$$

$$= \cos 2\pi f_0 n = \cos \omega_0 n$$

We will now derive some example transforms starting from $X_c(\Omega)$ to get $X(e^{j\omega})$ via $X_s(\Omega)$.

Example

$x_c(t) = \cos(\Omega_0 t)$

$X_c(\Omega) = \pi[\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)]$

$X_s(\Omega) = \frac{\pi}{T}[\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)] \qquad -\Omega_s/2 \le \Omega \le \Omega_s/2$

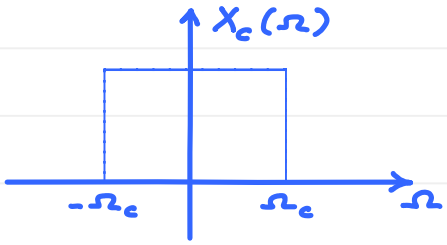
$$X(e^{j\omega}) = X_s(\Omega)\Big|_{\Omega \to \omega F_s} = \frac{\pi}{T}[\delta(\omega F_s - \Omega_0) + \delta(\omega F_s + \Omega_0)]$$

However, since $\delta(at+b) = \frac{1}{|a|}\delta(t + \frac{b}{a})$, we get $\frac{\pi}{T}\delta(\omega F_s - \Omega_0) = \frac{\pi}{T \cdot F_s}\delta(\omega - \frac{\Omega_0}{F_s}) = \pi\delta(\omega - \omega_0)$

Similar calculations give the final result $\cos \omega_0 n \leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$

Example

$$x_c(t) = \frac{\sin \Omega_c t}{\pi t}$$



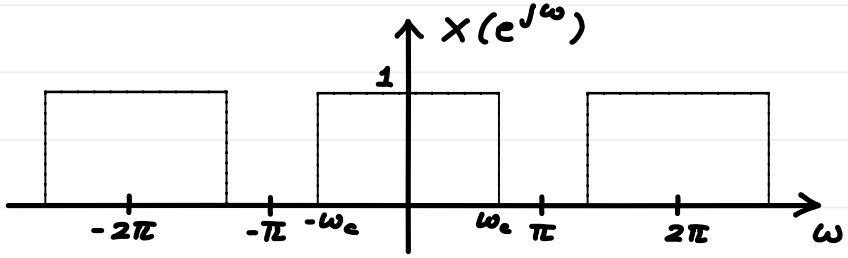
$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s)$$

$$x_c(nT) = \frac{\sin \Omega_c nT}{\pi nT} \Rightarrow X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \text{rect}(\Omega - k\Omega_s)$$

these cancel

Hence,

$$\frac{\sin \omega_c n}{\pi n} \leftrightarrow$$



where $\omega_c = \frac{\Omega_c}{F_s}$

Example

$$x_c(t) = \begin{cases} 1 & |t| < 1/2 \\ 0 & \text{otherwise} \end{cases} \quad \longleftrightarrow \quad \frac{\sin \Omega/2}{\Omega/2}$$

Since this signal is not bandlimited, sampling will cause aliasing.

Our goal is to derive the pair

$$x[n] = 1 \quad -N \le n \le N \quad \longleftrightarrow \quad \frac{\sin (2N+1)\omega/2}{\sin \omega/2}$$

starting from the given $x_c(t)$.

$$X_c(\Omega) = \frac{\sin \Omega/2}{\Omega/2}$$

$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \frac{\sin \frac{\Omega - k\Omega_s}{2}}{\frac{\Omega - k\Omega_s}{2}}$$

$$X(e^{j\omega}) = X_s(\Omega)\Big|_{\Omega \to \omega F_s}$$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} \frac{\sin\left(\frac{\omega F_s - k\Omega_s}{2}\right)}{\frac{\omega F_s - k\Omega_s}{2}}$$

$$= \sum_{k=-\infty}^{\infty} \frac{\sin\left(\frac{\omega - 2\pi k}{2T}\right)}{\frac{\omega - 2\pi k}{2}}$$

Thus,

$$\frac{\sin(2N+1)\omega/2}{\sin \omega/2} = \sum_{k=-\infty}^{\infty} \frac{\sin\left(\frac{\omega - 2\pi k}{2T}\right)}{\left(\frac{\omega - 2\pi k}{2}\right)}$$

Hence the **Dirichlet kernel** results from adding periodic repetitions of the **sinc** function. In deriving the above we have assumed that none of the samples fall at a discontinuity.

EE 2004 DSP Lecture 34

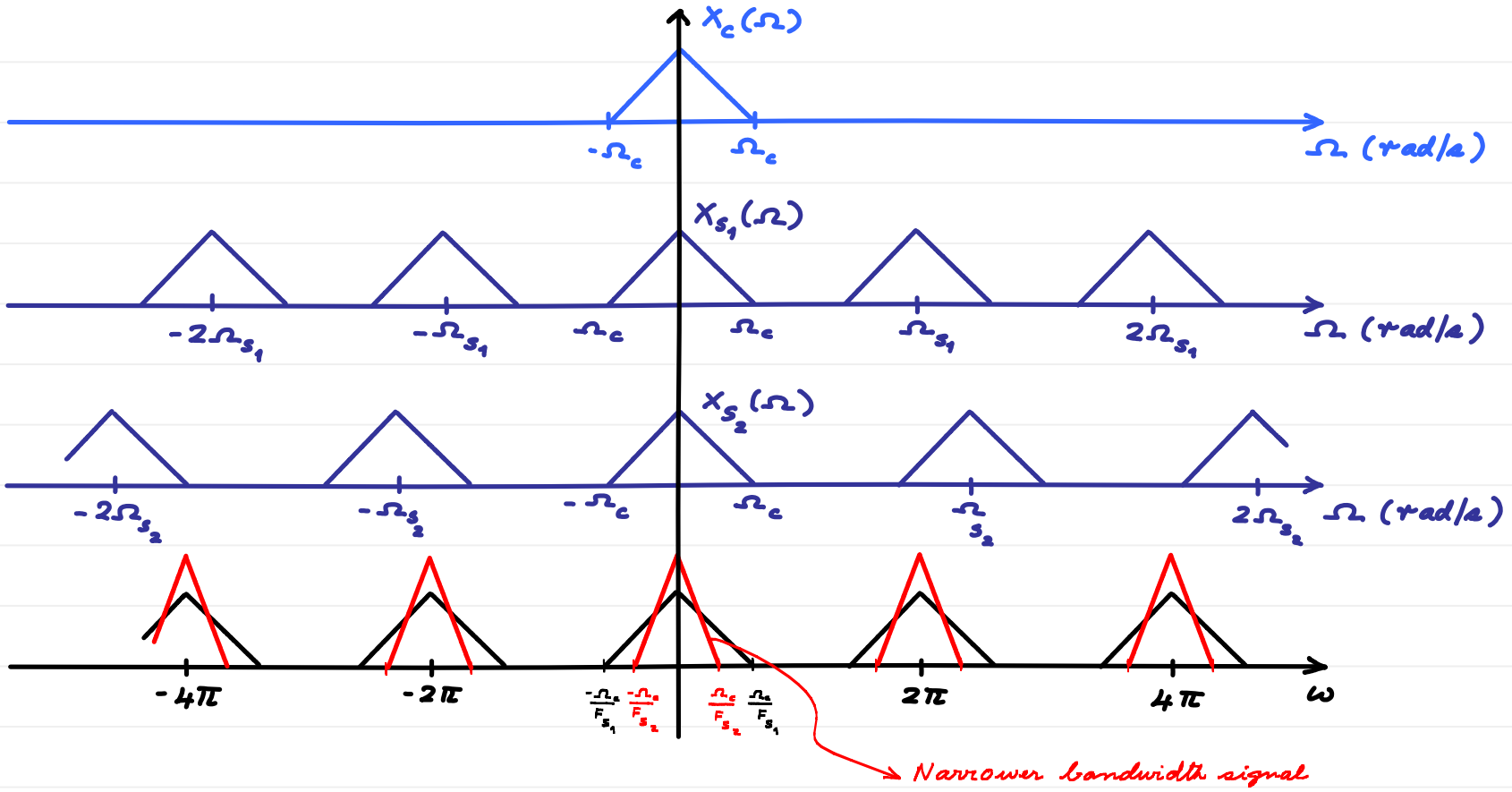
Note Title 04-04-2019

To minimize distortion, we need to increase Ω_c so as to reduce the information lost in the high frequency components. But this leads to larger Ω_s and hence to more storage and expensive sampling circuitry.

Does oversampling lead to a "closer representation" of $x_c(t)$?

Oversampling is bad because it needs to a narrower bandwidth signal in the sampled signal domain. This calls for more stringent specs on the filters needed, more storage for data, and faster processor requirements.

Hence oversampling must be avoided wherever possible.


$X_c(\Omega)$
$-\Omega_c$
Ω_c
Ω (rad/s)
$X_{s_1}(\Omega)$
$-2\Omega_{s_1}$
$-\Omega_{s_1}$
Ω_c
Ω_{s_1}
$2\Omega_{s_1}$
$X_{s_2}(\Omega)$
$-2\Omega_{s_2}$
$-\Omega_{s_2}$
$-\Omega_c$
Ω_c
Ω_{s_2}
$2\Omega_{s_2}$
-4π
-2π
2π
4π
ω
Narrower bandwidth signal

Another important point to note is that a frequency of F_0 gets mapped to $\frac{F_0}{F_s}$ in the DTFT domain. Consider the following two cases:

$x_c(t) = \cos 2\pi F_0 t$

$F_0 = 8\,kHz, \quad F_s = 24\,kHz$

$$x[n] = \cos 2\pi \frac{F_0}{F_s} n$$

$$= \cos 2\pi \frac{8 \times 10^3}{24 \times 10^3} n$$

$$= \cos \frac{2\pi}{3} n$$

$y_c(t) = \cos 2\pi F_0 t$

$F_0 = 16\,kHz, \quad F_s = 48\,kHz$

$$y[n] = \cos 2\pi \frac{F_0}{F_s} n$$

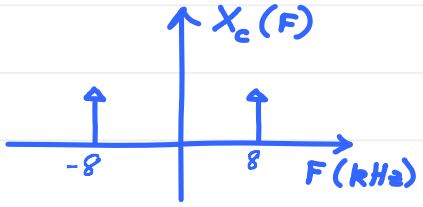
$$= \cos 2\pi \frac{16 \times 10^3}{48 \times 10^3} n$$

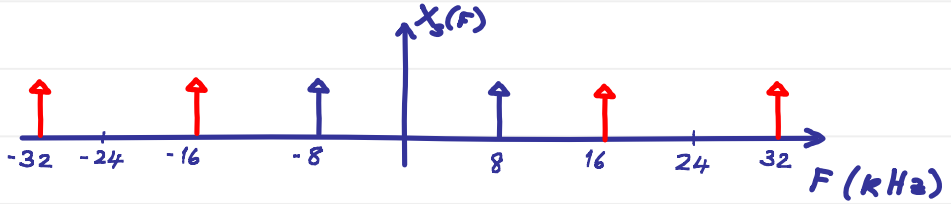
$$= \cos \frac{2\pi}{3} n = x[n]$$

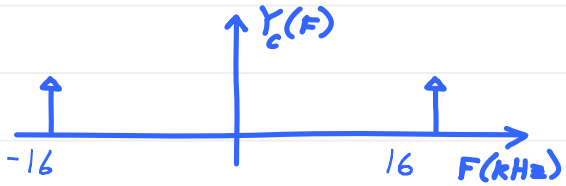
Thus given $x[n] = \cos \frac{2\pi}{3} n$, one cannot tell if this sequence is the result of sampling an 8 kHz component at 24 kHz or it is the result of sampling a 16 kHz component at 48 kHz.

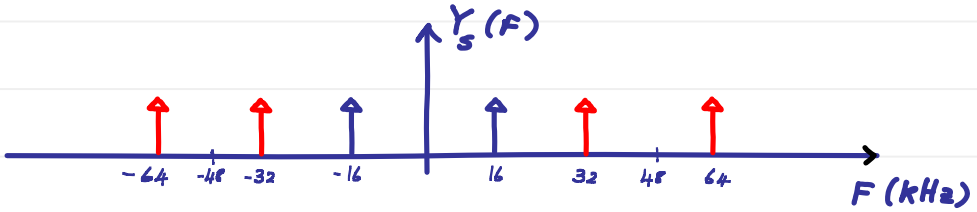
To map $X(e^{j\omega})$ to true frequency, we need sampling frequency information. Only then can we tell if $\omega_0 = \frac{2\pi}{3}$ corresponds to 8 kHz or 16 kHz.

It is easy to see that $X_s(\Omega)$ preserves information about actual frequency. It is the normalization by F_s that causes the loss of information. The true frequency can be ascertained if we know F_s.

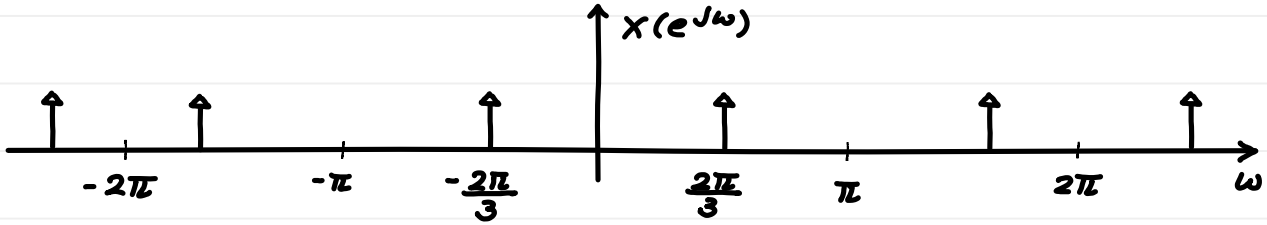








Both $X_s(F)$ and $Y_s(F)$ map to the same $X(e^{j\omega})$ after normalizing by their respective F_s:



What is the theoretical minimum F_s needed, i.e., is it $F_s \geq 2F_c$ or $F_s > 2F_c$, i.e., is the inequality strict?

Let $x_c(t) = \cos 2\pi F_0 t$ and $F_s = 2F_0$. Then

$$x[n] = \cos 2\pi \frac{F_0}{2F_0} n$$

$$= \cos \pi n$$

We can reconstruct $x_c(t)$

Let $x_c(t) = \sin 2\pi F_0 t$ and $F_s = 2F_0$. Then

$$x[n] = \sin 2\pi \frac{F_0}{2F_0} n$$

$$= \sin \pi n$$

$$= 0 \quad \forall n$$

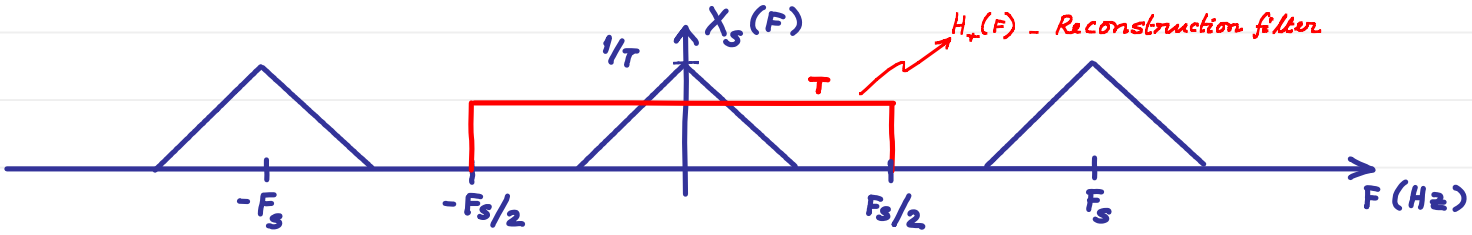
We cannot reconstruct $x_c(t)$

Hence, while $2F_0$ works of $\cos(2\pi F_0 t)$, it does not work for $\sin 2\pi F_0 t$ because we end up sampling at the zero-crossings. Hence $F_s > 2F_0$ guarantees full recovery.

In practice, since the non-ideal anti-aliasing front-end lowpass filters have a certain transition band, the actual F_s is slightly higher to account for this.

Reconstruction

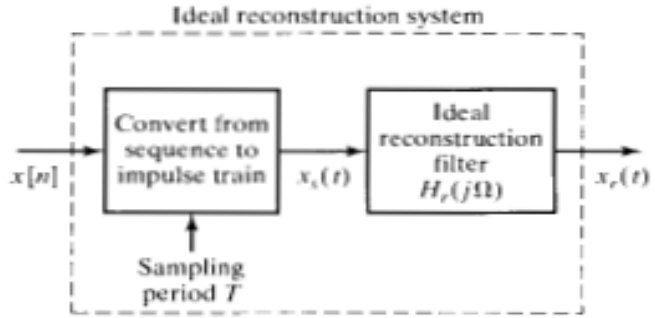
If $x_s(t)$, which is the *impulse train sampled signal* with spectrum $X_s(F)$, is passed through an *ideal LPF* with *gain T* (to counteract the $1/T$ scale factor introduced during sampling) over the frequency range $[-F_s/2, F_s/2]$, then the copies of the spectrum outside this range get eliminated, *resulting in* $X_c(F)$. Thus, the filter output becomes $x_c(t)$. *This corresponds to exact reconstruction.*



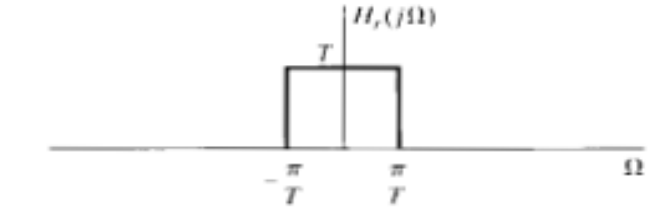
In the time domain, $H_r(F)$ has impulse response $h_r(t) = \text{Sinc}\left(\frac{t}{T}\right) = \frac{\text{Sin}\frac{\pi t}{T}}{\frac{\pi t}{T}}$

Hence **exact reconstruction is possible** in the following system:

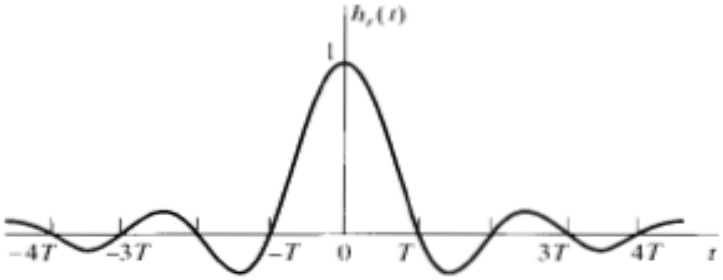
$x_r(t) = x_c(t)$ if there is no aliasing and $h_r(t)$ is chosen as above.

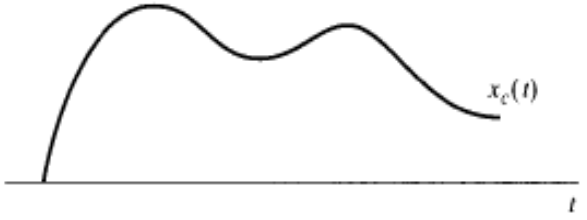


(a)

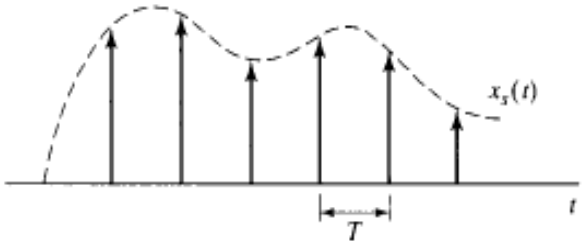


(b)

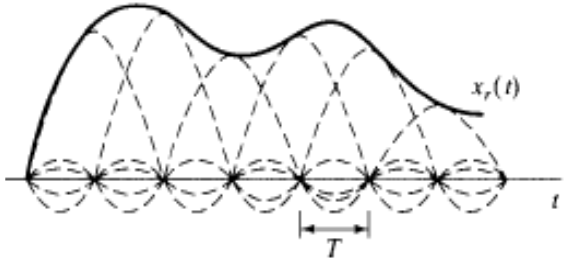




(a)



(b)



From "Discrete-Time Signal Processing" by Oppenheim and Schafer

In terms of equations, the reconstruction steps are as follows:

$$x_r(t) = x_s(t) * h_r(t)$$

$$= \sum_{n=-\infty}^{\infty} x_c(nT)\,\delta(t-nT) * \frac{\sin\frac{\pi t}{T}}{\frac{\pi t}{T}}$$

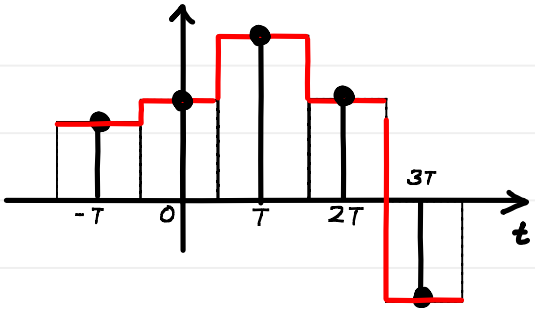
$$\frac{T}{\pi}\sum_{n=-\infty}^{\infty} x_c(nT)\,\frac{\sin\frac{\pi}{T}(t-nT)}{t-nT}$$

$$= x_c(t)$$

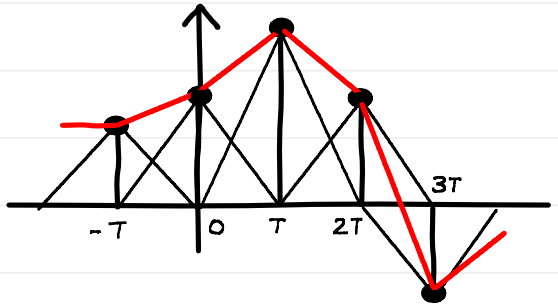
since $h_r(t)$ has been chosen as required. Note that $x_r(nT) = x_c(nT)$ is the result of the contribution due to sinc function located at $t = nT$ only.

Sinc interpolation is not realizable in practice because it is not only non-causal but also requires all past and future samples. In practice, non-ideal interpolation is used.

(1) Sample and Hold :



(2) Linear Interpolation :



The discontinuities and sharp corners are smoothed by smoothing with a lowpass filter. The sample-and-hold and the linear interpolators are made causal in an actual implementation.

EE 2004 DSP Lecture 35

Note Title 08-04-2019

Introduction to the Discrete-Fourier Transform:

First let us review the various Fourier representations seen so far.

(1) Continuous-Time Fourier Series:

$$x(t+T) = x(t) \Rightarrow x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\Omega_0 t} \quad \text{where} \quad a_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t) e^{-jk\Omega_0 t}\, dt$$

$$\Omega_0 = \frac{2\pi}{T}$$

Since $k \in \mathbb{Z}$, the spectrum is discrete ("line spectrum")

(2) Continuous-Time Fourier Transform:

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t}\, dt \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t}\, d\Omega$$

Both $x(t)$ and $X(\Omega)$ are not periodic;

$X(\Omega)$ is a continuous spectrum.

(3) Discrete-Time Fourier Transform:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \qquad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} \, d\omega$$

DTFT is nothing but the CTFS applied to the 2π-periodic $X(e^{j\omega})$

(4) Discrete-Time Fourier Series:

$$x[n+N] = x[n] \Rightarrow x[n] = \sum_{k=0}^{N-1} a_k e^{j \frac{2\pi k}{N} n} \quad \text{where} \quad a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi k}{N} n}$$

Note that the set $\{ \exp(j \frac{2\pi}{N} k) : k \in \mathbb{Z} \}$ has only N distinct complex exponentials. From this it follows that $a_{k+N} = a_k$. That is, the DTFS coefficients are also periodic with period N. Thus, both the signal and the transform are periodic.

As a general remark, periodicity in one domain implies a discrete variable in the other domain.

Let $x[n]$ be known for $0 \le n \le N-1$. No assumptions are made about $x[n]$ outside this interval. The Discrete Fourier Transform (DFT) is defined as follows:

$$X[k] \triangleq \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n}$$

From the definition it should be clear that $X[k+N] = X[k]$, i.e. the DFT coefficients are periodic with period N. Hence it is enough to consider $X[k]$ for $k = 0, 1, 2, \ldots, N-1$.

The inverse discrete Fourier transform is defined as $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}n}$

Clearly,

$$x[n+N] = x[n]$$

The DFT framework imposes periodicity on $x[n]$ even though no assumptions were made about

the data outside $[0, N-1]$. Comparing $X[k]$ with the formula for a_k in the DTFS, we see that

$$X[k] = N a_k$$

Thus, the DFT is not a new tool; it is nothing but a slightly different form of the DTFS. In this light, the periodicity of $x[n]$ should not come as a surprise.

DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi k}{N} n} \qquad k = 0, 1, 2, \ldots, N-1$$

IDFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi k}{N} n} \qquad n = 0, 1, 2, \ldots, N-1$$

An N-point sequence gives rise to an N-point transform

EE 2004 DSP Lecture 36

Note Title 09-04-2019

The DFT coefficients for all 'k' can be expressed succintly in a matrix equation form:

$$\begin{bmatrix} X_0 \\ X_1 \\ \vdots \\ X_{N-1} \end{bmatrix} = \begin{bmatrix} \leftarrow e^{-j\frac{2\pi k}{N}n} \rightarrow \\ \text{'k' varies} \downarrow \quad \uparrow \text{'n' varies} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}$$

where $X_k \equiv X[k]$

and $x_k \equiv x[k]$

i.e. $\underset{N\times 1}{\underline{X}} = \underset{N\times N}{\underline{W}}\ \underset{N\times 1}{\underline{x}}$

where $\underline{x}$ is the $N \times 1$ data vector, $\underline{W}$ is the $N \times N$ **DFT matrix**, and $\underline{X}$ is the $N \times 1$ transform vector.

It can be shown that $\underline{W}$ is **non-singular** and hence **invertible**. Therefore,

$$\underline{x} = \underline{W}^{-1}\underline{X}$$

$$W^{-1} = \frac{1}{N}\begin{bmatrix} \leftarrow e^{j\frac{2\pi k}{N}n} \rightarrow \\ \text{'n' varies} \downarrow \quad \uparrow \text{'k' varies} \end{bmatrix}$$

Thus, $$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\, e^{j\frac{2\pi k}{N}n}$$

It is easy to verify that the inversion formula gives back $x[n]$:

$$\frac{1}{N}\sum_{k=0}^{N-1} X[k]\, e^{j\frac{2\pi k}{N}n} = \frac{1}{N}\sum_{k=0}^{N-1}\left[\sum_{l=0}^{N-1} x[l]\, e^{-j\frac{2\pi k}{N}l}\right] e^{j\frac{2\pi k}{N}n} = \sum_{l=0}^{N-1} x[l]\left\{\underbrace{\frac{1}{N}\sum_{k=0}^{N-1} e^{j\frac{2\pi k}{N}(n-l)}}_{= \begin{cases} 1 & l=n \\ 0 & \text{otherwise} \end{cases}}\right\}$$

$$= x[n]$$

DFT as Samples of the DTFT:

Another view of the DFT is to think of it as the samples of the DFT. For this, we make the assumption that $x[n]$ is zero outside $[0, N-1]$.

Therefore,

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$= \sum_{n=0}^{N-1} x[n]\, e^{-j\omega n}$$

Let us now sample $X(e^{j\omega})$ at *N uniformly spaced points*, i.e., at $\omega_k = \frac{2\pi}{N} k$

$$X(e^{j\omega})\Big|_{\omega=\omega_k} = X(e^{j\omega_k}) \triangleq X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\omega_k n}$$

$$= \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi k}{N} n}$$

Since sampling in one domain causes periodic repetition in the other domain, with the repetition period being $2\pi/(\text{sampling interval})$, $x[n]$ repeats with period $\frac{2\pi}{\left(\frac{2\pi}{N}\right)} = N$.

This is consistent with the fact that the inversion formula gives a sequence that is periodic with period N.

If we sample the DTFT at L ($>N$) points, then $x[n+L] = x[n]$, i.e., the sequence repeats with a longer period. The samples of $x[n]$ between N and $L-1$ are zero. Alternately, if we zero-pad the N-point sequence $x[n]$ with $L-N$ zeros, so that the zero-padded sequence is of length L, and we take L-point DFT, then that results in sampling the underlying DTFT at a finer set of points.

$$y[n] = \begin{cases} x[n] & 0 \le n \le N-1 \\ 0 & N \le n \le L-1 \end{cases}$$

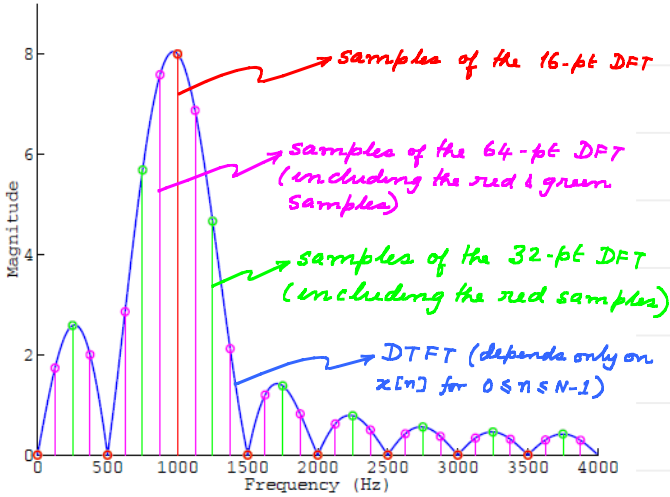
$$Y[k] = \sum_{n=0}^{L-1} y[n] e^{-j\frac{2\pi k}{L} n} \qquad k = 0, 1, 2, \ldots, L-1$$

$$= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{L} n} \qquad k = 0, 1, 2, \ldots, L-1$$

Note that the underlying DTFT remains unchanged:

$$Y(e^{j\omega}) = \sum_{n=0}^{L-1} y[n] e^{-j\omega n}$$

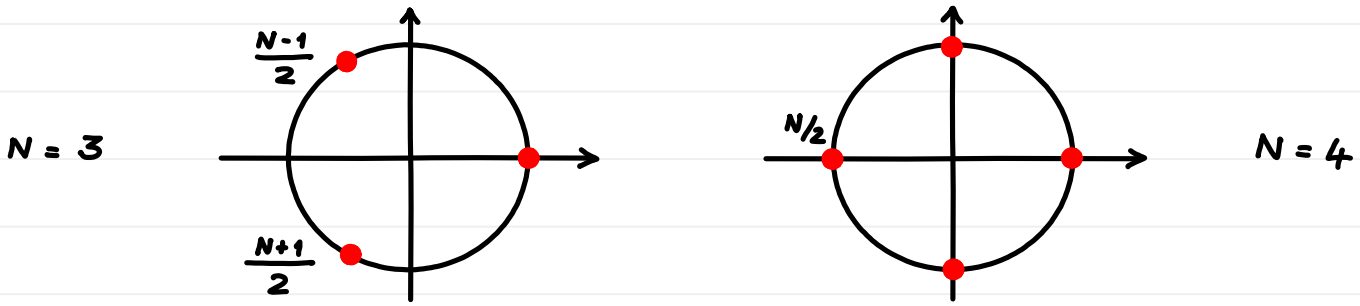
$$= \sum_{n=0}^{N-1} x[n] e^{-j\omega n} = X(e^{j\omega})$$



DTFT and 64-pt DFT

$$x[n] = \sin\frac{2\pi n}{8} \qquad n = 0, 1, 2, \ldots, 15$$

When N is even, a sample point falls at $\omega = \pi$ (when $k = \frac{N}{2}$); otherwise there is no sample point at $\omega = \pi$. Note that $\omega = 0$ (when $k = 0$) is always a sample point.



It is easy to convert the DFT index 'k' to actual frequency if F_s is known. Since $X_s(F)$ is periodic with period F_s, i.e. $X_s(F+F_s) = X_s(F)$, and since N samples are uniformly taken in the interval $[0, 2\pi)$, it is easy to see that an index k corresponds to the true frequency $\frac{k}{N} \times F_s$ (where the index is zero-based).

EE 2004 DSP Lecture 37

Note Title 11-04-2019

Recovering the DTFT from the DFT:

If $X(e^{j\omega})$ is sampled uniformly at N points or more, the repetitions in the time domain do not overlap. Hence we can recover $X(e^{j\omega})$ from $X[k]$.

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n} = \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi kn}{N}} \right] e^{-j\omega n}$$

$$= \sum_{k=0}^{N-1} X[k] \left[\frac{1}{N} \sum_{n=0}^{N-1} e^{-j(\omega - \frac{2\pi k}{N})} \right]$$

$$= \sum_{k=0}^{N-1} X[k] P(\omega - \tfrac{2\pi k}{N})$$

where $P(\omega) = \frac{1}{N} \sum_{n=0}^{N-1} e^{-j\omega n} = \frac{1}{N} e^{-j\omega \frac{N-1}{2}} \frac{\sin N\omega/2}{\sin \omega/2}$

That is, knowing $X(e^{j\omega})$ at N uniformly spaced points helps us reconstruct $X(e^{j\omega})$ over all 'ω'. This is similar to recovering $x_c(t)$ from $x_c(nT)$. The main differences are: (i) the number of points is finite (N, to be precise), and (ii) the interpolating function is $P(\omega)$ [the Dirichlet kernel] rather than $\mathrm{Sinc}\left(\frac{t}{T}\right)$. Since $P(0) = 1$, $X(e^{j\omega})$ at $\omega = \frac{2\pi k}{N}$ equals $X[k]$. Moreover, $P\left(\frac{2\pi k}{N}\right) = 0$ when $k \neq 0$.

Since both $x[n]$ and $X[k]$ are periodic, it is enough if we restrict ourselves to one period. Hence the indices are limited to the range $0, 1, 2, \ldots, N-1$. That is, $x[n] = x[\langle n \rangle_N]$ and $X[k] = X[\langle k \rangle_N]$ where $\langle n \rangle_N = n \bmod N$. As an illustration, $x[-1] = x[N-1]$, and so on. A right shift of 1, i.e., $y[n] = x[n-1]$ is equal to a left shift of $N-1$ since $x[n-1] = x[n+N-1]$, and so on.

Properties of the DFT

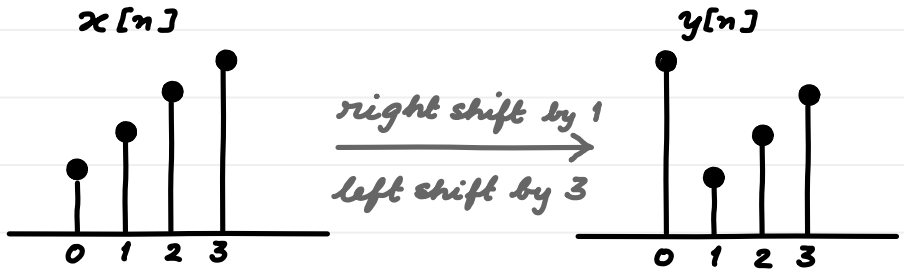
(1) $a_1 x_1[n] + a_2 x_2[n] \leftrightarrow a_1 X_1[k] + a_2 X_2[k]$

(2) $x[n-n_0] = x[\langle n-n_0 \rangle] \leftrightarrow e^{-j\frac{2\pi k}{N} n_0} X[k]$

(3) $e^{j\frac{2\pi \ell}{N} n} x[n] \leftrightarrow X[k-\ell] = X[\langle k-\ell \rangle]$

(4) $x[n] \circledast y[n] \leftrightarrow X[k]\,Y[k]$

(5) $x[n] \cdot y[n] \leftrightarrow \frac{1}{N} X[k] \circledast Y[k]$



$X(e^{j\omega})$ (assuming other samples are 0) is not related to $Y(e^{j\omega})$ but the DFTs are related as per property (2).

(6) $x^*[n] \longleftrightarrow X^*[-k] = X^*[N-k]$

If $x[n] \in \mathbb{R}$, then $X[k] = X^*[N-k]$. Therefore, $X[N-1] = X^*[1]$, and so on. Hence it is enough if only $X[k]$, $k = 0, 1, \ldots, \lfloor \frac{N}{2} \rfloor$ are stored because the missing samples can be generated from the given $X[k]$.

(7) $$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$

Proof:

$$\frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 = \frac{1}{N} \underline{X}^H \underline{X} = \frac{1}{N} (\underline{W}\underline{x})^H (\underline{W}\underline{x})$$

$$= \frac{1}{N} \underline{x}^H \underline{W}^H \underline{W} \underline{x} = \underline{x}^H \underline{x} \qquad (\text{since } \underline{W}^H \underline{W} = N \cdot \underline{I})$$

EE 2004 DSP Lecture 38

Note Title 15-04-2019

Circular or Periodic Convolution:

Circular or periodic convolution can be defined for both CT and DT signals.

Both signals have to be periodic with the same period.

If $\tilde{x}[n+N] = \tilde{x}[n]$ and $\tilde{y}[n+N] = \tilde{y}[n]$.

(identical period)

$$\tilde{x}[n] \circledast \tilde{y}[n] \triangleq \sum_{k=0}^{N-1} \tilde{x}[k]\,\tilde{y}[n-k]$$

Similarly, if $\tilde{x}(t+T) = \tilde{x}(t)$ and $\tilde{y}(t+T) = \tilde{y}(t)$.

(identical period)

$$\tilde{x}(t) \circledast \tilde{y}(t) \triangleq \int_0^T \tilde{x}(\tau)\,\tilde{y}(t-\tau)\,d\tau$$

Note that since the signals are periodic, $\tilde{x}[n-k] \equiv \tilde{x}[\langle n-k \rangle_N]$. That is, $\tilde{x}[-1] = \tilde{x}[N-1]$, $\tilde{x}[-2] = \tilde{x}[N-2]$, and so on.

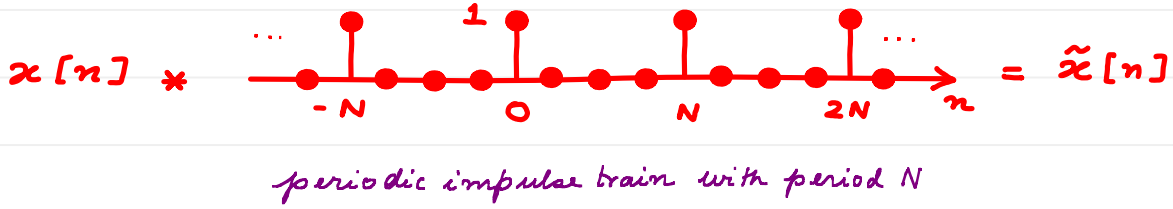
If $x[n]$ and $y[n]$ are *finite duration sequences*, then **circular convolution is NOT defined**, as the sequences are *not periodic*. If we want to carry out circular convolution, we form

$$\tilde{x}[n] = \begin{cases} x[n] & 0 \le n \le P-1 \\ 0 & P \le n \le N-1 \end{cases} \quad \text{and} \quad \tilde{y}[n] = \begin{cases} y[n] & 0 \le n \le Q-1 \\ 0 & Q \le n \le N-1 \end{cases}$$

where we have assumed that the lengths of $x[n]$ and $y[n]$ are P and Q, respectively, and $N \geq \max\{P, Q\}$. Now, $\tilde{x}[n+N] = \tilde{x}[n]$ and $\tilde{y}[n+N] = \tilde{y}[n]$.

Convolving an aperiodic sequence $x[n]$ with an impulse train with period N converts $x[n]$ to $\tilde{x}[n]$.

If $x[n]$ is of *infinite duration*, there will be *aliasing distortion* because of the overlap between the repetitions.

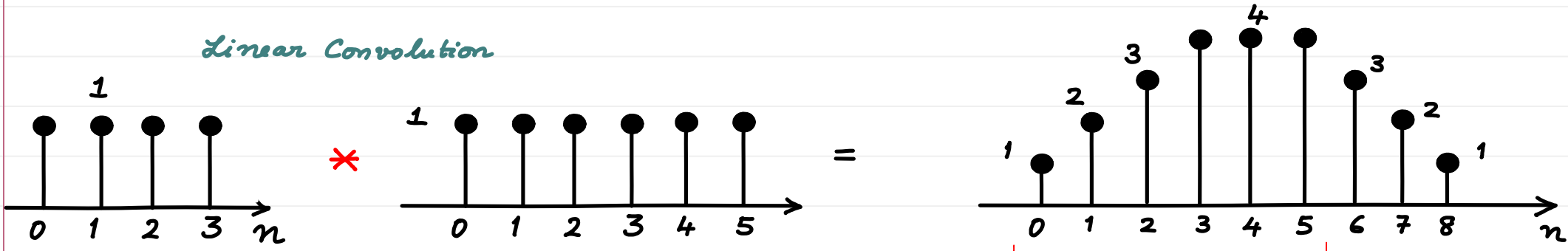


Note that $w_n[m] = \tilde{x}[m]\,\tilde{y}[n-m]$ is periodic with period N, i.e., it is really $\tilde{w}_n[m]$. Hence $\sum_{m=0}^{N-1} \tilde{w}_n[m]$ is a sum over one period that is a function of 'n' (shift).

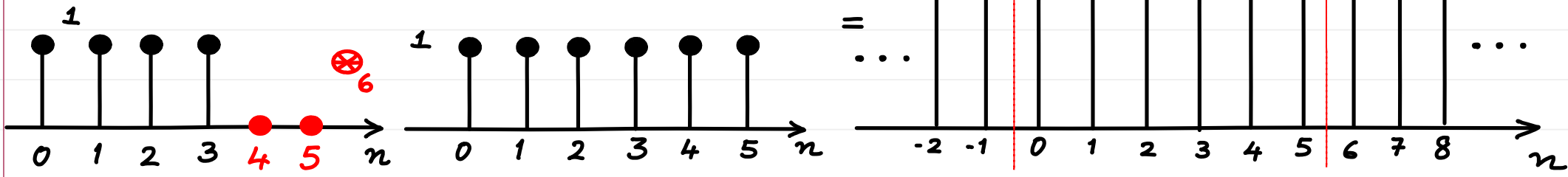
Since $\tilde{x}[n]$ and $\tilde{y}[n]$ are periodic, it is enough to consider shifts belonging to the set $\{0, 1, 2, \ldots, N-1\}$.

$x[n] * y[n]$ is of length $P+Q-1$, whereas $\tilde{x}[n] \circledast \tilde{y}[n]$ is of length N.
Sometimes the notation $\circledast_N$ is used to make the underlying periodicity explicit.

Linear Convolution



Circular Convolution



```
x = ones(4,1);  y = ones(6,1);
z = conv(x,y)                    % linear convolution
x1 = [x; zeros(2,1)];            % two zeros added to make the length 6
y1 = y;                          % same as y
z1 = cconv(x1,y1,6);             % circular convolution of length 6
```

Are $x[n] * y[n]$ and $\tilde{x}[n] \circledast \tilde{y}[n]$ related ? YES!

$x[n]$: length P

$y[n]$: length Q

$x[n] * y[n]$: length $P+Q-1$

Let $N \geq \max\{P, Q\}$.

$$\tilde{x}[n] \circledast \tilde{y}[n] = \sum_{k=0}^{N-1} \tilde{x}[\langle n-k \rangle_N] \, \tilde{y}[\langle k \rangle_N]$$

$$= \sum_{k=0}^{N-1} \tilde{x}[\langle n-k \rangle_N]\, y[k] \qquad \text{since } y[k] = \tilde{y}[\langle k \rangle_N] \text{ for } k = 0, 1, 2, \ldots, N-1$$

$$= \sum_{k=-\infty}^{\infty} \tilde{x}[\langle n-k \rangle_N]\, y[k] \qquad \text{since } y[k] = 0 \text{ outside } k = 0, 1, 2, \ldots, N-1$$

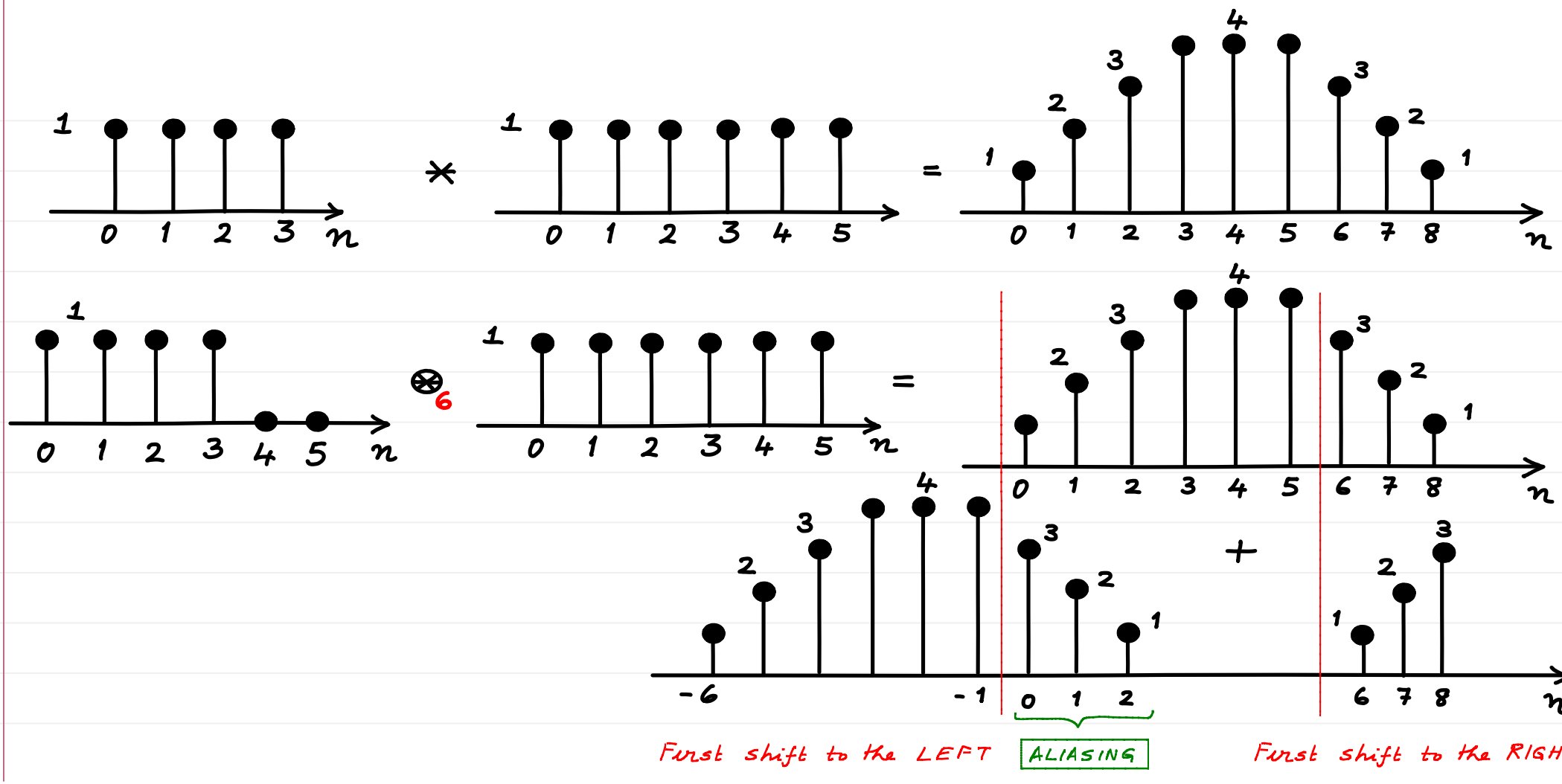
$$= \tilde{x}[n] * y[n] \qquad \text{LINEAR convolution !}$$

$$= (x[n] * p[n]) * y[n] \qquad p[n] \text{ is the periodic impulse train}$$

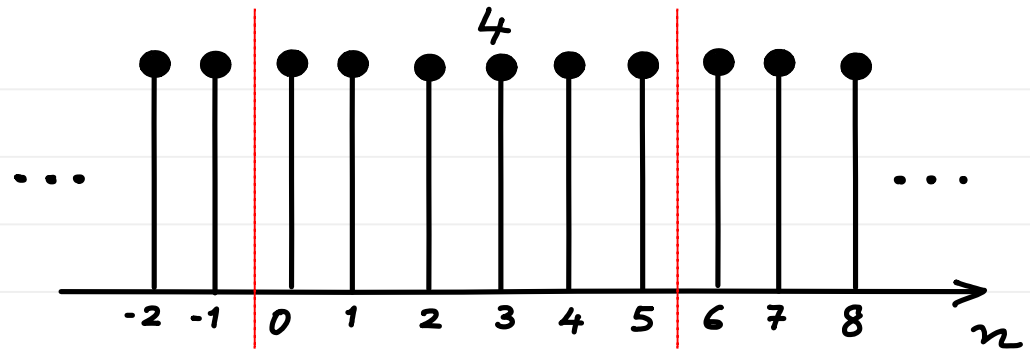
$$= p[n] * \underbrace{(x[n] * y[n])}_{z[n]}$$

Thus, $\tilde{z}[n] = \tilde{x}[n] \circledast \tilde{y}[n]$ is the same as $p[n] * z[n]$, i.e. it is the periodic repetition of linear convolution! Clearly, if $N \geq P+Q-1$, then the repetitions of $z[n]$ don't overlap. Hence

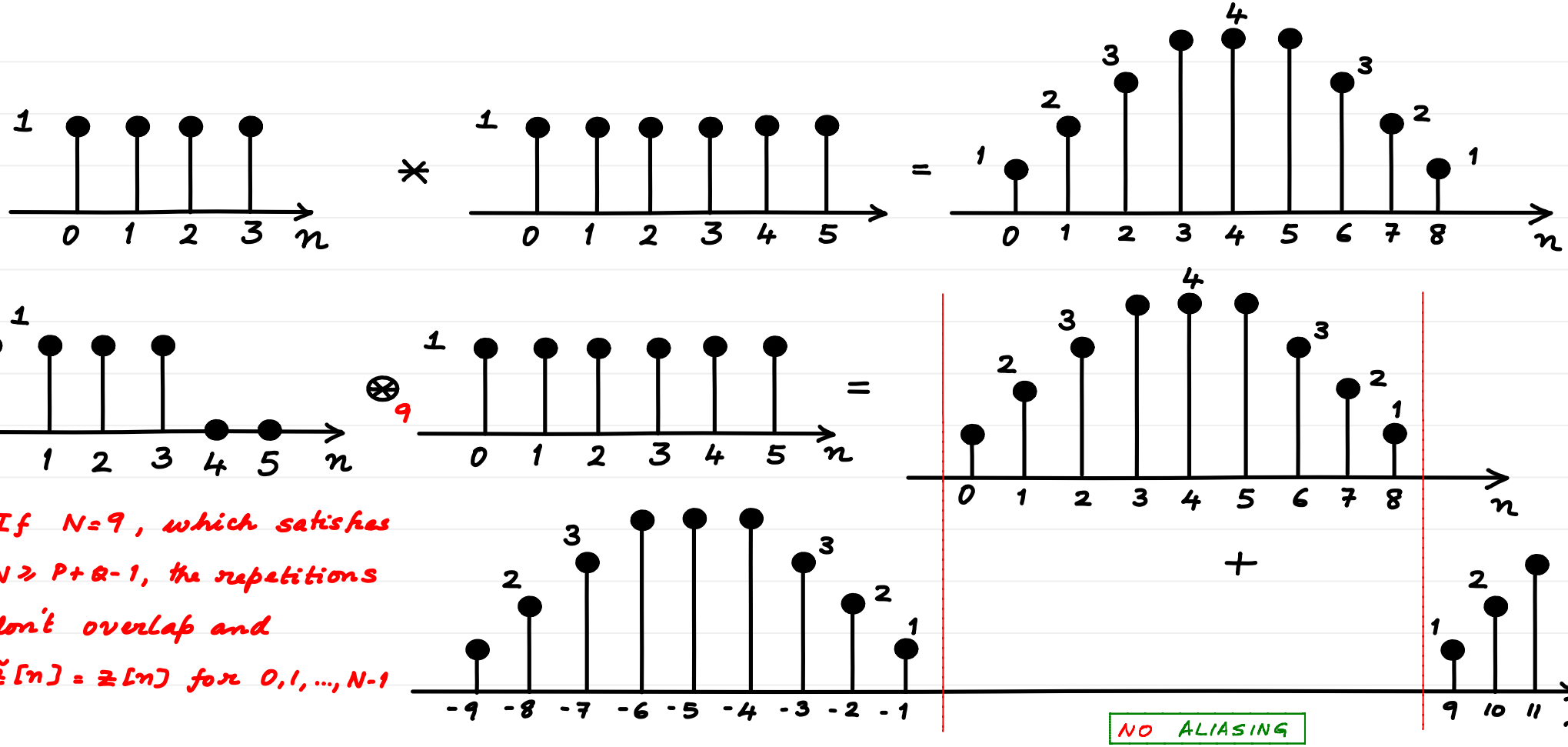
$$\tilde{z}[n] = z[n] \quad \text{for} \quad 0 \leq n \leq N-1 \qquad\qquad N \geq P+Q-1$$


1
0 1 2 3 n
*
1
0 1 2 3 4 5
=
1 2 3 4 3 2 1
0 1 2 3 4 5 6 7 8 n
1
0 1 2 3 4 5 n
⊛6
1
0 1 2 3 4 5 n
=
1 2 3 4 3 2 1
0 1 2 3 4 5 6 7 8 n
2 3 4
-6 -1
3 2 1
0 1 2
+
1 2 3
6 7 8 n
First shift to the LEFT
ALIASING
First shift to the RIGHT

Circular convolution result for $N = 6$.



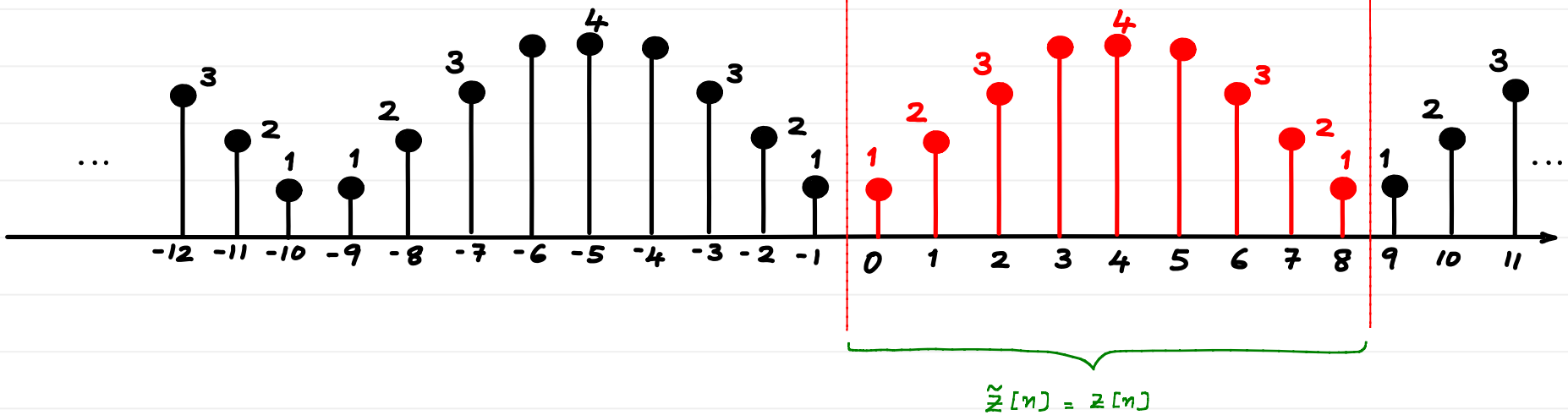
Now consider the result of circular convolution with $N = 9$. This satisfies the requirement of $N \geq P + Q - 1$, i.e. $N \geq 9$ for this example. In this case, $z[n] * p[n]$ causes the repetitions of $z[n]$ to not overlap in the interval $0 \leq n \leq N-1$. Hence there is no time-domain aliasing. Thus, $\tilde{z}[n] = z[n]$ for $0 \leq n \leq N-1$.



If $N=9$, which satisfies $N \geq P+Q-1$, the repetitions don't overlap and $\tilde{z}[n] = z[n]$ for $0, 1, \ldots, N-1$

NO ALIASING

=


3
2
1
1
2
3
4
3
2
1
1
2
3
4
3
2
1
1
2
3
...
...
-12 -11 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1
0 1 2 3 4 5 6 7 8 9 10 11
$\tilde{z}[n] = z[n]$